

SECTION 1: GENERAL PHYSICS

1.1: PHYSICAL QUANTITIES AND UNITS

Physical quantities

A physical quantity consists of a numerical magnitude and an appropriate unit.

They are two types of physical quantities

-Base quantities

-Derived quantities

A base quantity is a physical quantity that is independent of any other quantity for its measurement or derivation

A base unit is a unit of a base quantity that does not rely on other units for its derivation.

They are seven (fundamental) quantities

Base quantities and their units

The seven base quantities and their units.

Base quantity	Base unit	
	Name	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
electric current	ampere	A
Temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Derived quantities and their units

A derived quantity is one which is dependent on base quantities for its measurement and derivation.

They are products and/or quotient of base quantities.

Derived units are products and/or quotients of base units, depending on them for their derivation.

Worked Example 1.1

Express the following derived units as products or quotients of base units.

(a) newton (N);

(b) volt (V).

Solution

(a) The newton (N) is a unit of force.

Force is defined by the equation:

$$\text{force (N)} = \text{mass (kg)} \times \text{acceleration (ms}^{-2}\text{)}.$$

$$\text{Therefore, the newton (N)} = \text{kgms}^{-2}.$$

(b) The volt (V) is a unit of voltage.

Voltage is defined by the equation:

$$\text{voltage} = \frac{\text{power}}{\text{current}} = \frac{\text{work} \div \text{time}}{\text{current}} = \frac{\text{force} \times \text{displacement}}{\text{current} \times \text{time}}.$$

In terms of base units, the unit of:

force = kgms^{-2} ;

displacement = m ;

current = A ; and

time = s .

$$\text{The volt} = \frac{\text{kgms}^{-2} \times m}{A \times s}.$$

Therefore, the volt (V) = $\text{kgm}^2 \text{s}^{-3} \text{A}^{-1}$.

Homogeneity of equations

A physical equation is said to be homogeneous, when the base units on each term of its left hand side (LHS) are identical to the base units on each term of its right hand side (RHS).

Worked Example

Use base units to check the homogeneity of the following physical equations:

(a) $F = ma$;

(b) $v = u + at$;

where F is the resultant force, m is the mass, v is the final velocity, u is the initial velocity, a is the acceleration, and t is the time.

Solution

(a) LHS: The base unit of force, $F = \text{newton} = \text{kgms}^{-2}$.

RHS: The base unit of $ma = \text{kgms}^{-2}$.

The equation $F = ma$ is homogeneous since the base units on both sides are the same, i.e. kgms^{-2} .

(b) LHS: The base unit of $v = \text{ms}^{-1}$.

RHS: The base unit of $u = \text{ms}^{-1}$ and the unit of $at = \text{ms}^{-2} \times s = \text{ms}^{-1}$.

Therefore, the equation $v = u + at$ is homogeneous since the base units on the LHS side are matching with the base units on the RHS i.e. ms^{-1} .

Limitations of using base units

- (a) Constants cannot be checked
- (b) Missing terms cannot be checked
- (c) Coefficients cannot be checked

NB It is possible for an equation to be homogeneous and yet being incorrect

Prefixes and their symbols

Main prefixes and their symbols.

Prefix	Symbol	Multiplying factor
Tera	T	10^{12}
Giga	G	10^9
Mega	M	10^6
Kilo	k	10^3
Deca	da	10
Deci	d	10^{-1}
Centi	c	10^{-2}
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}

Scalar quantities and vector quantities

Scalar quantities are physical quantities with magnitude only.

For example speed, distance, temperature, energy, density, time, mass, volume, pressure and charge.

Vector quantities are physical quantities with both magnitude and direction. Examples of vectors include: force, acceleration, velocity, displacement, momentum, torque and electric current.

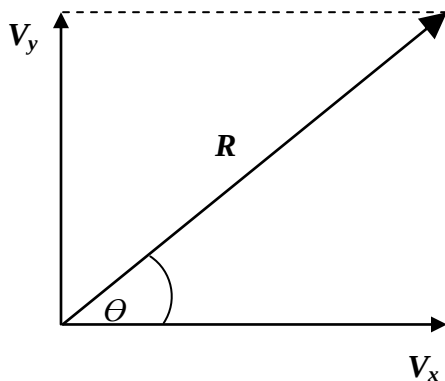
Case 2: Perpendicular vectors

When finding the resultant of two perpendicular vectors V_x and V_y , we draw the vectors as adjacent sides of a rectangle. The resultant is the diagonal of the rectangle, as illustrated in fig below

The resultant vector R can be found by:

(i) the calculation method; or

(ii) the scale drawing method.

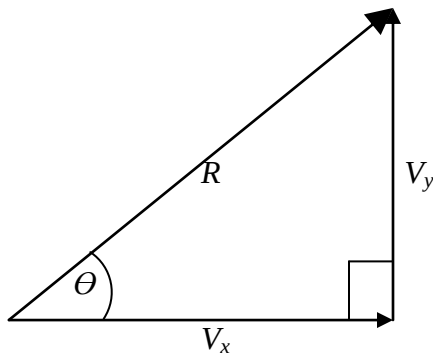


The resultant R of two perpendicular vectors V_x and V_y .

Note: In this module we shall look at the calculation method only. Work on the scale drawing method is left as research work for the student.

(i) The calculation method

The 'tail' of V_y is joined to the 'head' of V_x as illustrated in **Figure below**



The resultant R of two perpendicular vectors V_x and V_y after joining the tail of V_y onto the head of V_x .

Applying Pythagoras' theorem gives the magnitude of the resultant vector R .

$$R = \sqrt{V_x^2 + V_y^2}$$

To find the direction of the resultant, we use: $\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{V_y}{V_x}$

$$\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$$

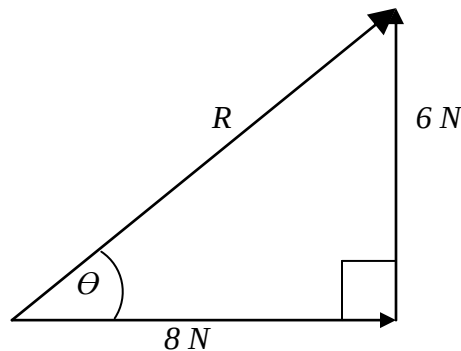
(ii) The scale drawing method

Both the magnitude and direction of the resultant can be found by using an accurate scale drawing.

Worked Example

A force of 6 N acts at 90° to a force of 8 N. Find the magnitude and direction of their resultant, R .

Solution



Applying Pythagoras' theorem gives the magnitude of the resultant vector R .

$$R = \sqrt{8^2 + 6^2} = 10 \text{ N.}$$

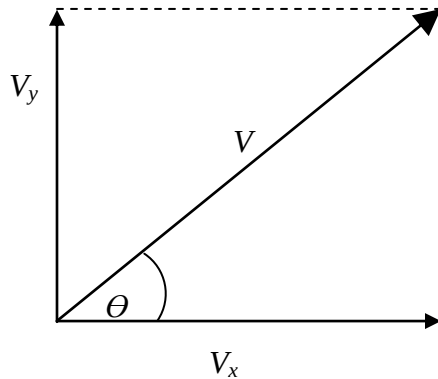
To find the direction of the resultant, we use: $\tan \theta = \frac{6}{8} = \frac{3}{4}$

$$\text{Therefore: } \theta = \tan^{-1}\left(\frac{3}{4}\right) = 37^\circ.$$

The resultant is therefore, a force of 10 N acting at 37° to the 8 N force and at 53° to the 6 N force.

Resolving a vector into two mutually perpendicular components

The component of a vector is the effective value of the vector along a particular direction. In case 2, we saw how to combine two perpendicular vectors so as to get a single resultant vector. Now we are looking at the reverse process which involves resolving a single vector into two components at right angles to each other (**Figure 1.4**).



Resolving a vector V into two perpendicular components.

To find the horizontal component V_x , we use: $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{V_x}{V}$

Therefore, $V_x = V \cos \theta$.

To find the vertical component V_y , we use: $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{V_y}{V}$

Therefore, $V_y = V \sin \theta$.

x

Resolving a displacement, s .

Resultant from sum of components

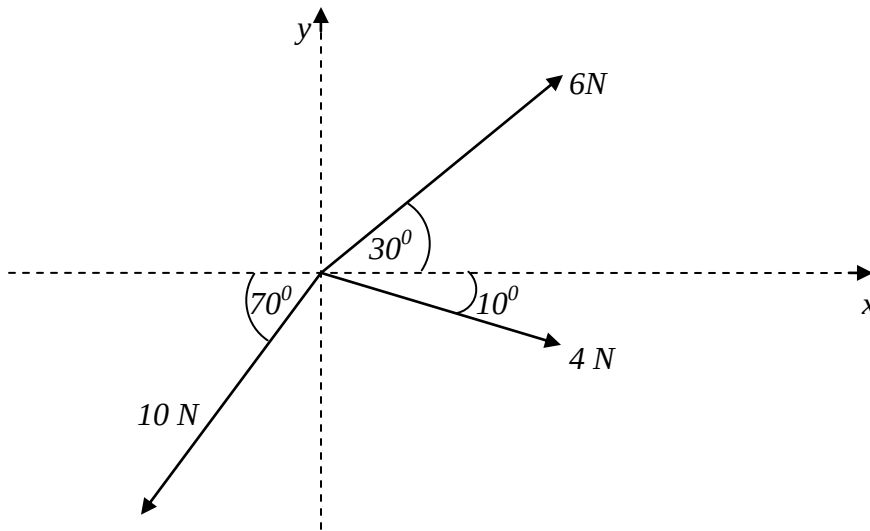
A number of vectors which all lie in one plane are said to be coplanar. When given a system of coplanar vectors, the resultant of the system can be found in two steps:

Step1: Choose two mutually perpendicular directions and find the components of all the vectors in these two directions.

Step 2: Find the algebraic sum of the components and then apply the method of finding the resultant of two perpendicular vectors (see worked example 1.6).

Worked Example

Find the resultant of three force vectors given in **Figure below**

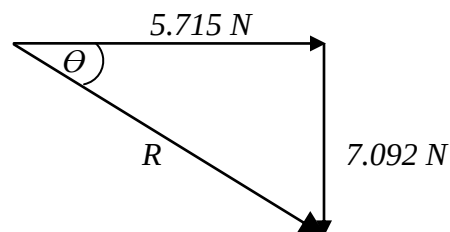


Three forces acting on a particle.

Solution

Components in the horizontal direction: $6\cos 30^\circ + 4\cos 10^\circ - 10\cos 70^\circ = 5.72 \text{ N}$.

Components in the vertical direction: $6\sin 30^\circ - 4\sin 10^\circ - 10\sin 70^\circ = -7.09 \text{ N}$.



Magnitude of $R = \sqrt{5.715^2 + 7.092^2} = 9 \text{ N}$.

Direction of $R = \theta = \tan^{-1}\left(\frac{7.092}{5.715}\right) = 51.1^\circ$ below the x-axis.

The resultant is therefore a 9 N force acting at 51.1° below the positive x-axis.

1.2: MEASUREMENTS TECHNIQUES

Errors and Uncertainties

- Whenever a measurement is taken there is a chance of making an error. An error may arise from the instrument and/ or experimental procedure used (called systematic error) or from the experimenter (called random error)

Systematic Errors

- Is one which results in all readings being faulty in one direction/ an error due to the instrument?
- Readings produce a consistent difference and are centred about a value other than the true value.
- One which affects all readings by same magnitude

Example of systematic errors.

1. Instrumental malfunction

The most common of these is error the zero; this is when an instrument has a reading even before a measurement has been taken. In other words the pointer of a balance is not on zero before the measurement is made. Other examples of this error are an oversensitive thermometer and a fast stopwatch.

2. Incorrect assumption

- For example assuming $\pi=4$ instead of $22/7$ or $g=10$ instead $g=9.81$ will give an incorrect answer when calculating.

3. Erroneous experimental procedure

- If we omit a step or rearrange a series of steps in an experiment, the result may be different from the expected.

4. Wrong method of assembly of apparatus

- The misplacement or omission of a piece of apparatus may give faulty measurements. For example fractional distillation without the fractionating column will not produce pure liquids.

NB Systematic errors are constant. They cannot be corrected by repeating the experiment. This means that the final result will miss the actual answer by the same value each time the experiment is carried out. They can only be resolved by

- *Recalibrating the instrument.*
- *Changing the instrument*
- *Taking error into consideration during final recordings and calculations (zero error)*
- *Change the method/procedure*
- *Plotting a graph of results*

RANDOM ERRORS

- *Result in a scatter of readings about a mean value caused by the observer.*
- *Measured value deviates about the true value.*
- *Arises when a repeated reading gives different values.*

Examples of random errors

- *Parallax error*
 - *This is the most common random error, it is introduced into a measurement when the experimenter takes readings at an angle to the meniscus or level of the reading rather than in line with it. This means that the reading taken at an angle will give a false recording.*
- *Fluctuations in external physical conditions*
 - *Changes in wind speed and direction, humidity, atmospheric pressure, temperature and light intensity may affect the results of an experiment*
- *Limitations experienced by the experimenter/observer*
 - *The student may experience a delayed reaction time, lack of sensitivity, colour blindness and even experimental approach.*

NB Random errors change constantly throughout the experiment and can be minimised by repeating the experiment and determining the average.

Uncertainty, precision and accuracy

As you have noticed by now, all measurements are estimated to the nearest. This is because regardless of how well a measurement is taken, a degree of uncertainty remains due to the instrument used. Eg measuring the thickness of a book using a metre rule would bring a large uncertainty into the reading. This is because the metre rule has a low precision. A student will be more comfortable measuring the thickness using a vernier or a micrometre screw gauge. The uncertainty of a measurement is simply a doubt that arises from a reading. The more precise an instrument, the smaller the uncertainty in the measurement

Precision is the measure of the size of the smallest measurement that can be made by an instrument.

- ❖ Precise readings are not scattered but rather are close to each other and this depends primarily on the instrument.

Measurements are considered accurate if their average (arithmetic mean) coincides with the true value; accuracy of a measurement principally lies with the experimenter and the method.

NB It is possible to have readings taken with great precision which are not accurate. This will happen if there is a systematic error.

It is also possible to have readings which are accurate but not very precise. This occurs if there is only small random error in the readings taken.

- An accurate experiment is one for which the systematic error is relatively small.
- A precise experiment has a small random error

Accuracy is when the mean of readings coincide with the true value

A precision is when readings are close to their mean.

Combining errors

1.SUM

Suppose the quantity g we require is related to quantities a and b , we have measured g by the equation $g = a + b$

Given that $a = (5.1 \pm 0.1)\text{cm}$ and $b = (3.2 \pm 0.1)\text{cm}$
 g with its uncertainty is given by $g = (8.3 \pm 0.2)\text{cm}$

2.DIFFERENCE

$$Q = a - b$$

Q with its uncertainty is given by $Q = (1.9 \pm 0.2)\text{cm}$

3.PRODUCT AND QUOTIENT

We add the fractional or percentage error

$$(i) \text{ If } Q = \frac{a}{b}$$

$$\frac{\Delta Q}{Q} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$$

$$(ii) \text{ If } P = a b$$

$$\frac{\Delta P}{P} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$$

$$(iii) \text{ If } P = a^{1/2} b^2$$

$$\frac{\Delta P}{P} = \frac{1}{2} \frac{\Delta a}{b} + 2 \frac{\Delta b}{b}$$

(iv) If $Q = ab^2/c^{1/2}$

$$\frac{\Delta Q}{Q} = \frac{\Delta a}{a} + 2 \frac{\Delta b}{b} + \frac{1}{2} \frac{\Delta c}{c}$$

Error rounding off

When performing calculations on errors (combination of errors) we could calculate to any number of decimal places or significant figures. How many are meaningful or significant?

The whole purpose of error/uncertainty calculation is to find out the accuracy or reliability of the mean value for the quantity measured so that we can then assume with reasonable certainty that the true value of the quantity is likely to lie somewhere in the range $x + p$ and $x - p$. Where x is the mean and p is the uncertainty

For example we found $p = 0.00090060\text{mm}$. If there is an error in the 4th decimal place, there is **no point** in quoting any more figures.

Thus the error/uncertainty is rounded off to the first non-zero digit. Thus the measured value is rounded off to the same number of decimal places as the error or uncertainty.

Examples

(a) $4.3682\ldots\text{m}$ and $p = \pm 0.06508\text{m}$

$(4.37 \pm 0.07)\text{m}$

(b) 44567.13cm and $p = 106,21\text{cm}$

$(4500 \pm 100)\text{cm}$

SECTION 2: NEWTONIAN MECHANICS

2.1 KINEMATICS

Rectilinear motion

Rectilinear motion refers to the motion moved in or formed from a straight line.

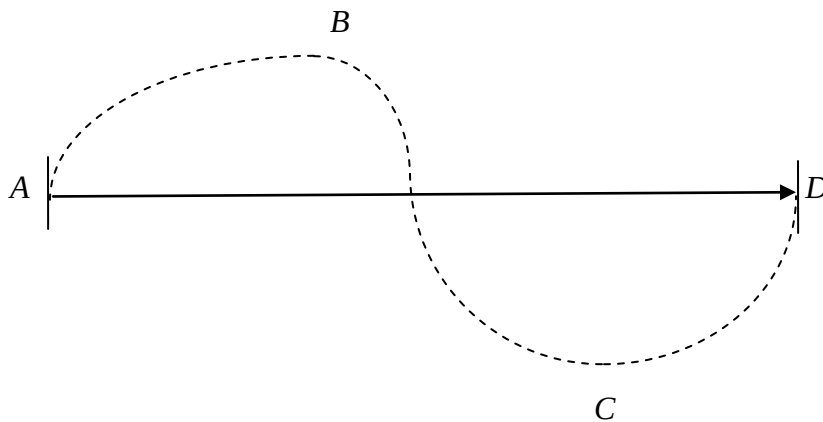
Distance and displacement

Distance is the length measured between any given reference points. It is a scalar quantity. The SI unit of distance is the metre (m).

Displacement is the distance moved in a specific direction. It is a vector quantity. The SI unit of displacement is the metre (m).

In **Figure 2.1**, Calvin moves from A to D along an irregular path ABCD. The distance moved is the length of the dotted line ABCD.

Calvin's displacement has a magnitude equal to the length of the straight line AD and has a direction along this line AD.



Distance and displacement.

Speed and velocity

Speed is the rate of change of distance.

$$\text{Average speed, } v = \frac{s}{t};$$

where s is the distance moved in metres and t is the time taken in seconds.

Speed is a scalar quantity and is measured in ms^{-1} .

Velocity is the rate of change of displacement.

$$\text{Average velocity, } v = \frac{s}{t};$$

where s is the total displacement in metres and t is the time taken in seconds.

Acceleration

Acceleration is the rate of change of velocity.

$$\text{Acceleration, } a = \frac{dv}{dt};$$

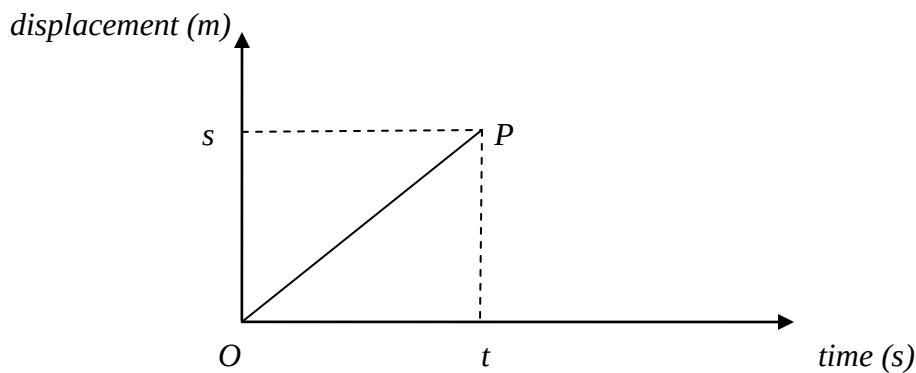
where dv is the change in velocity (ms^{-1}) and dt is the time taken (s).

Acceleration is a vector quantity and its SI unit is ms^{-2} .

Displacement-time graphs**(a) Straight line graphs (motion with uniform/constant velocity)**

Velocity from O to P = $\frac{ds}{dt}$, where ds is the total displacement and dt is the time taken

(Figure below)

**Linear displacement – time graph.**

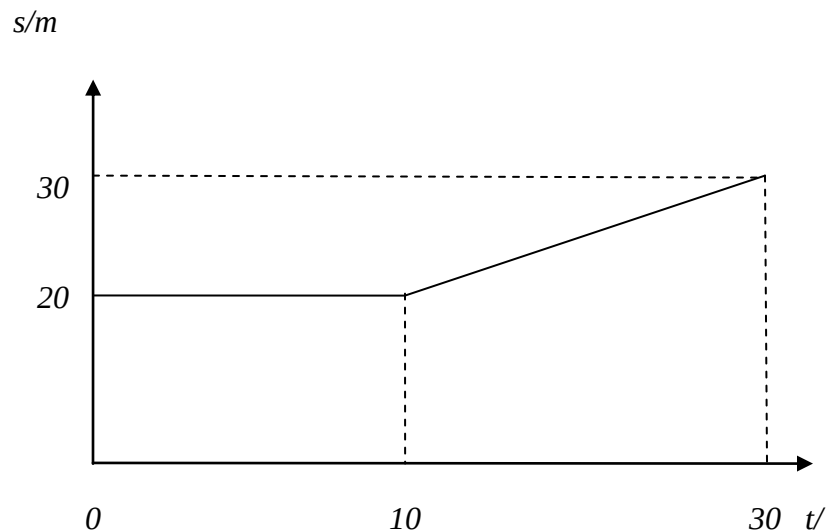
$\therefore \text{velocity} = \text{gradient of line OP.}$

Therefore, velocity is calculated from the gradient of a displacement-time graph. The steeper the gradient, the greater the velocity. In **Figure above**, the gradient is constant implying that the velocity is also constant.

In the case of distance-time graphs, the gradient would give speed.

Worked Example

Use **Figure below**, to find the velocity of the car during (a) the first 10 s and comment on your answer, and (b) the last 20 s.



Displacement-time graph.

Solution

(a) Velocity of the car during the first 10 s = gradient of the graph during the first 10 s.

$$\text{velocity, } v = \frac{20 - 20}{10 - 0}.$$

$$\therefore v = 0 \text{ ms}^{-1}.$$

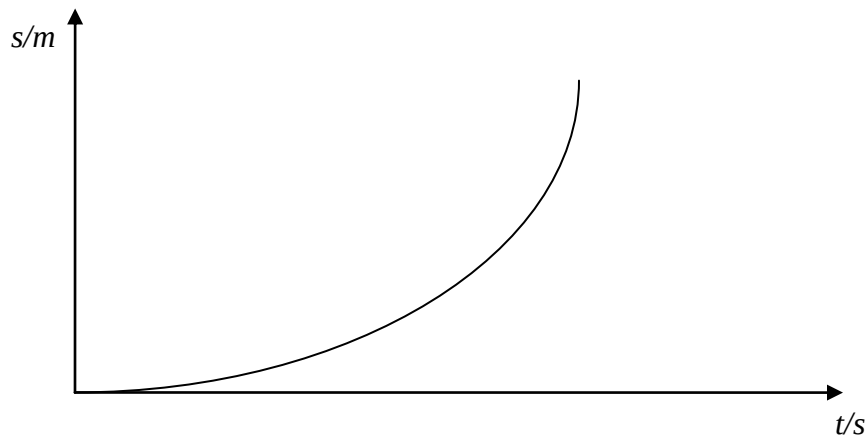
Therefore, the car is stationary. A horizontal line has a zero gradient, and hence a zero velocity.

(b) Velocity of the car during the last 20 s = gradient of the graph during the last 20 s.

$$\text{velocity, } v = \frac{30 - 20}{30 - 10}.$$
$$\therefore v = 0.5 \text{ ms}^{-1}.$$

(b) Curves (motion with non-uniform velocity)

In **Figure below(a)**, the gradient is not constant but is gradually increasing with time. This shows that the velocity is increasing with time, so there is an acceleration.



(a): Velocity is increasing with time.

Figure below (b) shows a gradient which is gradually decreasing with time. This implies that the velocity is decreasing with time, so there is a negative acceleration i.e. there is a deceleration/retardation.

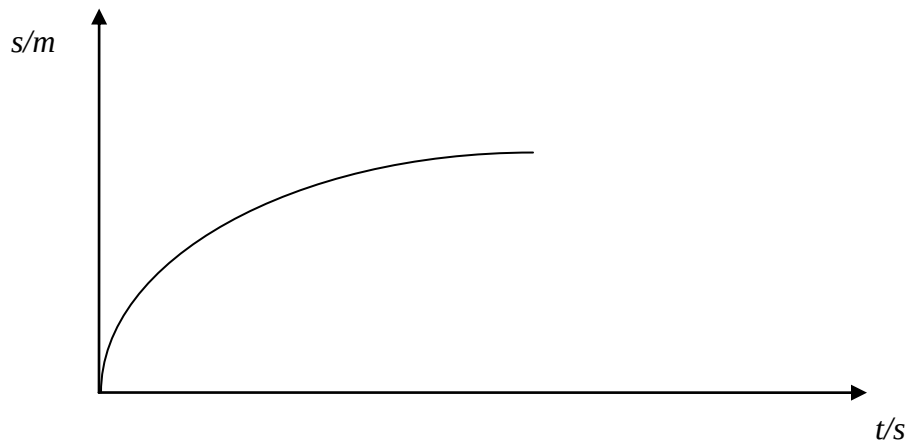


Figure 2.4 (b): Velocity is decreasing with time.

The actual (instantaneous) velocity at any point is equal to the gradient at that point. This is found by drawing a tangent to the curve at that point and calculating its gradient.

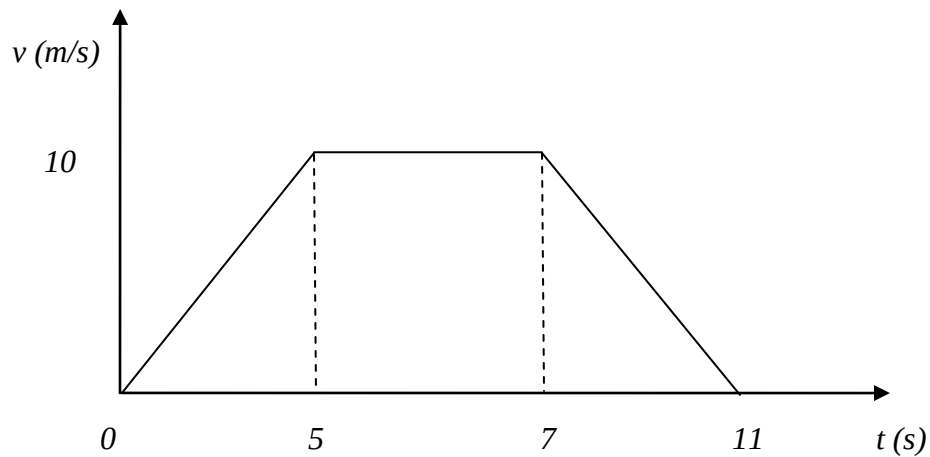
Velocity-time graphs

(a) Linear velocity-time graphs

Acceleration = gradient of a velocity-time graph.

Worked Example

Figure below shows the velocity-time graph of a trolley which starts from rest and reaches a velocity of 10 m/s in 5 s. It then maintains this velocity for a further 2 s until it is finally brought to rest in a further 4 s.



Velocity-time graph.

Find the acceleration of the trolley:

- (a) during the first 5 s;
- (b) between 5 s and 7 s;
- (c) during the last 4 s.

Give a brief comment to each of your answers to the above questions.

Solution

(a) acceleration = gradient of a velocity-time graph.

$$\text{acceleration} = \left(\frac{10 - 0}{5 - 0} \right).$$

$$\therefore \text{acceleration} = 2 \text{ ms}^{-2}.$$

The gradient is constant during the first 5 s indicating that the acceleration is also constant/uniform during the first 5 s.

(b) acceleration = gradient of a velocity-time graph

$$\text{acceleration} = \left(\frac{10 - 10}{7 - 5} \right).$$

$$\therefore \text{acceleration} = 0 \text{ ms}^{-2}.$$

A horizontal line between 5 s and 7 s implies that the velocity is constant and hence there is no acceleration during that time interval.

(c) *acceleration = gradient of a velocity-time graph*

$$\text{acceleration} = \left(\frac{0 - 10}{11 - 7} \right).$$

$$\therefore \text{acceleration} = -2.5 \text{ ms}^{-2}.$$

The trolley has an acceleration of -2.5 ms^{-2} . This is a deceleration of 2.5 ms^{-2} .

Note: A positive value for the gradient implies that there is an acceleration and a negative gradient value indicates that there is a deceleration/retardation.

Displacement = Area under a velocity-time graph.

Distance moved = Area under a speed-time graph.

In **Figure below (a)**, the gradient is not constant but is gradually increasing with time. This shows that the acceleration is increasing with time.

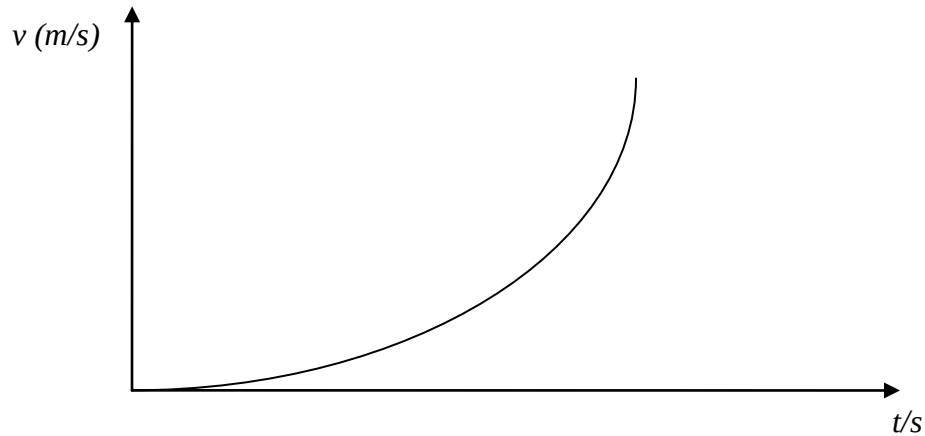


Figure above (a): Acceleration is increasing with time.

Figure below (b) shows a gradient which is gradually decreasing with time. This implies that the acceleration is decreasing with time

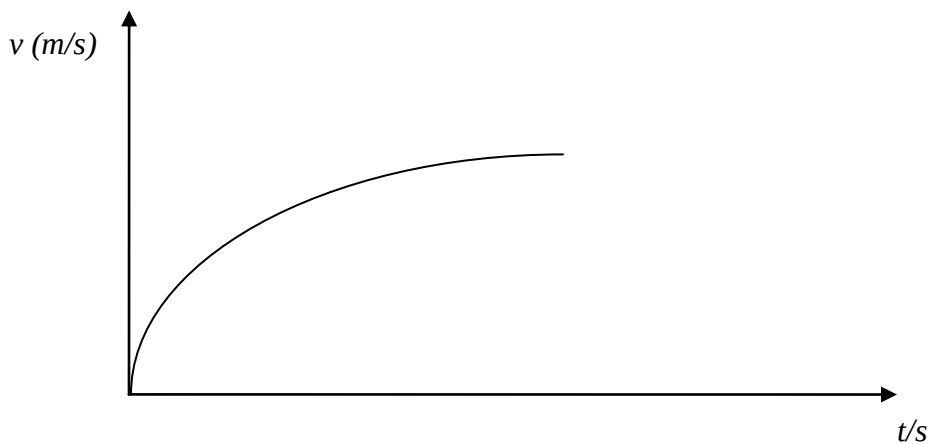


Fig 2.6 (b): Acceleration is decreasing with time.

Equations of uniformly accelerated linear motion

Four very important formulae can be used when a body travels with constant acceleration in a straight line.

$$v = u + at...(i)$$

$$s = \frac{1}{2}(v + u) t...(ii)$$

$$s = ut + \frac{1}{2}at^2...(iii)$$

$$v^2 = u^2 + 2as...(iv)$$

Where u is the initial velocity (m/s),

v is the final velocity (m/s),

a is the acceleration (m/s^2),

t is the time taken (s),

s is the displacement (m).

Derivations of the equations which represent uniformly accelerated motion in a straight line, from the definitions of velocity and acceleration.

Derivation of $v = u + at$.

Acceleration, $a = \frac{dv}{dt}$ where dv is the change in velocity (m/s) and dt is the time taken (s).

$a = \frac{(v - u)}{t}$, where v is the final velocity, u is the initial velocity and t is the time taken.

Rearranging yields: $v = u + at$.

Derivation of $s = \frac{1}{2}(v + u)t$.

$$\text{Average velocity} = \frac{1}{2}(\text{final velocity} + \text{initial velocity}) = \frac{1}{2}(v + u).$$

Also, average velocity $= \frac{s}{t}$ where s is the total displacement and t is the time taken.

$$\text{Equating the above two equations for average velocity yields: } \frac{s}{t} = \frac{1}{2}(v + u).$$

$$\text{Rearranging gives: } s = \frac{1}{2}(v + u)t.$$

Derivation of $s = ut + \frac{1}{2}at^2$.

$$\text{Average velocity} = \frac{1}{2}(\text{final velocity} + \text{initial velocity}) = \frac{1}{2}(v + u).$$

Also, average velocity $= \frac{s}{t}$ where s is the total displacement and t is the time taken.

$$\text{Equating the above two equations for average velocity yields: } \frac{s}{t} = \frac{1}{2}(v + u).$$

$$\text{Rearranging yields } s = \frac{1}{2}(v + u)t.$$

$$\text{From equation (i) } v = u + at, \text{ therefore substituting for } v \text{ yields: } s = \frac{(u + u + at)}{2}t.$$

$$\text{This simplifies to: } s = ut + \frac{1}{2}at^2.$$

Derivation of $v^2 = u^2 + 2as$.

$$\text{Average velocity} = \frac{1}{2}(\text{final velocity} + \text{initial velocity}) = \frac{1}{2}(v + u).$$

Also, average velocity = $\frac{s}{t}$ where s is the total displacement and t is the time taken.

Equating the above two equations for average velocity yields: $\frac{s}{t} = \frac{1}{2}(v + u).$

This simplifies to $s = \frac{1}{2}(v + u)t$, but from equation (i) $t = \frac{(v - u)}{a}$. Therefore substituting

$$\text{for } t \text{ yields: } s = \frac{(u + v)}{2} \times \frac{(v - u)}{a}.$$

This gives: $2as = v^2 - u^2$ and finally: $v^2 = u^2 + 2as$.

Worked Example 2.4

A trolley travelling downhill accelerates uniformly at 1.5 ms^{-2} . If its initial velocity at the top of the hill is 3 ms^{-1} , find: (a) how far it travels in 8 s, (b) how far it travels before reaching a velocity of 7 ms^{-1} .

Solution

(a) data

$$a = 1.5 \text{ ms}^{-2}$$

$$u = 3 \text{ ms}^{-1}$$

$$t = 8 \text{ s}$$

$$s = ?$$

$$s = ut + \frac{1}{2}at^2 = 3(8) + \frac{1}{2}(1.5)8^2 = 72 \text{ m}.$$

In 8 s, the trolley travels 72 m.

(b) data

$$a = 1.5 \text{ ms}^{-2}$$

$$u = 3 \text{ ms}^{-1}$$

$$v = 7 \text{ ms}^{-1}$$

$$s = ?$$

Using $v^2 = u^2 + 2as$ gives $7^2 = 3^2 + 2(1.5)s$.

$$49 - 9 = 3s \Rightarrow s = 13\frac{1}{3} \text{ m.}$$

Therefore, the distance travelled is $13\frac{1}{3} \text{ m}$.

Motion under gravity

Freefall

A body is said to be in freefall when the only force acting on it is due to the pull of gravity. This implies that if a body is in freefall, air resistance is negligible.

Galileo Galilei was the first person to demonstrate that all objects on or near the earth fall with the same acceleration which is known as the acceleration due to gravity (g). The acceleration due to gravity is generally taken as a constant equal to 9.81 ms^{-2} , although this value varies slightly from place to place.

Terminal velocity

Consider a body falling when the gravitational pull is opposed by air resistance.

The body will not accelerate at the value of 'g' because air resistance increases with speed. As the speed of the body increases, so does the air resistance until the air resistance balances out with the pull of gravity.

Once the air resistance becomes equal to the gravitational pull, there will be no net force acting on the body, so the body stops to accelerate. Thus, the body continues to fall at its maximum possible constant speed called its terminal velocity.

Projectile motion

A projectile is any particle which is given a non-zero initial velocity such that it moves in a parabolic path under the influence of gravity only.

Projectile launched at an angle to the horizontal

Consider the example of a projectile which is launched from ground level with an initial velocity u at an angle α to the horizontal.

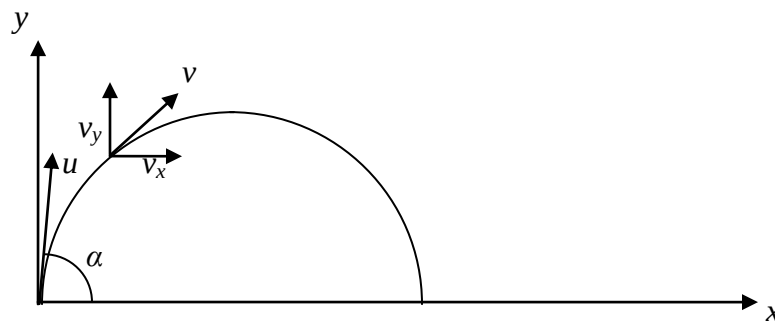


Figure 2.7: Projectile launched at an angle to the horizontal

(i) The initial horizontal velocity $= u \cos \alpha$.

(ii) The initial vertical velocity $= u \sin \alpha$.

Gravitational force is the only force acting on the projectile, so its vertical motion is under a constant acceleration g . However, the acceleration due to gravity has no component in the horizontal direction i.e. $a = 0 \text{ ms}^{-2}$ when considering motion in the horizontal direction. Consequently, the horizontal velocity of the projectile remains constant since there is no horizontal acceleration.

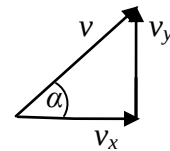
(iii) The velocity, v of the projectile at any time, t is found as follows:

Horizontal velocity at any instant $= v_x = u \cos \alpha = \text{constant}$.

Vertical velocity at any instant $= v_y = u \sin \alpha - gt$.

Therefore, the magnitude of the velocity at any given instant, $v = \sqrt{v_x^2 + v_y^2}$ and the

direction of the velocity at that time is given by: $\alpha = \tan^{-1} \left(\frac{v_y}{v_x} \right)$.



(iv) The position of the projectile at time, t seconds $= \sqrt{x^2 + y^2}$,

where $x = ut \cos \alpha$ and $y = ut \sin \alpha - \frac{1}{2}gt^2$.

(v) The time of flight is the time that the projectile is in the air, from the time it is launched to the time when it first hits the ground. It is found by calculating the time when the vertical displacement, y is zero i.e. the time $t > 0$ s, that the projectile is in the air until it reaches the ground.

$$y = ut \sin \alpha - \frac{1}{2}gt^2; \text{ from (iv).}$$

$$0 = t \left(u \sin \alpha - \frac{1}{2}gt \right)$$

$$\therefore t = 0 \text{ or } t = \frac{2u \sin \alpha}{g}.$$

The projectile is initially at ground level when $t = 0$, so the required time of flight is

$$\text{given by: } t = \frac{2u \sin \alpha}{g}.$$

(vi) The time, t taken by the projectile to reach the maximum height is found by calculating the time when the vertical component of the velocity of the projectile is zero i.e. $v_y = 0$.

$$v_y = u \sin \alpha - gt ; \text{ from (iii).}$$

$$0 = u \sin \alpha - gt \Rightarrow t = \frac{u \sin \alpha}{g}.$$

(vii) The maximum height is obtained by calculating the value of the vertical displacement, y when v_y is zero.

$$\text{Using } v_y^2 = u^2 \sin^2 \alpha - 2gy$$

$$0 = u^2 \sin^2 \alpha - 2gh_{\max} ; \text{ where } y = h_{\max} \text{ is the maximum height reached by the projectile.}$$

$$\therefore h_{\max} = \frac{u^2 \sin^2 \alpha}{2g}.$$

(viii) The range is the horizontal distance travelled by the projectile. It is found by calculating the value of x {from (iv)} with $t = \frac{2u \sin \alpha}{g}$, the time of flight.

$$x = ut \cos \alpha = \frac{2u^2 \sin \alpha \cos \alpha}{g} = \frac{u^2 \sin 2\alpha}{g}; \text{ where } \sin 2\alpha = 2 \sin \alpha \cos \alpha.$$

(ix) The path followed by a projectile is called the trajectory. The equation of the trajectory is found as follows:

Starting from (iv), the position of the projectile at time, t seconds is:

$$y = ut \sin \alpha - \frac{1}{2}gt^2 \dots (1)$$

$$x = ut \cos \alpha \Rightarrow t = \frac{x}{u \cos \alpha} \dots (2)$$

Substituting for t in equation (1) with its equivalence in equation (2) yields:

$$y = \frac{x(u \sin \alpha)}{u \cos \alpha} - \frac{1}{2}g \left(\frac{x}{u \cos \alpha} \right)^2 = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha}.$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha.$$

$$\therefore y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha); \text{ (Trajectory equation).}$$

Worked Example 2.6

A ball is kicked from a point on a horizontal ground so that it moves off with a velocity of 20 ms^{-1} at an angle of 30° above the horizontal. Assuming that the ball moves only under the influence of gravity, find:

(a) The time of flight of the ball;

(b) The range of the ball;

(c) The maximum height reached by the ball.

Solution

$$u = 20 \text{ ms}^{-1}; \alpha = 30^\circ \text{ and } g = 9.81 \text{ ms}^{-2}.$$

(a) The time of flight, $t = \frac{2u \sin \alpha}{g}$, from (v).

$$\therefore t = \frac{2(20) \sin 30^\circ}{9.81} = 2.04 \text{ s.}$$

(b) The range $x = ut \cos \alpha = \frac{u^2 \sin 2\alpha}{g} = \frac{20^2 \sin 60^\circ}{9.81} = 35.3 \text{ m.}$

(c) The maximum height $h_{\max} = \frac{u^2 \sin^2 \alpha}{2g} = \frac{20^2 \sin^2 30^\circ}{2(9.81)} = 5.10 \text{ m.}$

Horizontal projection

Consider the example of a particle which is projected horizontally from a height, h above the horizontal ground with an initial velocity u as shown in **Figure 2.8**

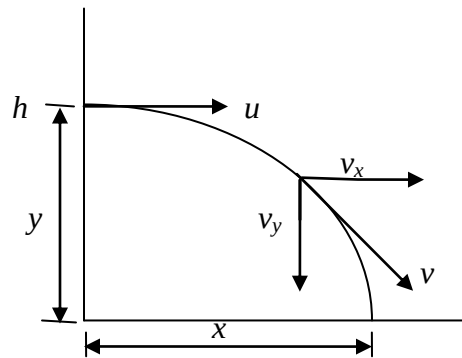


Figure 2.8: Horizontal projection

(i) The initial horizontal velocity = u .

(ii) The initial vertical velocity is zero i.e. $u_y = 0$, since the particle is given only an initial horizontal velocity.

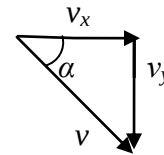
(iii) The velocity, v of the projectile at any time, t is found as follows:

Horizontal velocity at any instant = $v_x = u = \text{constant}$.

Vertical velocity at any instant = $v_y = gt$.

Therefore, the magnitude of the velocity at any given instant, $v = \sqrt{v_x^2 + v_y^2}$ and the

direction of the velocity at that time is given by: $\alpha = \tan^{-1}\left(\frac{v_y}{v_x}\right)$.



(iv) The position of the projectile at time, t seconds is given by $\sqrt{x^2 + y^2}$,

where $x = ut$ and $y = \frac{1}{2}gt^2$.

(v) The time of flight, t is found by calculating the time when the vertical displacement, y is equal to h .

$y = \frac{1}{2}gt^2$; from (iv) $\Rightarrow t = \sqrt{\frac{2h}{g}}$; where $y = h$.

(vi) The height, h is obtained by making h in (v) the subject of formula.

$t = \sqrt{\frac{2h}{g}} \Rightarrow h = \frac{1}{2}gt^2$.

(vii) The range is found by calculating the value of x with t being the time of flight.

$\therefore x = ut$.

(viii) The equation of the trajectory is found as follows:

Starting from (iv), the position of the projectile at time, t seconds is:

$y = \frac{1}{2}gt^2 \dots (1)$

$x = ut \Rightarrow t = \frac{x}{u} \dots (2)$

Substituting for t in equation (1) with its equivalence in equation (2) yields:

$$y = \frac{1}{2}g\left(\frac{x}{u}\right)^2 = \frac{gx^2}{2u^2}.$$

$$\therefore y = \frac{gx^2}{2u^2}; \text{ (Trajectory equation).}$$

Worked Example 2.7

A particle is projected horizontally at 40 ms^{-1} from a point 2 km above a horizontal surface as shown in **Figure 2.9**. Find the time taken by the particle to reach the surface and the horizontal distance travelled in that time. (Take $g = 10 \text{ ms}^{-2}$.)

Solution

$$2 \text{ km} = 2000 \text{ m.}$$

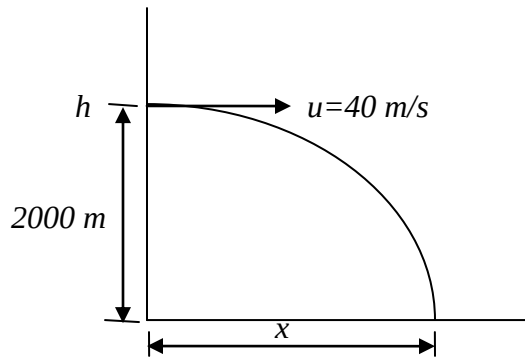


Figure 2.9: Horizontal projection

$$\text{The time taken by the particle to reach the surface, } t \text{ is: } t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(2000)}{10}} = 20 \text{ s.}$$

$$\text{The horizontal distance travelled is: } x = ut = 40 \times 20 = 800 \text{ m.}$$

Question 2.6

A stone is thrown over the edge of a cliff with a horizontal velocity of 25 ms^{-1} . The cliff is 400 m high. Ignoring air resistance, calculate:

- the time it takes the stone to reach the ground;
- the magnitude of the stone's velocity when it hits the ground.

2.2: DYNAMICS

Dynamics is the study of motion without neglecting what causes the motion.

Newton's laws of motion

Newton's first law

Everybody continues in its state of rest or of uniform velocity in a straight line unless it is compelled to change that state by some external force.

Newton's first law implies that all bodies have a built-in reluctance to change their state of rest or motion i.e. reluctance to start moving or to stop moving once it has started.

This reluctance to change the velocity is called inertia. Inertia depends on the mass of an object i.e. the greater the mass, the larger the inertia. Thus, the mass of a body is a measure of its inertia.

Newton's first law also provides the definition of force as something which changes the velocity of an object. Therefore, if a body moves with a constant velocity then the resultant force will be zero.

Newton's second law

The rate of change of momentum of a body is directly proportional to the resultant force acting on the body and takes place in the direction of that resultant force.

$F \propto$ rate of change of momentum

$$F \propto \frac{d}{dt}(mv) \Rightarrow F = k \frac{d}{dt}(mv).$$

Note: $\frac{d}{dt}(mv) =$ change in momentum/ time taken.

$$F = k \frac{(mv - mu)}{t} = km \left(\frac{v - u}{t} \right) = kma.$$

But 1 N is a force which is required to give a mass of 1 kg an acceleration of 1 ms^{-2} .

This implies that $k = 1$.

Therefore, $F = ma$;

where F = resultant force

m = mass

a = acceleration.

Newton's third law

To every action, there is an equal but opposite reaction.

This law implies that forces always occur in pairs. However, the forces do not cancel each other out because they act on different objects. For example if a car is accelerating forward, the car tyres will be pushing backward on the road and the reaction to this is a forward pushing force exerted by the road on the tyres.

Mass and weight

Mass is the amount or quantity of matter in an object. It is the property of a body which resists change in motion. The mass of a body is a measure of its resistance to acceleration i.e. mass is a measure of the inertia of a body. It is a scalar quantity. The mass of body is always constant and its S.I. unit is the kg.

The weight of a body is the force of gravity acting on its mass. Like any other forces, weight is a vector quantity and is measured in newtons. Mathematically, weight is given by:

$$W = mg ;$$

where W is the weight of the object,

m is the mass of the object,

g is the acceleration due to gravity.

Unlike the mass which is always constant, the weight of a body varies slightly from place to place.

Linear momentum and its conservation

Linear momentum

The linear momentum of a body is the product of its mass and velocity. Linear momentum is a vector quantity and is given by the equation:

$p = mv$, where p is the momentum;

m is the mass;

v is the velocity.

The units of momentum are kg ms^{-1} or alternatively Ns .

Impulse

From Newton's second law: $F = \frac{d}{dt}(mv)$.

$Fdt = d(mv) = \text{change in momentum} = \text{impulse}$.

The impulse of a force is the product of the force and the time for which it acts. This implies that the greater the force acting on an object and the longer it acts for, the larger the impulse of that force.

On a force time graph impulse, impulse is the area under the graph.

Worked Example 3.3

Calculate the impulse produced by a force of 250 N acting for 4 s.

Solution

$\text{Impulse} = \text{force} \times \text{time} = 250\text{N} \times 4\text{s} = 1000\text{Ns}$.

Therefore, the impulse = 1000Ns.

The principle of conservation of linear momentum

The total linear momentum of a system of interacting objects in any direction remains constant provided that no external forces act on the objects in that direction.

Consider two bodies X and Y that are involved in a collision as illustrated by **Figure 3.2**.



Figure 3.2: Two bodies X and Y, (i) before , and (ii) after collision.

Where:

m_1 and m_2 are the masses of two bodies X and Y respectively.

u_1 and u_2 are the initial velocities of two bodies X and Y respectively.

v_1 and v_2 are the final velocities of two bodies X and Y respectively.

During collision: Force on X, $F_x = \frac{m_1(v_1 - u_1)}{t}$ and Force on Y, $F_y = \frac{m_2(v_2 - u_2)}{t}$.

Newton's third law states that $F_x = -F_y$.

Therefore, $\frac{m_1(v_1 - u_1)}{t} = -\frac{m_2(v_2 - u_2)}{t}$

Rearranging gives: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

This is the law of conservation of momentum i.e. Total momentum before collision = Total momentum after collision (if there is no external resultant force acting on the system).

Worked Example 3.4

A bullet of mass 20 g is fired at 500 ms^{-1} into a block of wood of mass 290 g resting on a smooth surface (**Figure 3.3**). If the bullet remains embedded in the wood, calculate the velocity v , that the block moves off at.

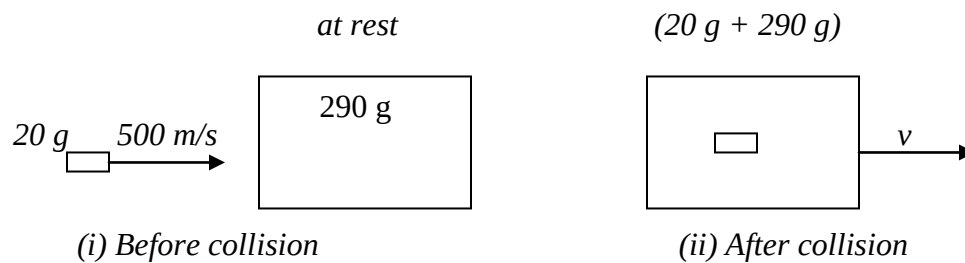


Figure 3.3: The bullet and block, (i) before , and (ii) after collision.

Solution

Taking velocities to the right as positive and applying the principle of conservation of momentum: Total momentum before collision = Total momentum after collision.

$$(0.020)(500) + 0 = (0.020 + 0.290)v \Rightarrow 10 = 0.31v \Rightarrow v = 32 \text{ ms}^{-1} \text{ to the right.}$$

Elastic and inelastic collisions

An elastic collision is a collision in which kinetic energy is conserved.

Total kinetic energy before collision = Total kinetic energy after collision. The relative speed of approach = the relative speed of separation.

An inelastic collision is a collision in which some or all of the kinetic energy is lost and is transferred to other forms usually internal/heat energy.

Total kinetic energy before collision > total kinetic energy after collision. The relative speed of approach > the relative speed of separation.

Worked Example 3.5

A car of mass 1000 kg moving at 50 ms^{-1} collides with a stationary bus of mass 2000 kg as shown in **Figure 3.4**.

- (a) Find the velocity, v of the bus after the collision, given that the car rebounds at 5.0 ms^{-1} . (b) Is this collision elastic or inelastic?

$$50 \text{ ms}^{-1} \qquad \text{at rest} \qquad 5.0 \text{ ms}^{-1} \qquad v$$

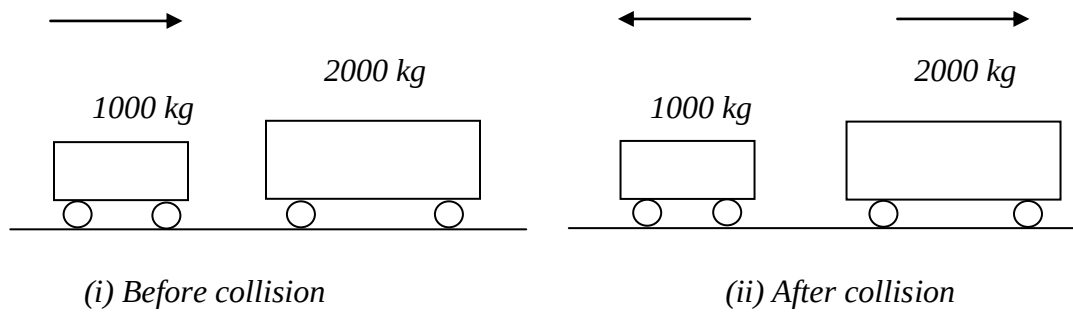


Figure 3.4: The bus and car, (i) before , and (ii) after collision.

Solution

(a) Taking velocities to the right as positive and applying the principle of conservation of momentum: Total momentum before collision = Total momentum after collision.

$$1000(50) + 0 = 1000(-5.0) + (2000)v \Rightarrow v = 27.5 \text{ ms}^{-1}.$$

$$(b) \text{ Total } E_k \text{ before collision} = \frac{1}{2}(1000)50^2 = 1.25 \times 10^6 \text{ J}.$$

$$\text{Total } E_k \text{ after collision} = \frac{1}{2}(1000)(5.0)^2 + \frac{1}{2}(2000)(27.5)^2 = 7.69 \times 10^5 \text{ J}.$$

Since the total E_k before collision $>$ total E_k after collision, the collision is inelastic.

RELATIVE VELOCITY RULE

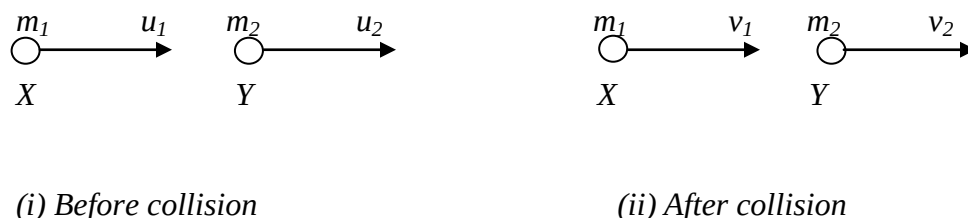
It states that in a perfectly elastic collision

$$(\text{Relative velocity before collision}) = (\text{relative velocity after collision})$$

Can be proved using the principle of conservation of momentum and energy.

Suppose bodies of masses m_1 and m_2 are moving in the same direction with velocities u_1 and u_2 before collision

After collision their velocities are v_1 and v_2 still to the right



By the principle of conservation of momentum $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

This simplifies to $m_1(u_1 - v_1) = m_2(v_2 - u_2)$ (1)

By the principle of conservation of energy

$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$. This equation can be re arranged to.

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2)$$

Applying difference of two squares.

$$m_1(u_1 + v_1)(u_1 - v_1) = m_2(v_2 + u_2)(v_2 - u_2)$$
(2)

Combining equation (1) and equation (2)

$$\text{Yields } u_1 - u_2 = -(v_1 - v_2)$$

That's the velocity of m_1 relative to m_2 before collision equals minus the velocity of m_1 relative to m_2 after collision.

For a perfectly elastic, the relative speed of approach is equal to the relative speed of separation.

Energy transfers in collisions

If a moving body has a perfectly elastic collision with a stationary body of the same mass, we can show that the moving body comes to rest and the stationary one travel on with the velocity of the moving one. That is there is a complete transfer of kinetic energy from the moving to the stationary body. Thus since $m_1 = m_2$ and $u_2 = 0$

$$\text{From } m_1(u_1 - v_1) = m_2(v_2 - u_2)$$

$$\therefore u_1 - v_1 = v_2$$

$$\text{Therefore } u_1 = v_1 + v_2 \text{(4)}$$

$$\text{using equation } u_1 - u_2 = -(v_1 - v_2) \text{ since } u_2 = 0$$

$$\text{yields } u_1 = v_2 - v_1 \text{(5)}$$

Adding equation (4) and (5)

$$2u_1 = 2v_2 \text{ this yields } u_1 = v_2$$

$$\text{From equation (4) } u_1 = v_1 + v_2$$

$$\therefore u_1 = v_2 \text{ .It follows that } v_1 = 0$$

The energy transfer is therefore a maximum when colliding masses are equal.

In the cases of a very small mass colliding elastically with a very large one at rest. It can similarly be proved that the small one rebounds with the same velocity while the large one remain at rest.

In the opposite case of a moving body of very large mass colliding elastically with a stationary body of very small mass, the greatest velocity of the small mass can have is twice that of the large mass. Velocity of large mass remains almost unchanged .

In both cases the energy transfer is negligible.

LIFT PROBLEMS

(a) Upwards acceleration.

T(tension) must be greater than weight W for the mass to accelerates upwards , so the resultant force is upwards.

$$T - W = ma$$

$$T - mg = ma$$

$$T = mg + ma$$

If the lift moves upwards with acceleration a its floor must push upwards on the body inside to give the same acceleration also. Hence new weight of the body in Newton

$$= m(g + a)$$

(b) Down wards acceleration

In this case T must be less than W as the resultant force must be downward.

$$W - T = ma$$

$$T = mg - ma$$

Passenger in a lift which comes to a halt while ascending or starts to descend experiences a brief decrease in the contact force between their feet and the floor.

(c) When lift is moving with a constant velocity. The acceleration is zero.

$$T - mg = 0$$

$$T = mg$$

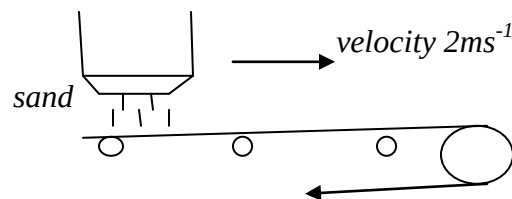
The body is in equilibrium under the action of these two forces and the reading on the spring balance is equal to the weight of the mass.

This condition applies if the lift is stationary or moving with a constant velocity.

Passengers in the lift experience only a steady contact force at the floor, equal to their own weight.

THE CONVEYOR BELT (Continuous flow of mass)

Flow rate
 5kg s^{-1}



Tensile force in belt

Sand falls at a rate of 5kg s^{-1} conveyors velocity $= 2\text{ms}^{-1}$. Calculate the horizontal force Needed to drive the belt

Solution

We use the rate of change of momentum. Consider what happens in unit time. That is

Mass deposited = (flow rate $\times$ time)

$$= 5\text{kg s}^{-1} \times 1\text{s}$$

$$= 5\text{kg}$$

Initial velocity of the sand in the horizontal direction is zero.

Change of momentum $= 5\text{kg} \times 2\text{ms}^{-1}$

$$= 10\text{kgms}^{-1}.$$

Rate of change of momentum = change of momentum

Time

$$= 10\text{kgms}^{-1}/1\text{s}$$

$$= 10\text{kgms}^{-2}$$

Resultant force = rate of change of momentum

$$= 10\text{N}.$$

2.3: FORCES

Introduction

In this unit, we will study forces. A force is generally a push or pull on an object. In the previous unit, we saw that Newton's first law of motion can be used to define a force as something which changes the velocity of an object. This unit's content consists of types of force, equilibrium of forces, centre of mass, centre of gravity and turning effects of forces.

Force is the rate of change of momentum.

Types of force

The three major types of force are: gravitational, electromagnetic and nuclear forces.

Gravitational, electromagnetic and nuclear forces

(a) Gravitational force is an attractive force that exists between all objects with mass. For example, if you drop a stone, it will fall towards the Earth because of the force of attraction between the stone and the Earth. The stone attracts the Earth with the same force that the Earth exerts on the stone but this attractive force has more effect on the stone because, the stone is much lighter than the Earth. As a result, the stone is pulled towards the Earth.

Planets are held in orbit around the Sun by the gravitational force. The Earth's gravitational pull is also responsible for keeping our feet on the ground and gives us weight.

(a) Electromagnetic force is responsible for the repulsion and attraction between negative and positive electric charges. It holds electrons to the nucleus. It also causes magnetic poles to attract and repel. Electromagnetic forces are also the binding forces of atoms and molecules.

(b) Nuclear forces are only experienced by sub-atomic particles and exist in two types namely: the strong nuclear force and the weak nuclear force.

The strong nuclear force holds protons together in a nucleus. (This explains why

the repulsion between positive protons in the nucleus does not force the nucleus apart). This is the strongest force of all, though it only acts over a very short range.

The weak nuclear force is responsible for radioactive decay of atoms

Upthrust / Bouyant force

Upthrust or buoyant force is the upward force resulting from an object being totally or partially immersed in a fluid. It arises from the pressure difference between different depths i.e. $p_2 > p_1$, so that there is a net upward force which is called an upthrust or a buoyant force (**Figure 4.1**).

Archimedes' principle states that, when a body is immersed in a fluid (totally or partially), it experiences an upthrust which is equal to the weight of the fluid displaced.

Suppose a cylinder of height h and area of cross-section A is at a distance h_1 below the surface of a fluid of density ρ as shown in **Figure 4.1**.

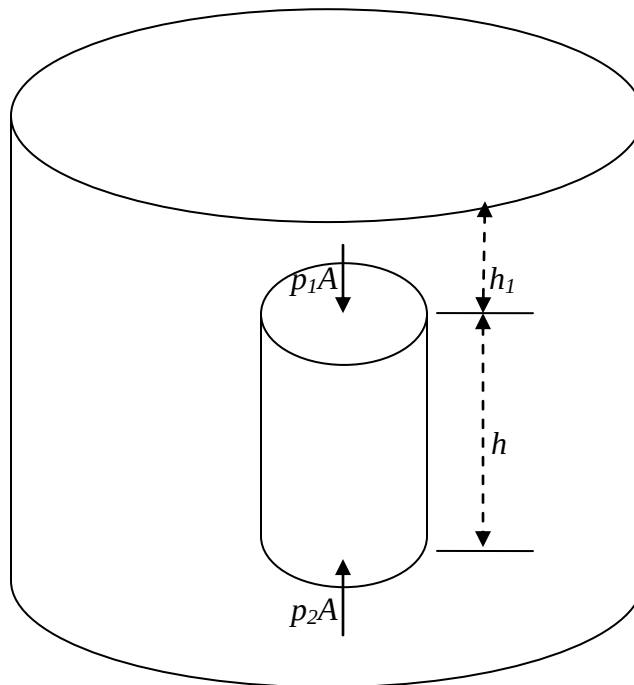


Figure 4.1: An upthrust (a buoyant) force i.e. $p_2 A - p_1 A$.

Volume of fluid displaced, V = volume of cylinder = Ah .

Mass of fluid displaced = density $\times$ volume of fluid displaced = ρAh .

Weight of fluid displaced = mass of fluid displaced $\times$ acceleration due to gravity
 $= \rho Ahg$,

but $V = Ah$. Therefore the weight of fluid displaced = ρVg ...(i)

On the other hand, the fluid exerts forces p_1A and p_2A on the top and bottom faces of the cylinder.

The resultant upward force due to the fluid = upthrust = $p_2A - p_1A$
 $= (h + h_1)\rho Ag - h_1\rho Ag$
 $= h\rho Ag$,

but $V = Ah$. Therefore the weight of fluid displaced = ρVg ...(ii)

By comparison of (i) and (ii) it is clearly visible that: upthrust = weight of fluid displaced. (This is Archimedes' principle).

Worked example 4.1

What is the buoyant force (upthrust) on a spherical balloon of diameter 160 cm in air of density $\rho_{\text{air}} = 1.20 \text{ kgm}^{-3}$.

Solution

$$\text{Upthrust} = \rho Vg = \rho \left(\frac{4}{3} \pi r^3 \right) g = 1.20 \left(\frac{4}{3} \pi (0.8)^3 \right) 9.81 = 25 \text{ N}.$$

Frictional force

Friction is force which opposes motion between two contact surfaces. It can cause the surfaces to heat up and eventually wear away.

Drag force

Both liquids and gases are fluids. The frictional force within a fluid is known as the drag force. Drag force tries to prevent the motion between objects and fluids. It depends on the viscosity of the fluid. Viscosity is a measure of how easily the fluid flows. For example, thick and sticky fluids are very viscous. Air resistance is a common example of drag force.

Viscosity

Friction is not only restricted to solids sliding over one another. Friction also occurs in fluids such as liquids and gases. Friction in fluids is known as viscosity. When an object moves through a fluid, a viscous drag opposes the motion of the object. The viscous drag depends on:

(a) The shape of the object

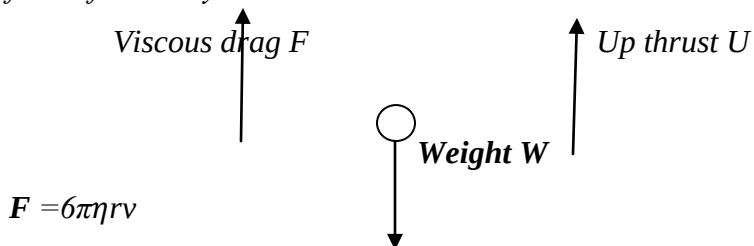
A streamlined object experiences less viscous drag.

(b) The velocity of the object

The viscous drag increases as the velocity increases.

(c) The viscosity of the fluid

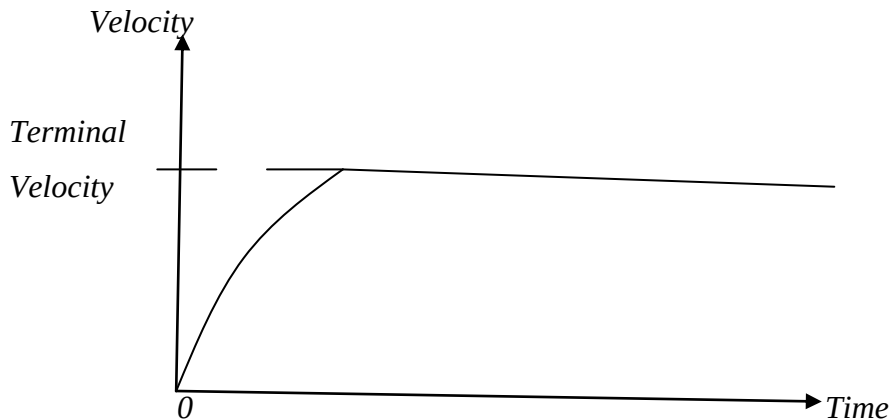
Stokes's law gives the viscous drag F on a sphere of radius r moving with a speed v in a fluid of viscosity η as



When a sphere falls from rest in a fluid, the forces acting on it are.

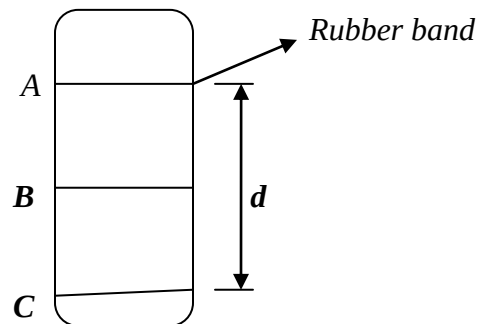
- (i) Its weight, W
- (ii) Viscous drag, F
- (iii) Up thrust, U

Initially, $U > F + U$ and the sphere accelerates downwards. As its velocity increases the viscous drag $F = 6\pi\eta rv$ increases until the terminal velocity is reached. When the terminal velocity is reached, $W = F + U$



Experiment based on the measurement of terminal velocity to determine the viscosity of a fluid

The viscosity of a liquid, such as glycerol, may be determined by measuring the terminal velocity of metal spheres as they fall through the liquid.



The glycerol is contained in a vertical tube about 50mm in diameter and about 0.5m tall. Steel spheres of diameter about 2mm. The density ρ_1 of the glycerol, and the density ρ of the material of the spheres, are first found, and the diameter $2r$ of the spheres is measured using a micrometer screw gauge. Three horizontal reference marks A, B and C are made on the tube (rubber bands stretched round the tube are suitable). Reference mark A should be several centimetres below the surface of the liquid, and the distances AB and BC should be equal. Measure the distance d between the marks at A and C. A steel sphere is dropped carefully from rest down the axis of the tube. The times t_1 and t_2 for it to cover the distances AB and BC are measured by observing by eye and timing with a stopwatch, or by using light gates and a timing unit. If t_1 and t_2 are equal, or very nearly so, then the terminal velocity has been attained. If they are unequal, adjust the

reference marks to the position further below the surface and repeat the check observation. A number of spheres should be timed over the known distance to obtain an average value for the terminal velocity. The terminal velocity v_T is given by $\frac{d}{(t_1+t_2)}$. The viscosity is obtained from $\eta = \frac{2r^2g(\rho-\rho_f)}{9v_T}$

NB It is important that the falling sphere should not carry with it a bubble of air: this may happen if the sphere is dropped into the glycerol from a height above the surface, so that it enters with a splash. To avoid this, moisten the sphere slightly with glycerol and release it carefully, close to the surface of the liquid. It is also important that the sphere should fall close to the axis of the cylinder, so that it is not too close to the wall of the cylinder. The constant 6π in Stokes's law equation applies for infinite liquid, rather than to one confined in a tube, but for spheres of diameter 2mm or less in a tube of diameter about 50mm, the error involved is not serious.

Force diagrams

Particle on a plane horizontal plane

Consider a particle of mass m kg, resting on a smooth horizontal plane. Since the particle does not fall, its weight mg is being balanced by an upward force called the normal reaction R , which acts perpendicular to the plane as shown in **Figure 4.2**.

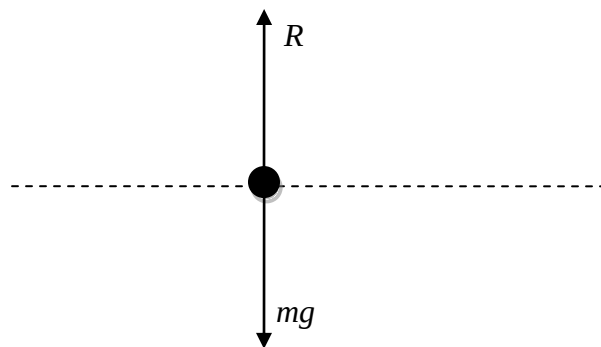


Figure 4.2: A particle of mass m kg, resting on a smooth horizontal plane.

Particle on a rough inclined plane

Now suppose the particle is at rest on a rough plane, inclined at θ to the horizontal. The frictional force will act up the slope.

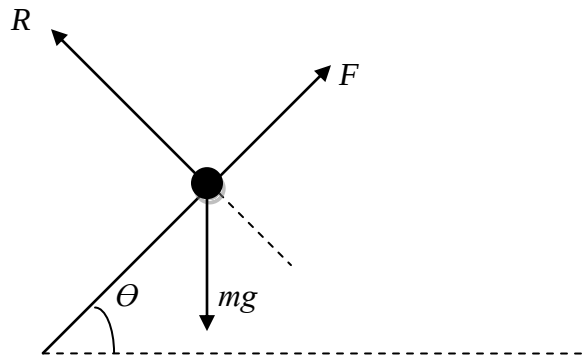


Figure 4.3: A particle at rest on a rough inclined plane.

Equilibrium

Conditions for equilibrium

A particle is said to be in equilibrium when:

1. There is no resultant force acting in any direction; and
2. There is no net turning effect i.e. no resultant torque about any point.

When a particle is in equilibrium, it will either remain at rest (static equilibrium) or move with a constant velocity (dynamic equilibrium).

Three forces in equilibrium

Lami's theorem

If a body is in equilibrium under the action of three coplanar forces P ; Q and R (**Figure 4.5 (a)**), then the forces can be represented in magnitude and direction by the sides of a triangle taken in order. This is known as the triangle of forces **Figure 4.5 (b)**.

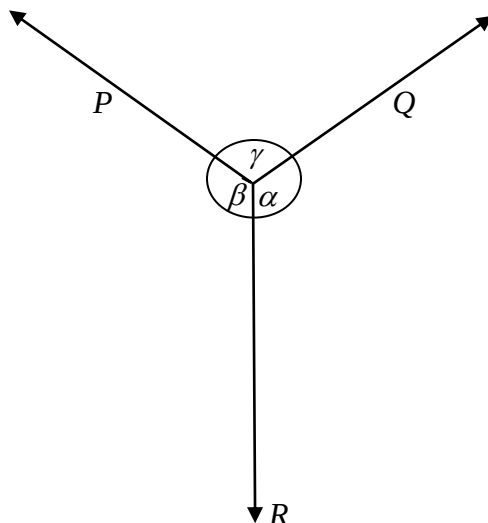


Figure 4.5 (a): Three forces in equilibrium.

Since the forces are in equilibrium, they form a triangle shown in **Figure 4.5 (b)**

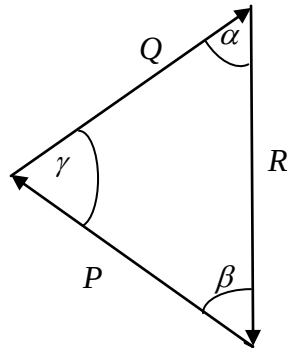


Figure 4.5 (b): The triangle of forces i.e. vector triangle.

Applying the sine rule:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}.$$

Worked Example 4.

A particle of mass 40 kg hangs in equilibrium suspended by two light inextensible strings, as shown in **Figure 4.6**. Find the tensions in the strings. (Use $g = 10 \text{ ms}^{-2}$)

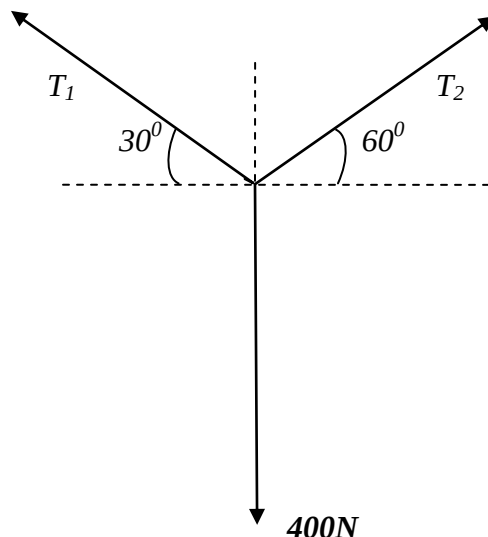
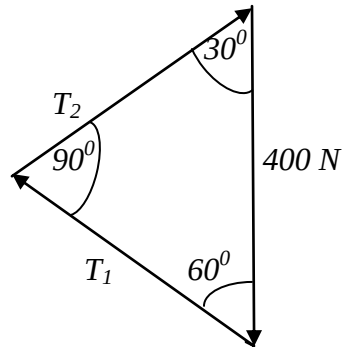


Figure 4.6: Three forces in equilibrium.**Solution**

Since the forces are in equilibrium, they form a vector triangle shown below.



Applying the sine rule:

$$\frac{T_1}{\sin 30^\circ} = \frac{T_2}{\sin 60^\circ} = \frac{400}{\sin 90^\circ} = 400.$$

$$T_1 = 400 \sin 30^\circ = 200 \text{ N. and } T_2 = 400 \sin 60^\circ = 346 \text{ N.}$$

Figure 4.7: Three forces in equilibrium.**More than 3 forces in equilibrium****Worked Example 4.4**

Find the values of F_1 and F_2 for which the following set of forces shown in **Figure 4.8**, is in equilibrium.

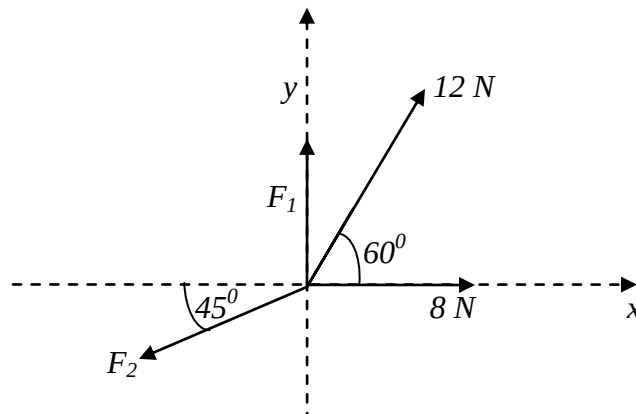


Figure 4.8: Four forces in equilibrium

Solution

Resolving parallel to the x-axis:

$$8 + 12 \cos 60^\circ = F_2 \cos 45^\circ$$

$$8 + 6 = \frac{1}{\sqrt{2}} F_2 \Rightarrow F_2 = 14\sqrt{2} \text{ N.}$$

Resolving parallel to the y-axis:

$$F_1 + 12 \cos 30^\circ = F_2 \cos 45^\circ$$

$$\text{Substituting for } F_2: F_1 + 12 \frac{\sqrt{3}}{2} = 14\sqrt{2} \times \frac{1}{\sqrt{2}} \Rightarrow F_1 = 2(7 - 3\sqrt{3}) \text{ N.}$$

Figure 4.9: Four forces in equilibrium

Turning effects of forces

Moments

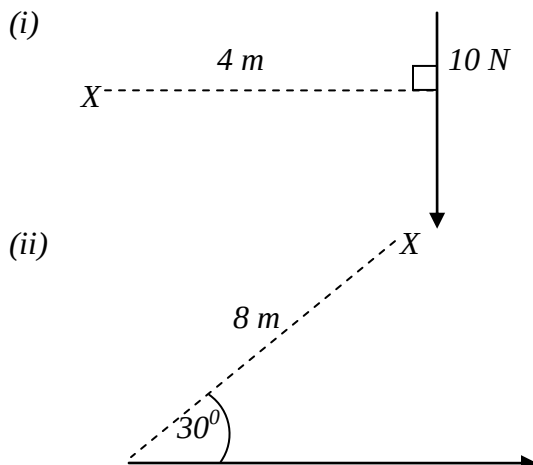
The moment of a force is the turning effect of that force.

Moment of force = force $\times$ perpendicular distance from the line of action of the force to that point

Moments are measured in Nm. A turning effect in the clockwise direction is called a clockwise moment and it is defined as negative. A turning effect in the anti-clockwise direction is called an anti-clockwise moment and it is defined as positive.

Worked Example 4.5

Find the moments of each of the given forces about X



15 N

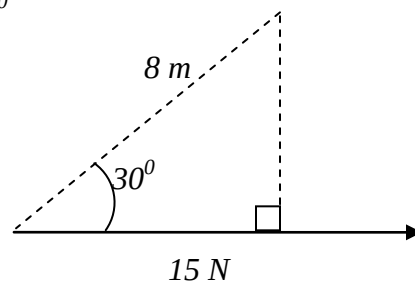
Figure 4.10: Moments of force.**Solution**

(i) Moment about X = force $\times$ perpendicular distance from the line of action of the force to that point.

$$= 10 \text{ N} \times 4 \text{ m} = 40 \text{ Nm in the clockwise direction.}$$

Since the force produces a clockwise moment, the answer could be written as -40 Nm , where the negative sign tells us that the direction of the turning effect produced is clockwise.

(ii) In this case, the perpendicular distance is found by resolving the 8 m distance vertically i.e. $8 \sin 30^\circ$



Moment about X = force $\times$ perpendicular distance from the line of action of the force to that point.

$$= 15 \times 8 \sin 30^\circ = 60 \text{ Nm in the anti-clockwise direction.}$$

This is an anti-clockwise moment, so the answer could be written as 60 Nm , where the positive sign tells us that the direction of the turning effect produced is anti-clockwise.

Couples

A couple is said to exist when 2 anti-parallel forces whose lines of action do not coincide act on a body. A couple has no resultant force. It only produces a turning effect i.e. A couple produces rotation only and no translation. For example, in **Figure 4.11** the forces in the driver's hands applied to a steering wheel can provide a couple to turn the wheel.

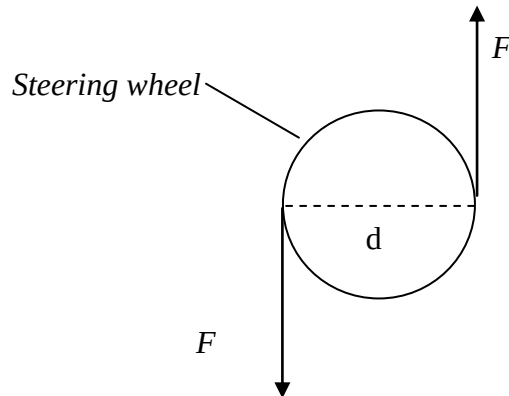


Figure 4.11 : Torque due a couple of forces in the driver's hands applied to a steering wheel.

The torque of a couple = magnitude of one force $\times$ perpendicular distance between the 2 forces forming the couple.

$$= Fd.$$

Self Assessment Question 4.5

Calculate the torque due to a couple of 50 N forces applied to a steering wheel (30 cm in diameter) by a driver's hands (see **Figure 4.12**).

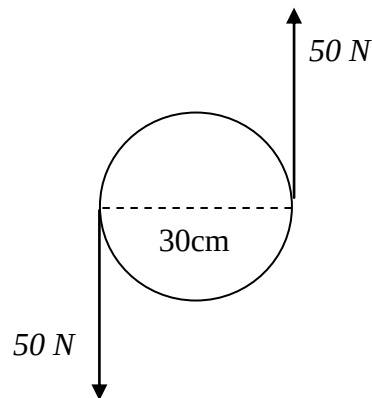


Figure 4.12 : Torque due a couple of forces in the driver's hands applied to a steering wheel.

Principle of moments

When an object is in equilibrium: The sum of clockwise moments about any point is equal to the sum of anti-clockwise moments about the same point.

Worked Example 4.6

Find the value of the weight W , for which the seesaw drawn in **Figure 4.13** is balanced.

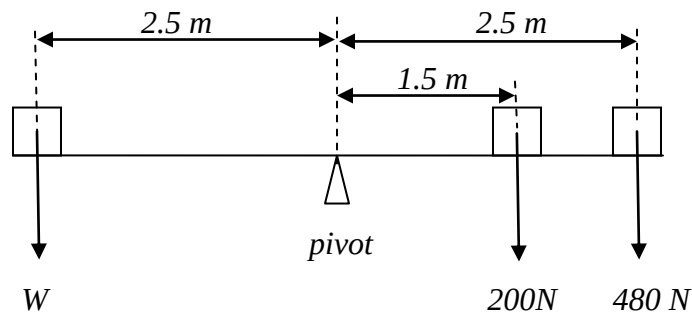


Figure 4.13 : A balanced seesaw.

Solution

When the seesaw is balanced i.e. in equilibrium:

The sum of clockwise moments about the pivot = the sum of anti-clockwise moments about the pivot.

$$W \times 2.5\text{ m} = (200\text{ N} \times 1.5\text{ m}) + (480\text{ N} \times 2.5\text{ m}) \Rightarrow W = 600\text{ N}.$$

Centre of gravity and centre of mass

The centre of gravity of an object is the single point at which its entire weight can be considered to act.

The centre of mass of an object is the single point at which its entire mass can be considered to act.

In places where the gravitational field strength is uniform, the centre of mass and the centre of gravity are at the same point. This is true for all objects near the earth's surface.

The centre of mass of a suspended lamina or body

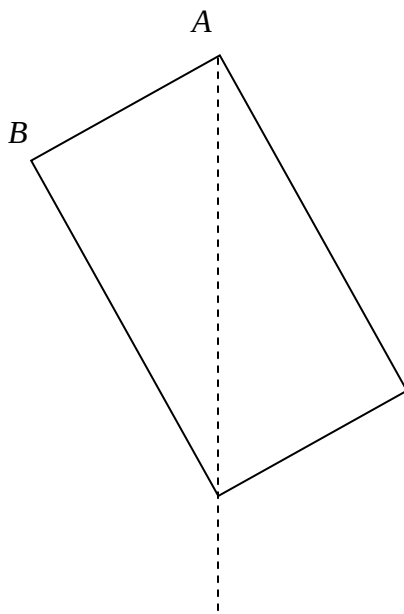
A suspended lamina or body, hangs with its centre of mass directly below the point of suspension.

For example (see **Figure 4.14**):

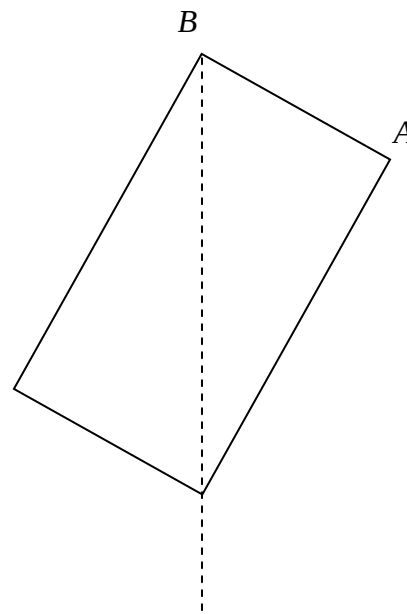
(i) If you suspend a rectangular disc from one of its corners i.e. corner A, it will hang with one of its diagonals vertical because the centre of mass is at a point on this diagonal.

(ii) If you then suspend the disc from a different corner B, again it hangs with the second diagonal vertical since the centre of mass is again at a point on the diagonal.

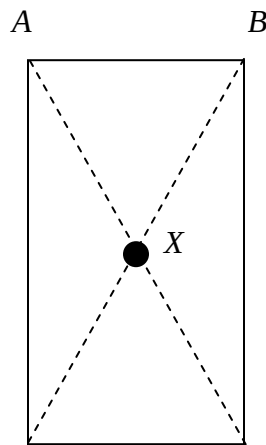
Actually, the centre of mass of the rectangular disc is at X, the point of intersection of the two diagonals as shown in case (iii).



(i)



(ii)



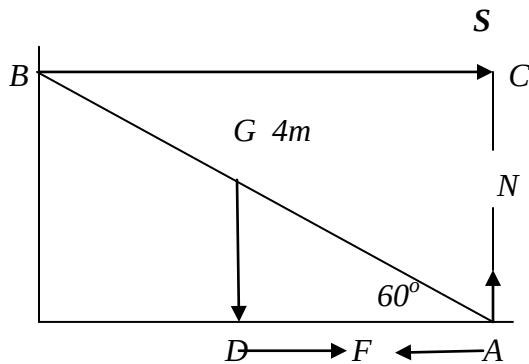
(iii)

Figure 4.14: The centre of mass of a suspended a rectangular disc.

Worked example

A uniform ladder 4.0m long of mass 25kg rests with its upper end against a smooth vertical wall and its lower end on a rough ground. What must be the least coefficient of friction between the ground and the ladder for it to be inclined at 60° to the horizontal without slipping $g=10\text{ms}^{-2}$

Solution



The weight (mg) of the ladder $=mg=25 \times 10= 250\text{N}$.

The wall is smooth and so $\mu=0$ Therefore the force S of the wall on the ladder is normal to the wall. As the ladder is uniform its weight acts at the midpoint G . Force exerted on it by the ground can be resolved into a normal force N and a limiting Frictional force $F = \mu N$.

μ is the required coefficient.

Resolving vertically $N=W=250\text{N}$

Resolving horizontally $F = \mu N=S$

Taking moments about A

$$S \times AC = W \times AD$$

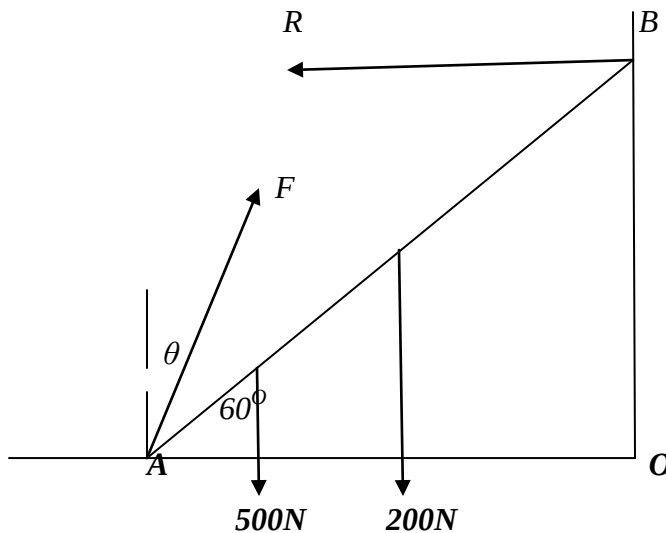
$$S \times 4 \cos 30^\circ = 250 \times 2 \sin 30^\circ$$

$$\mu = 0.29$$

Worked Example

A ladder of length 10m and weight 200N leans against a smooth wall such that it is at an angle of 60° to the horizontal. A boy of weight 500N stands on the ladder $\frac{1}{4}$ of the way from its lower end. Calculate the normal reaction at the wall and the magnitude and direction of the resultant force acting on the lower end of the ladder.

SOLUTION



Since the wall is smooth, the reaction R is normal to the wall. The forces acting on the lower end A of the ladder consists of the normal reaction vertically upwards and the friction in the direction AO . Therefore, the resultant force F at A is in the direction as shown. Since the ladder is in equilibrium, the algebraic sum of the moments of forces about any axis is zero.

Taking moments about A (so that the force F does not appear in the equation),

$$R(10 \sin 60^\circ) - 200(5 \cos 60^\circ) - 500(2.5 \cos 60^\circ) = 0$$

$$R = 130\text{N}$$

Also, the algebraic sum of the components of forces along any direction is zero.

Therefore, if θ is equal to the angle F makes with the vertical, resolving forces vertically,

$$F \cos \theta - 500 - 200 = 0$$

$$F \cos \theta = 700 \dots\dots\dots(1)$$

Resolving forces horizontally,

$$F \sin \theta - R = 0$$

$$F \sin \theta = R = 130 \dots\dots\dots(2)$$

$$(1)^2 + (2)^2 : F^2 \cos^2 \theta + F^2 \sin^2 \theta = 700^2 + 130^2 \quad \therefore [\cos^2 \theta + \sin^2 \theta] = 1$$

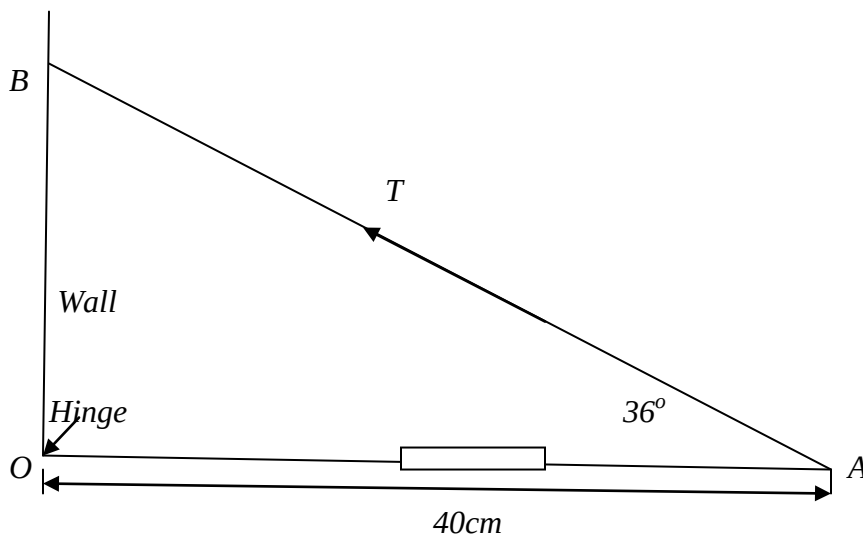
$$F = 712$$

$$(2)/(1) : \tan \theta = \frac{130}{700}$$

10° from the vertical

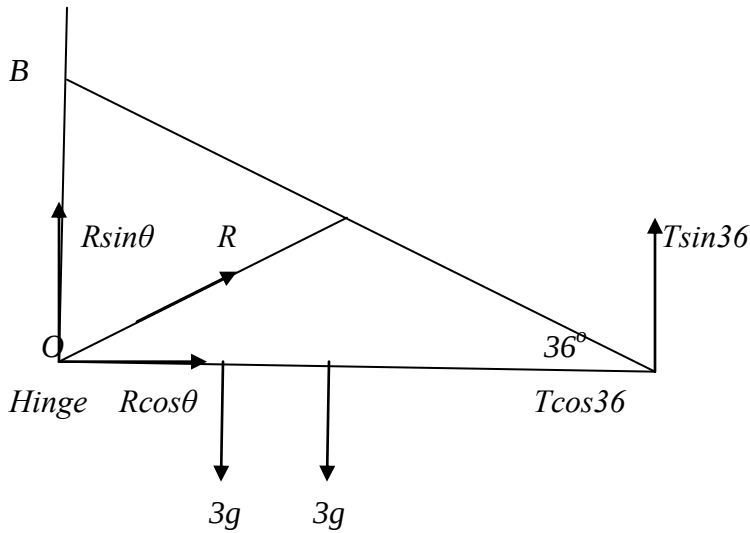
Worked example

Fig below shows a uniform wooden board of mass 3kg and length 40cm used as a book shelf. The board is supported by a light string from end A to the wall at an angle of 36° . Two books of mass 1.5 kg each are placed such that their centre of mass is 25cm from the hinge.



- Calculate (i) The tension in the string
(i) The force exerted by the hinge

Solution



Taking moments about O

(i) Clockwise moments = anticlockwise moments

$$0.2 \times 3 \times 9.81 + 0.25 \times 3 \times 9.81 = T \times 0.4 \sin 36$$

$$T = 56.3 \text{ N}$$

$$(ii) T \sin 36 + R \sin \theta = 6g$$

$$R \sin \theta = 25.77$$

$$R \cos \theta = T \cos 36 = 45.55 \text{ N}$$

$$R = 52.33 \text{ N} \quad \theta = 29.49^\circ$$

2.4: WORK, ENERGY AND POWER

Work

Work is said to be done when a body moves as a result of a force being applied to it.

Work done = force $\times$ distance moved in the direction of the force.

i.e. $W = Fs$; where F is the resultant force and s is the distance moved in the direction of the force.

The SI unit of work is the joule (J).

$$1 \text{ J} = 1 \text{ Nm}.$$

1 J of work is done, and 1 J of energy is converted from one form to another, when a 1 N force moves through a distance of 1m in the direction of the force.

Note: If a body upon which a force acts does not move, then no work is said to be done.

i.e. $s = 0 \text{ m} \Rightarrow W = 0 \text{ J}$.

Worked Example 5.1

Ephraim is 1.65 m tall. Calculate the work done:

(a) if he exerts a 400 N force in lifting his bag 1.5 m high.

(b) If he exerts a 400 N force in holding his bag above his head.

Solution

(a) Work done = force $\times$ displacement.

$$= 400 \text{ N} \times 1.5 \text{ m}$$

$$= 600 \text{ Nm}$$

$$= 600 \text{ J}.$$

(b) Work done = force $\times$ displacement.

$$= 400 \text{ N} \times 0 \text{ m}$$

$$= 0 \text{ Nm}$$

$$= 0 \text{ J}.$$

Work done by a force, F acting at an angle, α to the displacement, s .

When the force, F is exerted at an angle, α to the direction of the displacement, s (see **Figure 5.1**). We find the component of the force acting in the direction of the displacement.

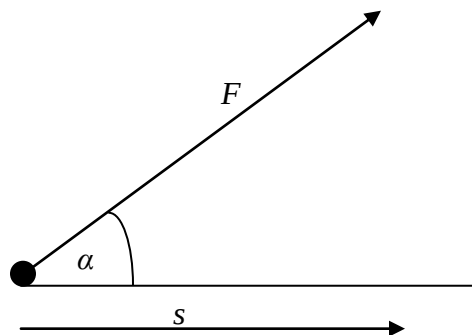


Figure 5.1: Work done by a force, F acting at an angle, α to the displacement, s .

Work done, $W = Fs \cos \alpha$.

(i) $\alpha = 0^\circ \Rightarrow W = Fs$.

(ii) $\alpha = 90^\circ \Rightarrow W = 0 \text{ J}$. For example: work done by the gravitational force acting on a moving object.

(iii) $\alpha = 180^\circ \Rightarrow W = -Fs$. For example: work done by frictional force acting on a moving object.

Worked Example 5.2

Find the work done by a force $F = 300 \text{ N}$, in pulling a luggage carrier in **Figure 5.2** at an angle $\alpha = 20^\circ$ for a distance $s = 45.0 \text{ m}$.

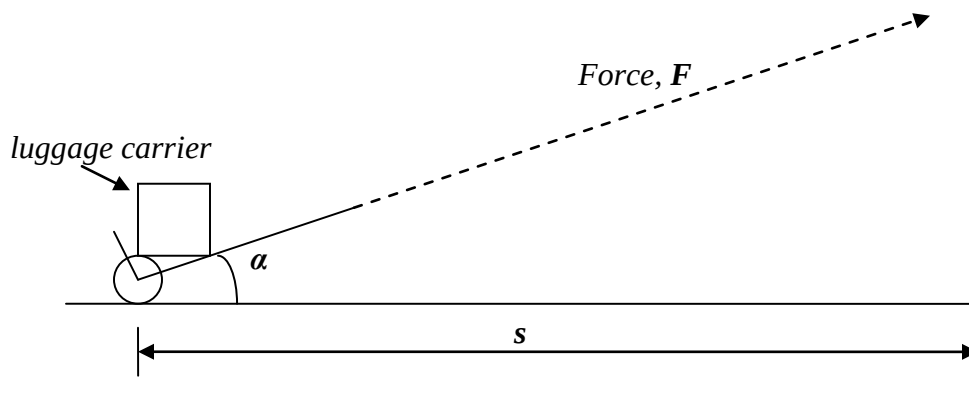


Figure 5.2: Work done by a force F , acting at an angle α , to the displacement s .

Solution

$$\begin{aligned} \text{Work done, } W &= Fs \cos \alpha \\ &= 300 \times 45.0 \times \cos 20^\circ \\ &= 12\,686 \text{ J.} \end{aligned}$$

Work done by a gas during expansion

Consider a gas enclosed in a cylinder by a frictionless piston of cross-sectional area A .

Let the pressure, p of the gas be constant (**Figure 5.3**).

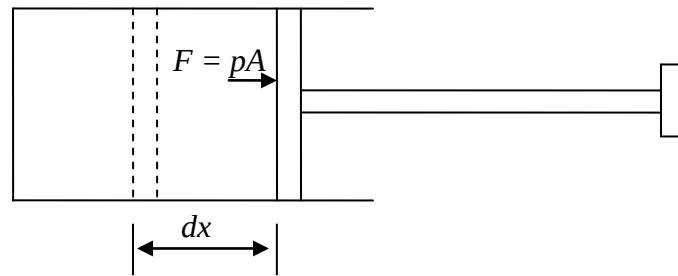


Figure 5.3: Work done by a gas during expansion

If the gas now expands and pushes the piston outwards through a distance dx , the volume of the gas is increased by dV . The work done dW by the gas during expansion is given by:

$$dW = \text{force} \times \text{distance moved by the piston} = Fdx.$$

But force = pressure $\times$ area = pA .

$$dW = pAdx, \text{ but also } (dV = Adx)$$

Therefore, $dW = pdV$.

Energy

Energy is the ability/capacity to do work. The energy used is equivalent to the work done.

Forms of energy

Energy can exist in different forms namely:

- (i) Mechanical energy { potential energy and kinetic energy };
- (ii) Heat energy;
- (iii) Light energy;
- (iv) Sound energy;
- (v) Chemical energy;
- (vi) Nuclear energy;
- (vii) Electrical energy;
- (viii) Magnetic energy, etc.

Basically, there are only two main types of energy namely: kinetic energy and potential energy.

Kinetic energy (E_k)

Kinetic energy is the energy possessed by an object by virtue of its motion.

Consider a constant force F acting on a body of mass m , which is initially at rest on a smooth horizontal surface. Suppose that, after moving a distance s across the surface, the body has a velocity v .

Work done on the body = $F \times s$, but $F = ma$ and $a = \frac{v^2 - 0}{2s}$.

Therefore, $F = \frac{mv^2}{2s}$

But work done on the body $\frac{mv^2}{2s} \times s = \frac{1}{2}mv^2$.

By definition, work done = E_k of the body = $\frac{1}{2}mv^2$.

Worked Example 5.3

Find the kinetic energy of a lorry of mass 3000 kg moving with a speed of 60 km/h.

Solution

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(3000)\left(\frac{60000}{60 \times 60}\right)^2$$

$$= 417 \text{ kJ.}$$

Potential energy (E_p)

Potential energy is the energy stored in a body by virtue of its position, state or shape.

Potential energy exists in several forms:

Table 5.1: Some common forms of potential energy.

Form of potential energy	An example of where it is stored
Chemical potential energy	Food and fuels
Gravitational potential energy	Stone on top of a cliff
Elastic potential energy	Stretched rubber band/catapult/spring
Nuclear potential energy	Particles in atomic nuclei
Elastic potential energy	Electric field

In mechanics, we are mainly concerned with gravitational potential energy which is the energy gained by an object when it is lifted up against the force of gravity.

Gravitational potential energy = mgh .

Where g = gravitational field strength;

m = mass;

h = height of the body above some reference level where $h = 0$, usually the surface of the earth.

To show that $E_p = mgh$, we consider a body of mass m which is lifted vertically through a height h (**Figure 5.4**).

The force acting vertically on the object = weight of the object (mg)

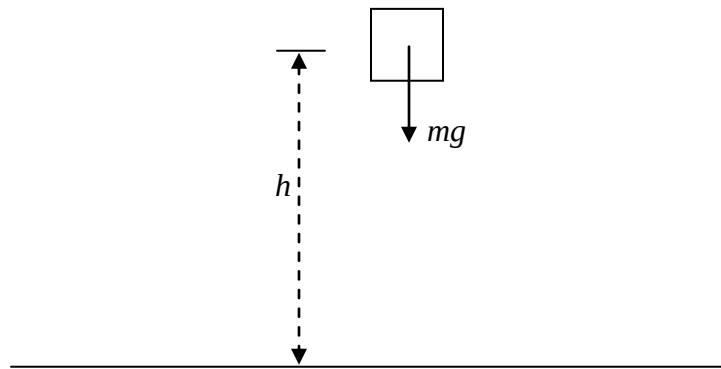


Figure 5.4: To show that $E_p = mgh$

Potential energy gained, E_p = work done against gravity

= force $\times$ distance moved

= $mg \times h$

= mgh .

$\therefore E_p = mgh$.

Internal energy

Internal energy is the sum of the kinetic and potential energies of all the molecules of a substance.

Principle of conservation of energy

Energy is never created or destroyed but is only transferred from one form to another i.e.

The total amount of kinetic and potential energies is constant.

$$E_p + E_k = \text{a constant.}$$

Interchange between E_p and E_k

Consider a body of mass m , which falls freely from rest at a height h in a vacuum (**Figure 5.5**).

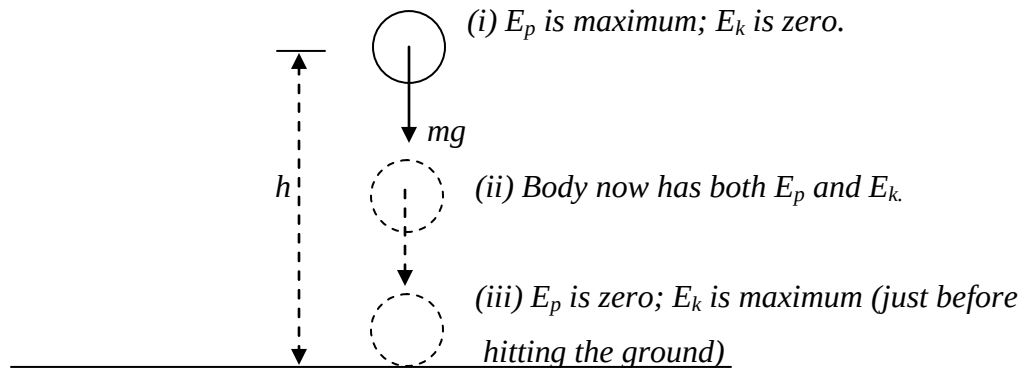


Figure 5.5: Interchange between E_p and E_k

As the body falls, it loses potential energy and gains kinetic energy. From the principle of conservation of energy:

$$E_k \text{ gained} = E_p \text{ lost.}$$

If v is the velocity of the body after it has fallen a distance h , then:

$$\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2gh}.$$

Energy losses

In our everyday lives we use energy but this energy is eventually transferred to internal energy in our surroundings. It is almost impossible to convert the internal energy back to any useful form of energy.

For example, a bus is powered by the chemical potential energy which is stored in diesel. The engine of the bus converts some of this chemical potential energy to kinetic energy but the rest is wasted as heat. Eventually, as the bus stops, its kinetic energy is also converted to heat due to friction at the brakes and between the tyres and the road. The heat energy produced is then transferred to the surrounding air. However, this heat

energy cannot be converted back to any useful form resulting in the so called 'energy loss'.

Efficiency

A machine is a device that enables work to be done more conveniently. When a machine does work, not all the energy used is transferred in a useful way. So the rest of the transferred energy is wasted as internal (heat) energy.

The efficiency of a machine is the ratio of the useful energy output to the total energy input. i.e.

$$\begin{aligned}\text{efficiency} &= \frac{\text{energy}_{\text{output}}}{\text{energy}_{\text{input}}}. \\ &= \frac{\text{power}_{\text{output}}}{\text{power}_{\text{input}}}.\end{aligned}$$

Efficiency is usually expressed as a percentage, by multiplying by 100 %.

Worked Example 5.4

An electric generator requires 20 kW of power so as to produce a power output of 17 kW.

What is the efficiency of the generator?

Solution

$$\begin{aligned}\text{efficiency} &= \frac{\text{power}_{\text{output}}}{\text{power}_{\text{input}}} \times 100\%. \\ &= \frac{17000}{20000} \times 100\% = 85\%.\end{aligned}$$

Power

Power is the rate at which work is done or the rate at which energy is transferred from one form to another.

$$\text{Power, } P = \frac{W}{t} = \frac{E}{t}.$$

Where W is the work done,

E is the energy transferred, and

t is the time taken.

The SI unit of power is the watt (W).

$$\text{Power, } P = \frac{W}{t} = \frac{\text{Force, } F \times \text{displacement, } s}{\text{time, } t}.$$

$$\text{Power, } P = \text{Force} \times \frac{\text{displacement}}{\text{time}}.$$

$$\therefore \text{Power, } P = \text{Force} \times \text{velocity} = Fv.$$

2.5: CIRCULAR MOTION

Introduction

In this unit, we shall deal with motion in a circle. Here, an accent is placed on kinematics of uniform circular motion and centripetal force.

Kinematics of uniform circular motion

Expressing angular displacement in radians

Angles can be measured in degrees as well as radians.

$360^\circ = 2\pi$ radians. Angles in radians (rad) are often expressed as multiples of π . For

example: $180^\circ = \pi$; $90^\circ = \frac{\pi}{2}$ and $30^\circ = \frac{\pi}{6}$.

Worked Example 7.1

Calculate how many degrees there are in 1 radian.

Solution

$$1\text{rad} = \frac{360^\circ}{2\pi} \approx 57^\circ.$$

Now, consider a circle with centre O and radius r (**Figure 7.1**).

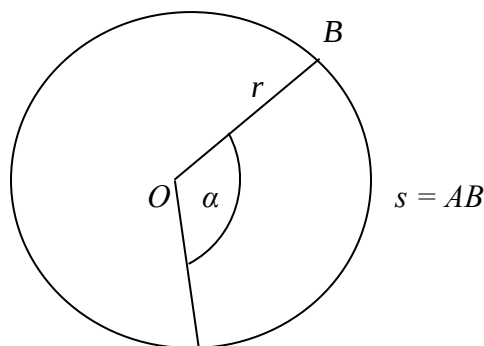


Figure 7.1 Motion in a circle.

If α is the angle subtended at the centre of a circle by arc AB, then

$$\frac{\text{arc-length}, s}{\text{circumference}, 2\pi r} = \frac{\alpha}{2\pi} \Rightarrow \text{arc length}, s = \frac{\alpha}{2\pi} \times 2\pi r. \text{ Therefore, } s = r\alpha.$$

If $s = r$, then $\alpha = \frac{s}{r} = 1$ rad. Therefore, one radian (i.e. 1 rad) is the angle subtended at the centre of a circle by an arc of the circumference equal in length to the radius of the circle.

Period (T)

Period is the time taken to complete one rotation/revolution. It is measured in seconds.

Frequency (f)

Frequency is the number of complete rotations/revolutions generated per unit time.

Frequency is measured in hertz (Hz).

$$\text{frequency} = \frac{N}{t}; \text{ where } N \text{ is the number of rotations and } t \text{ is the time taken.}$$

$$\text{Frequency, } f = \frac{1}{T}, \text{ where } T \text{ is the period. Therefore, } 1 \text{ Hz} = 1 \text{ s}^{-1}.$$

Angular velocity (w) is the rate of change of angular displacement.

Consider a particle (**Figure 7.1**) moving from point A to B along arc s at a constant speed in a time interval t . The angular velocity, w of the particle is given by:

$$w = \frac{\alpha}{t};$$

where α is the angle turned through (rad) and t is the time taken (s).

Angular velocity is measured in rad s^{-1} . One revolution is 2π radians. This takes T

$$\text{seconds. } \therefore w = \frac{\alpha}{t} = \frac{2\pi}{T} = 2\pi f.$$

w is also called the angular frequency.

From **Figure 7.1**, in a time t , the distance moved by the particle is the arc length s .

Therefore, the linear speed v of the particle is given by:

$$\text{speed, } v = \frac{\text{distance, } s}{\text{time, } t} = r \frac{\alpha}{t} = rw.$$

$$\therefore v = wr.$$

Worked Example 7.2

Suppose particle is moving in a circular path of radius 2 m at a speed of 24 ms^{-1} . What is the speed of the particle in (a) rad/s, (b) rev/min?

Solution

Using $v = wr$ with $v = 24 \text{ ms}^{-1}$ and $r = 2 \text{ m}$, gives:

$$(a) w = \frac{v}{r} = \frac{24}{2} = 12 \text{ rad/s.}$$

$$(b) 1 \text{ rev} = 2\pi \Rightarrow 12 \text{ rad/s} = \frac{12}{2\pi} \text{ rev/s} = \frac{12}{2\pi} \times 60 \text{ rev/min} = \frac{360}{\pi} \text{ rev/min.}$$

Centripetal acceleration

When an object moves at a constant speed in a circle, it will accelerate because its direction keeps changing and so does its velocity.

$$\text{Centripetal acceleration, } a = \frac{v^2}{r} = w^2 r.$$

Centripetal acceleration is directed towards the centre of the circle.

DERIVATION OF $a = \frac{v^2}{r}$

Fig below shows a particle in a circular motion. At A, its velocity is v_A at a tangent to the circle at A. After a short time Δt , it is at B and its velocity is v_B at a tangent to the circle at B. The angular displacement is $(\Delta\theta)$ is small

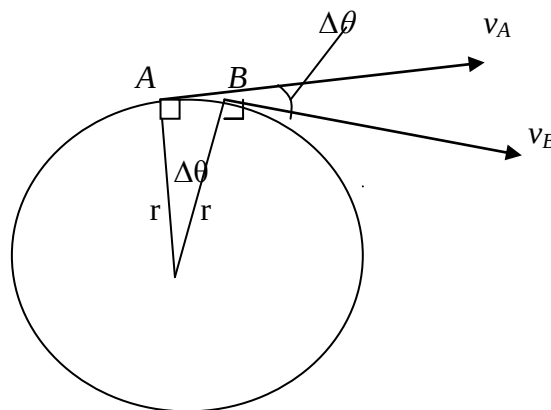
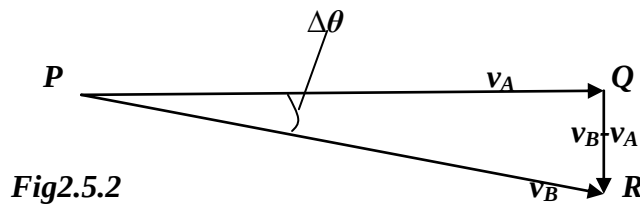


Fig 2.5.1



The change in velocity, $\Delta v = v_A - v_B$ is represented by QR . Since the angle $(\Delta\theta)$ is very small, then the direction of (Δv) is almost perpendicular to both v_A and v_B

In Fig 2.5.2, since $(\Delta\theta)$ is small, the length of $QR = \Delta v = v(\Delta\theta)$; where v is the magnitude of v_A and v_B , which are of equal magnitudes. The magnitude of the centripetal acceleration is

$$a = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a = \frac{\Delta v}{\Delta t} \text{ but } \Delta v = v(\Delta\theta)$$

$$a = \frac{v(\Delta\theta)}{\Delta t} \quad \text{But } \frac{\Delta\theta}{\Delta t} = \omega$$

$$a = v\omega \quad \text{But } v = r\omega$$

$$\text{Therefore } a = r\omega^2 = \frac{v^2}{r}$$

Centripetal force

From Newton's first law, a body continues to move in a straight line unless it is acted upon by a resultant force. So to make a body move along a circular path, you always need a resultant force directed towards the centre of the circular path. This is called the centripetal force.

From Newton's second law: $F = ma$.

$$\text{Centripetal force, } F = \frac{mv^2}{r} = m\omega^2 r.$$

Motion in a horizontal circle

Examples of motion in a horizontal circle:

- (i) If a body of mass m is whirled in a horizontal circle on one end of a piece of string, then the centripetal force is provided by the tension T in the string.

- (ii) When a car is rounding a bend, the centripetal force is provided by the frictional force between the car tyres and the road.

Worked Example 7.3

What is the centripetal force needed to keep a bicycle of mass 20 kg moving around a bend of radius 40 m at 10 ms^{-1} ?

Solution

$$F = \frac{mv^2}{r} = \frac{20(10)^2}{40} = 50 \text{ N}.$$

Motion in a vertical circle

Consider the motion of an object in a vertical circle as illustrated in **Figure 7.2** (at the top of the circle) and **Figure 7.3** (at the bottom of the circle).

At the top of the circle

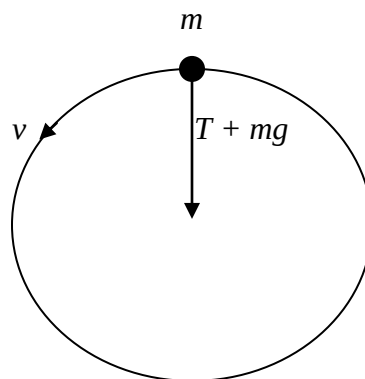


Figure 7.2: At the top of the circle, the centripetal force is provided by both the tension T in the string and the weight mg .

$$\text{i.e. } T + mg = \frac{mv^2}{r}.$$

Therefore, the tension T in the string can be calculated using the equation:

$$T = \frac{mv^2}{r} - mg.$$

At the bottom of the circle

At the bottom of the circle, the centripetal force is provided by $T - mg$, because the tension and the weight will be acting in opposite directions.

$$\text{i.e. } T - mg = \frac{mv^2}{r}.$$

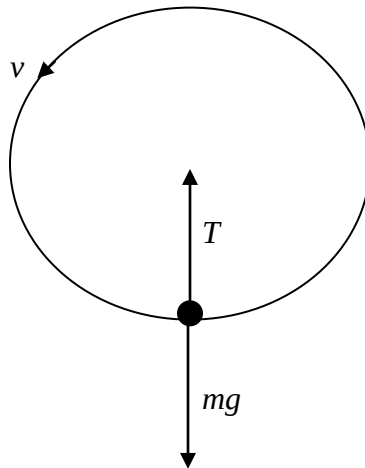


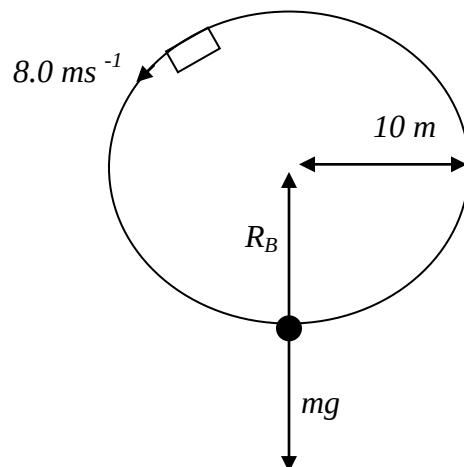
Figure 7.3: At the bottom of the circle, the centripetal force is provided by $T - mg$.

Worked Example 7.4

A roller coaster has a vertical loop of radius 10 m. Given that the cars hurtle round the loop at 8.0 ms^{-1} . Find,

- (a) The centripetal force exerted on a rider of mass 80 kg,
- (b) The contact force exerted by the car seat on the rider at the bottom of the loop.

Solution



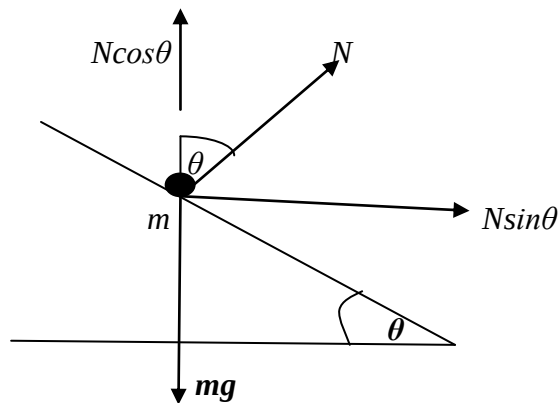
$$(a) F = \frac{mv^2}{r} = \frac{80(8.0)^2}{10} = 512 \text{ N.}$$

(b) At the bottom of the loop, the contact force R_B is given by:

$$R_B - mg = \frac{mv^2}{r} \Rightarrow R_B = \frac{mv^2}{r} + mg.$$

$$\therefore R_B = 512 + (80)(9.81) = 1296.8 \text{ N.}$$

BANKED ROAD



The problem is to the angle θ at which a bend should be banked so that the centripetal force arises entirely from a component of the normal force N of the road. Treating the car as a particle and resolving N vertically and horizontally we have;

$N\sin\theta$ provides the centripetal force of the car which is moving round a banked road.

$$\text{Therefore } N\sin\theta = \frac{mv^2}{r} \quad (1)$$

$$N\cos\theta = mg \quad (2)$$

Dividing (1) by (2) yields

$$\tan\theta = \frac{v^2}{rg}$$

Satellite orbits

A satellite is an object held in an orbit around a relatively larger object by the gravitational attraction.

Natural satellites

The Moon is an original Earth satellite. The Earth itself is a satellite of the Sun.

Artificial satellites

Observation satellites which monitor weather, vegetation cover, land use or communication etc. are good examples of artificial satellites.

A centripetal force is required to keep a satellite moving along a circular path. This centripetal force is provided by the Earth's gravitational force.

Consider a satellite of mass m in an orbit of radius r and velocity v about a planet of mass M .

Centripetal force = gravitational force.

$$\frac{mv^2}{r} = \frac{GMm}{r^2} \Rightarrow v = \sqrt{\frac{GM}{r}}.$$

The orbital period is given by

$$T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{GM}}.$$

Both G and M are constants; therefore both the speed and orbital period of an Earth satellite depend only on the orbital radius.

Geostationary (synchronous) orbit

A satellite in a geostationary orbit will be approximately at a distance of 36 000 km above the Earth's equator and takes only 24 hours to complete one orbit. The satellite appears to remain fixed in the sky, as the Earth spins. Satellites in geostationary orbits can be used to relay television signals and telephone messages from one point on the Earth's surface to another. Examples include Hot bird, Syncom 2 etc. Also METEOSAT

weather satellites are placed in geostationary orbits to continuously monitor large areas of the globe.

2.6: GRAVITATIONAL FIELD

Gravitational field and field strength

Gravitational field

The gravitational field is a region in which the gravitational force is experienced. It can be represented by gravitational field lines whose arrows indicate the direction of the field (i.e. the direction in which a mass in the gravitational field would be pulled). The spacing of the gravitational field lines gives information about the magnitude of the gravitational field i.e. the gravitational field strength. The closer the gravitational field lines, the larger the gravitational field strength and stronger the gravitational force. In **Figure 6.1**, points far away from the Earth have relatively weaker gravitational field strengths than those close to the Earth i.e. the gravitational field strength decreases as we move away from the Earth's surface as illustrated by the spacing of gravitational field lines in **Figure 6.1**.

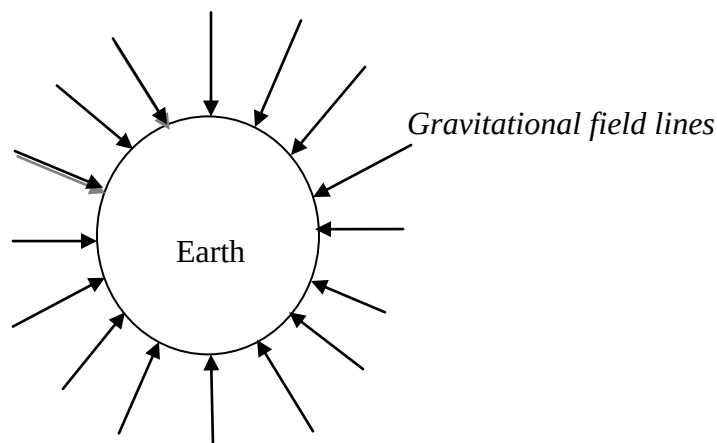


Figure 6.1: Earth's gravitational field

Close to the Earth's surface, the gravitational field is considered as uniform although it actually varies slightly from place to place. This is illustrated by the uniform or equal spacing of the lines of force near or on the surface of the Earth (**Figure 6.1**).

Gravitational field strength

Gravitational field strength is the force acting on a unit mass placed in the gravitational field.

Gravitational field strength, $g = \frac{\text{Force}}{\text{mass}}$.

Gravitational field strength is a vector quantity and is measured in Nkg^{-1} .

Newton's law of universal gravitation

Newton's law of universal gravitation states that, every particle of matter in the universe attracts every other particle with a force which is directly proportional to the product of their masses and inversely proportional to the square of their separation distance (Figure 6.2).

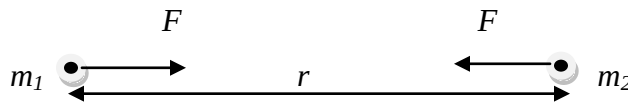


Figure 6.2: Any two point masses or particles (m_1 and m_2) are gravitationally attracted to each other with a force $F \propto \frac{m_1 m_2}{r^2}$. i.e. $F = \frac{G m_1 m_2}{r^2}$.

Where F is the gravitational attraction force between two particles of masses m_1 and m_2 , which are at a distance r apart. G is a constant of proportionality known as the universal gravitational constant ($= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$).

This type of relationship is called an inverse square law (Figure 6.3) since

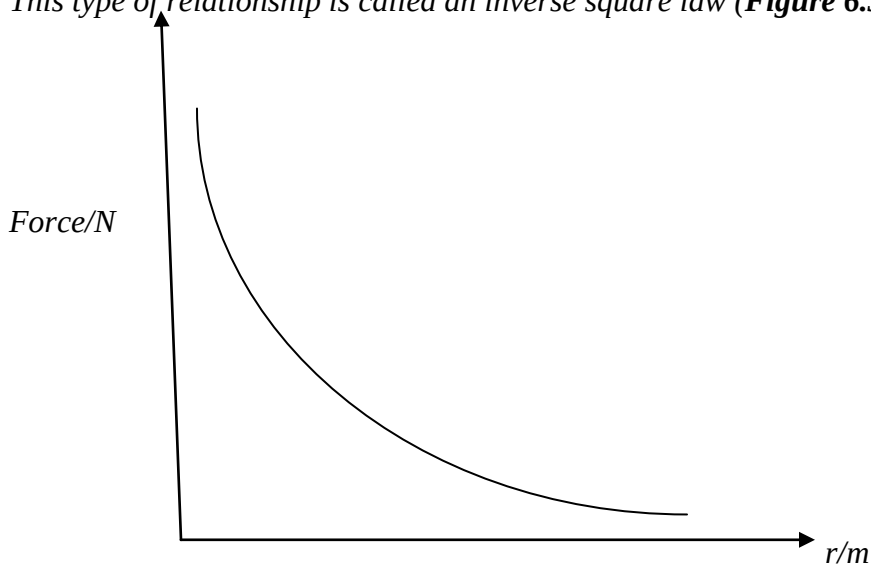


Figure 6.3: Variation of the force F between masses with the separation distance r . i.e.

$$F \propto \frac{1}{r^2}.$$

Worked Example 6.1

Assuming that the Earth and the Moon are uniform spheres, find the magnitude of the gravitational force between the Earth and the Moon.

($M_E = 6.0 \times 10^{24}$ kg, $m_m = 7.4 \times 10^{22}$ kg and $r_{EM} = 3.79 \times 10^8$ m.)

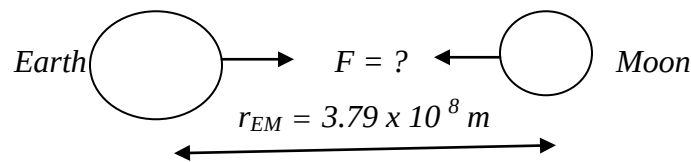


Figure 6.4: The gravitational force between the Earth and the Moon.

Solution

$$F = \frac{GM_E m_m}{r_{EM}^2} = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(7.4 \times 10^{22})}{(3.79 \times 10^8)^2} = 2.1 \times 10^{20} \text{ N}.$$

.

Deriving, from Newton's law of gravitation and the definition of gravitational field strength, the equation $g = Gm/r^2$ for the gravitational field strength of a point mass.

- Newton's law of gravitation: $F = \frac{Gmm_1}{r^2}$
- Definition of gravitational field strength: $g = \frac{F}{m_1}$.

$$\text{Therefore: } g = \frac{Gmm_1}{r^2} \times \frac{1}{m_1} \Rightarrow g = \frac{Gm}{r^2}.$$

This equation implies that taking g to be constant everywhere on the Earth's surface is facilitated by making the assumption that the Earth is a homogeneous sphere, (i.e. $r_E =$

constant and also the Earth has a uniform mass distribution), which is not true.

Therefore, taking g to be constant is only an assumption since $g \propto \frac{1}{r^2}$.

At a height h above the surface of the Earth, $g = \frac{GM_E}{(r_E + h)^2}$.

Worked Example 6.2

Estimate the gravitational field strength at the surface of the Earth given that its radius is $6.37 \times 10^6 \text{ m}$ and its mass is $5.98 \times 10^{24} \text{ kg}$ (**Figure 6.6**). ($G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$).

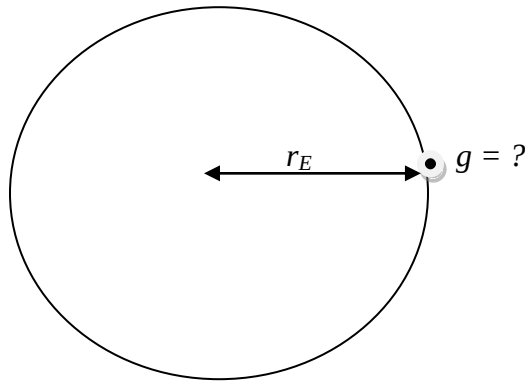


Figure 6.6: The gravitational field strength at the Earth's surface.

Solution

$$g = \frac{GM_E}{r^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.37 \times 10^6)^2} = 9.83 \text{ Nkg}^{-1}.$$

Worked Example 6.3

Estimate the gravitational field strength at a height of $2.0 \times 10^6 \text{ m}$ above the surface of the Earth given that the radius of the Earth is $6.4 \times 10^6 \text{ m}$ and its mass is $5.98 \times 10^{24} \text{ kg}$ (**Figure 6.7**). ($G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$).

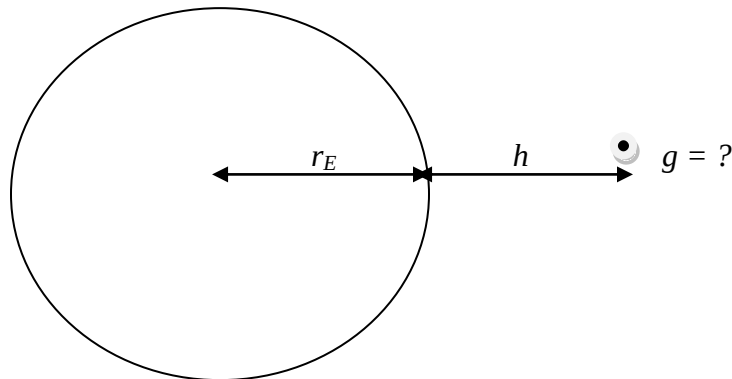


Figure 6.7: The gravitational field strength at a height h above the Earth's surface.

Solution

$$g = \frac{GM_E}{(r_E + h)^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.4 \times 10^6 + 2.0 \times 10^6)^2} = 5.65 \text{ Nkg}^{-1}.$$

Gravitational Potential

The gravitational potential (ϕ) at a point in a gravitational field is the work done against gravity in bringing a unit mass from infinity to that point.

$$\phi = -\frac{GM}{r}.$$

Where: ϕ = the gravitational potential due to a body of mass m at a point outside the body and at a distance r from its centre.

The potential at infinity is zero, so the negative sign for ϕ indicates that the gravitational potential at that point is less than that at infinity. The attractive nature of gravity pulls masses together, so we only do work against gravity if the 1 kg mass is pulled away, thus ϕ must be negative.

The unit for ϕ is the J/kg.

Worked Example 6.4

Calculate the gravitational potential, ϕ at 7000 km from the centre of the Earth of mass $5.98 \times 10^{24} \text{ kg}$ (**Figure 6.8**). ($G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$).

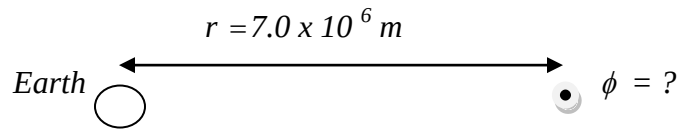


Figure 6.8: The gravitational potential, ϕ at 7000 km from the centre of the Earth.

Solution

$$\phi = -\frac{GM}{r} = -\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{7.0 \times 10^6} = -5.7 \times 10^{-53} \text{ Jkg}^{-1}.$$

Figure 6.9: The gravitational potential, ϕ at $2.20 \times 10^{29} \text{ m}$ from the centre of a star.

Gravitational potential energy

Gravitational potential energy, E_p is the work done in bringing the mass from infinity to that point.

$$E_p = m\phi = -\frac{GMm}{r}$$

Worked Example 6.5

Calculate the gravitational potential energy of a body of mass 35000 kg at a height of $1.6 \times 10^6 \text{ m}$ above the surface of the Earth.

($M_E = 6.0 \times 10^{24} \text{ kg}$, $r_E = 6.4 \times 10^6 \text{ m}$ and $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$).

Solution

$$E_p = m\phi = -\frac{GM_E m}{(r_E + h)} = -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(35000)}{(6.4 \times 10^6 + 1.6 \times 10^6)} = 1.75 \times 10^{12} \text{ J}.$$

WHY GRAVITATIONAL POTENTIAL IS ALWAYS NEGATIVE

1. Potential at infinity is taken as zero/reference of potential
2. Gravitational force is attractive, attractive forces are associated with negative potential
3. Bodies do not fall away from the earth, but towards earth/potential decreases as one moves towards the earth. This means earth is at a lower potential than at infinity.

Relationship between the gravitational field strength g and the gravitational potential ϕ .

$$g = \frac{GM}{r^2} \text{ and } \phi = -\frac{GM}{r}.$$

Figure 6.10 illustrates the relationship between the gravitational field strength g and the gravitational potential ϕ .

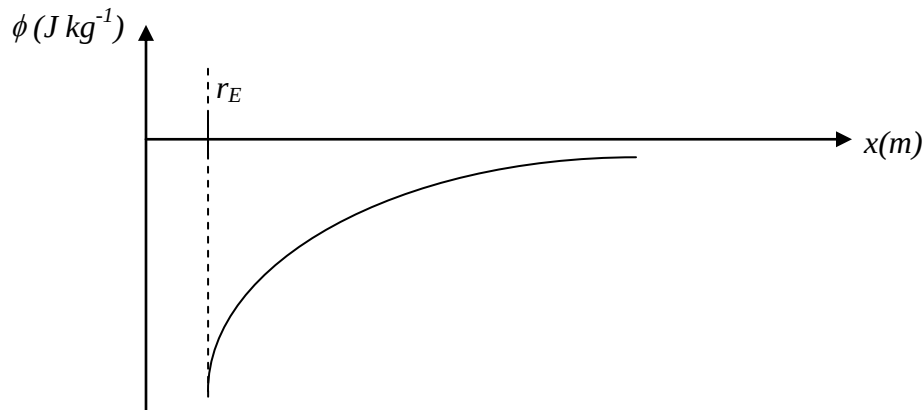


Figure 6.10: The value of the gravitational field strength = the value of the gradient at any point.

Therefore, $g = -\frac{\Delta\phi}{\Delta x} = \text{potential gradient}.$

VARIATION OF g WITH DISTANCE BELOW THE EARTH'S SURFACE

The gravitational field of the earth exists both inside and outside the earth.

Consider a mass m at P below the earth's surface at a distance r , ($r < R$) from the centre of earth. The gravitational attraction on the mass m is only due to that part of the earth centre O and radius r . assuming the earth to be a uniform sphere of density ρ , the mass M^1 of the part of the earth of radius r is

$$M^1 = \frac{4\pi\rho r^3}{3}$$

Weight of mass = gravitational attraction.

$$mg^1 = \frac{GmM^1}{r^2} \quad g^1 = \text{gravitational field.}$$

$$g^1 = \frac{G}{r^2} \left(\frac{4}{3} \pi r^3 \rho \right) \text{-----Eqn (1)}$$

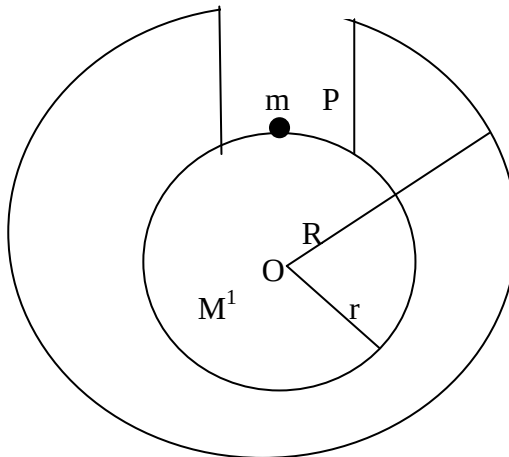
$$\text{On the earth's surface } g = \frac{GM_E}{R}.$$

$$R = \text{radius of earth.} \quad g = \frac{G}{R^2} \left(\frac{4}{3} \pi R^3 \rho \right)$$

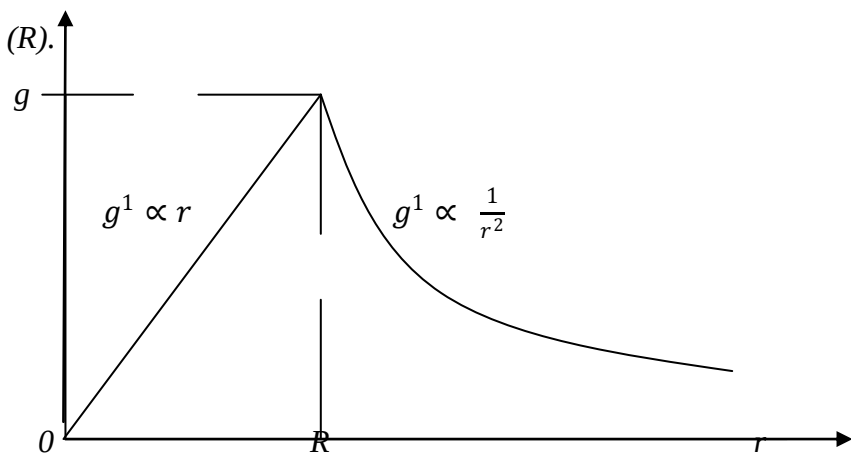
$$\frac{g^1}{g} = \frac{r}{R}$$

$$g^1 = \frac{r}{R} g$$

$$g^1 \propto r \quad (r < R)$$



Therefore inside the earth, the gravitational field strength (g^1) is directly proportional to radius (r) but outside the earth gravitational field strength, (g) is inversely proportional to the square of the radius (R).



ASSUMPTION

1. The density of the earth is constant.

SECTION 3: OSCILLATIONS AND WAVES

3.1: OSCILLATIONS

Simple Harmonic motion

Is the motion of an object whose acceleration?

- (i) Is proportional to its displacement (x) from a fixed point and
- (ii) Is always directed towards that point.

Examples of simple harmonic motion are

1. Simple pendulum
2. Vibrating loaded spring
3. Mercury in U-tube
4. Trolley between two springs

Oscillatory motion involves periodic or vibration about a fixed position.

Periodic oscillation describes the repeated retracing of a path in a fixed time interval. The direction changes abruptly, velocity varies widely and acceleration is far from constant.

Simple harmonic motion is a motion in which the acceleration of the body is directly proportional to its displacement from a fixed point and is always directed towards that point.

That is the body is oscillating equal distances either side of the fixed point.

Further the object is from the point the greater its acceleration.

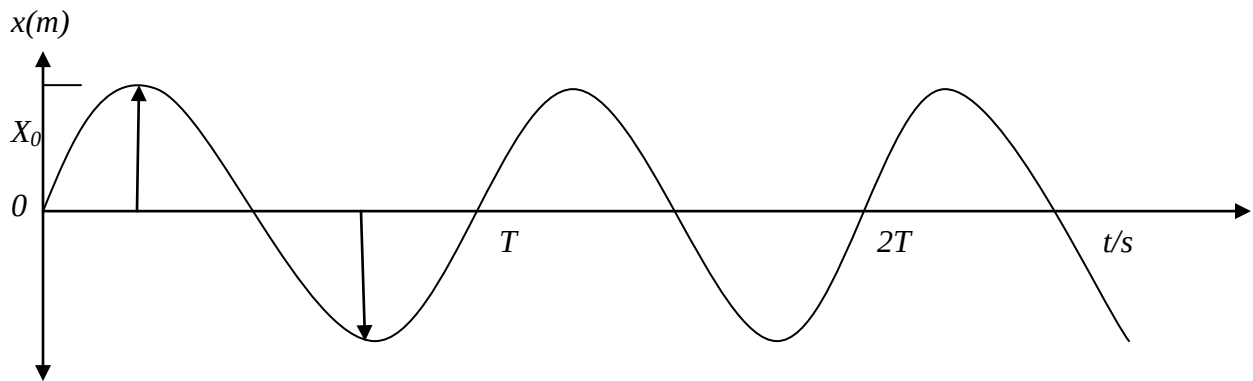
This implies that the resultant force exerted on the body must increase with distance from this point (equilibrium point) and always be acting in a direction towards that point, so as to restore the body to its undisturbed position.

In this position, the resultant force will be zero, it will be in equilibrium.

For a simple harmonic motion then variation of displacement with time has the same shape as a sine wave. That is the oscillations are sinusoidal and the equation is

$$x = x_0 \sin \omega t$$

The sine curve



$$x = x_0 \sin \omega t.$$

x = displacement at any time t

x_0 = maximum displacement from the equilibrium position (amplitude)

ω = angular frequency (in rads^{-1})

t = the time in seconds

Definition of terms

Amplitude (m)

Amplitude is the maximum displacement from the point of equilibrium.

Period (T)

Period is the time taken to complete one oscillation

Frequency (f)

Frequency is the number of complete oscillations per unit time.

Angular frequency

Angular frequency is the rate of change of angular displacement

Phase difference

Phase difference is the difference in angular displacement between particles oscillating.

Phase difference = $\frac{2\pi \Delta x}{\lambda}$ where Δx = small distance between two oscillating particles, λ

= the wavelength. OR

Phase difference = $\frac{2\pi \Delta t}{T}$

Starting with the equation $x = x_0 \sin \omega t$

Differentiating the equation for the first time yields

$v = x_0 \omega \cos \omega t$ where v is the velocity

Using the identity $\sin^2 \theta + \cos^2 \theta = 1$

Displacement $x = x_0 \sin \omega t$ then $x^2 \omega^2 = \omega^2 X_0^2 \sin^2 \omega t$. (1)

Velocity $v = x_0 \omega \cos \omega t$ then $v^2 = X_0^2 \omega^2 \cos^2 \omega t$ (2)

Adding equation (1) and (2)

$$v^2 + x^2 \omega^2 = \omega^2 X_0^2 \sin^2 \omega t + X_0^2 \omega^2 \cos^2 \omega t$$

$$v^2 + x^2 \omega^2 = \omega^2 X_0^2 [\sin^2 \omega t + \cos^2 \omega t]$$

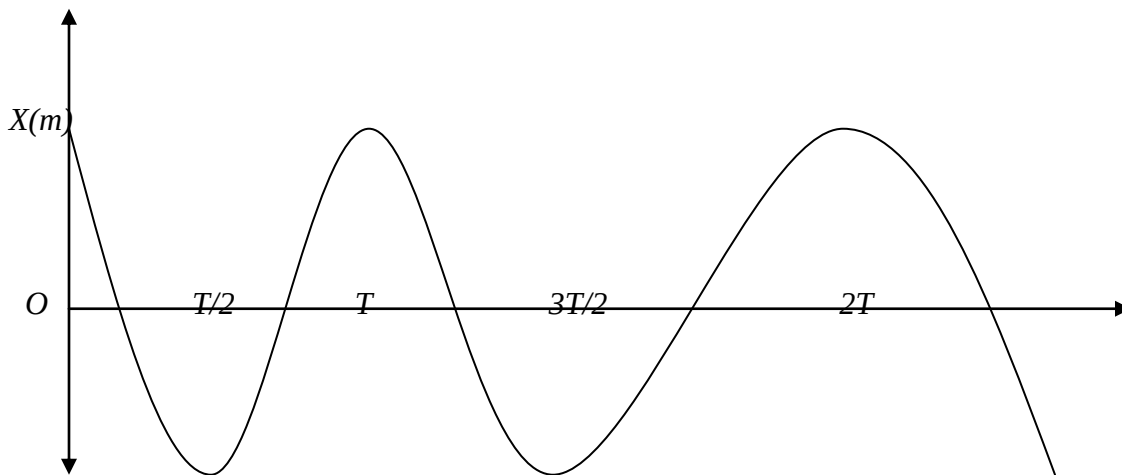
$$v^2 + x^2 \omega^2 = \omega^2 X_0^2 \text{ using identity } \sin^2 \omega t + \cos^2 \omega t = 1$$

$$v^2 = \omega^2 X_0^2 - x^2 \omega^2$$

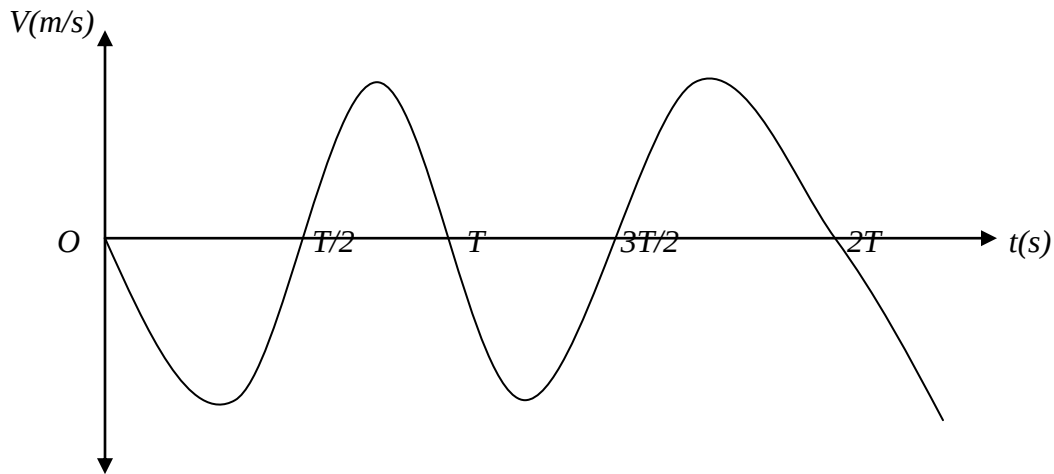
$$V = \pm \omega \sqrt{x_0^2 - x^2}$$

The equation enables speed to be calculated by any value of displacement.

Graphs for s.h.m



Displacement time graph for an object moving with s.h.m



Velocity time curve for an object moving with s.h.m

The velocity is zero when the displacement is greatest and greatest when the displacement is zero.

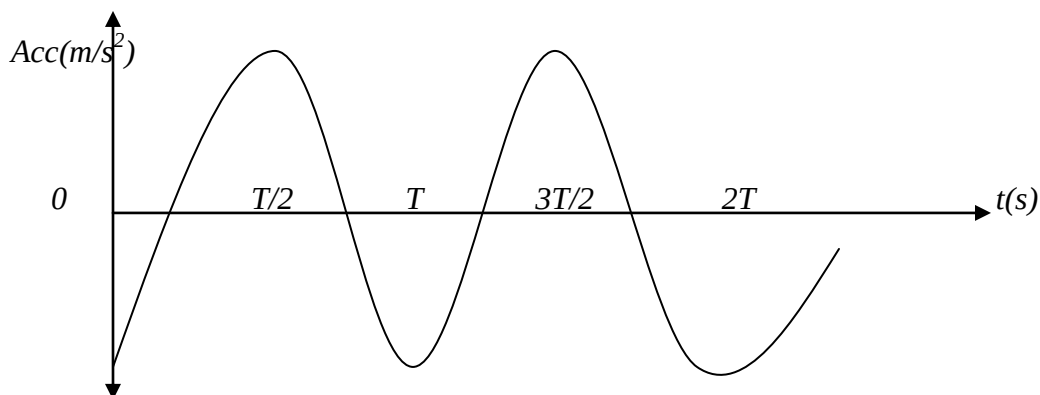
Velocity

Velocity is given by the gradient of the displacement time graph.

The gradient of the displacement time curve is zero initially, but then becomes more and more negative for a quarter cycle.

So given that the displacement-time curve is a cosine wave, then the velocity-time curve is an inverted sine wave.

Acceleration



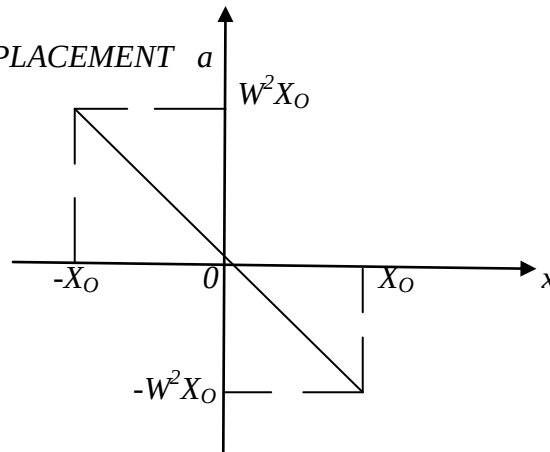
Acceleration is given by the gradient of the velocity time graph.

The acceleration time curve is an inverted cosine curve (ie an inverted displacement time curve).

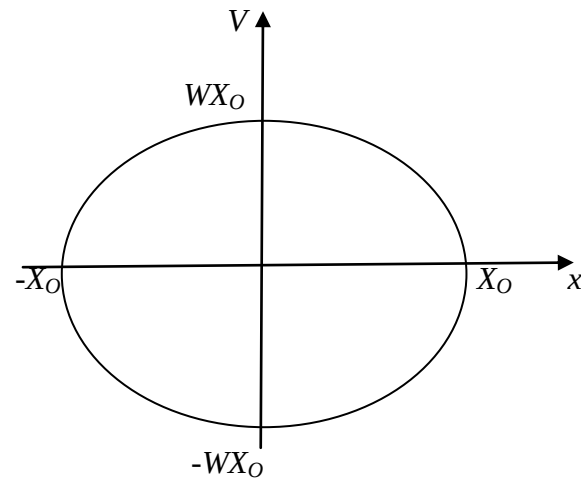
When displacement is positive the acceleration is negative. The acceleration is proportional to the displacement.

GRAPHS FOR VELOCITY, ACCELERATION AGAINST DISPLACEMENT

ACCELERATION AGAINST DISPLACEMENT



VELOCITY AGAINST DISPLACEMENT



FORCES AND S.H.M

The motion of an object is determined by the forces acting on it, so when an object moves with S.H.M, the force on it is the cause of that motion.

The resultant force F on any object causes an acceleration given by $F=ma$ so for s.h.m, the resultant force must be given by.

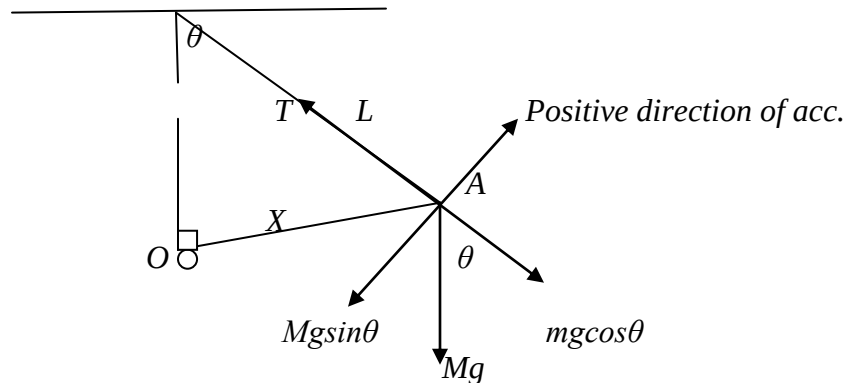
$$F = -m\omega^2 x.$$

The forces acting on the object must therefore give a resultant force which is always directed towards equilibrium (ie so as to try to restore the object to equilibrium) always proportional to the displacement.

The simple pendulum

The simple pendulum consists of a small bob of mass m , suspended by a light inextensible thread of length L from a fixed point.

If the bob is drawn slightly it oscillates to and fro in a circle.



Force at A are the weight mg and the Tension T .

Resolving mg radially and tangentially at A, we see that the tangential component $Mg \sin \theta$ is unbalanced restoring force acting towards O. The component $mg \cos \theta$ balances to T .

If a is the acceleration of the bob along the arc at A due to $mg \sin \theta$, then the equation of motion of the bob is $-mg \sin \theta = ma$

The negative sign indicates that the force is towards O while distance x is measured along the arc from O in the opposite direction.

Where θ is small $\sin \theta$ is approximately equal to θ ($\sin \theta \approx \theta$)

Therefore $x = L\theta$

$$\text{Hence } -mg\theta = m \frac{x}{L} = ma$$

$$\therefore a = -\frac{g}{L}x = \omega^2 x \quad (\text{where } \omega^2 = \frac{g}{L})$$

$$\text{Period } T = \frac{2\pi}{\omega}$$

$$T = 2\pi\sqrt{g/L}$$

$$2\pi\sqrt{\frac{L}{g}}$$

Assumptions

1. The string should be light and inextensible.
2. The angle θ should be very small and does not exceed 10°

OSCILLATIONS OF LOADED VERTICAL SPRING

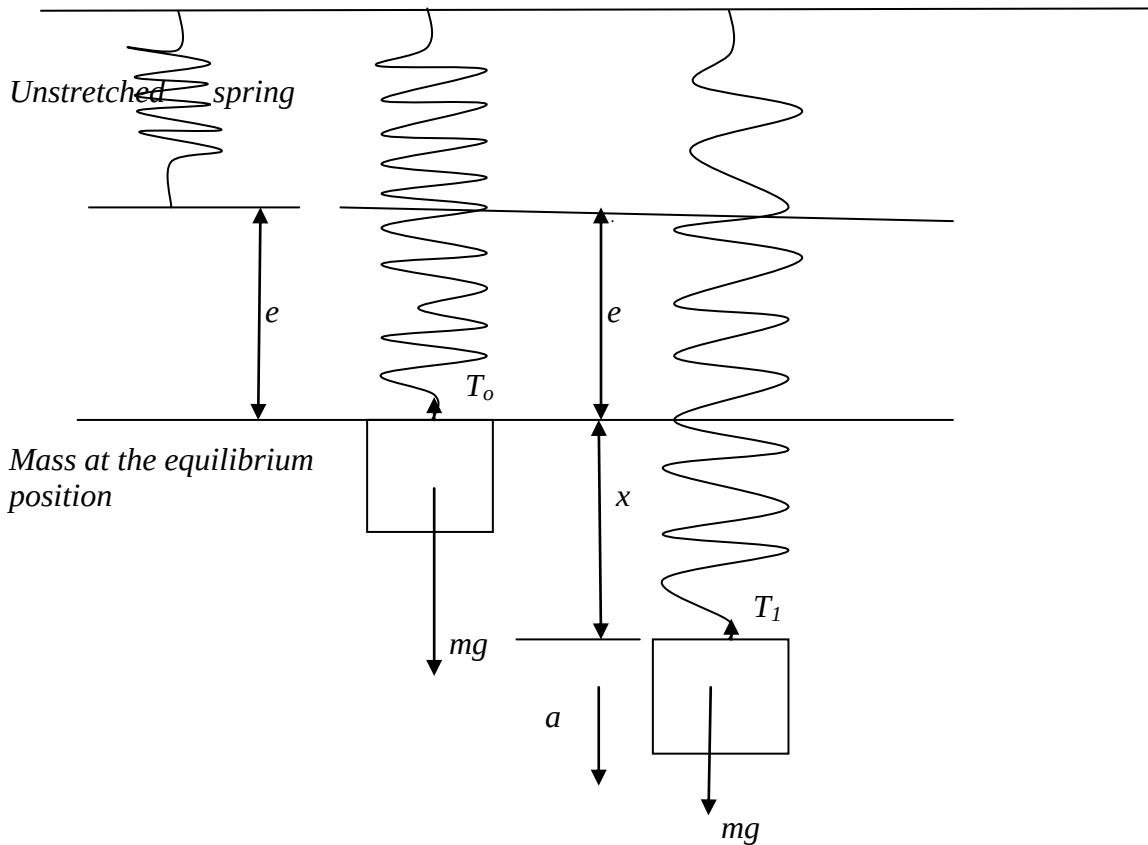


Fig above shows a spring being stretched by an amount e when mass m hangs in equilibrium from the end of it.

When in equilibrium, the tension in the spring is given by the expression:

$T_0 = mg$. If the spring is not stretched beyond its elastic limit, using Hooke's law,

$T_0 = ke$ k = spring constant.

Therefore $mg = ke$.

The mass m is now pulled downwards a further distance x_0 ($x_0 < e$) and then released. The mass vibrates vertically with amplitude x_0 . Suppose that at any time t , the displacement of the mass from its equilibrium position is x .

Using

$$F = ma$$

$$mg - T_1 = ma$$

Using Hooke's law $T_1 = k(x + e)$

$$mg - k(x + e) = ma$$

$$ke - k(x + e) = ma \quad : \quad ma = -kx$$

Therefore acceleration, $a = -\frac{k}{m}x \equiv -\omega^2 x$

Therefore the mass m moves in S.H.M with angular frequency $\omega = \sqrt{\frac{k}{m}}$

Period $T = 2\pi\sqrt{\frac{m}{k}}$

$$2\pi\sqrt{\frac{m}{k}}$$

THE CONDITIONS FOR THE MASS m TO MOVE IN S.H.M:

1. The spring should not be stretched beyond its elastic limit, so that Hooke's law is obeyed.
2. The amplitude of vibration $x_0 < e$, the equilibrium extension.
3. The mass of the spring is negligible.

Energy of oscillating systems

In an oscillating system there is a constant interchange of energy between the kinetic and potential forms, and if the system does no work against resistive forces (ie is undamped) its total energy is constant.

(a) Kinetic energy

The velocity is given by

$$V = \pm \omega \sqrt{x_0^2 - x^2}$$

$$E_k = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$$

(b) Potential energy

As an oscillating particle move away from the equilibrium position, work is done against the force (eg the tension in a stretched spring) trying to restore to the equilibrium position.

The particles lose kinetic energy and gains potential energy.

When $x=0$ the restoring force is zero, at a displacement x , the force is $m\omega^2 x$

Since acceleration has a magnitude $\omega^2 x$. Hence

Average force on a particle while moving to displacement x

$$= \frac{1}{2} m \omega^2 x$$

Work done = average force $\times$ displacement in the direction of force

$$= \frac{1}{2} m \omega^2 x \cdot x = \frac{1}{2} m \omega^2 x^2$$

$$E_p \text{ at displacement } x = \frac{1}{2} m \omega^2 x^2$$

Another formula for E_p : $E_p = \text{average force} \times \text{displacement} = \frac{T}{2} \cdot x$ but $T = \frac{2\pi}{\omega}$

$$E_p = \frac{1}{2}kx^2$$

(c) Total energy

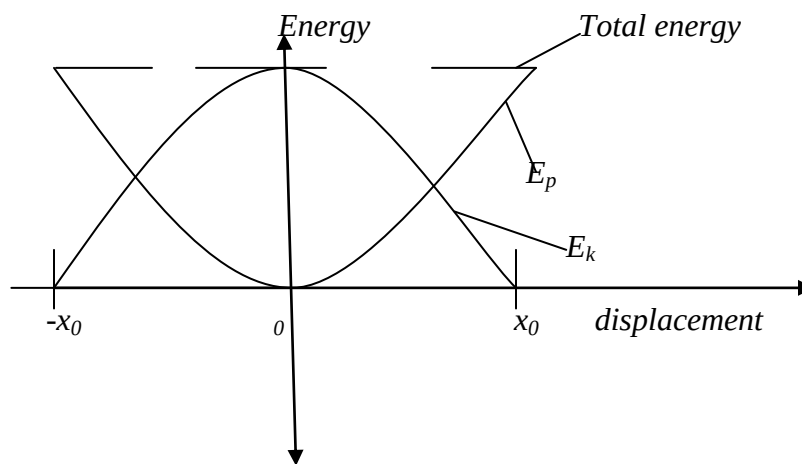
At displacement x we have

$$\begin{aligned} \text{Total energy} &= E_k + E_p \\ &= \frac{1}{2}mw^2(x_0^2 - x^2) + \frac{1}{2}mw^2x^2 \\ &= \frac{1}{2}mw^2x_0^2 \end{aligned}$$

This constant, does not depend on x and is directly proportional to the product of

- (i) The mass
- (ii) The square of frequency
- (iii) The square of amplitude

NB In a simple pendulum all energy is kinetic as the bob passes through the centre of oscillation and at the top of the swing it is all potentials.

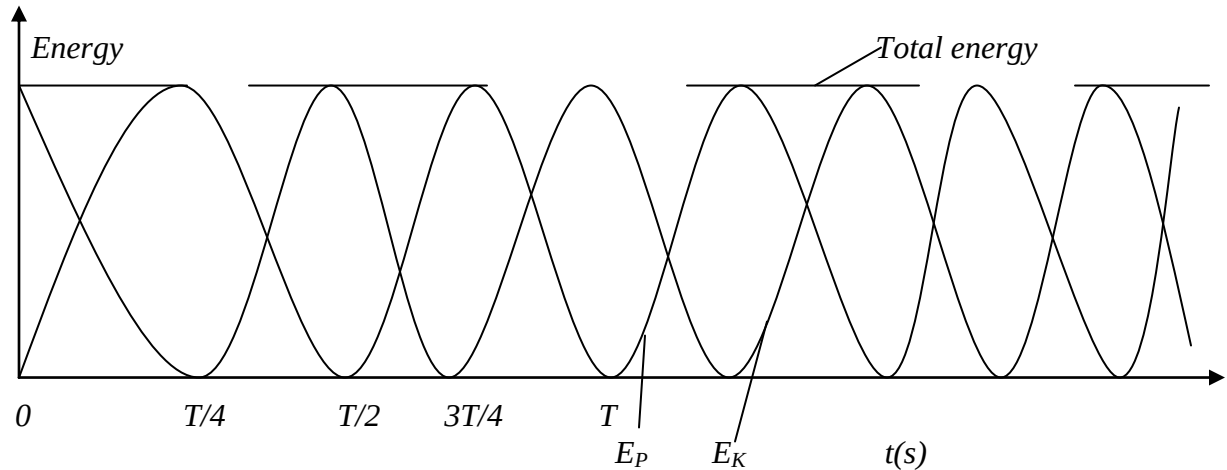


(c) Variation of E_k and E_p with time

These vary in time as shown by fig below

$$\text{Since } E_k = \frac{1}{2}mv^2 = \frac{1}{2}mw^2x_0^2\sin^2\omega t$$

$$E_p = \frac{1}{2}mw^2x^2 = \frac{1}{2}mw^2\cos^2\omega t$$



DAMPED OSCILLATIONS

Free oscillations describe a system in which the total energy of the system does not decrease with time.

The amplitude of vibration remains constant.

In real cases such as a simple pendulum, the amplitude of the oscillation will decay to zero as the system loses energy to the surrounding through friction and viscous forces. This action is described as damping.

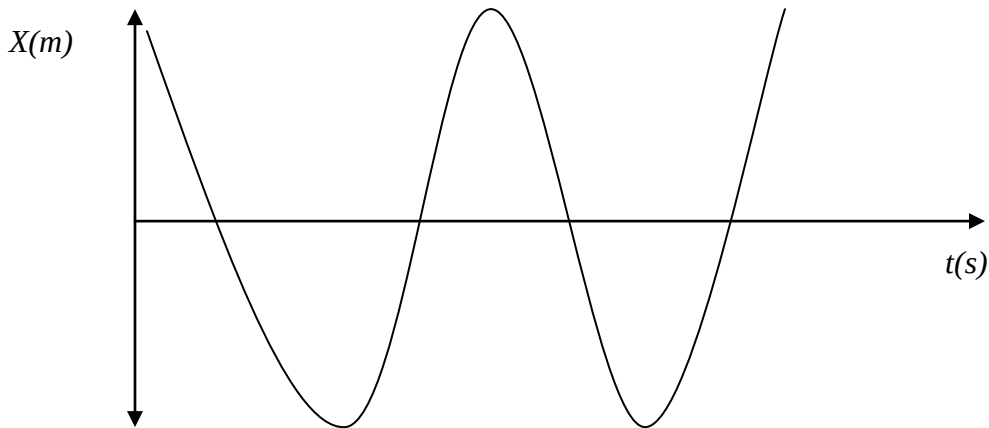
There are situations where damping is desirable eg in shock absorbers (dampers) on vehicles.

Suspension systems and springs are usually used to absorb shocks from bumps in the road and to allow wheels to follow an even surface while the vehicle follows a more or less horizontal path.

The paper cone of the loudspeaker is heavily damped so that it dissipates energy to the air as sound.

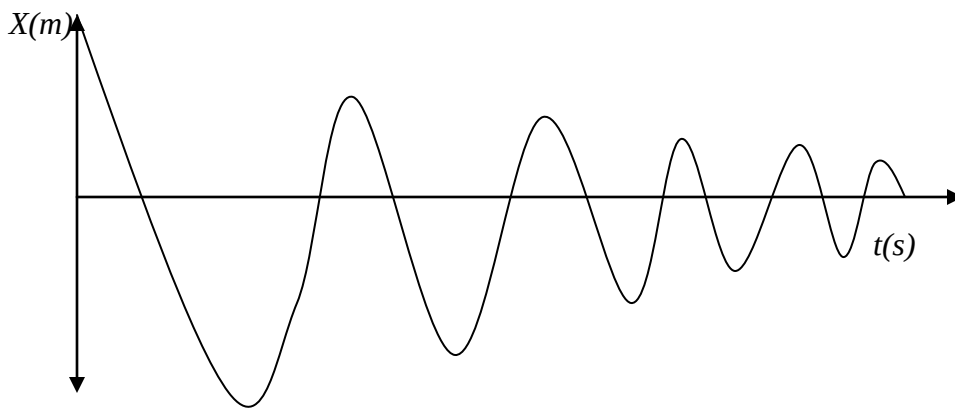
Types of damping

(a) Free oscillation



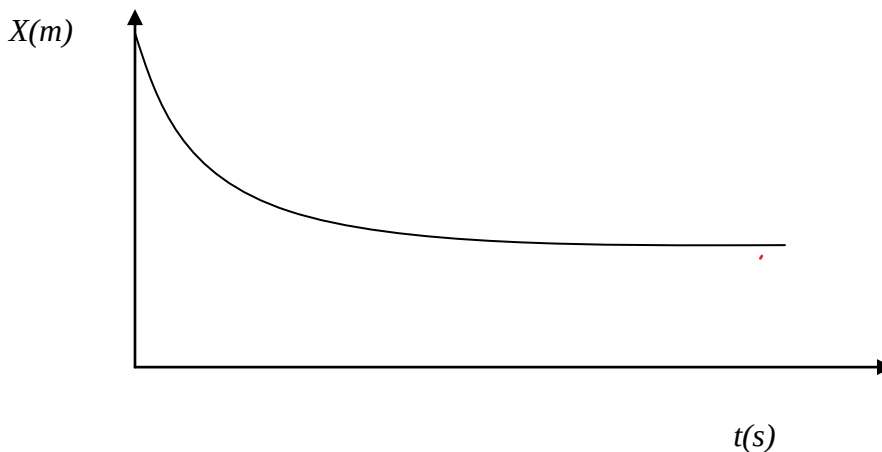
(b) Lightly damped

If a system is slightly damped, oscillations of decreasing amplitude occur.



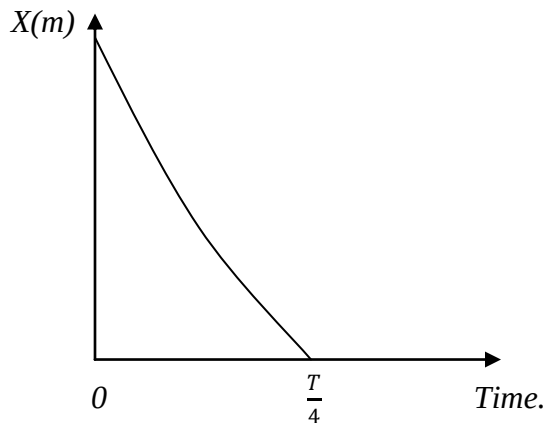
(c) Heavily damped

When heavily damped no oscillations occur and the system returns very slowly to its equilibrium position.



(d) Critically damped

When the time taken for the displacement to become zero is a minimum, the system is said to be critically damped. This time is $\frac{T}{4}$ where T is the period of free oscillation.



Application of damping

Critical damping- instruments such as balances and electrical analogue meters are critically damped. So that the pointer moves quickly to the correct position without oscillating. (the damping is often produced by electromagnetic force); car suspension system.

Light damping- (level meters) on the tape recorders

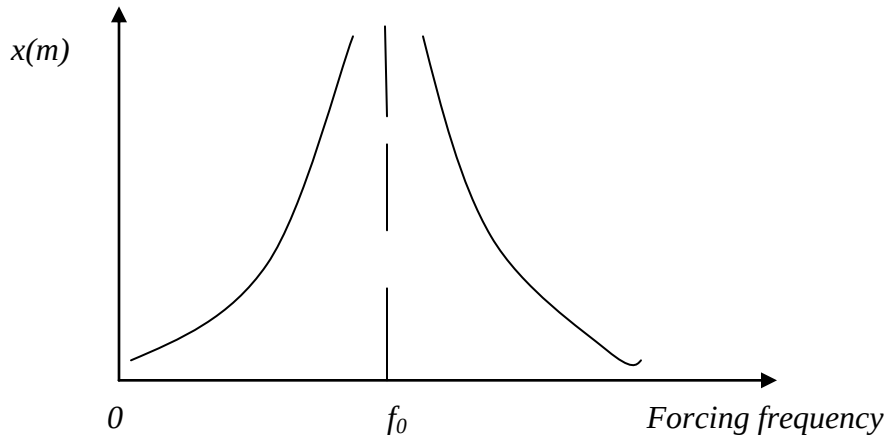
Heavy damping- fuel gauges

FORCED OSCILLATIONS AND RESONANCE

If a system is stimulated by a periodic force, it will be forced to vibrate at the forcing frequency.

If the forcing (driving) frequency is equal to the natural frequency of the system, then the amplitude of vibration will be very large.

If there were no damping to dissipate energy, then the amplitude of vibration would increase continuously as the system absorbed energy from whatever was stimulating it.



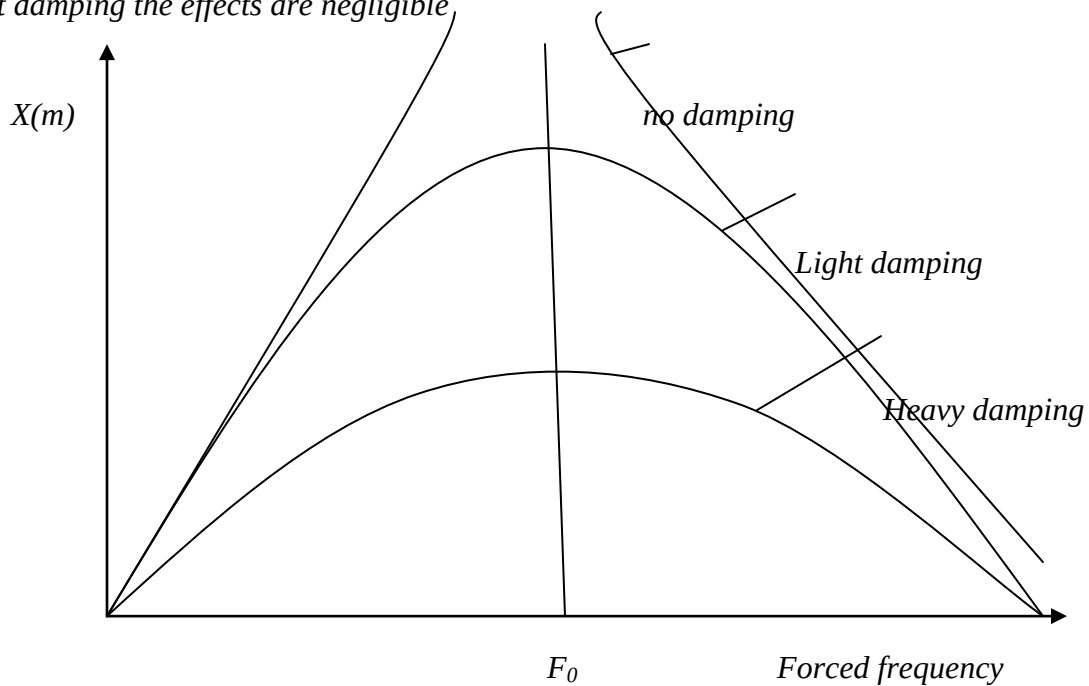
The frequency, at which maximum response occurs, f_0 , is known as the resonant frequency of the system. If the system is undamped, it will be equal to the natural frequency of the system.

Effects of damping on natural frequency

(a) Damping decreases the amplitude of the peak oscillation

(b) The frequency of which maximum response occurs also decreases

The decrease in amplitude and frequency are more marked with heavier damping. With light damping the effects are negligible.



EXAMPLES OF RESONANCE

1. A child on a swing
2. A diver bouncing on a spring board
3. Electrical resonance occurs when a radio circuit is tuned by making its natural frequency for electrical oscillations equal to that of the incoming radio signal.

RESONANCE CAN BE A PROBLEM

At certain speeds, car body panels can resonate with the engine vibrations from the road wheels.

Soldiers marching in step with the same frequency as a natural frequency of a bridge in France in 1880 fell to their deaths when the large amplitude oscillations they caused collapsed the bridge.

3.2: WAVES

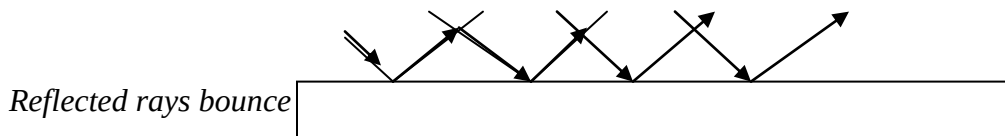
REFLECTION AND REFRACTION OF LIGHT

Definition of terms

- (1) Ray a line which represents the direction of travel of wave
- (2) Normal- an imaginary line that makes a right angle with the plane surface or a boundary between two media.
- (3) Incident ray- a ray that approaches (is incident upon) a plane surface or a boundary between media.
- (4) Reflected ray- a ray that bounces off a plane surface or boundary between media and returns to the original medium.
- (5) Refracted ray- a ray that is transferred between media
- (6) Angle of incidence- the angle that the incident ray makes with the normal at the point of incidence.
- (7) Angle of reflection- the angle that the reflected ray makes with the normal at the point of incidence.
- (8) Angle of refraction- the angle that the refracted ray makes with the normal at the point of incidence.

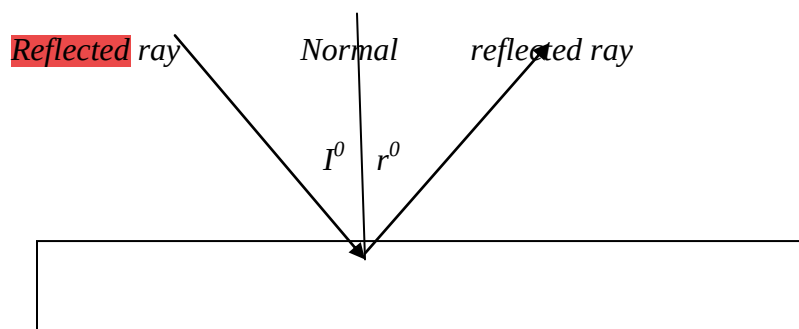
REFLECTION

Occurs when a wave encounters a plane surface and bounces back into the original medium through which it is travelling. There are two types of reflection. Specular reflection occurs on a smooth plane. While Diffuse reflection occurs on uneven surface.



LAWS OF REFLECTION

- 1) The incident ray, the reflected ray and the normal, at the point of incidence, lie in the same plane.
- 2) The angle of incidence is equal to the angle of reflection



Angle of incidence i = angle of reflection r

PLANE (FLAT) MIRRORS

A plane mirror is simply a flat thin glass with a silvered black surface. The silver surface reflects all the light and the glass is thin to avoid refraction occurring at the surface. The reflected light rays constitute the image that we see.

PROPERTIES OF THE IMAGE IN A PLANE MIRROR

The reflection in a plane mirror has the five main properties

1. Virtual-This means that the image.
 - (i) is not produced by real rays crossing
 - (ii) It cannot be formed on a screen
 - (iii) the rays appear to come from the image

2) The image and object are equidistant from the mirror line. This explains why if an object is brought near the mirror surface, the reflection also seems to approach the mirror.

3) Not magnified-The image is rotated so that the left side is on the right and the right is on the left

4) Upright

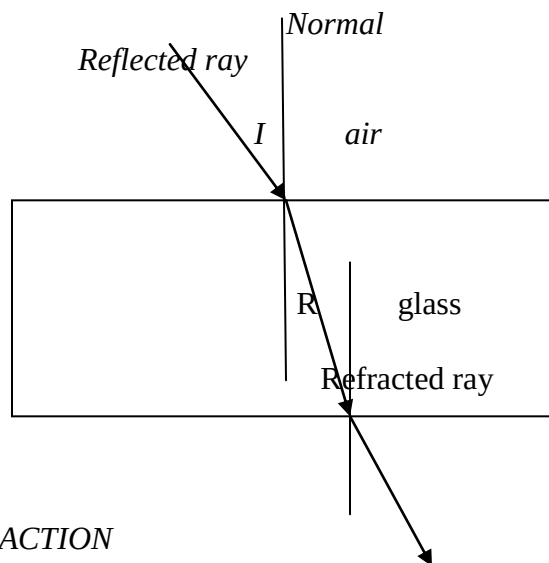
5) Laterally inverted- the image is rotated so that the left side is on the right and the right is on the left.

REFRACTION

When a ray is transmitted from one medium to another of a different optical density, refraction occurs.

In general refraction causes a change in the speed of a wave, bending it.

In monochromatic light, refraction is dependent on the wavelength of the wave.



LAWS OF REFRACTION

1) The incident ray, the refracted ray and the normal, at the point of incidence, lie in the same plane.

2) The ratio of the sine of the angle of incidence to the ratio of the sine of angle of refraction is a constant of two media.

$\frac{\sin i}{\sin r} = n$ (constant) $n_1 \sin \theta_1 = n_2 \sin \theta_2$. n_1 and n_2 are refractive indices for medium 1 and 2 respectively.

This is called Snell's law.

The constant, given the symbol n is called the refractive index and is constant for a pair of media, for example air – water, air- glass or water – glass.

Since refraction causes a change in the wave speed, the ratio of the speed of the incident ray to the speed of the refracted ray also gives the refractive index. $n = \frac{v_i}{v_r}$

v_i – speed of the incident ray.

v_r – speed of the refracted ray.

If we consider the speed of light in a vacuum the equation can be modified to $n = \frac{c}{v}$
 $c = 3 \times 10^8 \text{ ms}^{-1}$ and v is the speed of light in the medium.

NB if light travels from an optically less dense medium to an optically dense medium, the refracted ray bends towards the normal. If it travels from dense to less dense, the refracted ray bends away from the normal.

CRITICAL ANGLE AND TOTAL INTERNAL REFLECTION

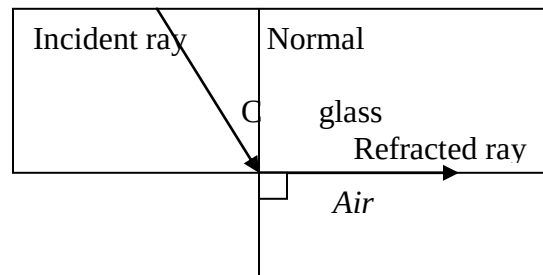
CRITICAL ANGLE

When light move from denser to less dense medium.

Critical angle of a ray is the greatest angle of incidence for which refraction can occur.

Or Is the angle of incidence above which total internal reflection occurs.

Or Is the incidence angle for which the refracted angle is equal to 90° .



TOTAL INTERNAL REFLECTION

Total internal reflection is an optical phenomenon that occurs when a ray of light moving from a dense medium to a less dense medium, strikes the medium boundary at an angle of incidence larger than the critical angle, with respect to the normal to the surface. The ray of light is reflected back to the dense medium.

APPLICATION OF TOTAL INTERNAL REFLECTION

Optical fibre-used for communication

Endoscope –used for image internal structure eg alimentary canal.

ADVANTAGES OF FIBRE OPTICS OVER COPPER CABLES.

1. Cheap
2. Fast-data is transferred at the speed of light.
3. Large data volumes and bandwidth (range of frequencies for information transmission)
4. Conserve energy compared to copper cables because they are insulators (copper cables lose energy as heat)
5. Immune to changes in physical conditions, for example rain, wind or snow.
6. Secure- no cross talk or compromised transmission.
7. Very little intensity is lost- data is completely intact throughout the transmission, for example no missing or damaged data.
8. Viewing images produced in inaccessible location, for example the oesophagus and the stomach when checking for ulcers or examining the colon without performing surgery.
9. Little maintenance over a long period.

WAVE

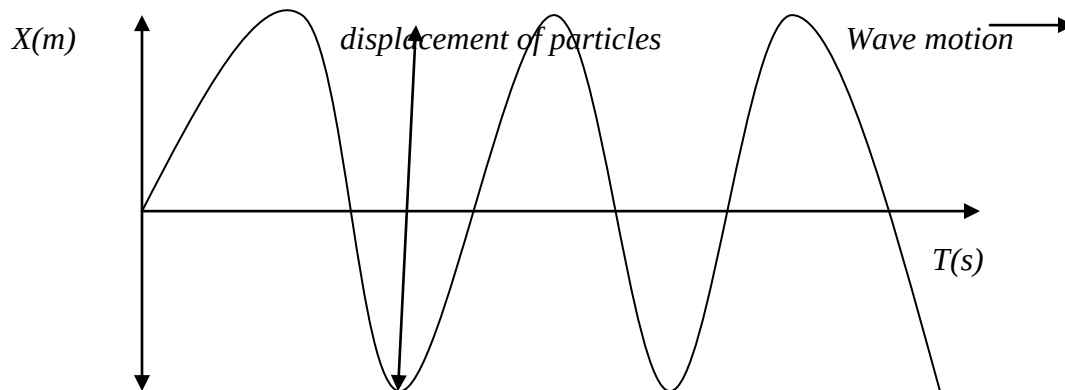
Wave is a disturbance that propagates from one place to another.

Or is a self propagating disturbance in a medium which changes either in form or position or both.

Wave is also able to carry energy, momentum, information from one point to another without the net transfer of matter.

TRANSVERSE WAVES

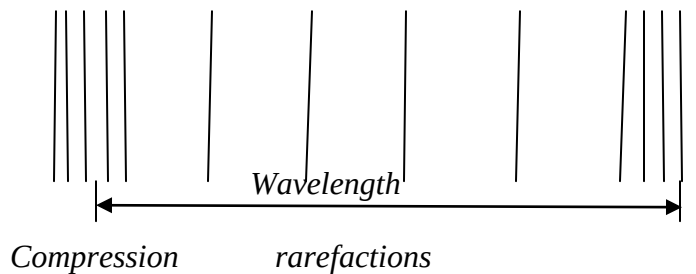
The displacement of individual particles is perpendicular to the direction of propagation of the wave.eg all electromagnetic waves



LONGITUDINAL WAVE

The displacement of individual particles is in the same direction as the direction of propagation of the wave eg sound

The particles of air move back and forth such that a series of compression and rarefaction are observed.



MECHANICAL WAVES

Require a medium for its propagation eg sound.

USEFUL QUANTITIES IN DESCRIBING WAVES

(a) Wavelength- the distance over which a wave repeats, eg the distance between successive crests and the distance between successive troughs.

(b) Angular frequency- is the rate of change of angular displacement.

(c) Frequency is the number of complete oscillations per unit time. $\omega = 2\pi f$

(d) Period- is the time taken to complete one oscillation.

(e) Velocity is the distance that the wave travels per unit time

(f) Amplitude is the maximum displacement of a particle from the equilibrium position

(g) Intensity the intensity of a wave motion at a point is defined as the power per unit area at that point.

The energy associated with a wave is directly proportional to the square of the amplitude of the wave $I = k a^2$.

(h) Phase difference is the difference in angular displacement between particles oscillating.

(i) Angular wave number $k = \frac{2\pi}{\lambda}$ λ is the wavelength.

DERIVATION OF $V = f\lambda$

If a wave starting from rest at time $t=0$, travels a complete wavelength in time t , the speed V of the wave can be defined by the equation.

$$\text{Speed} = \frac{\text{distance travelled in one wave}}{\text{time taken to travel one wave}} = \frac{\text{wavelength } \lambda}{\text{period } T}$$

$$V = \lambda/T \text{ but } \frac{1}{T} = f. \text{ Therefore } V = f \cdot \lambda$$

WAVE EQUATION

Of a progressive wave; both transverse and longitudinal are progressive waves
The wave profile moves along with the speed of the wave.

$Y = A \sin(kx - \omega t + \phi)$ for a wave propagating to the right. A is the amplitude, k – wave number, ϕ phase difference, ω - angular frequency, x at any distance x , t at any time t .

Or $Y = A \cos(kx - \omega t + k\phi)$ that is $\cos$ and $\sin$ affect the origin but does not affect direction.
The equation can be given as $Y = A \sin k(x + ct)$ c is the speed.

Qn The wave function for a harmonic wave in a string is $Y(x,t) = 0.001 \sin(62.8x + 314t)$ where y and x are in metres and t , in seconds.

(a) In what direction does the wave travel and what is its speed.

(b) Find the frequency, wavelength, period of the wave.

(c) What is the maximum displacement of any segment

Solution

(a) Negative x direction or to the left. $Y = A \sin k(x+ct)$; $Y = 0.001 \sin 62.8(x + \frac{314t}{62.8})$

$Y = 0.001 \sin 62.8(x+5t)$; $k = \frac{2\pi}{\lambda}$ but $k = 62.8$ therefore (wavelength = 0.1m)

$C = f\lambda$ $f = 50\text{Hz}$ and $T = 0.02$ seconds.

(b) Amplitude = 0.001m.

$$Y = \gamma \sin \left(\omega t - \frac{2\pi x}{\lambda} \right)$$

Qn $Y(x,t) = 0.02 \sin(0.4x - 50t + 0.8)\text{m}$

Find (a) amplitude (b) wavelength (c) period (d) initial phase (e) wave speed.

Solution

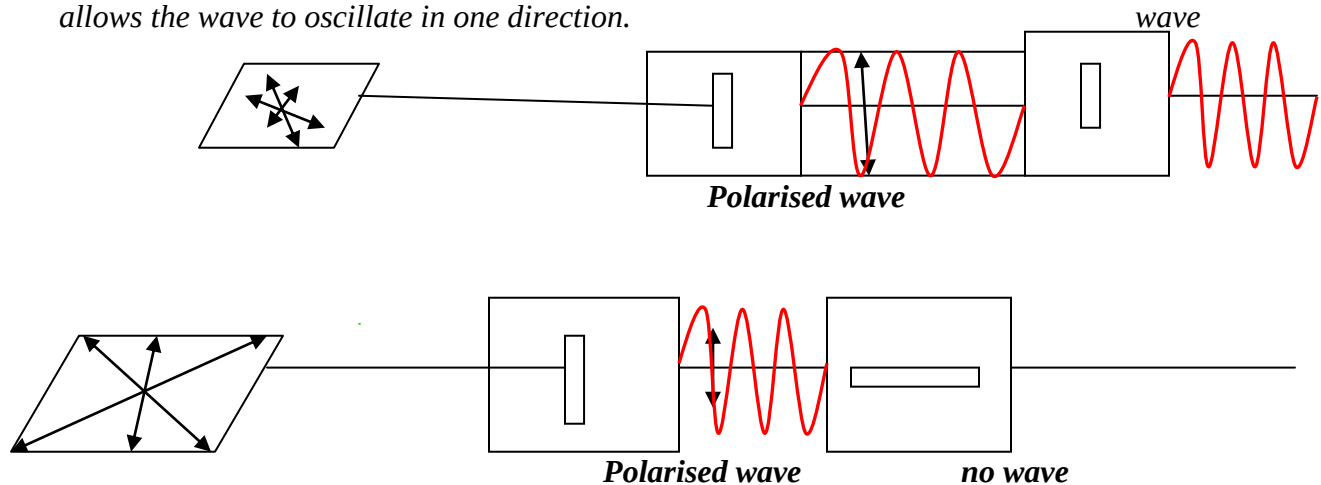
$Y = A \sin(kx - \omega t + \phi)$

(a) amplitude = 0.02m (b) $k = \frac{2\pi}{\lambda}$ but $k = 0.4$: $\lambda = 15.7\text{m}$ (c) $\omega = 2\pi f$ but $\omega = 50$; $f = \frac{2\pi}{50}$ and

$T = \frac{50}{2\pi}$ (d) $c = \frac{\omega}{k} = 50/0.4 = 125\text{m/s}$.

POLARISATION

Is a property which distinguishes a transverse from a longitudinal wave? Polarisation allows the wave to oscillate in one direction.



It is important to note that no polarization can be obtained with longitudinal waves.

ELECTROMAGNETIC SPECTRUM

Electromagnetic waves are a family of waves which travel in a vacuum at the same speed.

The electromagnetic spectrum

Radio, microwave, infrared, visible light, ultraviolet, x rays and gamma rays in order of increasing frequency.

USES OF ELECTROMAGNETIC WAVES

WAVE	USES
Radio	Radio and television communication systems. Amplitude modulation and frequency modulation are examples of radio signals.
Microwave	Satellite television and telephone networks, microwave ovens, telemetry(satellite communications)
Infrared	Electrical appliances eg cooking stoves, remote controllers for TV, intruder alarms, physical therapy, infrared photography(used to check for cancer growths and tumour)m, fibre optics.
Visible light	Fibre optics in medicine and telecommunications
UV	Sun beds, fluorescent tubes, sterilisation of medical and surgical equipment.
X-rays	In medicine as diagnostic tool, eg to check for broken bones, treatment of cancer and security.
Gamma rays	Medical treatment of cancer, sterilisation of surgical or medical instruments, irradiation of food.

PROPERTIES

1. Electromagnetic waves consist of varying perpendicular electric and magnetic field.
2. They are unaffected by electric and magnetic field.
3. All travel with the same speed in vacuum (speed of light).
4. They travel in straight line.
5. They can be polarised
6. They continue to travel after the source is turned off.
7. They travel through empty space that is they do not require a medium.
8. They can travel in a vacuum.

X RAYS

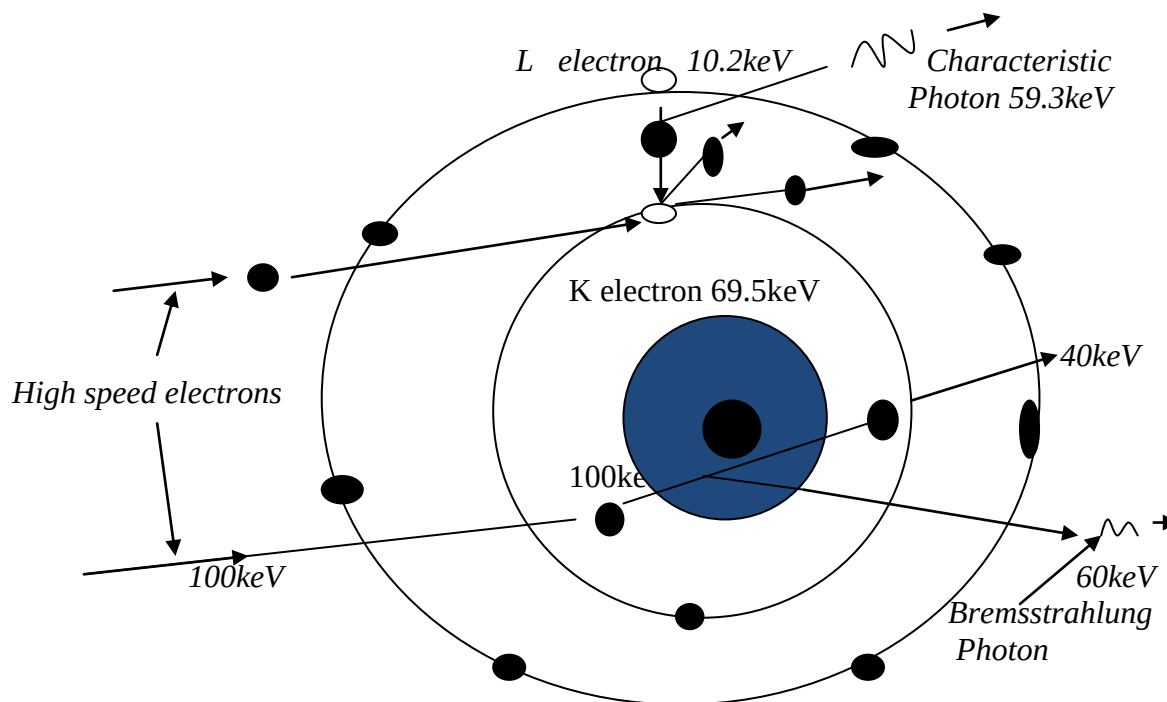
X rays are waves similar to light waves, but of much shorter wavelength; about 10^{-10} m or 0.1nm. They are produced when fast electrons, or cathode rays, strike a target, such as the walls or anode of a low pressure discharge tube.

The electrons are provided by thermionic emission from a white hot tungsten filament. X rays are produced by bombarding matter with electron particles. The production of X-rays is therefore the inverse process to the photo electric effect, where electrons (particles) are liberated from metals by incident light waves

PRINCIPLE OF X –RAY PRODUCTION

X-rays are produced by bombarding metal targets with high speed electrons. Two types of interactions with the target produce radiation.

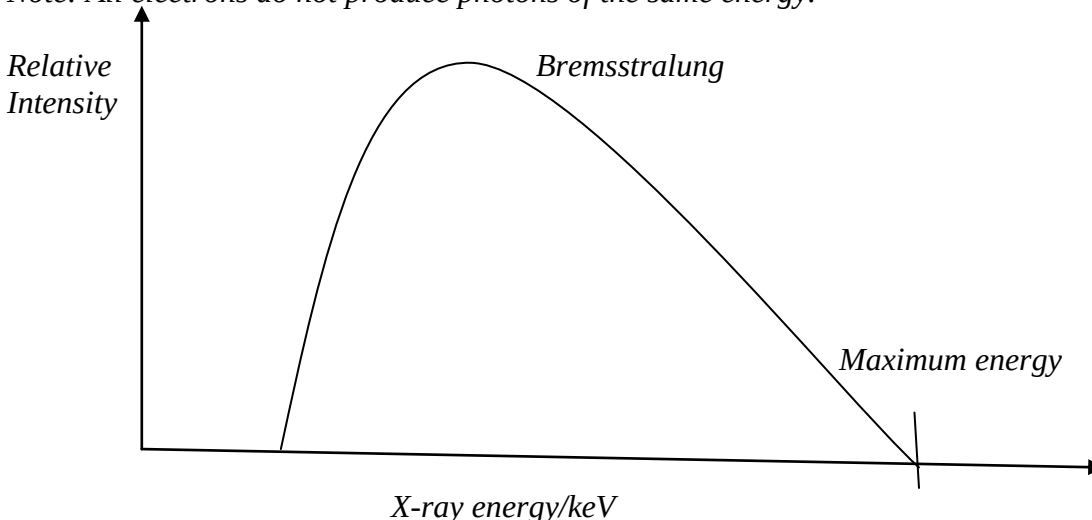
1. An interaction with electron shells produces characteristic X-ray photons.
2. Interactions with the atomic nucleus produce Bremsstrahlung X-ray photons



BREMSSTRAHLUNG

Electrons that pass close to a nucleus are deflected and slowed down by the attractive force from the nucleus. The energy lost by the electron during this encounter (deceleration) appears in the form of an X-ray photon. The interaction process that produces the most photons is the Bremsstrahlung process.

Note: All electrons do not produce photons of the same energy.



In reality, the Bremsstrahlung spectrum looks more like this due to effects of X-ray **tube filtration**

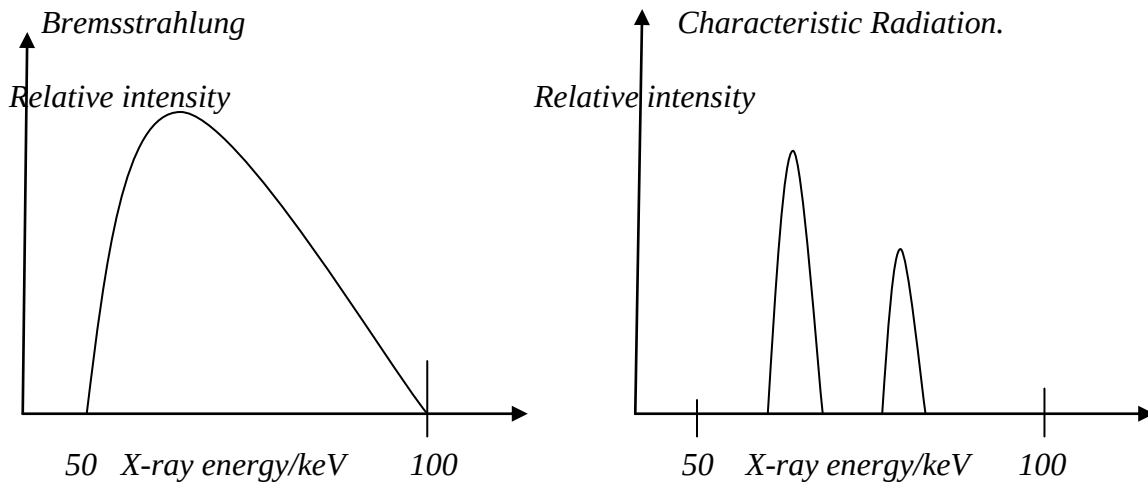
CHARACTERISTIC RADIATION

Interaction occurs only if incoming electron has kinetic energy (E_K) greater than the energy of orbital electron within the atom (E_B). Electron is dislodged from the atom and leaves a vacancy that is filled by an electron from a higher energy level. As filling electron moves down to fill the vacancy, it gives up energy emitted in the form of an X-ray photon. The energy of the photon is characteristic of the chemical element that serves as the anode material.

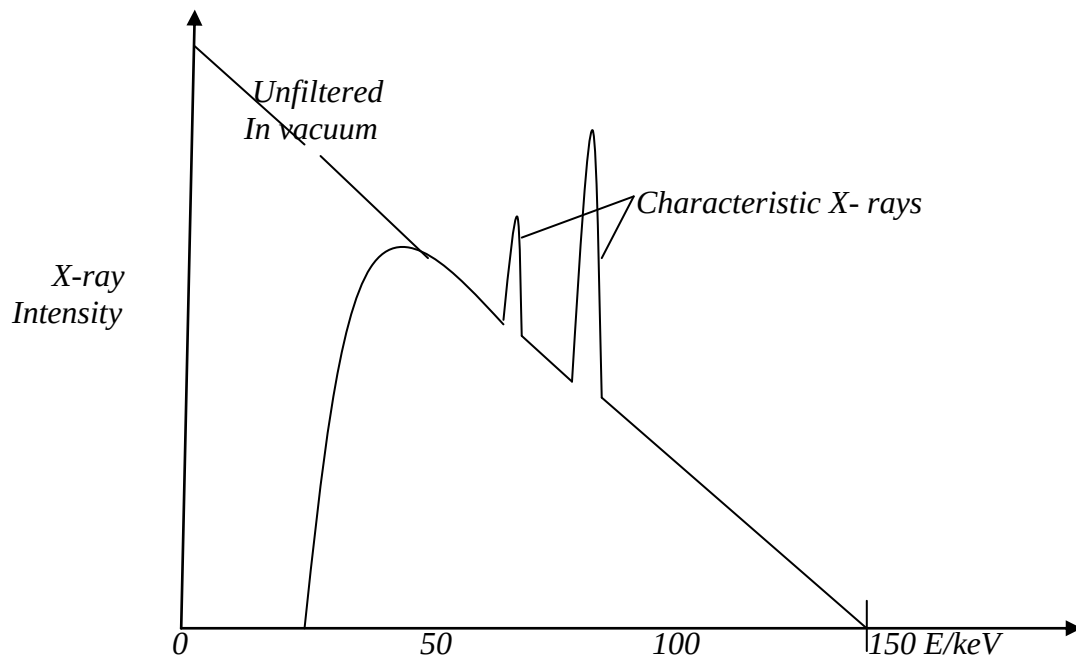
Example of Characteristic radiation (Tungsten Target)

Electron dislodge a tungsten K-shell electron ($E_B=69.5\text{keV}$). Vacancy is filled by an electron from the L-shell ($E_B=10.2\text{keV}$). The energy of the characteristic X-ray photon released from this transition is equal to the energy difference between these two levels (59.3keV)

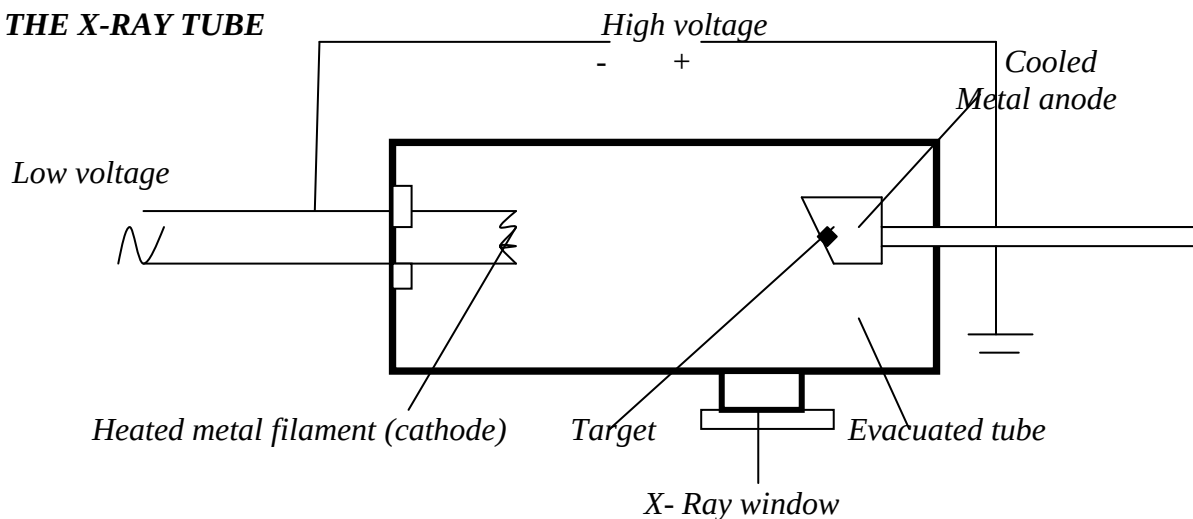
Characteristic radiation produces a line spectrum whereas, Bremsstrahlung produces photon energies over specific range.



THE TYPICAL SPECTRUM OF X-RAY PHOTONS PRODUCED BY BOMBARDING METAL TARGET WITH ELECTRONS.

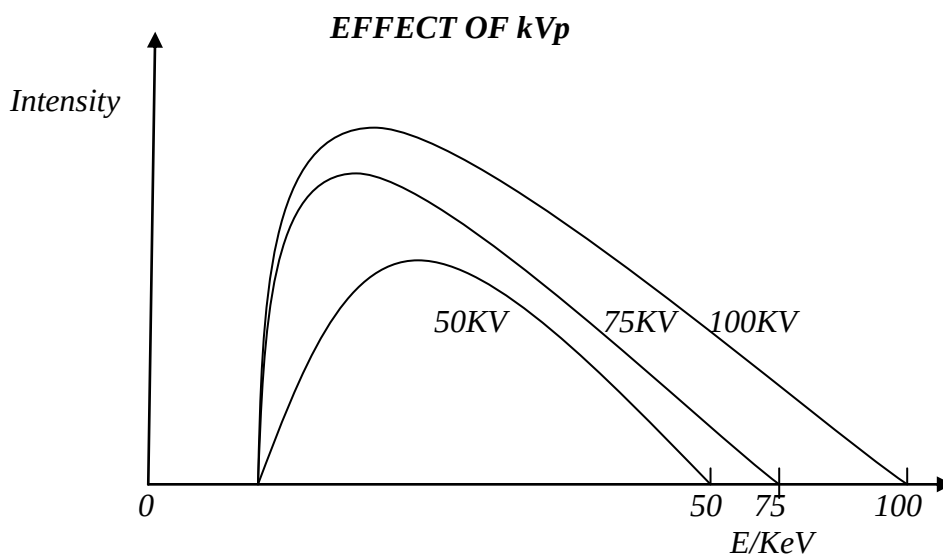


THE X-RAY TUBE



Note: Most X-ray tubes are tungsten, because it has a high melting point (3422°). The anode is rotated in order to aid with heat dissipation.

Energy used by the X-ray tube to produce X-rays is supplied by an electrical circuit from the generator. The high D.C voltage in kilo voltage (kV) KV establishes the energy of the electrons as they reach the anode. No X-ray photon can be created with energy greater than that of the electrons. Maximum photon energy, therefore in keV is numerically equal to the maximum applied potential in(kV). In most equipment maximum photon energy coincides with kVp.

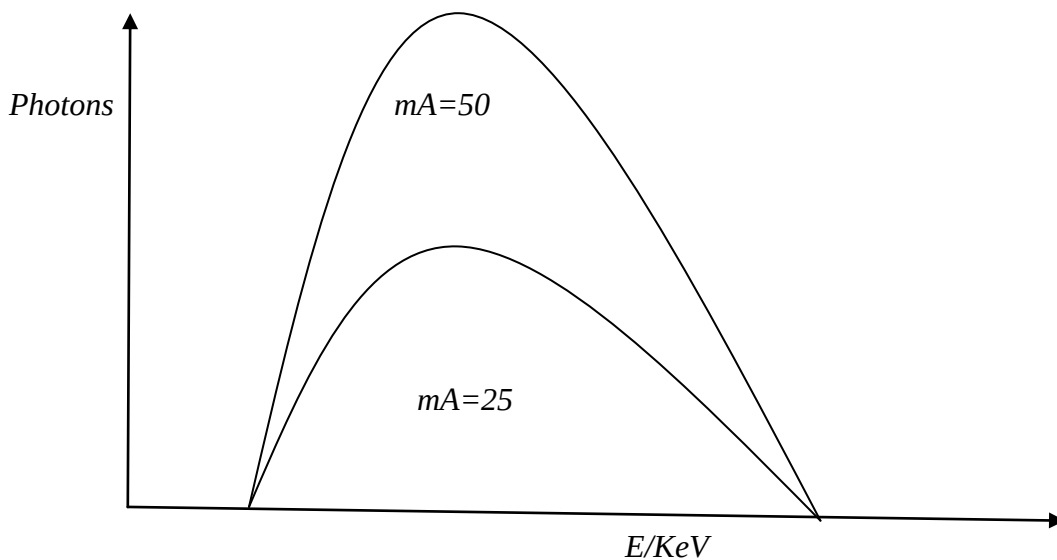


Decreasing KVp, decreases the intensity and the maximum energy of X-ray photons.

The electrical current that flows through the tube (mA), can be adjusted on the circuit.

EFFECTS OF mA

The tube current is controlled by the filament/ heater current. The greater the filament current, the greater the amount of thermo-electrons available for acceleration.



Decreasing the current (mA), decreases the number of photons, but the maximum energy of the photons remain the same.

X-RAY BEAM FILTRATION

Some X-ray photons are of such low energy, they would not be able to penetrate the subject. They contribute to radiation dose to the patient without any useful purpose. There is need to block off such photons from the X-ray spectrum before they reach the patient. Aluminium filter is frequently fitted across X-ray window so as to block off soft radiation.

IMAGE QUALITY

Quality of the image obtained depends on the sharpness and contrast of the image.

Sharpness is the ease with which edges of structures can be determined from image.

Contrast is the difference in optical density or the degree of blackening in image.

FACTORS THAT AFFECT SHARPNESS

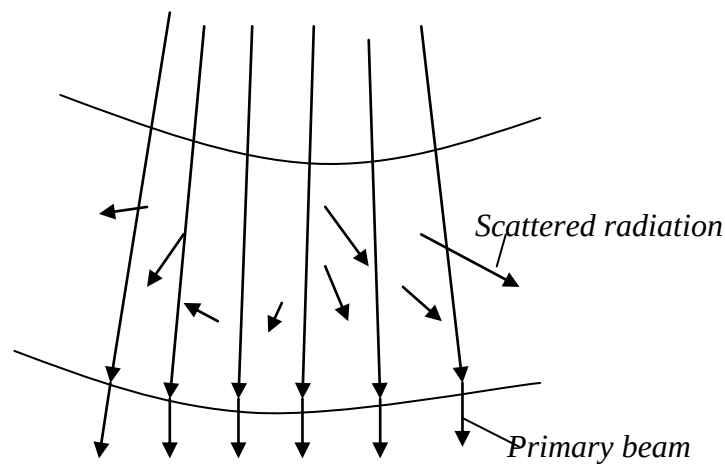
1. Movement of subject during data acquisition.
2. Nature of X-ray source (Not point source but infinite source)- Geometric sharpness
3. Distance of object from image receptor. (Geometric sharpness).
4. Nature of image receptor (Photographic sharpness)

FACTORS AFFECTING IMAGE CONTRAST

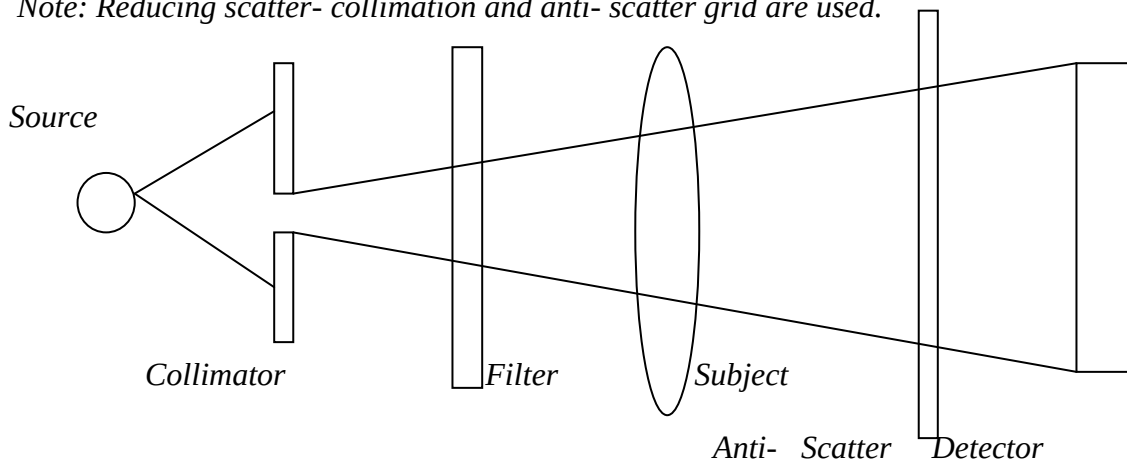
1. Energy of the X-ray beam (KV).

HIGH CONTRAST	VS	LOW CONTRAST
-Short scale		-long scale
-50kV(800mA)		-110kV(20mA)

2. SCATTERED RADIATION



Note: Reducing scatter- collimation and anti- scatter grid are used.



3. Body part.

Contrast agents can be used to enhance contrast in body parts of low inherent contrast. For example Barium is a good absorber of X-ray photons. As a result, when the barium sulphate solution coats the inside of the stomach, the outline of the stomach will show up clearly on the image.

ATTENUATION OF X-RAYS IN MATTER

When X-ray photons radiate from a source in a vacuum, the intensity decreases in proportion to the inverse of the square of the distance from the source.

$I \propto \frac{1}{d^2}$. Approximately the same

In a medium where absorption processes are occurring. Intensity of a parallel beam decreases by a constant fraction when passing through equal small thickness of the medium. Gives rise to an exponential decrease in transmitted beam intensity.

Beer's law $I = I_0 e^{-\mu x}$

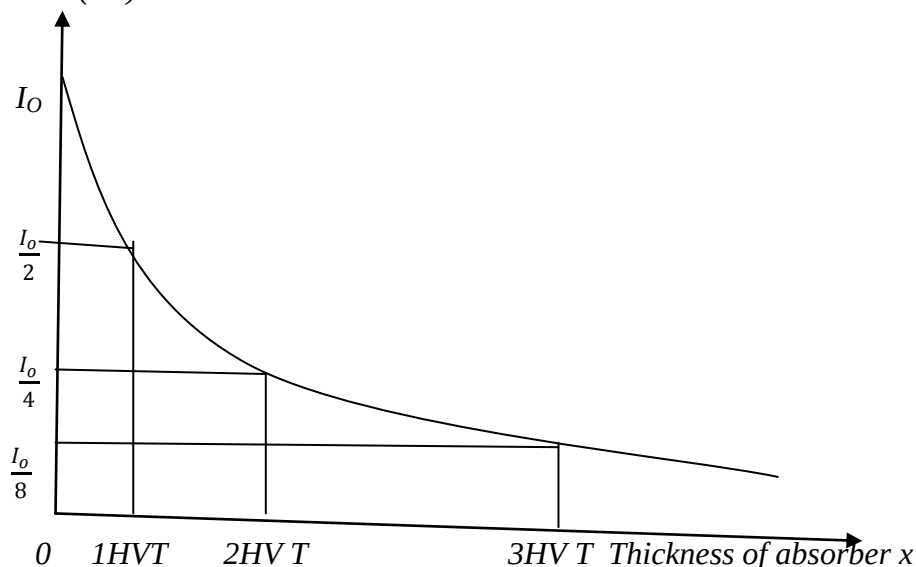
I = Transmitted intensity

I_0 = Initial intensity

μ = Linear attenuation coefficient (cm^{-1})

x = thickness of material (cm).

Percentage
Transmission
Intensity



HALF VALUE THICKNESS (HVT) $x_{\frac{1}{2}}$:

Half value thickness is the thickness of medium required to reduce the transmitted intensity to one half of its initial value.

$$x_{\frac{1}{2}} = \frac{\ln 2}{\mu}$$

EXAMPLE

The linear absorption coefficient of copper is 0.693cm^{-1} . Calculate

(a) the thickness of copper required to reduce the incident intensity by 50% (The HVT)

(b) The fraction of the incident intensity of a parallel beam that is transmitted through a copper plate of thickness 1.2cm

SOLUTION

$$(a) I = I_0 e^{-\mu x}$$

$$\frac{I}{I_0} = 0.5 = \exp(-0.693x_{\frac{1}{2}})$$

$$x_{\frac{1}{2}} = 1.0\text{mm}$$

$$(b) \frac{I}{I_0} = \exp(-0.693 \times 12) \\ = 2.4 \times 10^{-4} \text{cm.}$$

CT SCANNING

In 1972 Sir Godfrey Hounsfield developed the first CT machine that could image slice Slices were free of superimposition from over and underlying structures. This was achieved by obtaining various measurements at different angular positions. CT is a procedure which forms a 3 dimensional plane of the object.

ADVANTAGES OF CT SCANNING

1. Produce images that show three-dimensional relationships between different tissues.
2. Can distinguish tissues with quite similar densities.

QUESTION

(a) Distinguish between images produced by CT Scanning and X-ray imaging[3]

(b) By reference to the principle of CT Scanning suggest why CT Scanning could not be developed before powerful computers were available [5]

SOLUTION

(a)

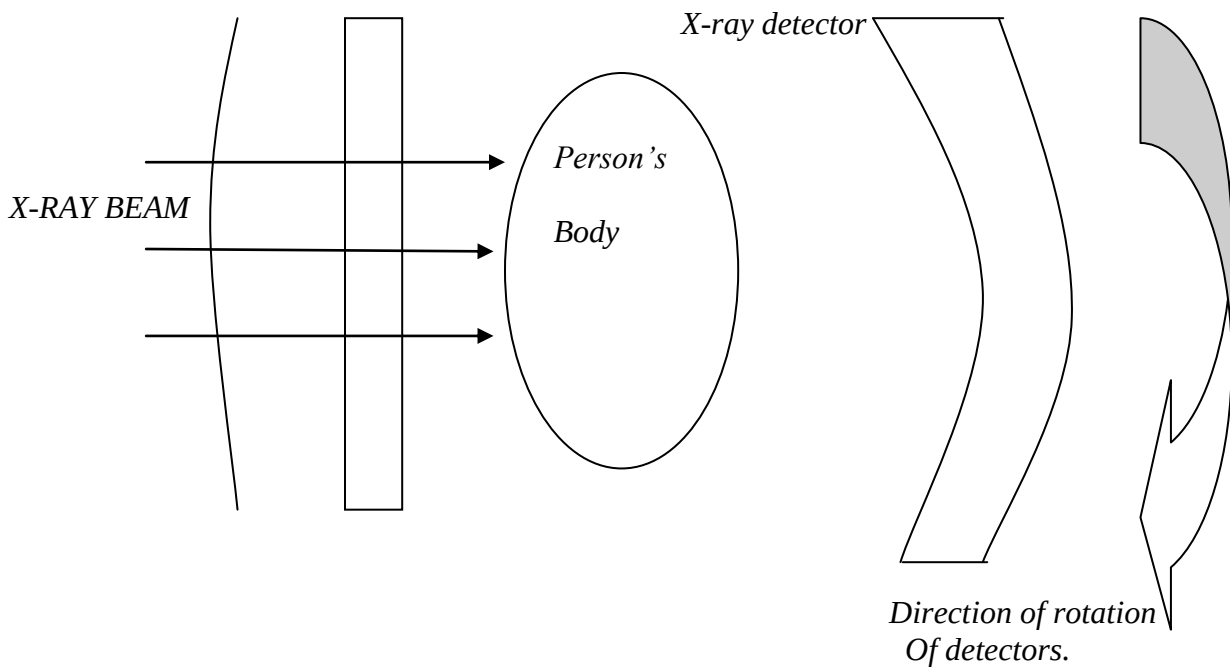
CT SCANNING	X-RAY IMAGE
Produce images that show three-dimensional relationships between different tissues	Produce two dimensional images
Can distinguish tissues with quite similar densities.	Produce shadow image of the whole structure

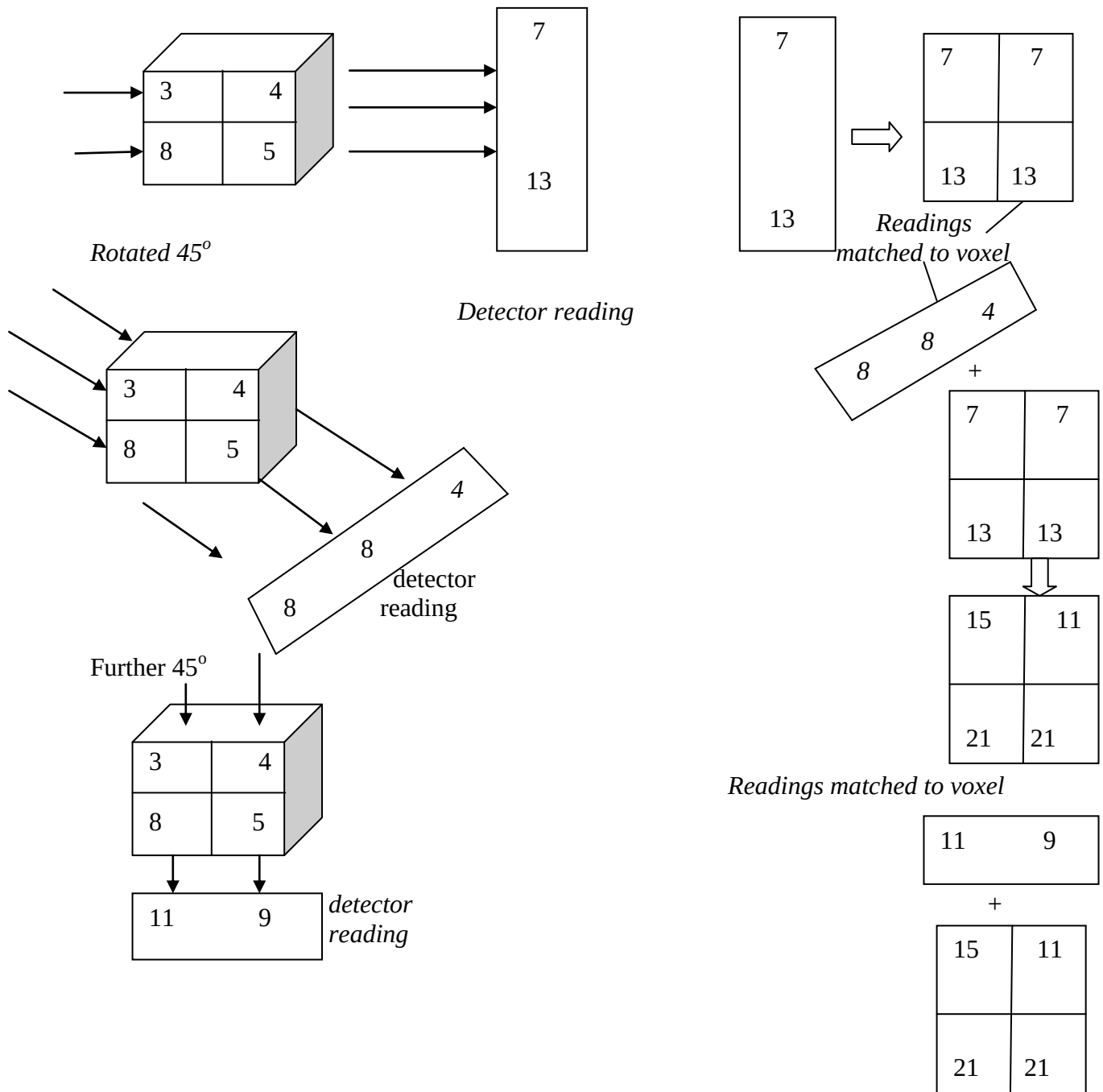
(b) X-Ray images of slices are taken from different angles.

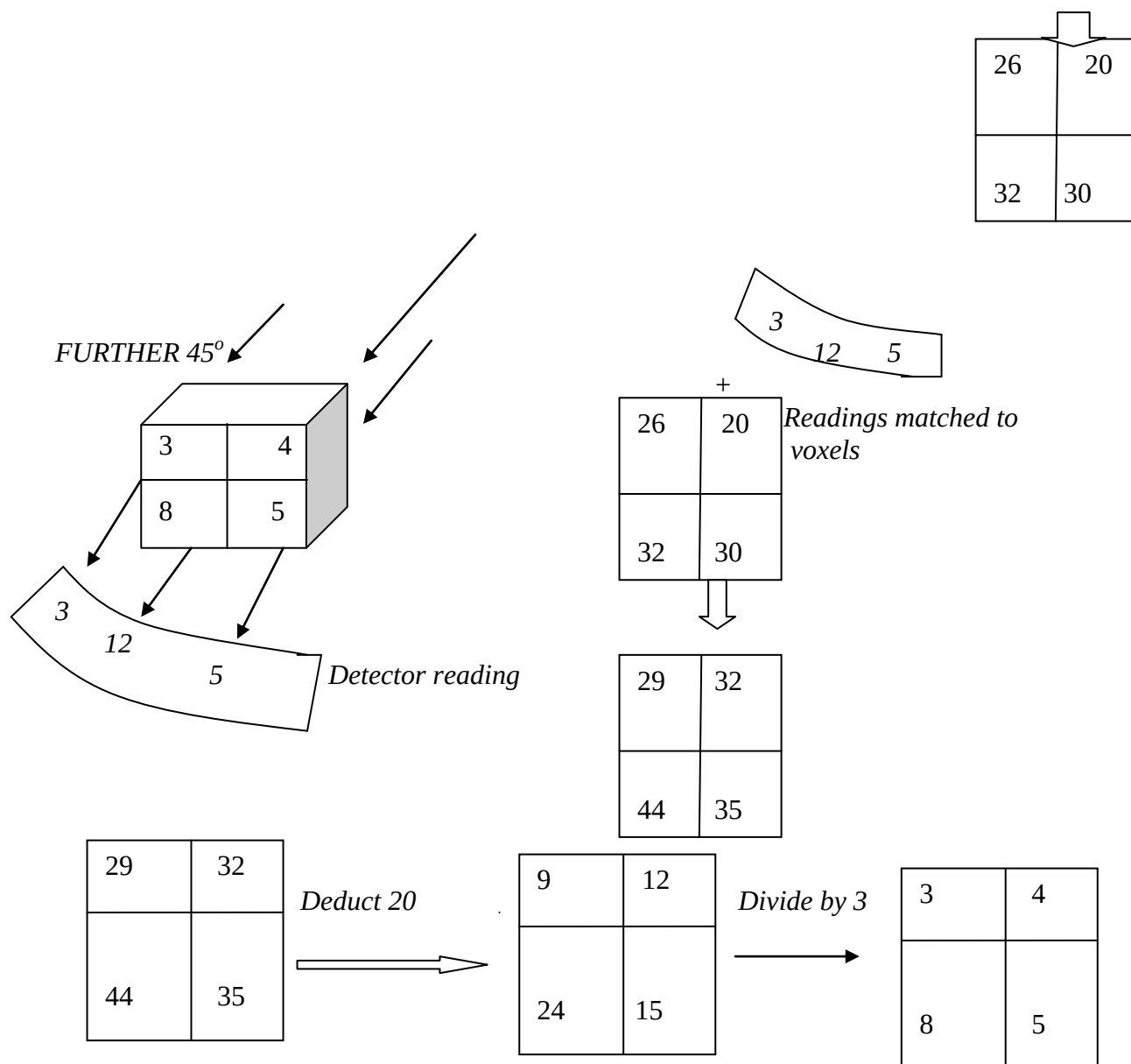
These images are combined and processed and repeated for many different slices. A three dimensional image is built, which can be rotated. A computer is required to store and process huge quantity of data.

VOXEL DEVELOPMENT IN (CT SCANS)

A voxel is a small cube in three dimensional image. The number in each square is the density that the computer will register for that section of the object. As the scanner goes around each part has a different density which the computer can model. The computer then put these together to form a three dimensional shape. The intensity transmitted through each voxel alone can be given a number referred to as a pixel. The various pixels are built up from measurement of the X-ray intensity along different directions through the section or slice.







Having summed all measurements, it remains to reduce the measurements to that of the original; this is achieved in two stages.

1. The background intensity must be removed. This background is equal to the sum of the detector readings for any one position. In this case, the sum is 20. This background is deducted from each pixel.
2. Allowance must now be made for the fact that more than one view was made of the section. In this example, there were four sets of readings and consequently each pixel reading is divided by 3.

EFFECTS AND USES OF X RAYS

X-rays are used in medicine as diagnostic tool, for example to check for broken bones, treatment of certain forms of cancer, and for security.

When X rays strike many minerals, such as zinc sulphide, they make them fluoresce.

If a human or other body is placed between an x ray tube and a fluorescent screen, the shadow of its bones can be seen on the screen, because they absorb X-rays more than flesh does.

Unusual objects, such as swallowed safety-pins, if are dense enough, can also be located.

X ray photographs can likewise be taken with the plate in place of the screen.

In this way cracks and flaws can be detected in metal castings. X rays are used to investigate the structure of crystals.

3.3: SUPERPOSITION**Stationary waves and superposition****Superposition**

Is the combination of two or more waves to form a resultant?

THE PRINCIPLE OF SUPERPOSITION

States that at a point where two or more waves meet, the instantaneous displacement is the vector sum of the individual displacements due to each wave at that point.

NB The wave must be of the same type.

STANDING OR STATIONARY WAVES.

Two sinusoidal travelling waves of the same wavelength, same frequency and the same amplitude, travelling in opposite direction overlapping in the same region of space making a standing wave.

Suppose the amplitudes of the waves are equal. $y_1 = a\sin(\omega t - kx)$ travelling from left to right and $y_2 = a\sin(\omega t + kx)$ travelling from right to left or a reflected wave.

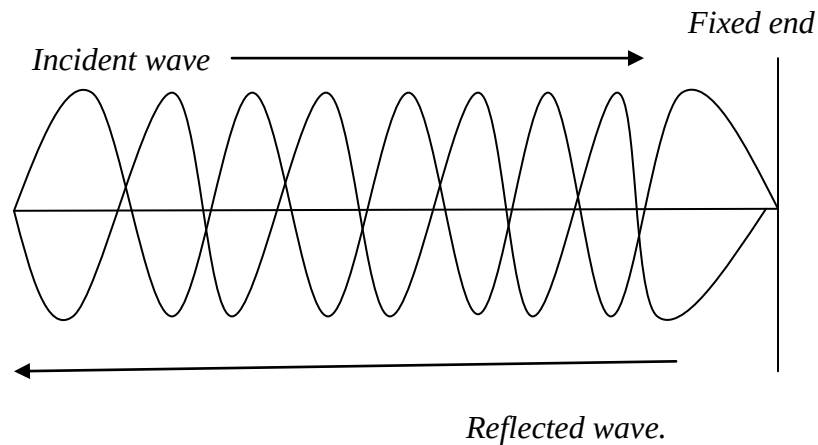
Now the combined wave $y = y_1 + y_2 = a\sin(\omega t - kx) + a\sin(\omega t + kx)$

Using the identity $\sin A + \sin B = 2\sin\frac{1}{2}(A+B)\cos\frac{1}{2}(A-B)$.

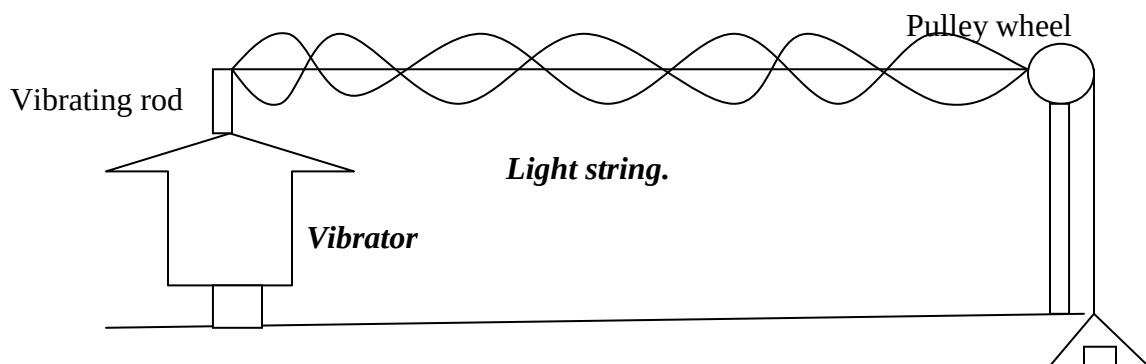
$$y = 2a\sin\omega t\cos kx$$

Amplitude = $2a$ and $\sin\omega t$ is the function of time and $\cos kx$ is the function of position.

If you pluck a guitar string, or blow across the mouth of a pop bottle to create a tone, you have generated standing waves. , In general a standing wave is one that oscillates with time, but remains in its location.

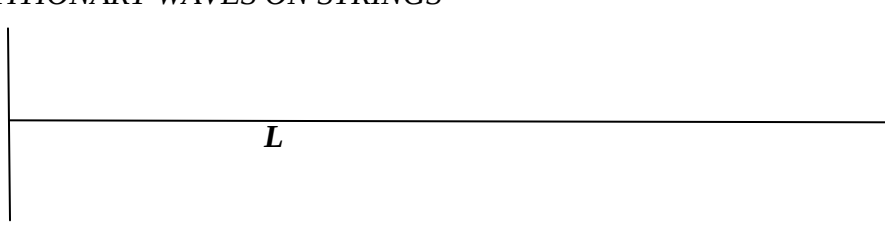


SIMPLE EXPERIMENT TO DEMONSTRATE STATIONARY WAVE

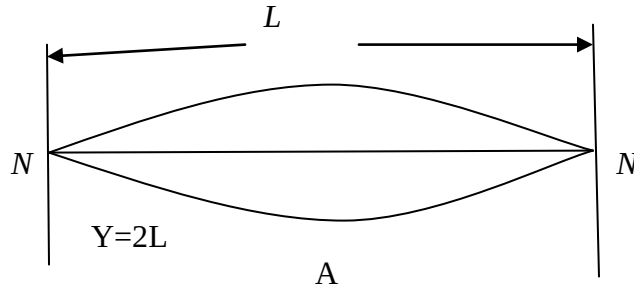


If the weights on the scale-pan are suitably adjusted, a number of stationary vibrating loops are seen on the string when one end is set vibrating. This time the wave-like profile on the string does not move along the medium which is the string, and the wave is therefore called a stationary (or standing) wave. A wave travelling is reflected. So the stationary wave is due to superposition of two waves of equal frequency and amplitude travelling in opposite direction along the string.

STATIONARY WAVES ON STRINGS



A string is tied down at both ends. If the string is placed in the middle a standing wave results. This is the fundamental mode of oscillation of the string. The fundamental consists of one half a wavelength between the two ends of the string. Hence its wavelength is $2L$ or $\lambda = 2L$.



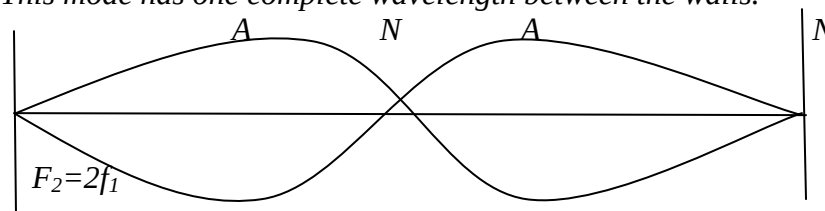
If the speed of waves on the string is V , it follows that the fundamental frequency f_1 is given by $V = f \lambda$ $f_1 = \frac{V}{\lambda} = \frac{V}{2L}$

Note that the fundamental frequency increases with the speed of the waves, and decreases as the string is lengthened. Other than the fundamental frequency, there are an infinite number of standing modes or harmonics for any given string.

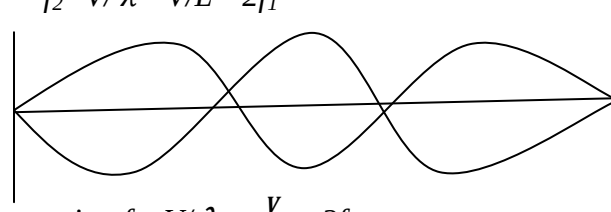
Remarks

Points on a standing wave that stay fixed are referred to as node (N). Half way between any nodes is a point on the wave of maximum displacement is called an anti-node (A).

The second harmonic can be constructed by including one more half wavelength in the standing wave. This mode has one complete wavelength between the walls.



$$f_2 = V / \lambda = V / L = 2f_1$$



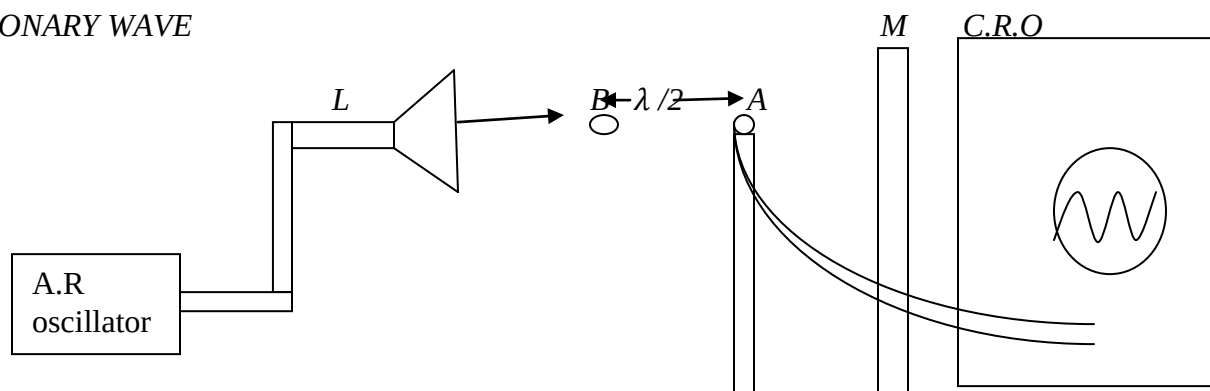
Third harmonic: $f_3 = V / \lambda = \frac{V}{2/3L} = 3f_1$.

In general $y_n = 2L/n$; $f_n = nV/2L$.

COMPARISON OF PROGRESSIVE AND STATIONARY WAVES

PROGRESSIVE WAVE	STATIONARY WAVE
Energy is transferred along the direction of propagation.	No energy is transferred along the direction of propagation
The wave profile moves in the direction of propagation	The wave profile does not move in the direction of propagation
Every point along the direction of propagation is displaced but at different instants.	There are points known as nodes where no displacement occurs.
Every point has the same amplitude	Points between two successive nodes have different amplitudes
Neighbouring points are not in phase	All points between two successive nodes vibrate in phase with one another.

EXPERIMENT TO DETERMINE THE WAVELENGTH OF SOUND USING STATIONARY WAVE



Sound waves of constant frequency, such as 1500Hz, travel from a loud speaker L towards a vertical board M. Here the waves are reflected and interfere with the incident waves. The two travelling in opposite directions produce a stationary wave between board M and L. A small microphone, positioned in front of both M and L is connected to the Y-plate of an oscilloscope. As the microscope is moved back from M towards L, the amplitude of the waveform seen on the screen increases to a maximum at the point A, as shown on the diagram above. This is an antinode of the stationary wave. When the microscope is moved on, the amplitude diminishes to a minimum (a node) and then increases to a maximum again at a position B, the next antinode. The displacement between successive antinodes is $\lambda/2$. Thus by measuring the average distance d between successive maxima, the wavelength can be found.

NOTE: Knowing the frequency f of the note from loudspeaker, the velocity v of the sound wave can be calculated from $V = f\lambda$

INTERFERENCE

In region where wave travels from a coherent source, superposition occur giving reinforcement of the waves at some points and cancellation at others. The resulting effect is called an interference pattern or a system of fringes. Coherent sources have a constant phase difference which means they must have the same frequency and wavelength. If their phase difference is not constant, at a certain point there maybe reinforcement at one instant and cancellation at the next.

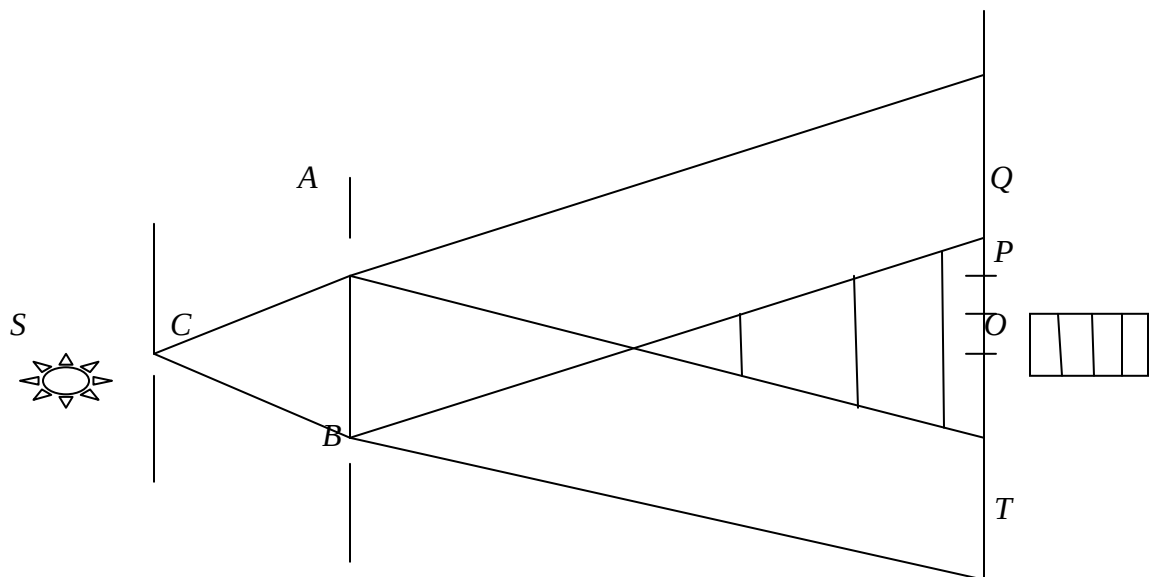
YOUNG'S DOUBLE SLIT EXPERIMENT

It can be understood that two conditions are essential to obtain an interference phenomenon.

- (i). Two coherent sources of light must be produced
- (ii). The coherent sources must be very close to each other as the wavelength of light is very small, otherwise the bright and dark pattern produced some distance away would be too fine and no interference pattern is obtained.

Young placed source S , of monochromatic light in front of a narrow slit C , and arranged two very narrow slits A, B close to each other, in front of C .

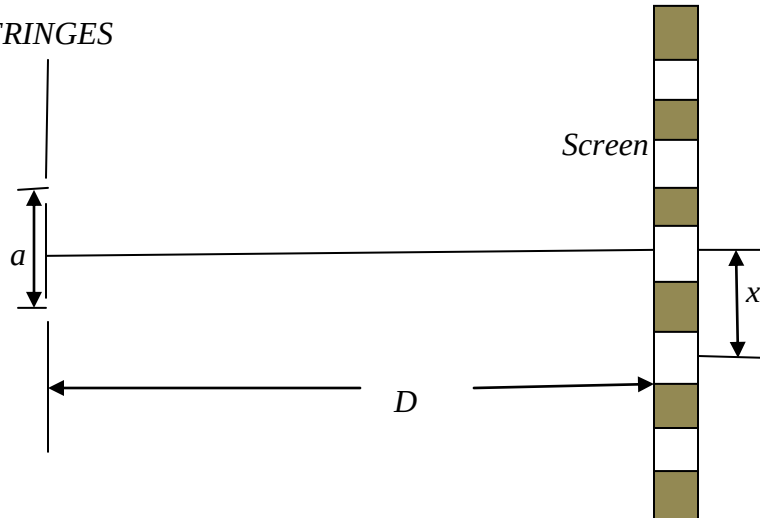
Young observed bright and dark bands on either side of O on a screen T , where O is on the perpendicular bisector AB



Light diverging from A has exactly the same frequency and is always in phase with the light from B . Therefore A and B act as close coherent sources. Interference takes place in the shaded region $AO = OB$, a bright band is obtained at O . At a point close to O , such

that $BP-AP=\lambda/2$ where λ is the wavelength of light from S, a dark band is obtained. At a point Q such that $BQ-AQ= \lambda$, a bright band is obtained, and so on for either side of O.

SEPERATION OF FRINGES



Slit separation – a . Is the distance between centres of slits? Can be measured by a travelling microscope.

x -fringe separation. This is the distance between centres of two adjacent bright or (dark) fringes

D - slits to screen distance. Is the distance from the midpoint of the two slits to the centre of the central bright fringe? Can be measured using a metre rule.

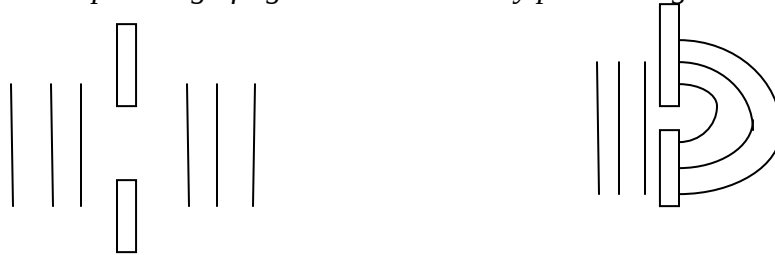
The wavelength can be found by $\lambda = \frac{ax}{D}$

CONDITIONS FOR A TWO SOURCE INTERFERENCE FRINGE

- (a) The sources must be coherent: Wave emitted are phase linked that have zero or constant phase difference. Same frequency. Coherent sources must be derived from a single source.
- (b) For good contrast between maximum and minimum of intensity, the wave amplitude must be approximately equal.
- (c) The distance between the sources must be comparable to the wavelength of the waves to ensure adequate separation of the maxima and minima

DIFFRACTION

Diffraction is the spreading of light waves when they pass through obstacles or apertures.



The extent to which the wave spread depends upon the gap or slit. If the size of the gap is of the same magnitude as the wavelength of the wave the spreading is more pronounced. Diffraction is greatest when the slit and the wave length are equal. Light and sound can also be diffracted

DIFFRACTION GRATING

Diffraction grating provides a very valuable means of studying spectra, and for measuring wavelength accurately, it has replaced a prism.

$$d \sin \theta = n \lambda$$

θ is the angle between the zeroth order fringe to the n^{th} order

λ is the wavelength. n is the order of the fringe from the maxima to the central fringe and d is the slit separation.

SECTION 4: ELECTRICITY AND MAGNETISM

4.1: ELECTRICITY

ELECTROSTATICS

Introduction

Static electricity is the study of charges at rest. There are only two types of charge: positive and negative.

Charging by friction

When two materials are rubbed against one another, electrons are transferred from one material to another. One material becomes negatively charged as a result of excess electrons. The one that loses electrons becomes positively charged. When electrons move from one medium to another, they leave spaces that are deficient of electrons called holes

which are regarded as positively charged. Charge is always conserved that is when electrons transfer they leave an equal number of holes (+ve charges).

Charging by induction

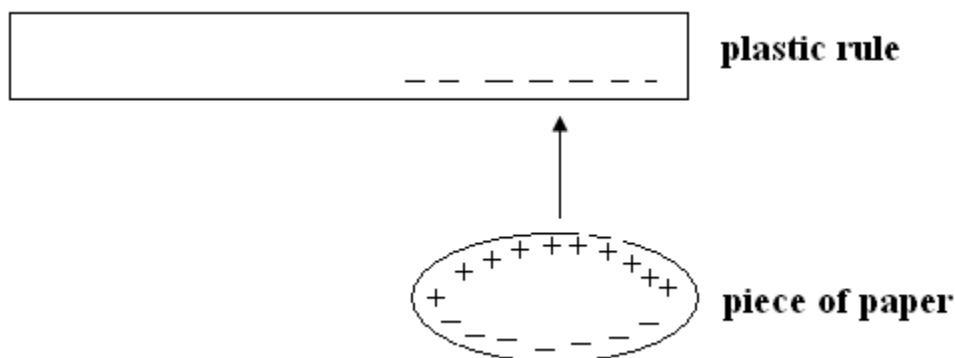


Fig. 1.1: Charging by Induction

Consider a plastic ruler which is negatively charged and a neutral piece of paper (uncharged). When the ruler is held close to the paper, positive charges within the paper are attracted by the negative charges on the ruler since unlike charges attract. Negative charges within the piece of paper are repelled to the far end of the paper since like charges repel. The positive and negative charges generated on the piece of paper are called induced charges. The charge separation is called a dipole.

Electric conductors and insulators

A conductor is a material which allows current to pass through because of the presence of delocalized (free) electrons which can be moved by a potential difference e.g. copper, iron and carbon.

An insulator is a material which does not allow current to pass through because of absence of delocalized electrons e.g. rubber, dry wood and plastic.

The Gold leaf electroscope

Charging the gold leaf electroscope positively

We use a positively charged material such as a positive rod as shown in **Fig. 1.2**.

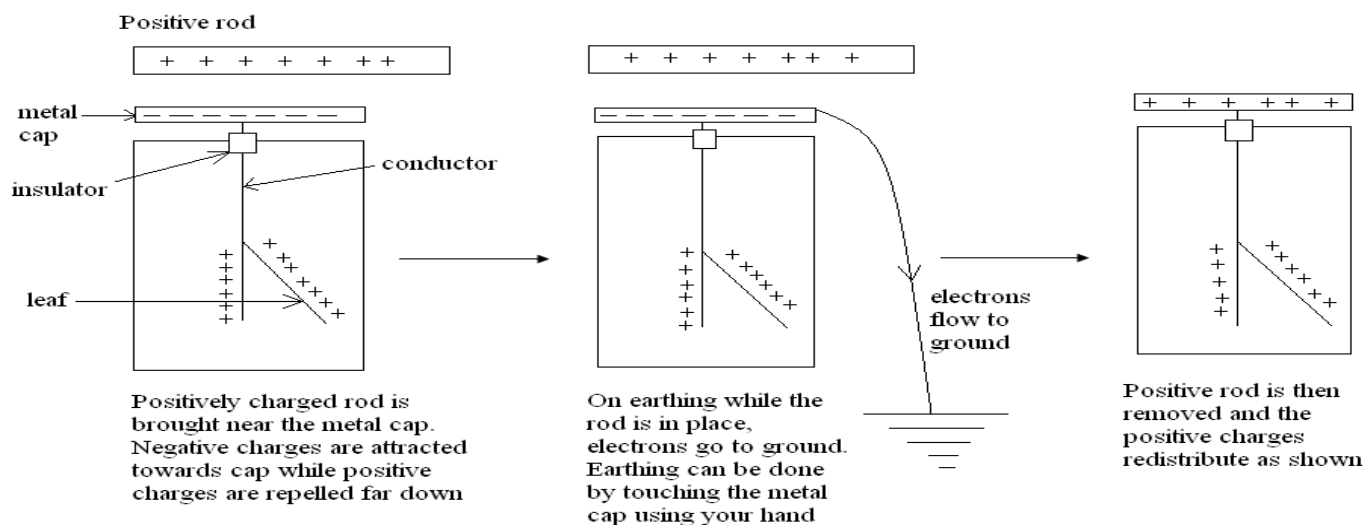


Fig. 1.2: Charging the Gold leaf Electroscope Positively

Charging the gold leaf electroscope negatively

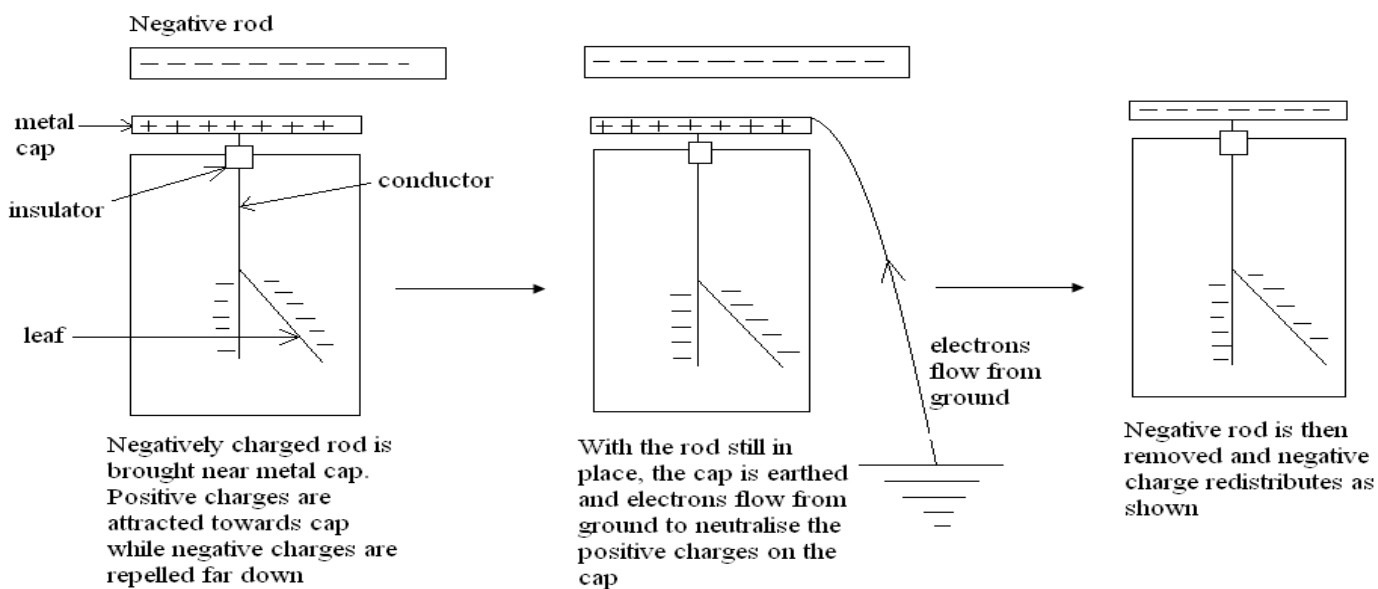


Fig.1.3: Charging the Gold leaf Electroscope Negatively

Electrostatic Hazards

Lightning (briefly describe how lightning takes place)

An aircraft flying rubs against air molecules and its body is charged by friction. If a passenger touches the body at touchdown he/she may be electrocuted. To reduce this, the wheels of aircrafts are made slightly conducting so that all charges go to ground at the moment the aircraft touches the ground.

Tankers that carry fuel over long distances rub against air molecules and the containers become charged by friction. There is some danger that a small spark may cause the containers to explode. To avoid built up of charge a small chain connects the lorry and the road so that any charges generated are immediately grounded.

Application of Electrostatics

Spray painting

Garden sprays

Electrostatic precipitators (dust extraction)

CURRENT ELECTRICITY**Introduction**

An electric current is the flow of electric charges between any points as a result of a difference in potential between the points. The unit of electric current is the Ampere (A) shortened Amp. The unit of charge is the Coulomb (C) and it is defined as the charge passing any point in an electric circuit when a steady current of 1A flows for 1s.

For any steady current (I) flowing for a time (t), the charge (Q) is given by

$$Q = It \dots\dots\dots (1)$$

The direction of current (convention current) is that of the motion of positive charges i.e. a direction opposite to that of the motion of negative charges.

*We define **potential difference (p.d.)** between two points on a conductor as the amount of electrical energy per unit charge converted to other forms of energy when the charge moves between the points. Potential difference is also called **voltage**. The unit of p.d. is the volt. Electrical energy W, charge Q and p.d. V are related by*

$$V = \frac{W}{Q} \dots\dots\dots (2)$$

where W is measured in joules (J)

The volt can be defined as one joule per coulomb.

$$\text{i.e. } (V \equiv JC^{-1})$$

Resistance

Resistance is opposition to the flow of current. Mathematically it is the ratio of the voltage across a material or conductor to the current through the conductor and its units are ohms (Ω)

Resistance R is defined by: $R = \frac{V}{I}$ (3)

We can thus define the ohm as one volt per ampere:

$$\text{i.e. } (\Omega \equiv VA^{-1})$$

Electric Power

It is the amount of electrical energy converted to other forms per unit time or alternatively it is the rate at which electrical energy is converted to other forms. The unit of power is the watt (W)

Power P is related to energy W and time t by:

$$P = \frac{W}{t}$$
 (4)

The watt is then defined as one joule per second.

Derivation of equations for calculating power

From (2) $W = QV$ (5)

From (1) $t = \frac{Q}{I}$ (6)

Substituting (5) and (6) in (4) gives

$$P = \frac{QV}{\frac{Q}{I}}$$

$$\Rightarrow P = IV \text{ (7)}$$

From (3) $V = IR$ (8)

Substituting (8) in (7) gives

$$P = IR \times I \triangleright$$

$$\Rightarrow P = I^2 R \dots\dots\dots(9)$$

$$\text{From (8) } I^2 = \frac{V^2}{R^2} \dots\dots\dots(10)$$

Substituting (10) in (9) gives

$$P = \frac{V^2}{R^2} \times R$$

$$\Rightarrow P = \frac{V^2}{R} \dots\dots\dots(11)$$

Heating effect

The amount of heat energy dissipated in a conductor is the product of the power supplied to it and the time over which the power is supplied. Heat H is calculated as:

$$H = Pt \dots\dots\dots(12)$$

Substituting (7) in (12) gives

$$H = VIt \dots\dots\dots(13)$$

Substituting (9) in (12) gives

$$H = I^2 Rt \dots\dots\dots(14)$$

Substituting (11) in (12) gives

$$H = \frac{V^2}{R} t \dots\dots\dots(15)$$

Ohm's law

It states that for a metallic conductor at constant temperature, the p.d. across it is directly proportional to the current through it i.e. $V \propto I$

$$\Rightarrow V = kI \dots\dots\dots(16)$$

The constant of proportionality is called the resistance of the conductor. Therefore Ohms' law can be written as

$$V = IR \dots\dots\dots(17)$$

Any conductor which obeys Ohms' law is called an ohmic conductor.

Resistivity

The resistance R of a conductor is directly proportional to its length L and inversely proportional to its cross sectional area A . The constant of proportionality is called the resistivity of the conductor.

$$R \propto L$$

$$R \propto \frac{1}{A}$$

$$\Rightarrow R = \frac{\rho L}{A}$$

The unit of ρ is the ohm meter ($\Omega \text{ m}$)

Example 1

An electric lamp is rated at 60 W, 240 V. Calculate the filament resistance when the lamp is operating normally.

Solution

$$P = \frac{V^2}{R}$$

$$\Rightarrow R = \frac{V^2}{P}$$

$$\Rightarrow R = \frac{(240)^2}{60}$$

$$\Rightarrow R = 960\Omega$$

Example 2

A resistor has a p.d of 40 V when the current in it is 120mA. Calculate the p.d. when the current is 600mA

Solution

We first calculate the resistance of the resistor.

$$V = IR$$

$$\Rightarrow R = \frac{V}{I}$$

$$\Rightarrow R = \frac{40}{120 \times 10^{-3}} \Omega$$

$$\therefore V = IR$$

$$\Rightarrow V = 600 \times 10^{-3} \times \frac{40}{120 \times 10^{-3}}$$

$$\Rightarrow V = 200V$$

I-V Characteristics

These are graphical representations of the variation of current with p.d. for a particular conductor

Metallic Conductor at Constant Temperature

It obeys Ohm's law therefore V is directly proportional to I

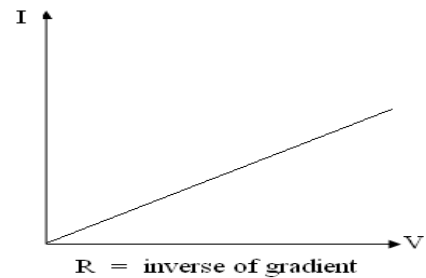
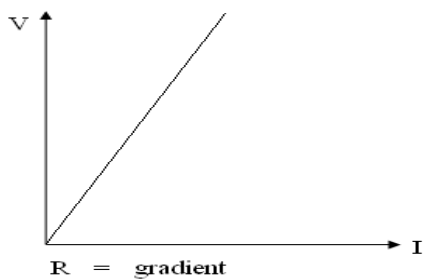


Fig. 2.1: I – V Characteristics of a Metallic Conductor

Filament lamp

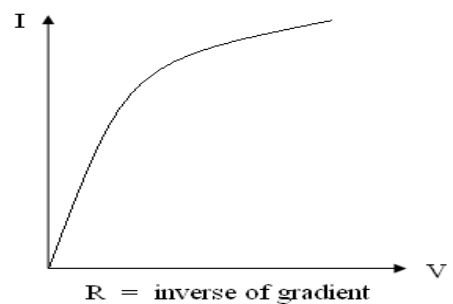
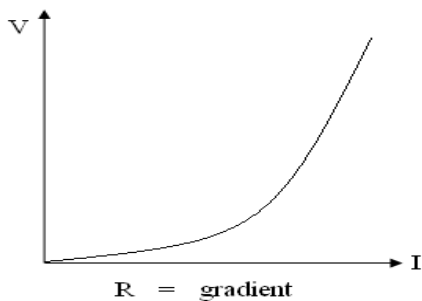


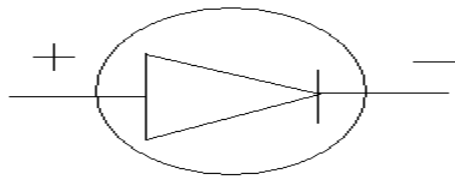
Fig. 2.2.: I – V Characteristics of a Filament Lamp

R is the resistance

The resistance of the filament increases with temperature. As the filament gets hotter and hotter due to increased current flow, atomic vibrations increase thereby obstructing the motion of charge carriers i.e. electrons and thus offering more resistance.

Semiconductor diode

A diode is a device whose resistance is low in one direction but very high in the opposite direction, as a result it almost conducts in one direction. Its symbol is shown in fig The + and – are the diode polarities or terminals

**Fig. 2.3: Symbol of a Diode**

Its I-V characteristic is shown in **Fig. 2.4**

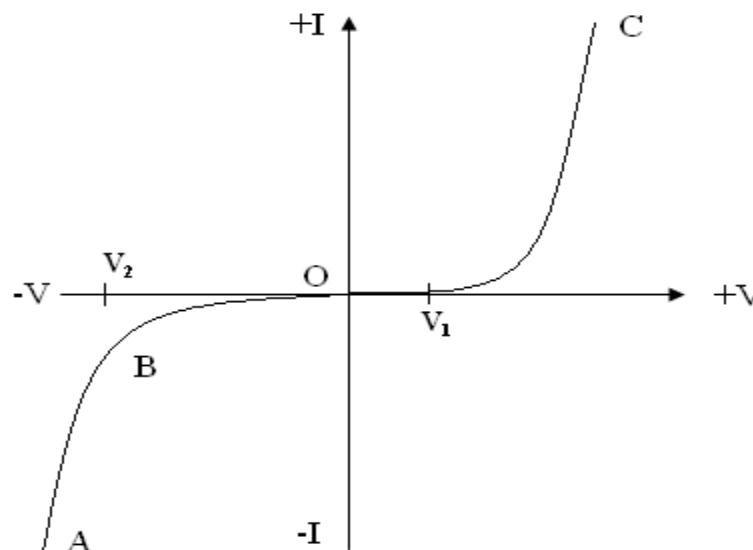
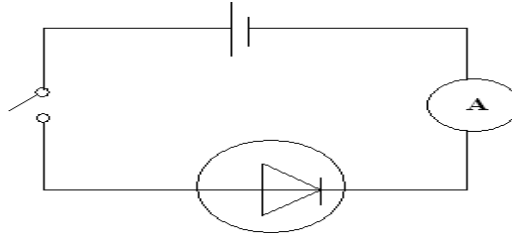


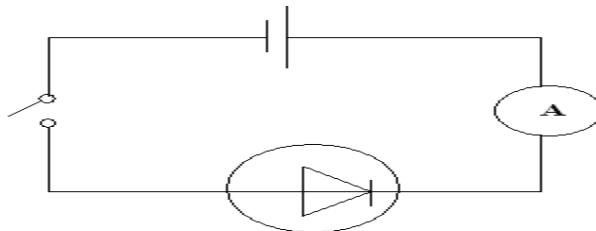
Fig. 2.4: I – V Characteristics of a Diode

When the positive terminal of a diode is connected to the positive terminals of the power supply and the negative of the diode is connected to the negative of the power supply, the diode is said to be **forward biased** and it conducts as shown in **Fig. 2.4**. Line OC in **Fig. 2.4** shows the forward bias curve

**Fig. 2.5: Forward Bias Connection of a Diode**

The diode however starts to conduct after a p.d. at least V_1 has been applied across its terminals. Even if the diode is forward biased, any p.d. below V_1 does not initiate conduction. The p.d. V_1 is called the switch on potential of the diode.

When the power supply or diode polarities are reversed, the diode is said to be **reverse biased** as shown in fig. Conduction is very little as zero as shown by line OB in **fig.2.6**. The small current observed in reverse bias called leakage current.

**Fig. 2.6: Reverse Bias Connection of a Diode**

At some value of p.d. V_2 , diode conduction suddenly increases in the negative direction. The negative p.d. at which this takes place is called the break down voltage. Semiconductor diodes are made of semi conducting material e.g. Silicon and Germanium.

Thermistor (thermal resistor)

A thermistor is a device whose resistance changes with temperature. Thermistors are of two types:

- (i) Negative temperature co-efficient of resistance thermistor (NTC) type.

In an NTC type thermistor, the resistance decreases with temperature.

(ii) Positive temperature coefficient of resistance thermistor (PTC) type.

Its resistance increases with temperature

The characteristics for the two types are shown in Fig. 2.7.

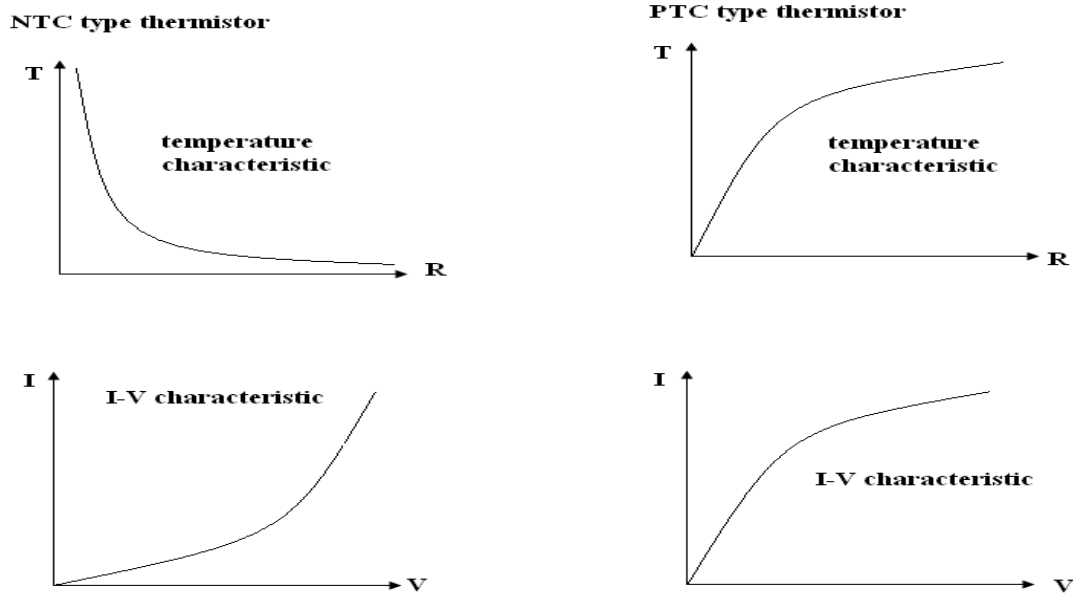


Fig. 2.7: I – V Characteristics of the NTC and PTC Thermistors

Electromotive force, Terminal p.d. and internal resistance

For a source of electrical energy, the electromotive force (e.m.f) of the source refers to the amount of electrical energy converted from other forms of energy per unit charge. Units of e.m.f. are volts. The p.d. measured across the terminal of a battery is called terminal p.d. When a battery is not supplying current, it is said to be an open circuit. The e.m.f. of the battery can be defined as its terminal p.d. on open circuit.

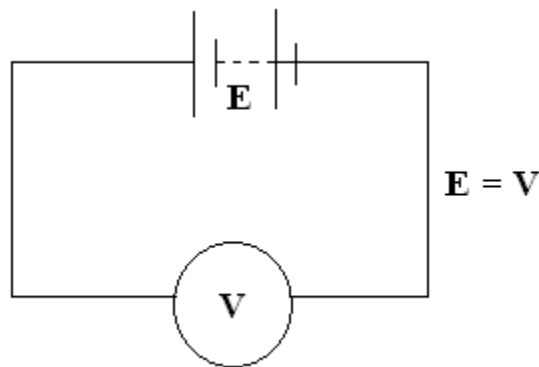
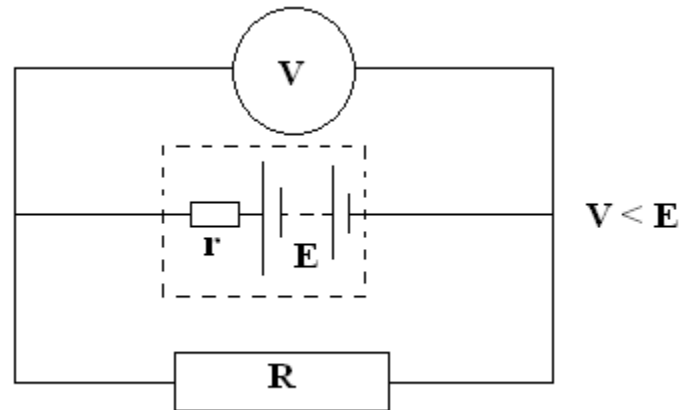


Fig. 2.8: A Battery on Open Circuit

Battery is on open circuit. Connecting wires are assumed to have negligible resistance while the voltmeter is assumed to have infinite resistance. The e.m.f. E of the battery equals the terminal p.d. V .

When connected to an external circuit, the battery is said to be on closed circuit as shown in **Fig. 2.9**.

**Fig. 2.9: A Battery on Closed Circuit**

The voltage V across the terminals of the battery is less than E . Not all the electrical energy supplied by the battery is used in the external circuit. This is due to the battery itself having internal resistance. Some of the voltage or p.d. is dropped across the internal resistance of the battery. We can then write that

$$E = V + V^*$$

Where V is the energy used per unit charge in the external circuit and V^* is the energy lost per unit charge on battery resistance. If the battery of internal resistance r supplies total current I to an external circuit of resistance R , we can write that

$$V = IR$$

$$V^* = Ir$$

$$\Rightarrow E = IR + Ir$$

$$\Rightarrow E = I(R + r)$$

Sources in series and in parallel

When identical sources each of e.m.f. E are connected in series the p.d. obtained across AB is NE , where N is the number of sources.

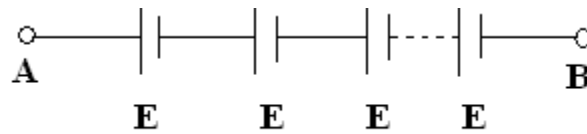


Fig. 2.10: Sources in Series.

If each source has internal resistance r then the total resistance is Nr

When connected in parallel, the p.d. across AB is E . The total resistance is $\frac{r}{N}$.

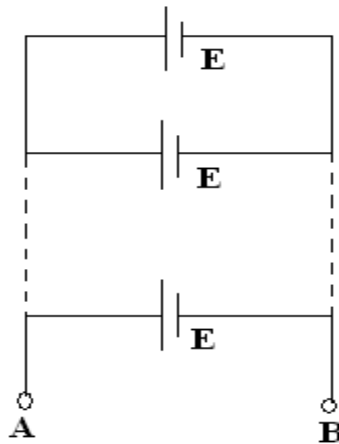
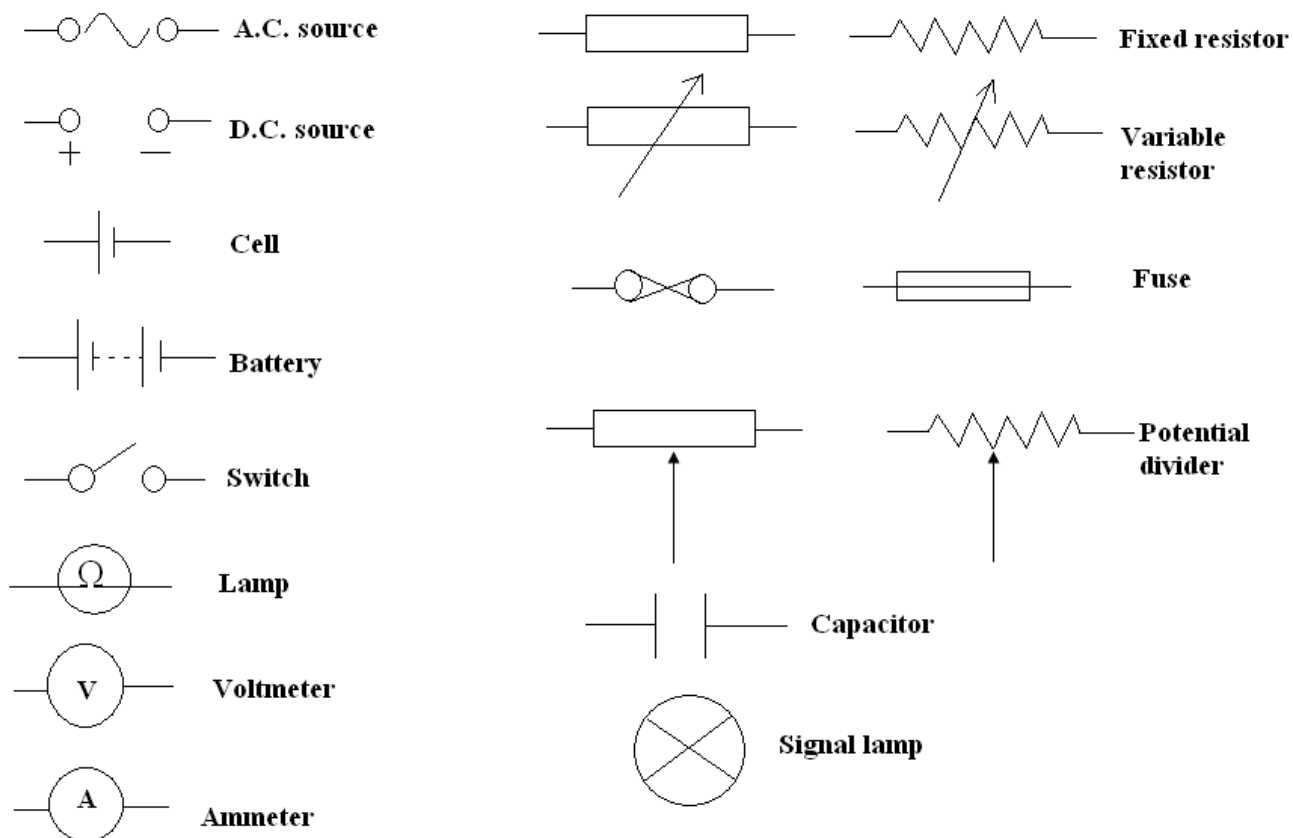


Fig. 2.11: Sources in Parallel

4.2: DIRECT CURRENT CIRCUITS [D.C CIRCUITS]

Circuit symbols



Kirchhoff's laws

First law / Current law

It states that at a junction in an electric circuit, the sum of the currents entering the junction is equal to the sum of the current leaving the junction

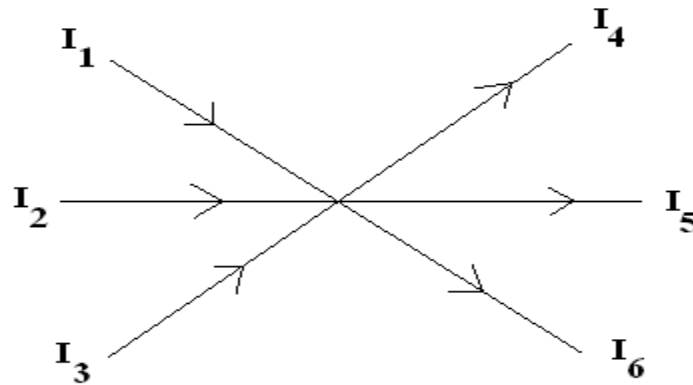


Fig. 3.1: Representation of Kirchhoff's First Law

$$I_1 + I_2 + I_3 = I_4 + I_5 + I_6 \dots\dots\dots(3.1)$$

This law is based on the principle of conservation of charge because charge does not accumulate or is not produced anywhere in an electric circuit.

$$\sum I = 0 \dots\dots\dots(3.2)$$

Second Law / Voltage Law

It states that round a closed loop, the algebraic sum of e.m.f.s is equal to the algebraic sum of the products of current and resistance.

$$\sum E = \sum IR \dots\dots\dots(3.3)$$

This law is based on the principle of conservation of energy since energy supplied by a source must be dissipated in the source resistance as well as in the external resistance.

Resistors in series

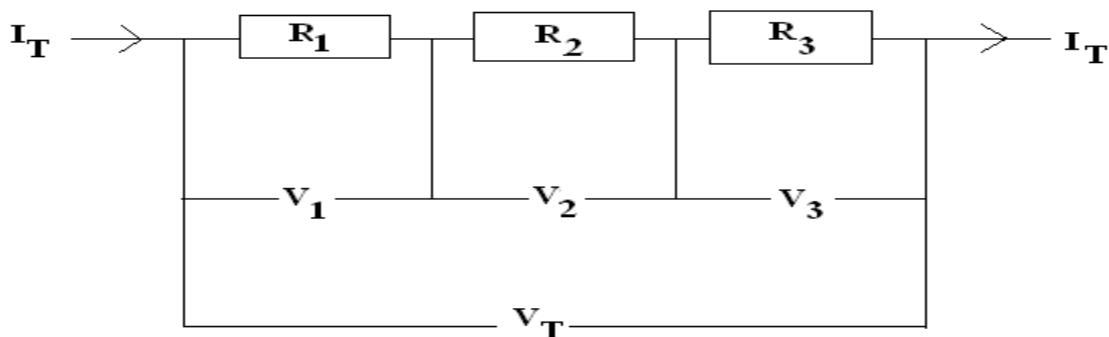
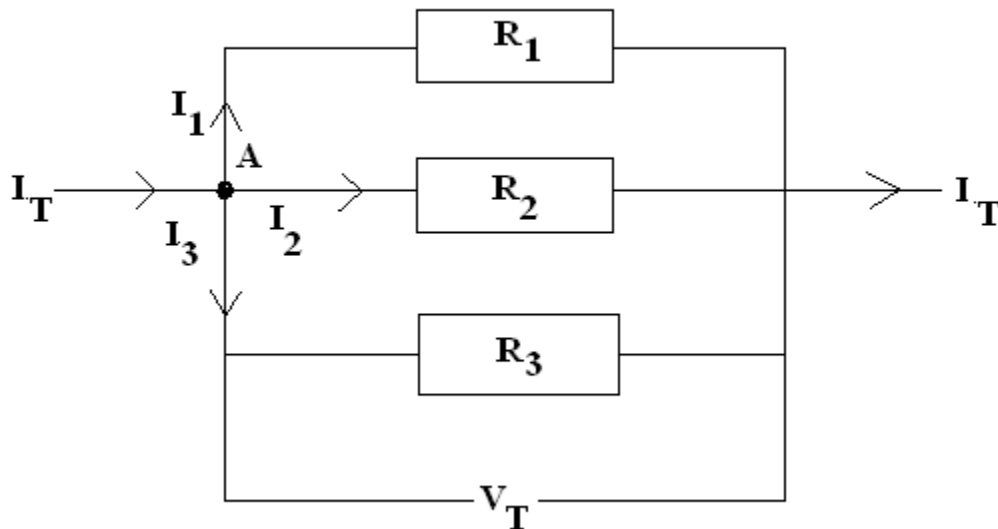


Fig. 3.2: Resistors in Series

When different resistors are connected in series, the same current passes through all, but the potential drop across them is different. Applying Kirchhoff's 2nd law then

$$\begin{aligned}
 V_T &= V_1 + V_2 + V_3 \\
 \Rightarrow I_T R_T &= I_T R_1 + I_T R_2 + I_T R_3 \\
 \Rightarrow I_T R_T &= I_T (R_1 + R_2 + R_3) \\
 \Rightarrow R_T &= R_1 + R_2 + R_3
 \end{aligned}$$

Resistors in parallel**Fig. 3.3: Resistors in Parallel**

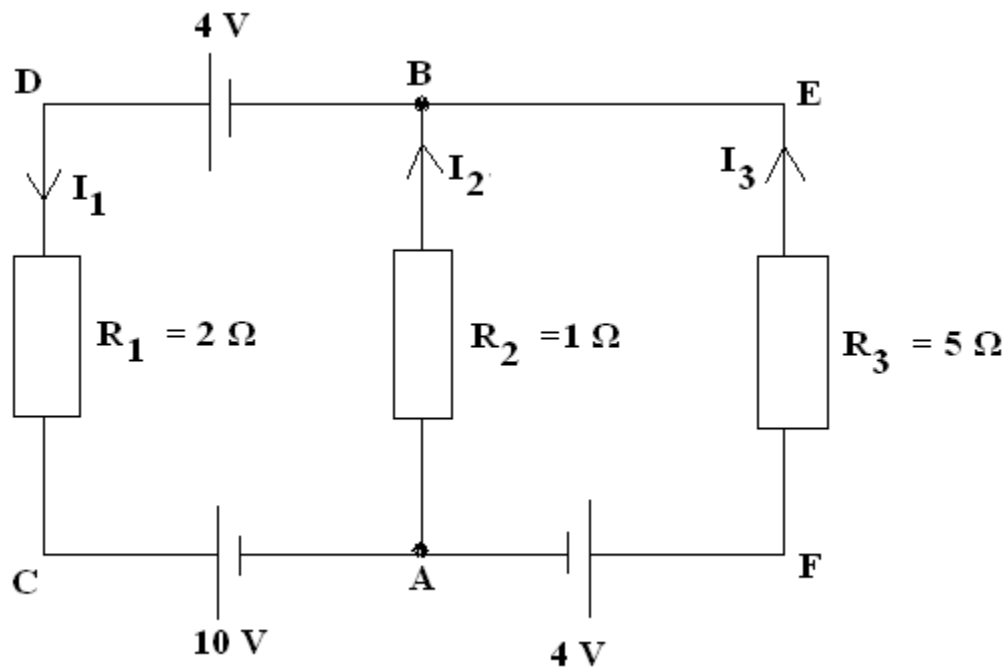
When different resistors are connected in parallel, they carry different currents but the potential drop across the combination all the same. Applying Kirchhoff's 1st law at the junction A:

$$\begin{aligned}
 I_T &= I_1 + I_2 + I_3 \\
 \Rightarrow \frac{V_T}{R_T} &= \frac{V_T}{R_1} + \frac{V_T}{R_2} + \frac{V_T}{R_3} \\
 \Rightarrow V_T \left[\frac{1}{R_T} \right] &= V_T \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right] \\
 \Rightarrow \frac{1}{R_T} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}
 \end{aligned}$$

Applications of Kirchhoff's law (The branch method of solving circuits)

Worked Example 1

In the circuit diagram below, find the value of the currents I_1 , I_2 and I_3 and comment on the values obtained.



Solution

We have three loops: ACDBA, ABEFA and the entire loop CDEFC. The issue of direction of current flow is important here. We can consider the clockwise direction to be positive any two loops we wish to consider. That means the 10 V source which is

driving current in the clockwise direction therefore acts as a positive source while the two 4 V sources act as negative sources. Currents flowing in the clockwise sense in a given loop are considered positive while those flowing in the counter clockwise direction are negative.

Considering loop ACDBA and applying Kirchhoff's 2nd law:

$$\begin{aligned} 10 - 4 &= -I_1 R_1 - I_2 R_2 \\ \Rightarrow 6 &= -I_1 R_1 - I_2 R_2 \dots \dots \dots (1) \end{aligned}$$

Considering loop ABEFA and applying Kirchhoff's 2nd law:

$$-4 = I_2 R_2 - I_3 R_3 \dots \dots \dots (2)$$

At junction A or B:

$$I_1 = I_2 + I_3 \dots \dots \dots (3)$$

Substituting (3) in (1):

$$\begin{aligned} \Rightarrow 6 &= -(I_2 + I_3) R_1 - I_2 R_2 \\ \Rightarrow 6 &= -I_2 R_1 - I_3 R_1 - I_2 R_2 \\ \Rightarrow 6 &= -2I_2 - 2I_3 - I_2 \\ \Rightarrow 6 &= -3I_2 - 2I_3 \dots \dots \dots (4) \end{aligned}$$

From (2):

$$\begin{aligned} -4 &= I_2 R_2 - I_3 R_3 \dots \dots \dots (2) \\ -4 &= I_2 - 5I_3 \dots \dots \dots (5) \end{aligned}$$

Now equation (4) and (5) are solved simultaneously

$$6 = -3I_2 - 2I_3 \dots\dots\dots(4) \times 1$$

$$-4 = I_2 - 5I_3 \dots\dots\dots(5) \times 3$$

$$6 = -3I_2 - 2I_3$$

$$-12 = 3I_2 - 15I_3$$

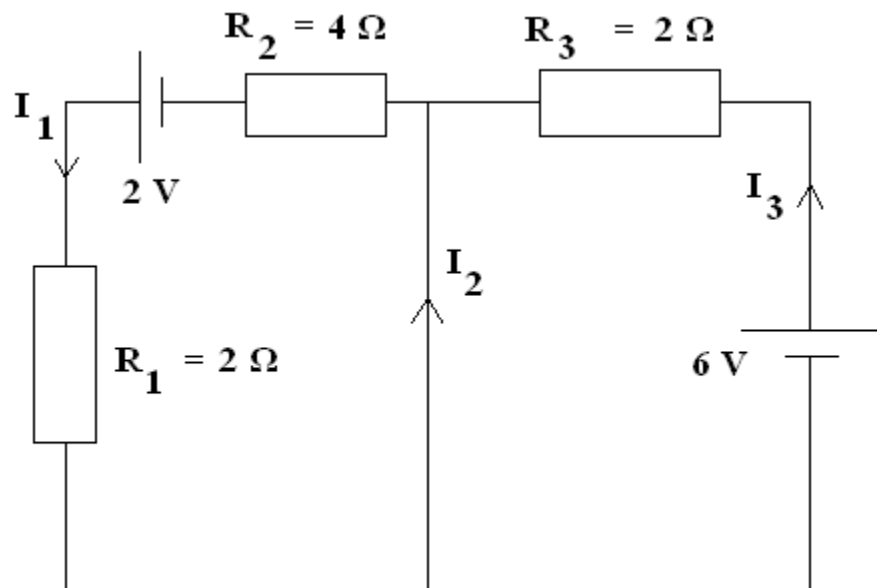
$$\Rightarrow I_3 = \frac{6}{17} A$$

$$\Rightarrow I_2 = \frac{-38}{17} A$$

Substituting for I_2 and I_3 in (3) gives

$$I_1 = \frac{-32}{17} A$$

In these results, I_1 and I_2 are negative. This shows that the actual direction of flow of these currents is opposite to the one indicated on the diagram

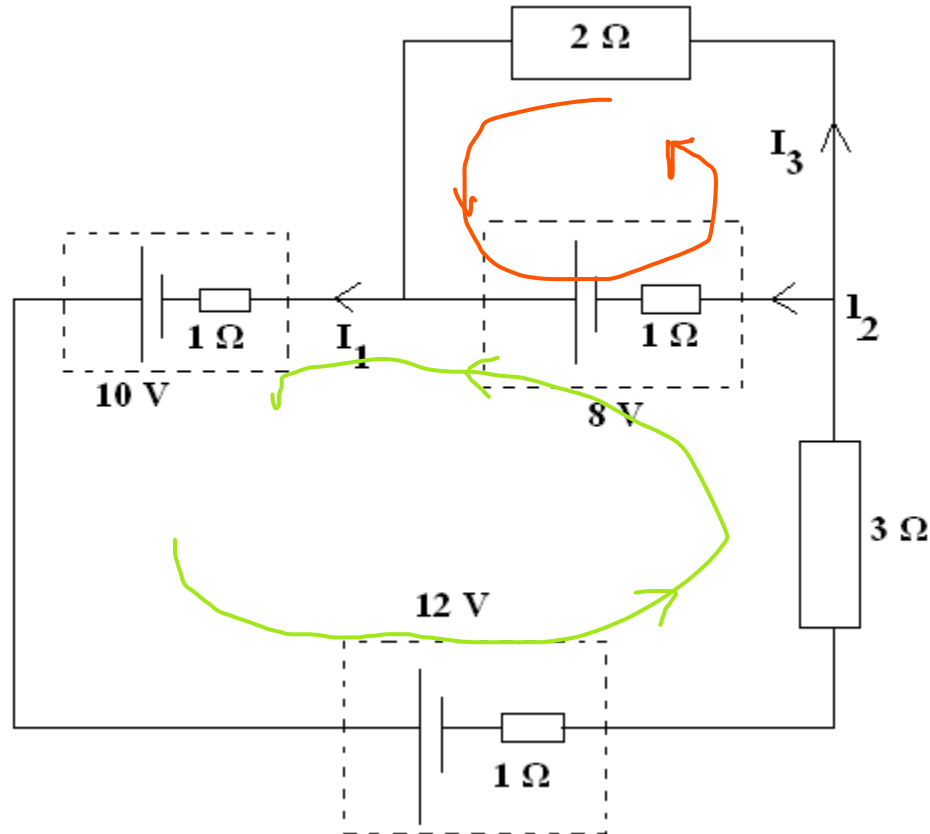


Answers

$$I_1 = \frac{1}{3} A \quad I_2 = \frac{-8}{3} A \quad I_3 = 3A$$

S.A.Q 2

In the diagram below calculate the currents through the $2\ \Omega$ and $3\ \Omega$ resistors. Each cell has an internal resistance of $1\ \Omega$.



Hint:

This problem involves internal resistances and these have to be considered just like the $2\ \Omega$ and $3\ \Omega$ resistors

Answers:

$$I_1 = \frac{10}{17} \text{ A}$$

$$I_2 = \frac{52}{17} \text{ A}$$

$$I_3 = \frac{-42}{17} \text{ A}$$

Potential Division

The potential divider

A potential divider is an arrangement of resistors that are used to obtain a fraction of the voltage or p.d. provided by a voltage supply.

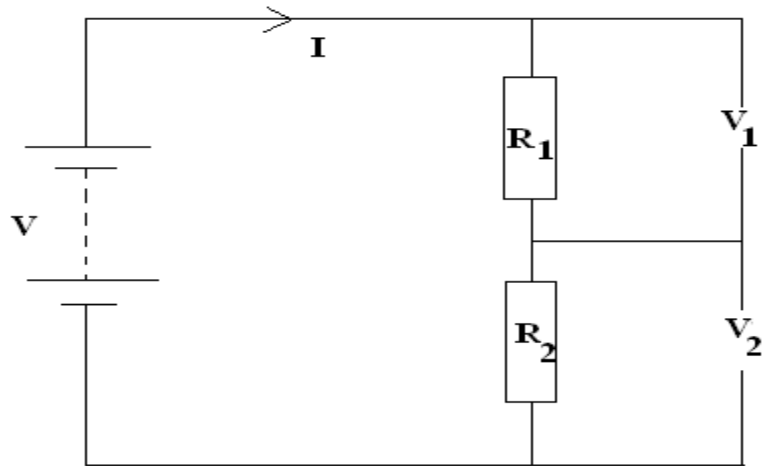


Fig. 3.4: Potential Divider Circuit

Consider 2 resistors connected in series with a voltage supply. As they are in series, they share the same current I but the p.d. across each resistor depends on its resistance value. We can write that

$$V_1 = IR_1 \dots \dots \dots (1)$$

$$V_2 = IR_2 \dots \dots \dots (2)$$

$$V = I(R_1 + R_2) \dots \dots \dots (3)$$

Dividing (1) by (3)

$$\begin{aligned} \frac{V_1}{V} &= \frac{R_1}{R_1 + R_2} \\ \Rightarrow V_1 &= \frac{R_1}{R_1 + R_2} V \dots \dots \dots (4) \end{aligned}$$

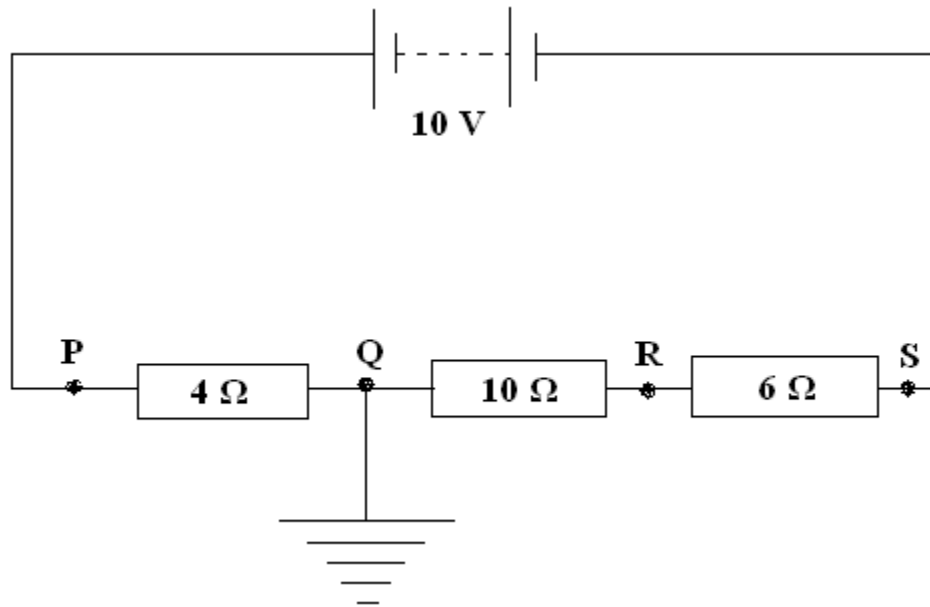
Dividing (2) by (3)

$$\begin{aligned} \frac{V_2}{V} &= \frac{R_2}{R_1 + R_2} \\ \Rightarrow V_2 &= \frac{R_2}{R_1 + R_2} V \dots \dots \dots (5) \end{aligned}$$

We can therefore calculate the fractions V_1 and V_2 in terms of the resistances and the total voltage. Expressions for V_1 and V_2 are called **potential division formulae**.

Application of the potential division formula**Worked Example 1**

The diagram show three resistors of resistances $4\ \Omega$, $10\ \Omega$ and $6\ \Omega$ connected in series. A p.d. of $10\ \text{V}$ is maintained across them.



Calculate the potential at points P, Q, R and S

Solution

Let $R_1 = 4\ \Omega$, $R_2 = 10\ \Omega$ and $R_3 = 6\ \Omega$

Point Q is grounded so its potential is zero volts

Using the potential division formula:

$$\frac{V_{R_1}}{V} = \frac{R_1}{R_1 + R_2 + R_3}$$

$$\Rightarrow V_{R_1} = \left[\frac{R_1}{R_1 + R_2 + R_3} \right] V$$

$$\Rightarrow V_{R_1} = \left[\frac{4}{4 + 10 + 6} \right] \times 10$$

$$\Rightarrow V_{R_1} = 2V$$

Now, V_{R1} is the potential difference across R_1 . One end of this resistor is held at zero potential (point Q) therefore the potential at point P is 2V.

$$\frac{V_{R_2}}{V} = \frac{R_2}{R_1 + R_2 + R_3}$$

$$\Rightarrow V_{R_2} = \left[\frac{R_2}{R_1 + R_2 + R_3} \right] V$$

$$\Rightarrow V_{R_2} = \left[\frac{10}{4 + 10 + 6} \right] \times 10$$

$$\Rightarrow V_{R_2} = 5V$$

V_{R2} is the potential difference across R_2 . One end of this resistor is held at zero potential (point Q) so the potential at point R is -5V. It is a negative potential because to the right of the earth connection we have positive potentials and to the left of this connection we have negative potentials.

$$\frac{V_{R_3}}{V} = \frac{R_3}{R_1 + R_2 + R_3}$$

$$\Rightarrow V_{R_3} = \left[\frac{R_3}{R_1 + R_2 + R_3} \right] V$$

$$\Rightarrow V_{R_3} = \left[\frac{6}{4 + 10 + 6} \right] \times 10$$

$$\Rightarrow V_{R_3} = 3V$$

V_{R3} is the potential difference across R_3 . One end of this resistor is held at -5 V (point R) so the potential at point S is -8 V

A device whose resistance is affected by physical conditions can be used as a potential divider e.g. a thermistor or a light dependent resistor (LDR). We have already looked at how the resistances of thermistors depend on temperature.

The resistance of an LDR decreases with increase in light intensity falling upon it. Its resistance is very high in the dark but very low in bright sunlight.

S.A.Q

The diagram shows a light dependent resistor (LDR) connected in series with a $10\text{ k}\Omega$ resistor and a 12 V d.c. supply.

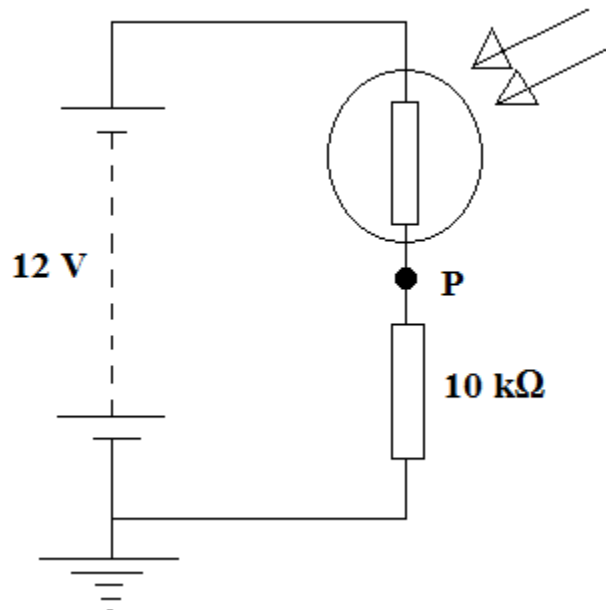


Figure 1

- (i) Find the potential at point P when the LDR is
 1. in the dark and has a resistance of $50\text{ k}\Omega$
 2. in bright sunlight and has a resistance of $2\text{ k}\Omega$
- (ii) What is the resistance of the LDR if the potential at P is 4 V ?

ANSWERS

Let V_{LDR} be the potential difference across the LDR

(i) 1.2 V

2. 10 V

(ii) 20 Ω

The Potentiometer

A potentiometer can be used to measure p.d.s and e.m.f.s to a very high degree of accuracy. Fig shows a circuit that incorporates a potentiometer

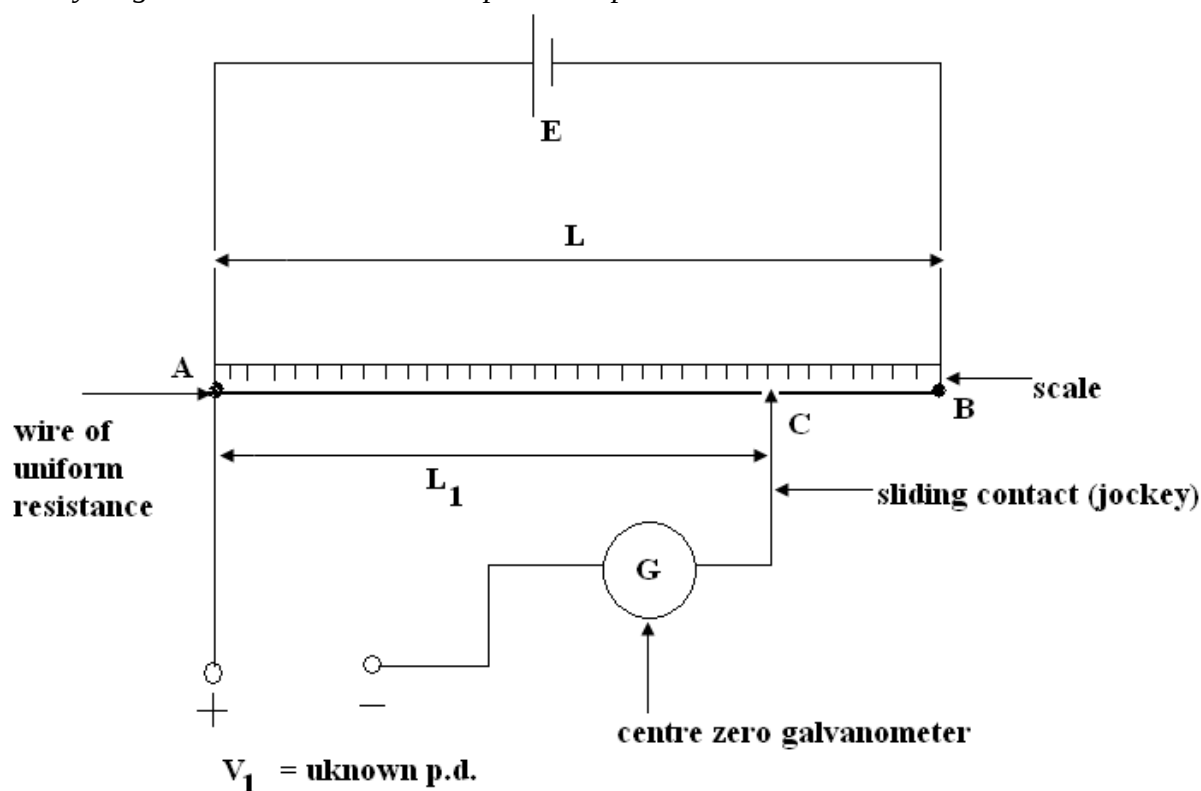


Fig. 3.5: The Potentiometer Circuit

The driver cell is used to maintain a constant p.d. across the ends of the wire.

The wire has a known resistance per unit length and acts as a potential divider.

When $V_{AC} > V_1$, the galvanometer deflects to the left of the zero mark.

When $V_{AC} < V_1$, the galvanometer deflects to the right of the zero mark.

The jockey or sliding contact is used to adjust the length of the wire until $V_{AC} = V_1$.

NOTE: In a potentiometer, the positive of the driver cell is connected to the positive of the unknown p.d.

When $V_{AC} = V_1$, no current flows through the galvanometer and the potentiometer is said to be balanced. The galvanometer shows zero deflection i.e. the pointer does not move.

Theory

The length of wire $AB = L$. If r is the resistance per unit length of the wire, then the total resistance of wire $AB = rL$. Assuming that the driver cell has zero internal resistance, then the p.d. across length L of the wire is equal to the e.m.f. of the cell. If the driver cell supplies current I then its e.m.f. E is given by

$$E = IrL \dots \dots \dots (1)$$

For the unknown p.d., the balance length is L_1 and the total resistance of the balance length is rL_1 . If the unknown p.d. is V_1 then

$$V_1 = IrL_1 \dots \dots \dots (2)$$

Dividing (2) by (1), then

$$\frac{V_1}{E} = \frac{IrL_1}{IrL}$$

$$\Rightarrow V_1 = \frac{L_1}{L} E$$

Therefore V_1 can be found (i.e. the unknown p.d.)

Advantages of the potentiometer

- (i) It uses the null method, i.e. its adjustment does not depend on the calibration of the galvanometer.
- (ii) Since no current flows at balance, the potentiometer does not draw current from the source whose p.d. is under investigation and so it does not affect the p.d. to be measured
- (iii) It is a voltmeter of infinite resistance

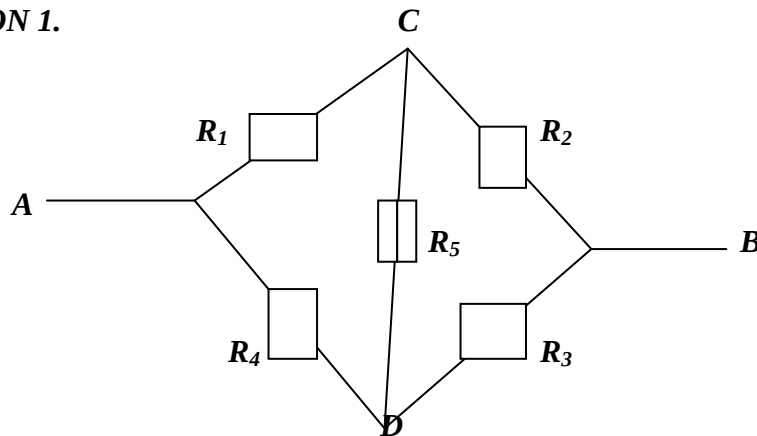
Disadvantages of the potentiometer

- (i) Since small distances cannot be measured on the scale, the potentiometer cannot be used to measure very small p.d.s

- (ii) It is a bulk instrument
- (iii) It is time consuming
- (iv) The wire may lose uniformity due to scratching by the jockey

D.C CIRCUITS TYPICAL EXAMINATION QUESTIONS

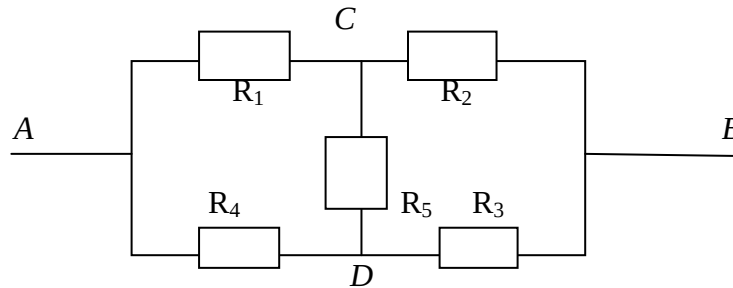
QUESTION 1.



If each resistor has \$10\Omega\$, what is the effective resistance between the points A and B.

SOLUTION

The diagram above can be redrawn to be like the diagram below.



When the points A and B are connected to a voltage supply, the potential at C and D are the same. Since there is no potential difference between C and D, no current flows through the resistor \$R_5\$. Therefore \$R_5\$ has no effect on the circuit. Hence \$R_1\$ and \$R_2\$ are in series, and so are \$R_3\$ and \$R_4\$. Each pair of resistors has an effective resistance of \$(10\Omega + 10\Omega) = 20\Omega\$ and these two pairs are in parallel. Therefore, the total effective resistance \$R\$ in the circuit is given by

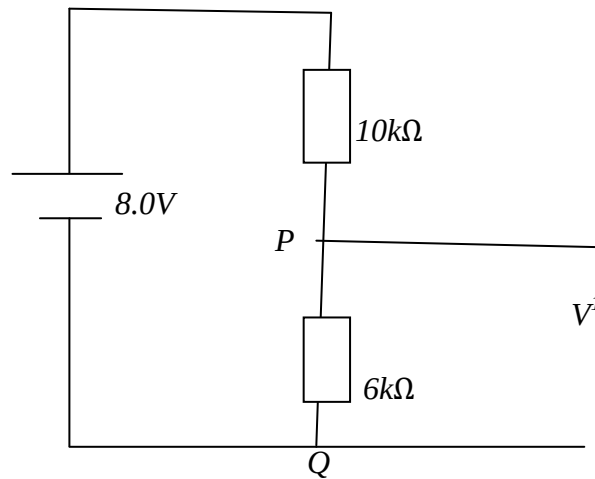
$$\frac{1}{R} = \frac{1}{20} + \frac{1}{20}$$

$$R = 10\Omega$$

QUESTION 2

A $10\text{k}\Omega$ and a $6\text{k}\Omega$ resistor are connected in series to a 8.0V battery. The potential across $6\text{k}\Omega$ resistor is measured by connecting a voltmeter of resistance $75\text{k}\Omega$ across it. Calculate the voltmeter reading.

SOLUTION



The actual voltage across the $6\text{k}\Omega$ resistor, using the potential divider rule

$$V^1 = \frac{6}{6+10} \times 8.0\text{V} = 3.0\text{V}.$$

With the voltmeter of $75\text{k}\Omega$ connected across the $6\text{k}\Omega$ resistor, the equivalence resistance R across PQ is given by

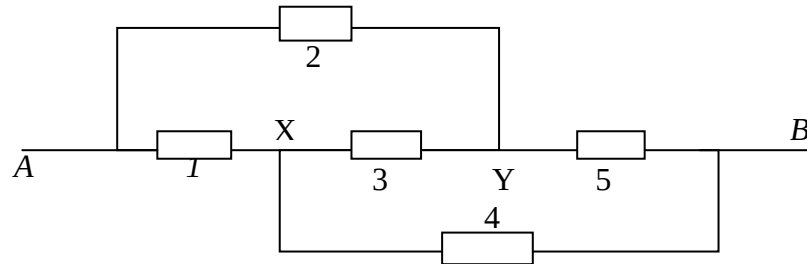
$$\frac{1}{R} = \frac{1}{6} + \frac{1}{75}$$

$$R = 5.56\text{k}\Omega$$

Therefore voltmeter reading $V^{11} = \frac{5.56}{5.56+10} \times 8.0\text{V} = 2.86\text{V}.$

NOTE: The voltmeter has resistance which is comparable to the resistance of $10\text{k}\Omega$ and $6\text{k}\Omega$, therefore the voltmeter draws current. The voltmeter must have infinite resistance so that it will give the same voltage reading to that calculated using the potential divider rule.

QUESTION 3



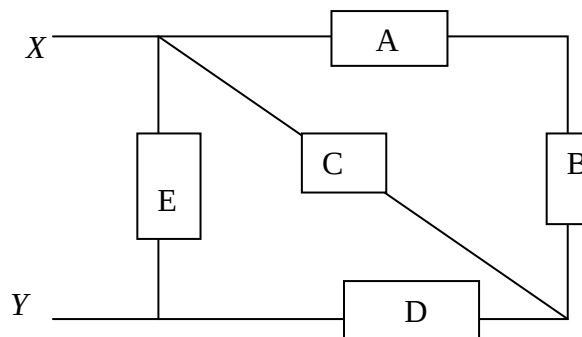
What is the effective resistance between points A and B, if the resistance of each resistor is 10Ω .

SOLUTION

This question is exactly the same as the one on QUESTION 1. Effective resistance = 10Ω .

NOTE: No current will pass through resistor 3, because points Y and X are at the same potential. Resistors 1 and 4 are in series and are in parallel to resistor 2 and 5 which are in series.

QUESTION 4



What is the effective resistance between points X and Y? If the resistance of each resistor is R .

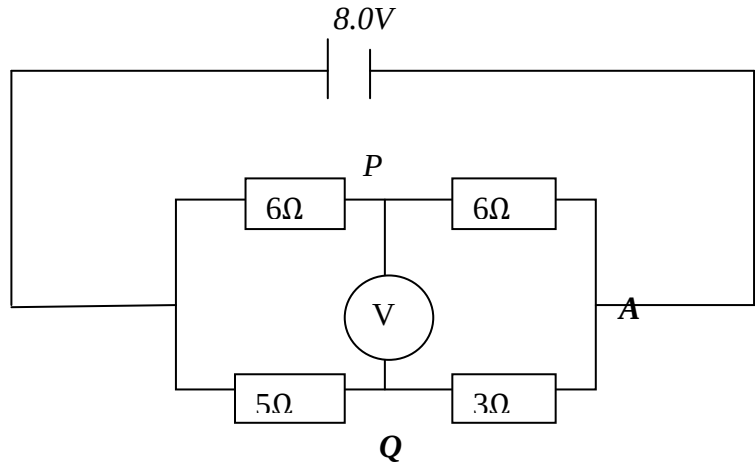
SOLUTION

STEPS

1. Resistors A and B are in series = $R + R = 2R$.
2. The total resistance from step 1: is in parallel to resistor C. $= \frac{2R \times R}{2R + R} = \frac{2R}{3}$
3. The total resistance from step 2: is in series with resistor D. $= \frac{2R}{3} + R = \frac{5R}{3}$
4. The total resistance from step 3: is in parallel to resistor E. $= \frac{\frac{5R}{3} \times R}{\frac{5R}{3} + R} = \frac{5R^2}{3} \div \frac{8R}{3} = \frac{5R}{8}$

Therefore the effective resistance for the circuit = $\frac{5R}{8}$

QUESTION 5



Show that the magnitude of the voltmeter reading is 1.0V.

SOLUTION

The first assumption is that the voltmeter is ideal (has infinite resistance), therefore it draws no current from the circuit.

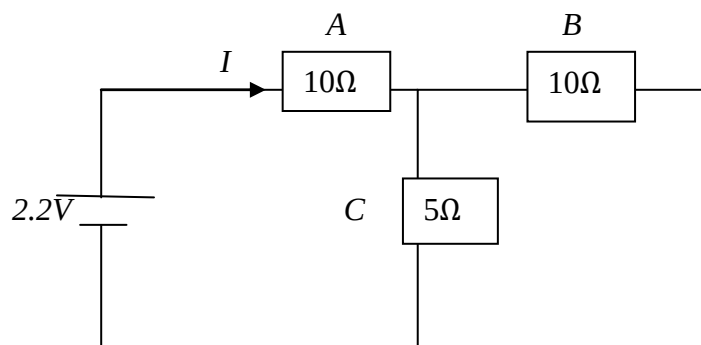
The potential difference between A and P is the same as the potential at P (point A is our reference point or earth), and the p.d between A and Q is the same as the potential at Q.

Now p.d between A and P (Using potential divider rule) = $\frac{6}{12} \times 8 = 4.0V$

Now p.d between A and Q = $\frac{3}{8} \times 8 = 3.0V$

The p.d between P and Q = $4.0V - 3.0V = 1.0V$

QUESTION 6



Calculate the effective resistance.

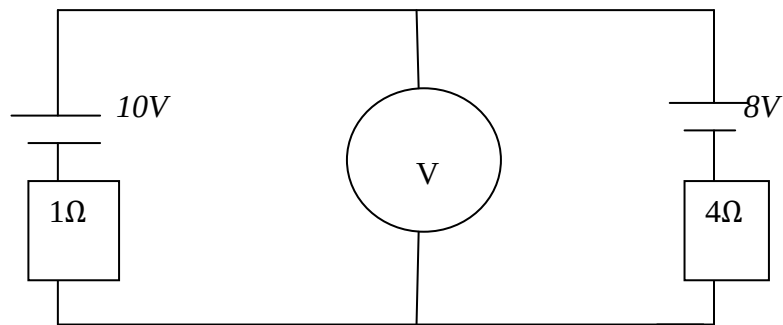
SOLUTION

Resistors B and C are in parallel and they are in series with resistor A.

$$R_T = \frac{5 \times 10}{5 + 10} + 10 = 13.3 \Omega$$

NOTE: To identify that resistors are in series or parallel. You may use the concept of current. The same amount of current will pass through resistors in series and different amount of current passes through resistors in parallel except when the current divides to branches with the same resistance. Current divides for resistors B and C, therefore they are in parallel. But the amount of current that passes through resistor A, is the same amount of current that passes through resistors C and B. Therefore resistors C and B are in series with resistor A.

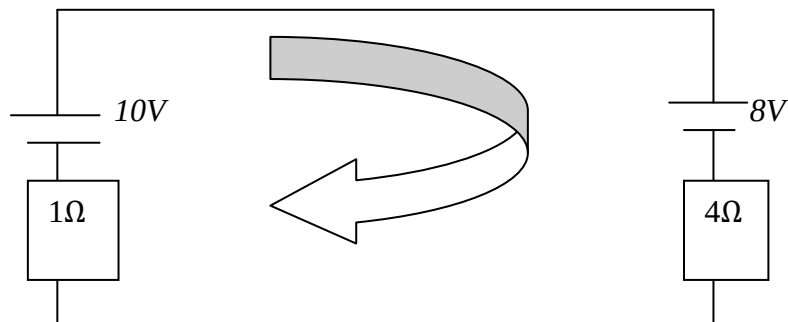
QUESTION 7



The batteries have negligible internal resistance and the voltmeter has very high resistance. What would be the voltmeter reading?

SOLUTION

Voltmeter has very high resistance. This means the voltmeter draws no current. It is an ideal voltmeter. Therefore the voltmeter can be removed for easy analysis of the circuit



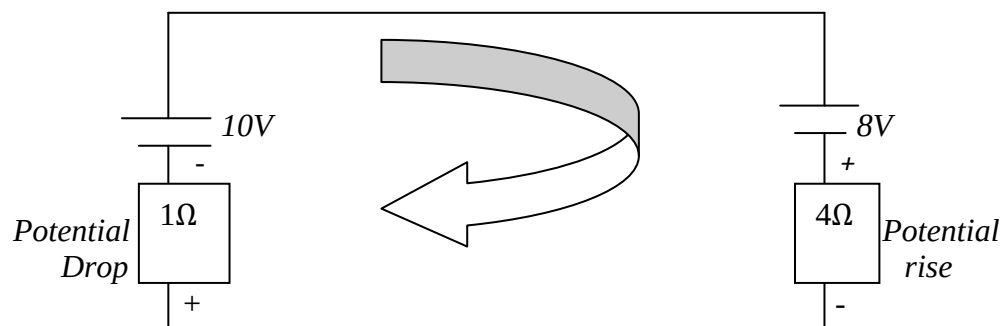
Considering the loop in clockwise direction as shown by the arrow on the diagram above.

Using Kirchhoff's loop rule, $10 - 8 = 4I + 1I$

$$I = 0.4A$$

Since the value of current is positive, it shows that the direction of current represented by the arrow is correct.

We are going to consider potential rises and drops in the circuit. Following the direction of the arrow (because it shows the actual direction of current in that circuit), mark + and – to every resistor in the circuit. Mark the first side of every resistor positive and the next side negative, following the direction of current in that order, as shown on the diagram below. This can be analysed to represent either a potential rise or drop, depending on the orientation of the cell or battery.



For the 8V battery, it can be easily seen that it's a potential rise. (If we are to consider the resistor as "another cell", we can see that the total p.d can be found by adding the two)

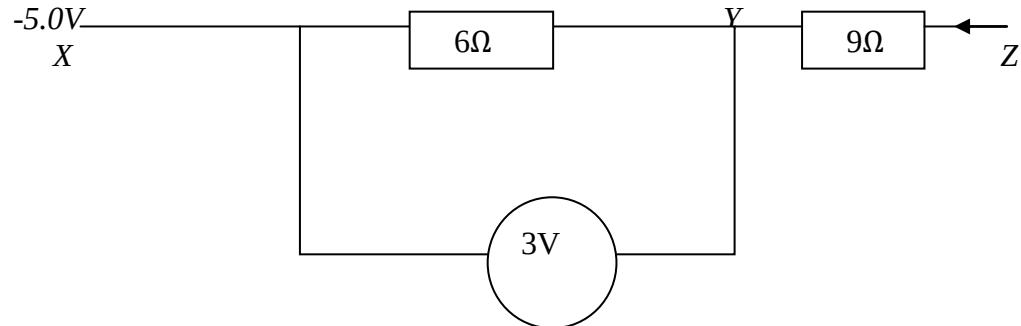
For the 10V battery it can also be seen that it's a potential drop, by making the same analysis as above.

The potential rise for the 4Ω resistor = current $\times$ resistance (IXR)
 $= 0.4 \times 4 = 1.6V$

The potential drop on the $1\Omega = 0.4 \times 1 = 0.4V$

The voltmeter reading: Using the 10V battery $= 10V - 0.4V = 9.6V$
 : Using the 8V battery $= 8.0V + 1.6V = 9.6V$

QUESTION 8



If current is moving flows from Z to X, what is the potential at Z. And at Y

SOLUTION

When current flows from Z to X, it means Z is at a higher potential than X.

The first step is to calculate potential difference between Z and X. $= [Z - (-5)] = Z + 5$

The voltmeter reading represents the potential difference across the 6Ω resistor $= 3V$

The potential difference across the 6Ω resistor can also be found by using the potential

divider rule $= \frac{6}{6+9} (Z + 5) = \frac{6Z+30}{15}$

These two potentials are equated together. $\frac{6Z+30}{15} = 3$
 $Z = 2.5V$

Therefore potential at Z $= 2.5V$

To calculate potential at Y:

Potential difference across 6Ω resistor $=$ potential at Y - Potential at X

This is because potential at Y is greater than potential at X

$= Y - (-5) = Y + 5$. This is equal to the potential difference recorded by the voltmeter.

Now: $Y + 5 = 3$

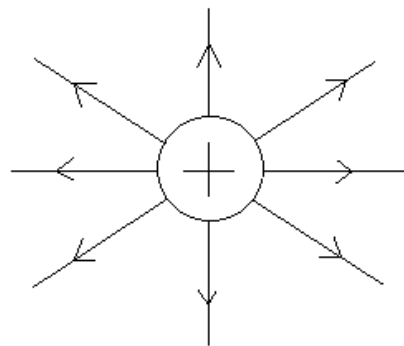
$$Y = -2V$$

Potential at Y $= -2V$.

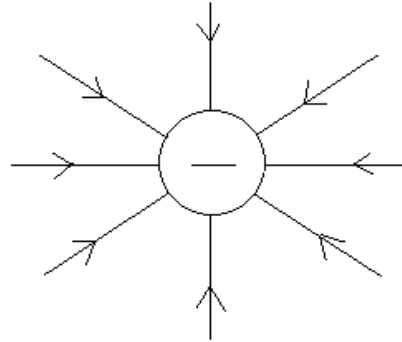
4.3: ELECTRIC FIELDS

Introduction

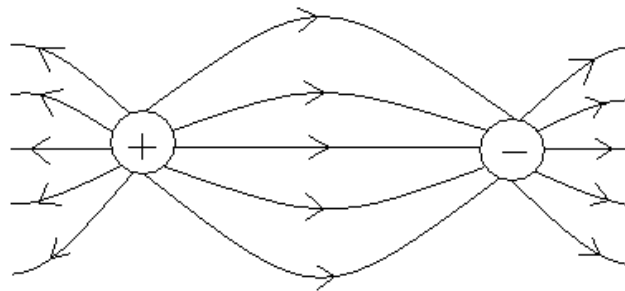
An electric field is a space around a charged particle in which the particle experiences a force. An electric field is represented by field lines or lines of force directed outwards on a positive charge and inwards on a negative charge.



electric field around a positive charge



electric field around a negative charge



electric field in the vicinity of two dissimilar charges

Fig. 4.1: Representation of Electric Fields

Coulomb's law

It states that any two charges exert a force on each other which is directly proportional to the product of their charges and inversely proportional to the square of the distance that separates them. For two charges Q_1 and Q_2 separated by distance r then the force between them is given by

$$F \propto \frac{Q_1 Q_2}{r^2}$$

The constant of proportionality is a quantity given by $\frac{1}{4\pi\epsilon_0}$ so that Coulomb's law can be written as

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$$

ϵ is called the permittivity of a medium containing the charges. ϵ is a property of the medium that determines the extent to which the electric force is experienced by the charges. The permittivity of vacuum is denoted by ϵ_0 and it has a known value.

Electric Field Strength

Electric field strength at a point in an electric field is defined as the force per unit charge placed at that point. It is represented by the symbol E . Electric field strength is also called electric field intensity. Consider two charges Q_1 and Q_2 . The electric field strength due to charge Q_1 acting per unit test charge Q_2 is given by

$$E = \frac{F}{Q_2}$$

$$\frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \times \frac{1}{Q_2}$$

$$E = \frac{Q_1}{4\pi\epsilon_0 r^2}$$

In general, the electric field strength due to a charge Q at a distance r away from it is given by

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

E is a vector quantity and has units NC^{-1} . The force experienced by a charge due to a system of charges is a vector sum of the individual forces.

Electric potential

Electric potential V at a point in the electric field is the work to move a unit positive charge from infinity to the point.

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

Electric Potential Energy

Electric potential energy U of charge q at a point in an electric field is the amount of work done in moving a positive charge from infinity to that point. It is represented by a symbol W . W is a scalar quantity and its units are joules.

$W = qV$ Then

$$W = \frac{qQ}{4\pi\epsilon_0 r}$$

Potential gradient

Potential gradient is defined as the rate of change of potential with distance.

$$\text{Potential gradient} = -\frac{dV}{dr}$$

The negative sign shows that potential decreases with distance. Potential gradient at a point is numerically equal to the electric field strength at that point

$$E = -\frac{dV}{dr}$$

The units of potential gradient are Vm^{-1}

Electric Field and Charge Density

Charge density is defined as charge per unit area denoted by δ

$$\delta = \frac{Q}{A} \dots\dots\dots (1)$$

where A is the area.

Consider a sphere of radius r with charge Q distributed at its surface, then the electric field strength at the surface of the sphere is given by

$$E = \frac{Q}{4\pi\epsilon r^2} \dots\dots\dots (2)$$

The area A a sphere is calculated as

$$A = 4\pi r^2 \dots\dots\dots (3)$$

Substituting (3) in (1)

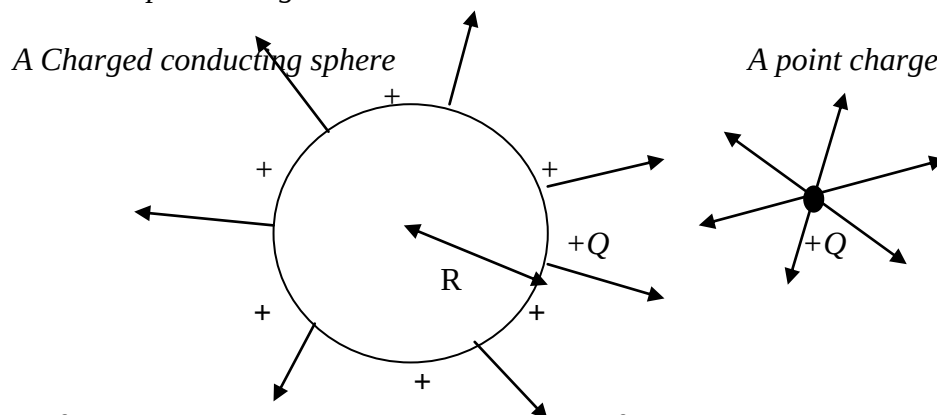
$$\delta = \frac{Q}{4\pi r^2} \dots\dots\dots (4)$$

Substituting (4) in (2) gives

$$E = \frac{\delta}{\epsilon} \dots\dots\dots (5)$$

This is an expression of the electric field strength at the surface of a conductor and it applies for all conductors.

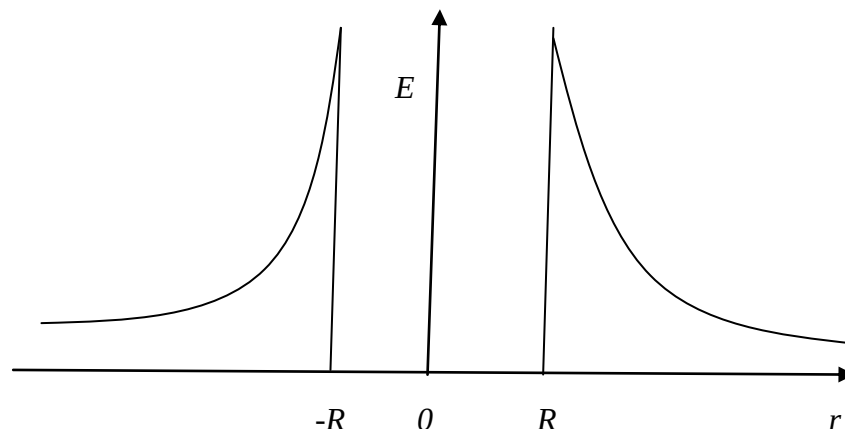
NOTE: If we compare the electric fields around a charged conducting sphere and a point charge, we will notice that outside the charged sphere, the electric field is identical to that around a point charge



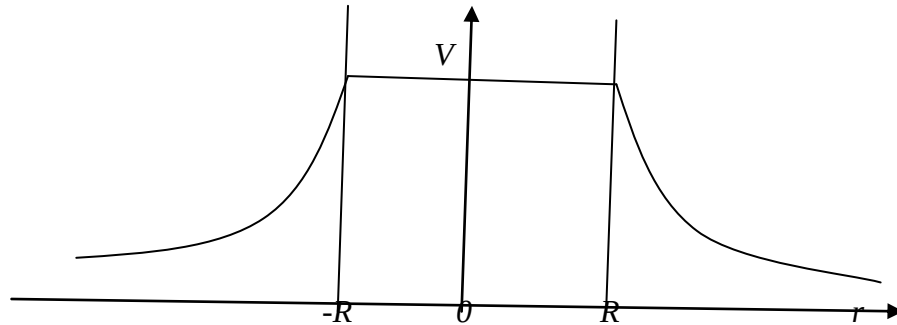
Hence for a point at a distance $r \geq R$, the radius of the charged sphere, the electric potential $V = \frac{Q}{4\pi\epsilon_0 r}$.

For points inside the sphere, the electric field intensity $E=0$. (No charge inside a charged sphere). Hence the electric potential is constant and is given by $V = \frac{Q}{4\pi\epsilon_0 R}$

GRAPH OF ELECTRIC FIELD STRENGTH OF A CHARGED SPHERE



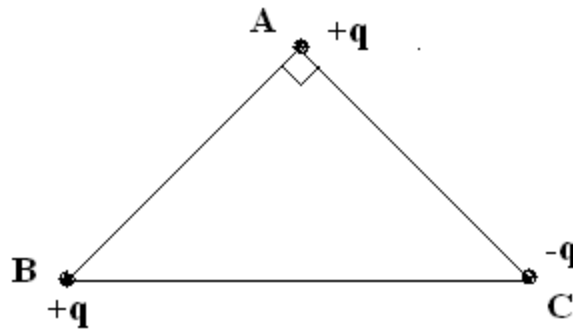
GRAPH OF ELECTRIC POTENTIAL OF A CHARGED SPHERE



Worked examples

Question 1

Three charges are at the vertices of a right angled triangle ABC where $AB = AC = 5 \text{ cm}$ and $q = 1.0 \times 10^{-7} \text{ C}$.



Find the force acting on the charge at A ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$)

Solution

$$F_B = \frac{q^2}{4\pi\epsilon_0 r^2}$$

$$\Rightarrow F_B = \frac{(1 \times 10^{-7})^2}{4\pi \times 8.85 \times 10^{-12} \times (0.05)^2}$$

$$\Rightarrow F_B = 3.59672 \times 10^{-2} \text{ N}$$

$$F_C = \frac{-q^2}{4\pi\epsilon_0 r^2}$$

$$F_C = -3.59672 \times 10^{-2} \text{ N}$$

We then calculate the resultant force by adding F_B and F_C vectorially

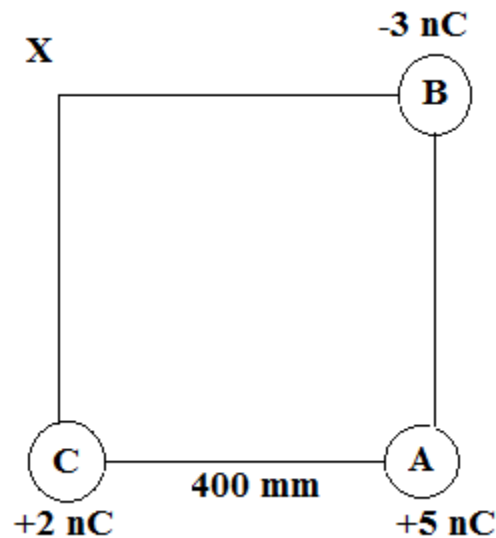
Resultant force F is given by

$$F = \sqrt{F_B^2 + F_C^2}$$

$$\Rightarrow F = 5.09 \times 10^{-2} \text{ N}$$

Question 2

Three charges A, B and C of magnitude +5 nC, 3 nC and 2 nC respectively are arranged at the corners of a square in vacuum as shown.



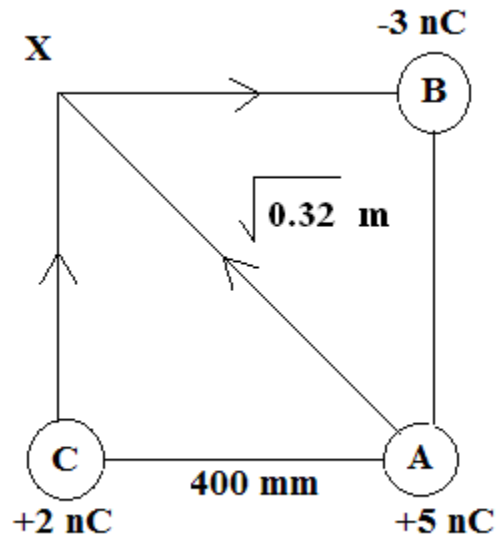
Calculate

- (i) the electric potential at X due to the three charges.

- (ii) the electric field strength and its direction at X due to the three charges

Solution

The direction of the electric field at X due to each charge is as shown in fig. We calculate the distance AX and show by the Pythagoras theorem that it is equal to $\sqrt{0.32}\text{m}$



$$V_C = \frac{Q_C}{4\pi\epsilon_0 r}$$

$$\Rightarrow V_C = \frac{2 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.4}$$

$$\Rightarrow V_C = 44.959V$$

$$V_B = \frac{Q_B}{4\pi\epsilon_0 r}$$

$$\Rightarrow V_B = \frac{-3 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.4}$$

$$\Rightarrow V_B = -67.439V$$

$$V_A = \frac{Q_A}{4\pi\epsilon_0 r}$$

$$\Rightarrow V_A = \frac{+5 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.4}$$

$$\Rightarrow V_A = 79.477V$$

$$V_{total} = V_C + V_B + V_A$$

$$\Rightarrow V_{total} = 44.959V + (-67.439V) + 79.477V$$

$$\Rightarrow V_{total} = 57V$$

$$E_C = \frac{\Delta V_C}{\Delta r}$$

$$\Rightarrow E_C = \frac{44.959}{0.4}$$

$$\Rightarrow E_C = 112.40Vm^{-1}$$

$$E_B = \frac{\Delta V_B}{\Delta r}$$

$$\Rightarrow E_B = \frac{-67.439}{0.4}$$

$$\Rightarrow E_B = -168.60Vm^{-1}$$

$$E_A = \frac{\Delta V_A}{\Delta r}$$

$$\Rightarrow E_A = \frac{79.447}{\sqrt{0.32}}$$

$$\Rightarrow E_A = 140.5Vm^{-1}$$

To find the total field strength, we add E_A , E_B and E_C vectorially
We need to resolve E_A vertically and horizontally because we have a system of three vectors.

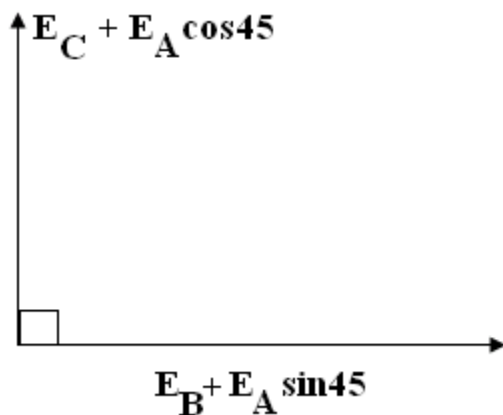
Resolving E_A vertically:

$$\begin{aligned} E_{A_{\text{vertical}}} &= E_A \cos 45 \\ \Rightarrow E_{A_{\text{vertical}}} &= 140.5 \cos 45 \\ \Rightarrow E_{A_{\text{vertical}}} &= 99.35 \text{Vm}^{-1} \end{aligned}$$

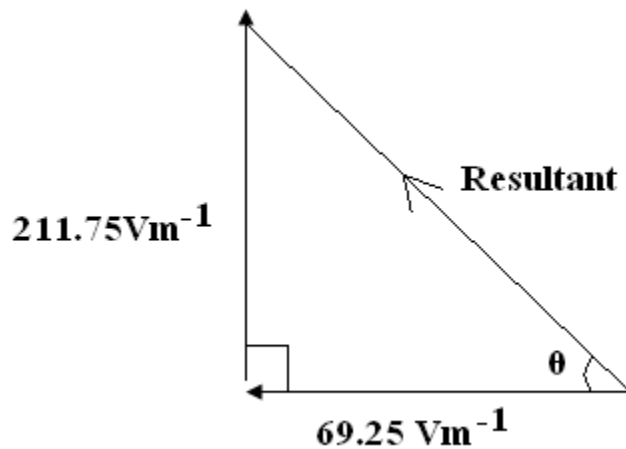
Resolving E_A horizontally:

$$\begin{aligned} E_{A_{\text{horizontal}}} &= E_A \sin 45 \\ \Rightarrow E_{A_{\text{horizontal}}} &= 140.4 \sin 45 \\ \Rightarrow E_{A_{\text{horizontal}}} &= 99.35 \text{Vm} \end{aligned}$$

Fig now shows a simplified diagram representing the electric fields and their directions



Now, $E_B + E_A \sin 45$ is negative so this sum is directed to the left as shown in fig



By using the Pythagoras theorem, the resultant E is calculated as

$$\begin{aligned}
 E^2 &= 211.75^2 + 69.25^2 \\
 \Rightarrow E &= \sqrt{211.75^2 + 69.25^2} \\
 \Rightarrow E &= 222.8 \text{ Vm}^{-1} \\
 \theta &= \tan^{-1} \left[\frac{211.75}{69.25} \right] \\
 \theta &= 71.9^\circ
 \end{aligned}$$

Therefore the electric field strength at X is 222.8 Vm^{-1} at 71.9° to the horizontal

Electric Potential due to a Conducting Sphere

When a sphere either hollow or solid is charged, all the charge uniformly distributes itself at the surface of the sphere. NB: no charges accumulate inside the sphere. Consider a sphere of radius a with total charge Q uniformly distributed at its surface. The sphere is assumed to behave as a point charge so that all the charge is concentrated at its centre. The electric field strength E at the surface of the sphere is given by

$$E = \frac{Q}{4\pi\epsilon a^2}$$

The electric potential at its surface is given by

$$V = \frac{Q}{4\pi\epsilon a}$$

Equipotentials

In electric fields, equipotentials are imaginary lines joining points of the same potential. The diagram shows equipotentials around a positive charge.

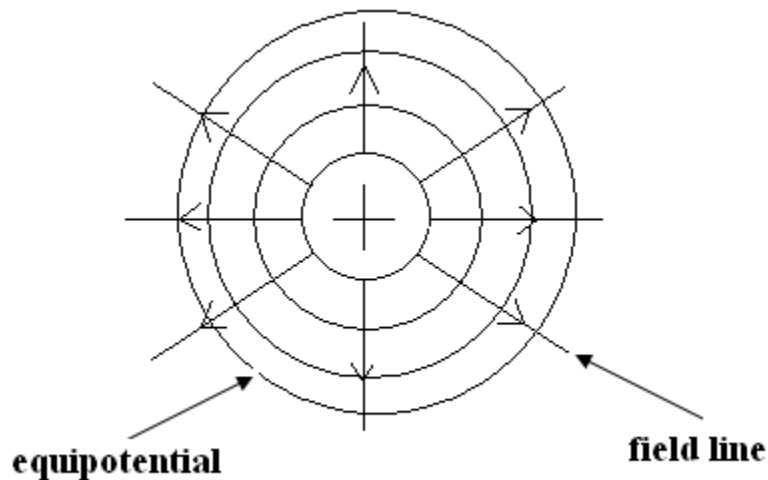
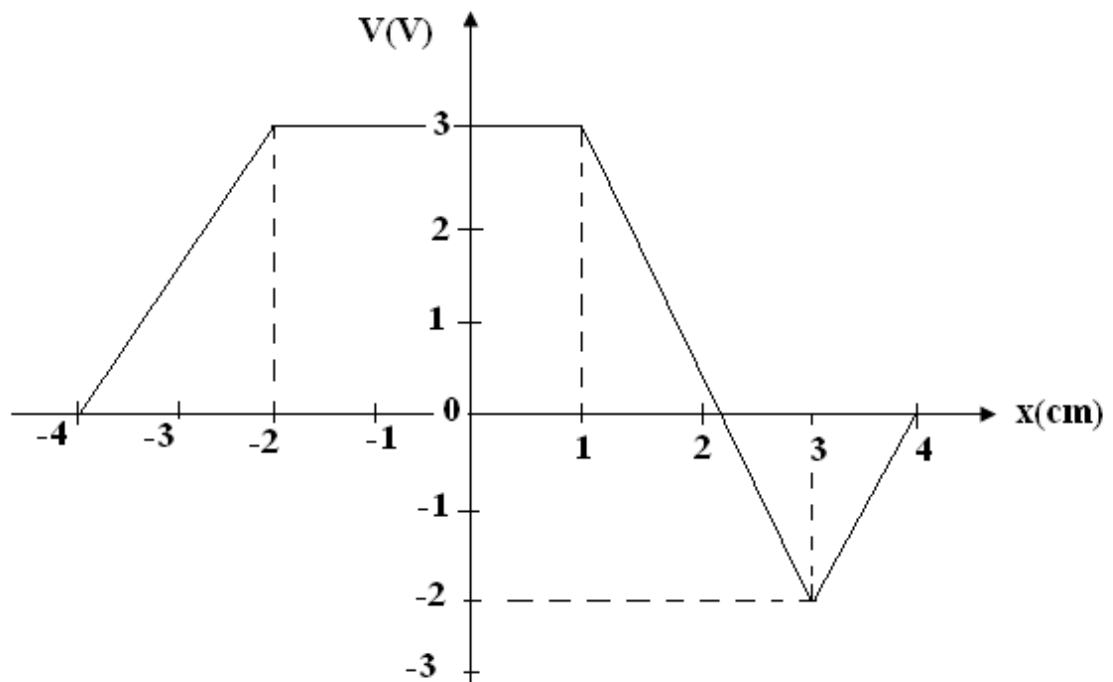


Fig. 4.2: Representation of Equipotentials

Worked Example

Question 1

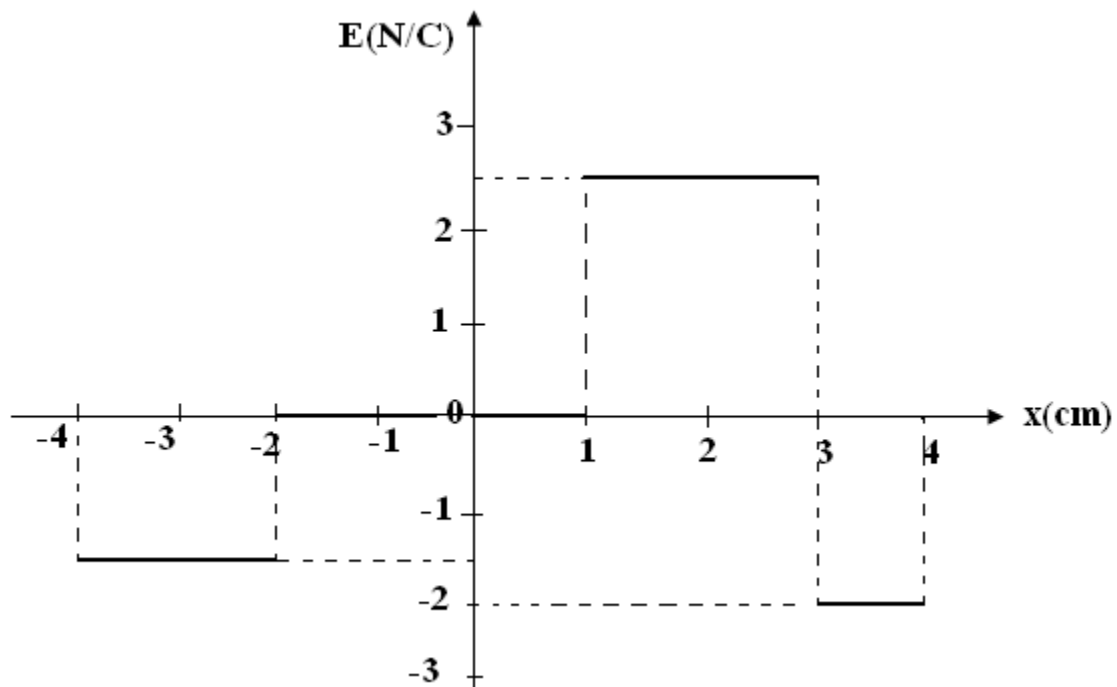
The diagram below shows the variation of the electric potential V with distance x from a fixed point along a straight line



Draw a sketch graph to show the variation of the electric field intensity with distance x

Solution

Since $E = -\frac{dV}{dr}$ we follow the variation of the gradient in fig but remembering that it is the negative of the gradient that we will plot.



Electric Field between Parallel Plates

The electric field strength E between the plates is given by $E = \frac{V}{d}$

Where V is the potential difference between the plates and d is the plate separation

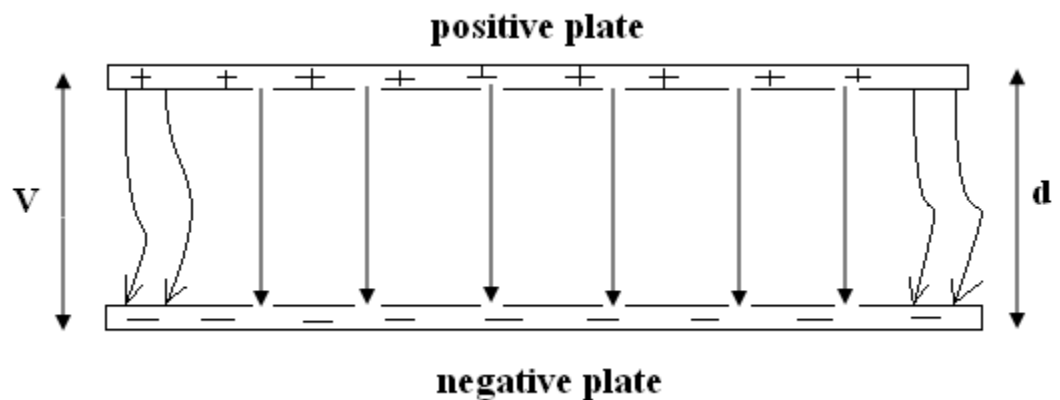


Fig. 4.3: Representation of Electric Field strength between Parallel Plates

Worked Example**Question 1**

An oil drop of mass $3.2 \times 10^{-14} \text{ kg}$ carries a net charge equal to that of 5 electrons. If the acceleration of free fall is 10 ms^{-1} , calculate the magnitude of the electric field required to keep the drop stationary

Solution

When the drop is stationary then the electric force must balance the gravitational force.

$$\begin{aligned}
 F_{\text{electric}} &= F_{\text{gravitational}} \\
 \Rightarrow EQ &= mg \\
 \Rightarrow E &= \frac{mg}{Q} \\
 \Rightarrow E &= \frac{3.2 \times 10^{-14} \times 10}{8.0 \times 10^{-19}} \\
 \Rightarrow E &= 4 \times 10^5 \text{ NC}^{-1}
 \end{aligned}$$

Effect of the Electric Field on the Motion of Charged Particles**Effect of the Electric Field on the Motion of (+)vely Charged Particle**

When a charged particle moves in the same direction as that of the electric field, the electric field does work on the particle and the particle gains electric potential energy. The particle however moves in a direction of decreasing electric potential therefore a gain in electric potential energy is accompanied by a loss in electric potential.

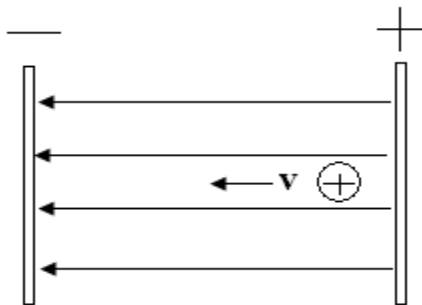


Fig. 4.4:

NOTE: When the positive charge moves to the opposite direction to that shown above. The particle itself does work against the electric field, so it loss electric potential energy.

The particle however moves in the direction of increasing potential. Therefore a loss in electric potential energy is accompanied by a gain in electric potential.

Effect of the Electric Field on the Motion of (-)vely Charged Particle

When a charged particle moves in the opposite direction to that of the electric field then the particle itself does work against the electric field so it loses electric potential energy. The particle however moves in a direction of increasing potential therefore a loss in electric potential energy is accompanied by a gain in electric potential.

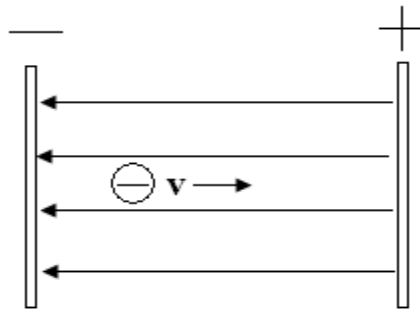


Fig. 4.5:

NOTE: When an electron move to the opposite direction to that shown above (that is to the negative), the electric field does work on the particle and the particle gain electric potential energy. The particle however moves in the direction of decreasing electric potential. Therefore a gain in electric potential energy is accompanied by a loss in electric potential.

COPMARISON OF ELECTRIC FIELD AND GRAVITATIONAL FIELD

QUALITATIVE ANALYSIS

SIMILARITIES

1. Inverse square law applies/ ($F \propto \frac{1}{r^2}$).
2. Involve particle, Newton's third law applies.
3. Involve force between particles.
4. Both can be represented by field lines.

DIFFERENCES

1. Deal with point masses for gravitational law/ deal with point charges for Coulomb's law.
2. Attractive only for gravitational, either attractive or repulsive for Coulomb's law.
3. Have different constants of proportionality.

4. Gravitational potential is always negative, whereas electric potential can be positive or negative.

QUANTITATIVE ANALYSIS

For example comparing quantitatively the electric force and gravitational force between an electron and a proton.

$$F_E = \frac{q_e q_p}{4\pi\epsilon_0 r^2} \quad F_G = \frac{G m_e m_p}{r^2} \quad \text{Where } F_E = \text{Electric force and } F_G = \text{Gravitational force.}$$

$$\text{Then } \frac{F_E}{F_G} = \frac{q_e q_p}{4\pi\epsilon_0 G m_e m_p} = \frac{1.6 \times 10^{-19} (-1.6 \times 10^{-19})}{4\pi (8.85 \times 10^{-12}) (6.67 \times 10^{-11}) (9.11 \times 10^{-31}) (1.67 \times 10^{-27})} = -2.27 \times 10^{39}$$

NOTE: Ignoring the negative sign. This shows that, the Electric force (F_E) is 2.27×10^{39} times greater than the gravitational force (F_G).

4.4: CAPACITANCE

Introduction

Capacitance is the ability of an insulator to store charge. Consider a conductor and suppose that upon receiving charge Q its potential difference increases by V , then mathematically we define the capacitance of the conductor by the equation

$$C = \frac{Q}{V}$$

The unit of capacitance is CV^{-1} and this unit is called the Farad (F). The farad is defined as one Coulomb per volt. The Farad is a very large unit and smaller units are commonly used e.g. microfarad, picofarad and nanofarad.

$$1\mu F = 1 \times 10^{-6} F$$

$$1nF = 1 \times 10^{-9} F$$

$$1pF = 1 \times 10^{-12} F$$

Capacitors

A capacitor is a charge storing device and there are several types.

Basically, a capacitor is formed from two parallel metal plates separated by an insulator called a dielectric

Operation of a capacitor

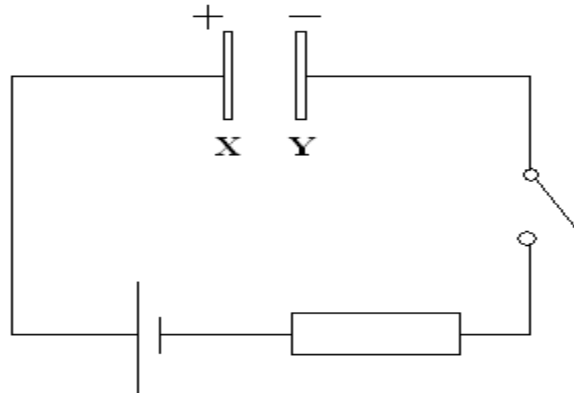


Fig. 5.1: Functional Principle of a Capacitor

It is possible for dielectric materials such as air or paper to hold an electric charge because free electrons cannot flow through an insulator. The charge however must be applied by some source. In **Fig. 5.1**, the battery can charge the capacitor as shown. With the dielectric contacting the two conductors connected to the potential difference V , electrons flowing from the negative of the voltage source accumulate on the Y-plate. The opposite side, X-plate loses electrons. As a result, the excess electrons produce negative charge on the Y-plate while the X-plate has positive charge. A potential difference is therefore developed on the parallel plates. The built up of charge continues until the potential difference between the plates is equal to the power supply voltage. At this point, no further current flows and the capacitor is said to be charged with charge Q that is $+Q$ on the X-plate and $-Q$ on the Y-plate.

The Parallel Plate Capacitor

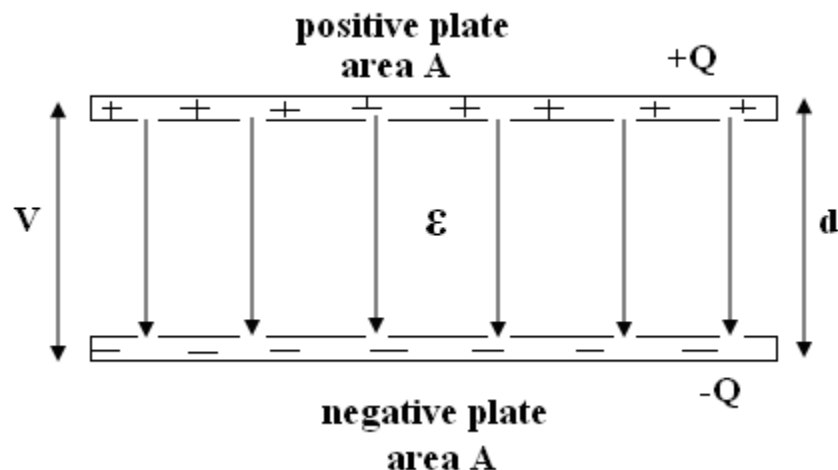


Fig. 5.2: Parallel Plate Capacitor

Consider two parallel plates each of area A separated by a distance d with a medium of permittivity ε between them. The electric field is therefore given by

$$E = \frac{V}{d} \dots\dots\dots(1)$$

Also, we have that $E = \frac{\delta}{\varepsilon} \dots\dots\dots(2)$

By definition, capacitance is given by

$$C = \frac{Q}{V} \dots\dots\dots(3)$$

and charge density is given by

$$\delta = \frac{Q}{A} \dots\dots\dots(4)$$

Substituting (4) in (2)

$$E = \frac{Q}{\varepsilon A} \dots\dots\dots(5)$$

From (3)

$$Q = CV \dots\dots\dots(6)$$

Substituting (6) in (5)

$$E = \frac{CV}{\varepsilon A} \dots\dots\dots(7)$$

From (1)

$$V = Ed \dots\dots\dots(8)$$

Substituting (8) in (7)

$$C = \frac{\varepsilon A}{d} \dots\dots\dots(9)$$

Relative permittivity

The relative permittivity of a medium is the ratio of the permittivity of that medium to the permittivity of vacuum ϵ_0 .

Capacitor combinations

Capacitors in Series

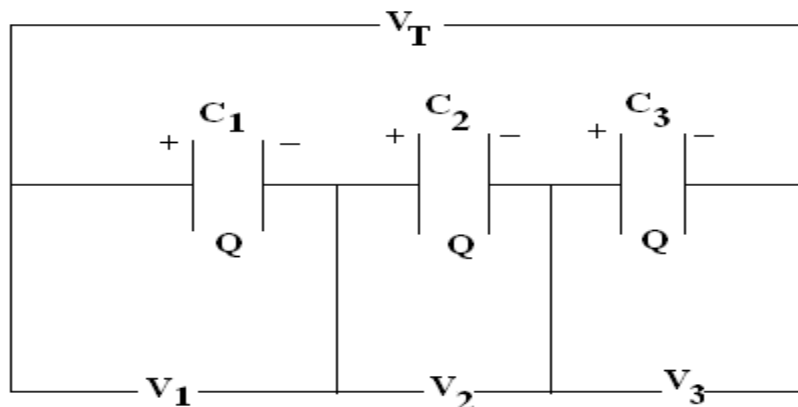


Fig. 5.3: Series Connection of Capacitors

When capacitors are arranged in series they carry equal charges. By definition,

$$C_1 = \frac{Q}{V_1}$$

$$C_2 = \frac{Q}{V_2}$$

$$C_3 = \frac{Q}{V_3}$$

$$V_{total} = V_1 + V_2 + V_3$$

$$\Rightarrow \frac{Q}{C_{total}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

$$\Rightarrow Q \frac{1}{C_{total}} = Q \left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right]$$

$$\Rightarrow \frac{1}{C_{total}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Capacitors in parallel

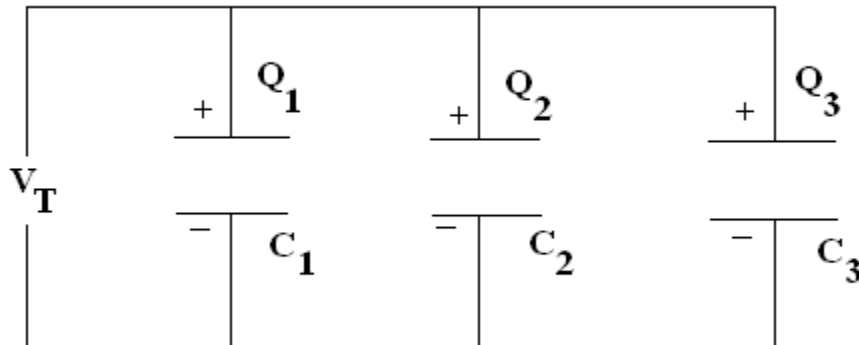


Fig. 5.4: Parallel Connection of Capacitors

When capacitors are arranged in parallel, the p.d. across the combination is the same but the charges may be different depending on the value of each capacitor.

$$C_1 = \frac{Q_1}{V}$$

$$C_2 = \frac{Q_2}{V}$$

$$C_3 = \frac{Q_3}{V}$$

$$Q_{total} = Q_1 + Q_2 + Q_3$$

$$\Rightarrow V_{total} C_{total} = C_1 V_{total} + C_2 V_{total} + C_3 V_{total}$$

$$\Rightarrow V_{total} C_{total} = V_{total} [C_1 + C_2 + C_3]$$

$$\Rightarrow C_{total} = C_1 + C_2 + C_3$$

Energy Stored In a Capacitor

$$C = \frac{Q}{V}$$

$$\Rightarrow V = \frac{Q}{C}$$

$$\Rightarrow V = \frac{1}{C} \times Q$$

A graph of V against Q is a straight line as shown in fig

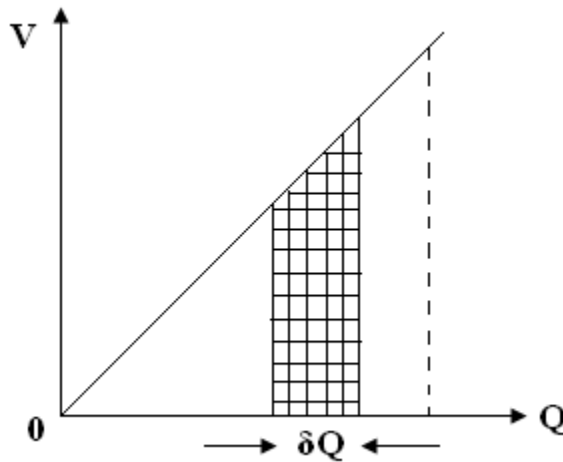


Fig. 5.5: Graphical representation of Energy Stored in Capacitor

Consider a capacitor that discharges in such a way that the charge δQ is so small that the potential difference V may be assumed constant, then the amount of work done in moving the charge is given by

$$\delta W = V \delta Q$$

This quantity is equal to the area of the shaded strip in fig. If the capacitor discharges completely then the total amount of energy lost W is equal to the sum of the areas of all similar strips.

The total area is equal to the area under the graph which is equal to the area of a triangle.

$$W = \frac{1}{2} QV \dots\dots\dots(1)$$

From

$$C = \frac{Q}{V}$$

$$\Rightarrow Q = CV \dots\dots\dots(2)$$

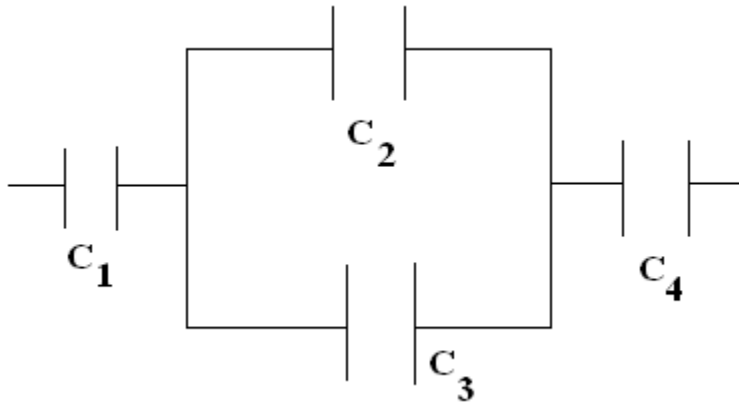
Substituting (2) in (1)

$$\Rightarrow W = \frac{1}{2} CV^2 \dots\dots\dots(3)$$

Worked examples

Question 1

Four $2\ \mu\text{F}$ capacitors are connected as shown in fig



Calculate the effective capacitance

Solution

C_2 and C_3 are in parallel so the total is

$$\begin{aligned}
 C_2 + C_3 &= C \\
 &= 2 + 2 \\
 &= 4 \\
 \frac{1}{C_{\text{total}}} &= \frac{1}{C_1} + \frac{1}{C_4} + \frac{1}{C} \\
 \Rightarrow \frac{1}{C_{\text{total}}} &= \frac{1}{2} + \frac{1}{2} + \frac{1}{4} \\
 \Rightarrow \frac{1}{C_{\text{total}}} &= \frac{5}{4} \\
 \therefore C_{\text{total}} &= \frac{4}{5} \mu\text{F}
 \end{aligned}$$

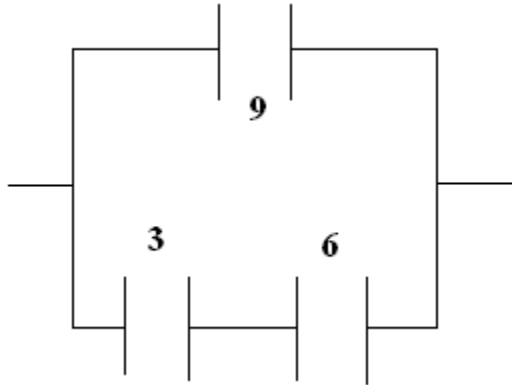
Question 2

How would you connect three capacitors measuring $3\ \mu\text{F}$, $6\ \mu\text{F}$ and $9\ \mu\text{F}$ to obtain an effect capacitance of $11\ \mu\text{F}$

Solution

$$11 = 9 + 2$$

We connect the $3\ \mu\text{F}$ and $6\ \mu\text{F}$ capacitors in series and their combination is connected in parallel with the $9\ \mu\text{F}$ capacitor as shown below.



Question 3

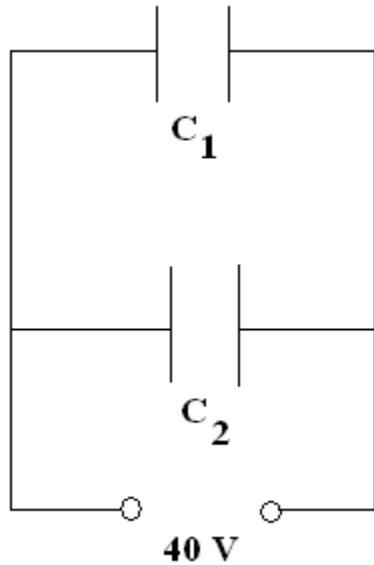
A $2\ \mu\text{F}$ capacitor is charged to a pd of $200\ \text{V}$ and then isolated. When it is connected in parallel with another capacitor which is initially uncharged, the common pd is $40\ \text{V}$. Calculate the capacitance of the second capacitor.

Solution

Let C_1 be the $2\ \mu\text{F}$ capacitor, C_2 be the unknown capacitor and V_1 be equal to $200\ \text{V}$. The $2\ \mu\text{F}$ capacitor acquires charge Q_1 given by

$$\begin{aligned} Q_1 &= C_1 V_1 \\ \Rightarrow Q_1 &= 2 \times 10^{-6} \times 200 \\ \Rightarrow Q_1 &= 4 \times 10^{-4} \text{ C} \end{aligned}$$

Fig shows the unknown capacitor now connected



We now calculate the charge that remains on the original $2 \mu\text{F}$ capacitor. If this charge is Q and 40 V is equal to V_2 then

$$\begin{aligned} Q &= C_1 V_2 \\ \Rightarrow Q &= 2 \times 10^{-6} \times 40 \\ \Rightarrow Q &= 8 \times 10^{-5} \text{ C} \end{aligned}$$

If Q_2 is the charge now on the unknown capacitor then

$$\begin{aligned} Q_2 &= Q_1 - Q \\ \Rightarrow Q_2 &= 4 \times 10^{-4} - 8 \times 10^{-5} \\ \Rightarrow Q_2 &= 3.92 \times 10^{-3} \text{ C} \end{aligned}$$

The value of the unknown capacitor C_2 is then calculated using

$$\begin{aligned} C_2 &= \frac{Q_2}{V_2} \\ \Rightarrow C_2 &= \frac{3.92 \times 10^{-4}}{40} \\ \Rightarrow C_2 &= 8 \mu\text{F} \end{aligned}$$

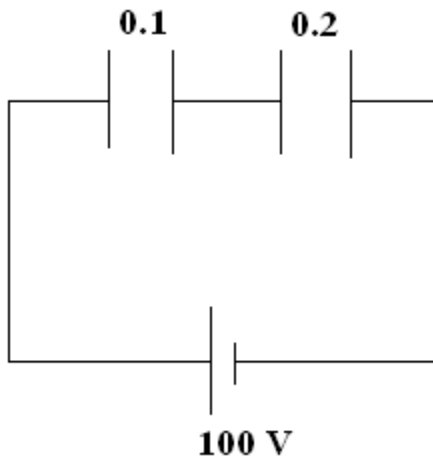
Question 4

Two capacitors of capacitance $0.1 \mu\text{F}$ and $0.2 \mu\text{F}$ respectively are connected in series with a 100 V d.c. supply.

- Calculate the charge on each capacitor.
- Without discharging, the capacitors are disconnected from the supply and connected in parallel so that those plates having the same charge are connected to each other. Calculate the potential difference across the capacitors

Solution

(a)



$$\frac{1}{C_{\text{total}}} = \frac{1}{0.1} + \frac{1}{0.2}$$

$$\Rightarrow C_{\text{total}} = \frac{1}{15} \mu\text{F}$$

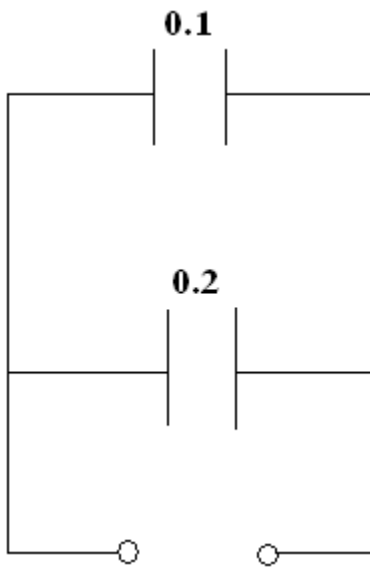
$$Q = C_{\text{total}} \times V$$

$$\Rightarrow Q = \frac{1}{15} \times 10^{-6} \times 100$$

$$\Rightarrow Q = 6.67 \times 10^{-6} \text{ C}$$

This is the charge on each capacitor

(b)



$$C_{total} = 0.1 \times 10^{-6} + 0.2 \times 10^{-6}$$

$$C_{total} = 0.3 \times 10^{-6} F$$

$$V = \frac{Q_{total}}{C_{total}}$$

$$\Rightarrow V = \frac{6.67 \times 10^{-6} + 6.67 \times 10^{-6}}{0.3 \times 10^{-6} F}$$

$$\Rightarrow V = 44.5V$$

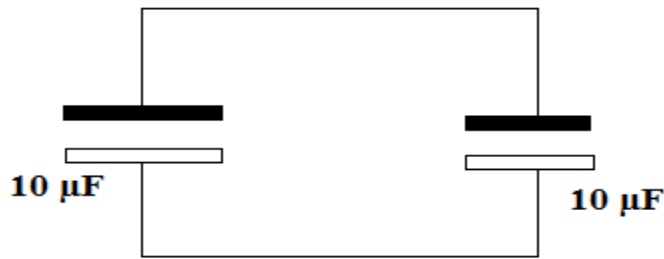
Question 5

(a) A $10\mu F$ capacitor is connected across the terminals of a 100 V d.c. power supply and allowed to charge fully.

(i) Calculate the charge on the capacitor.

(ii) Calculate the energy stored by the capacitor.

(b) The fully charged capacitor is disconnected from the power supply and connected via two wires across the terminals of an uncharged $10\mu F$ capacitor as shown below.



The charge on the original $10\mu\text{F}$ capacitor is shared equally between the capacitors in the combination. Calculate

- (i) the pd across the terminals of each capacitor;
- (ii) the total energy stored by the two capacitors.

(a) Account for the difference between the energy stored by the two capacitors and that stored by the original $10\mu\text{F}$ capacitor

Solution

(a)(i)

$$Q = CV$$

$$\Rightarrow Q = 10 \times 10^{-6} \times 100$$

$$\Rightarrow Q = 1 \times 10^{-3} \text{ C}$$

(ii)

$$W = \frac{1}{2} QV$$

$$\Rightarrow W = \frac{1}{2} \times 1 \times 10^{-3} \times 100$$

$$\Rightarrow W = 5 \times 10^{-2} \text{ J}$$

(b)(i) The charge on each capacitor is

$$Q = \frac{1 \times 10^{-3}}{2}$$

$$\Rightarrow Q = 5 \times 10^{-4} \text{ C}$$

Using one of the capacitors:

$$V = \frac{Q}{C}$$

$$\Rightarrow V = \frac{5 \times 10^{-4}}{1 \times 10^{-5}}$$

$$\Rightarrow V = 50 \text{ V}$$

(ii)

Total capacitance is $20\mu\text{F}$ for the capacitors in parallel

$$W = \frac{1}{2}CV^2$$

$$\Rightarrow W = \frac{1}{2} \times 20 \times 10^{-6} \times 50^2$$

$$\Rightarrow W = 2.5 \times 10^{-2} \text{ J}$$

(d) *The difference is due to energy dissipated as heat in the connecting wires*

5.1 Charging and discharging of capacitors

5.1.1 Charging

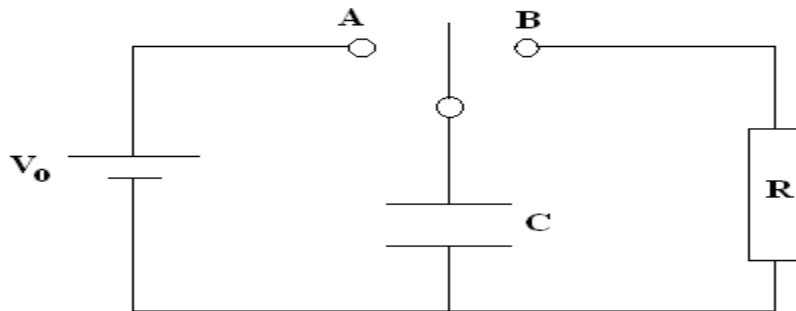


Fig. 5.6: Circuit for Charging a Capacitor

When the switch is closed at A, electrons flow from the negative of the battery onto the negative plate of the capacitor. The p.d across the plates increases exponentially as charge builds up. As the capacitor plates become more and more charged, the flow of electrons decreases and so the charging current decreases as charge builds up.

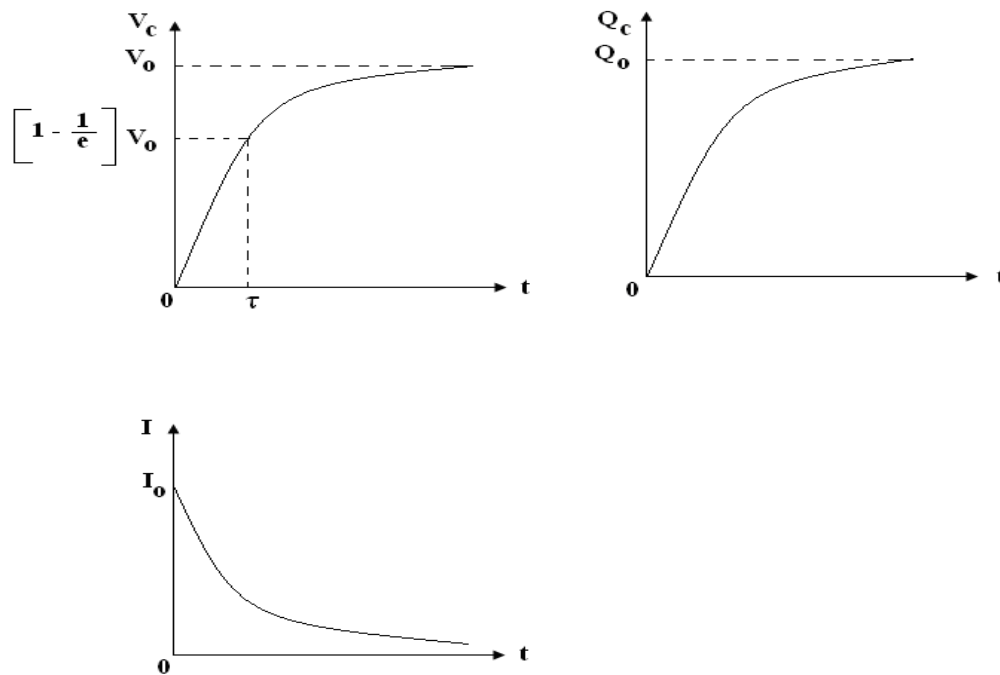


Fig. 5.7: Graphical Representation of Charging a Capacitor

At any time t the p.d. across the capacitor V_C , the charge on the capacitors Q_C and the charging current I are respectively given by

$$V_C = V_0 \left[1 - e^{-t/CR} \right]$$

$$Q_C = Q_0 \left[1 - e^{-t/CR} \right]$$

$$I = I_0 e^{-t/CR}$$

Where V_0 , Q_0 and I_0 are maximum values of voltage, charge and current

Discharging

To discharge the capacitor the contact at A is opened so that the switch is closed at B in figure

During discharge, electrons flow from the negative plate to pass through the resistor and move onto the positive plate. In this way, the plates become neutralized. The p.d. across

the capacitor decreases. The discharge current is initially high but it becomes less and less as the capacitor discharges.

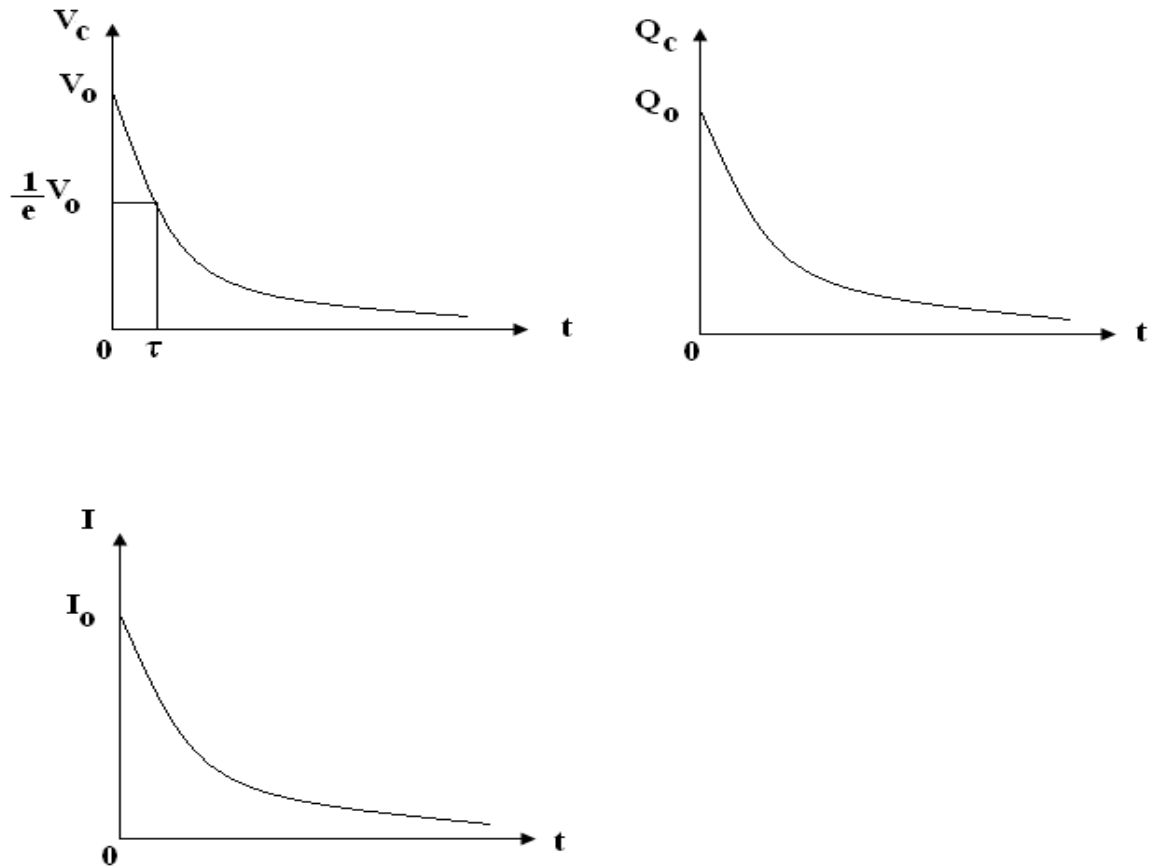


Fig. 5.8: Graphical Representation of Discharging a Capacitor

The variation of voltage, charge and current with time are shown in the equations.

$$V_c = V_0 e^{-t/CR}$$

$$Q_c = Q_0 e^{-t/CR}$$

$$I = I_0 e^{-t/CR}$$

Time constant of a capacitor (τ)

During charging, τ is defined as the time taken for the p.d. across the capacitor to increase to $1 - \frac{1}{e}$ of the charging p.d.

The time constant during discharge is the time taken for the p.d. across the capacitor to decrease to $\frac{1}{e}$ of the charging p.d.

Mathematically the time constant is given by

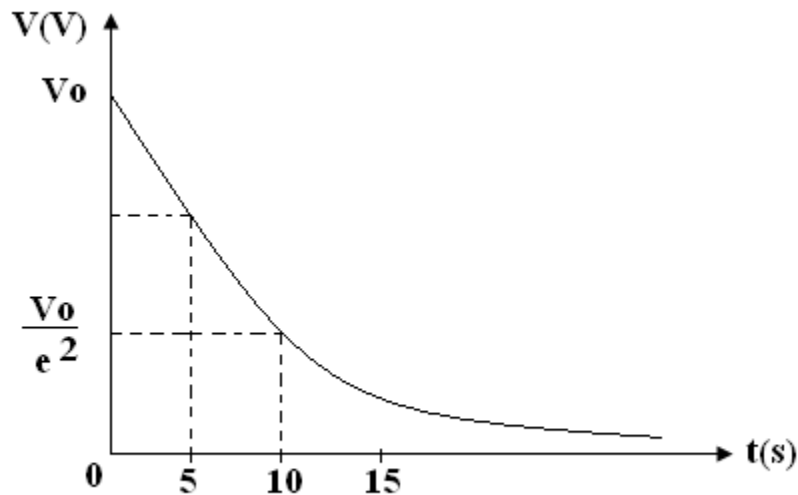
$$\tau = CR$$

where R is the resistance through which the capacitor discharges and C is the capacitance value of the capacitor

Worked example

Question

Figure shows an exponential decay of the p.d. across a charged capacitor which discharges through a resistor of resistance $2 \times 10^6 \Omega$



What is the capacitance of the capacitor?

Solution

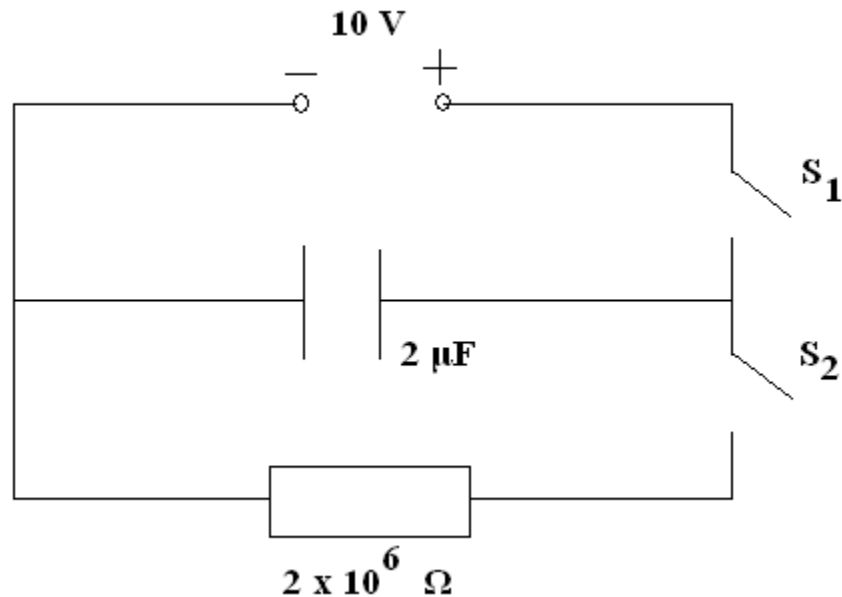
From the graph $\tau = 5s$ i.e. the time taken for the p.d. to reduce to $\frac{V_0}{e}$

Therefore using $\tau = CR$, the value of C is $2.5\mu F$.

S.A.Q

Question 1

Figure shows a $2\mu F$ capacitor connected to a pd of $10 V$ and a $2 \times 10^6 \Omega$ resistor



(a) Switch S_1 is closed and S_2 open, calculate the charge on the capacitor.
[3]

(b) Switch S_1 is then opened and S_2 closed, calculate the charge on the capacitor 8 seconds after switch S_2 is closed.
[3]

Answer

(a) $Q = 2 \times 10^{-5} C$

(b) $Q_C = 2.71 \times 10^{-6} C$

4.5: MAGNETIC FIELDS AND ELECTROMAGNETISM**Introduction**

A magnetic field is a space around a magnet where any other magnetic material experiences a force. The magnetic field is characterized by magnetic field lines or lines of force. Magnetic fields are also generated by current carrying conductors and moving

charged particles. A current carrying conductor or a moving charge may experience a force depending on its orientation when placed in a magnetic field

Definitions

Magnetic flux (ϕ)

This refers to the number of magnetic field lines in space. The unit of ϕ is the Weber (Wb). ϕ is a scalar quantity.

Magnetic flux density (B)

This is the number of magnetic field lines per unit cross sectional area. The unit of B is the Tesla (T). B is a vector quantity. ϕ and B are related by the equation $\phi = BA$ where

A is area. We can write the equation $\phi = BA$ as $B = \frac{\phi}{A}$. The Tesla can therefore be defined as one Weber per square metre i.e. $T \equiv \text{Wb m}^{-2}$

Other definitions of B

B is also called magnetic field strength, magnetic field intensity or magnetic field induction. We recall that the gravitational field and the electric field are forces per unit mass and charge respectively. The magnetic field can also be defined in a similar way. Consider a conductor of length L carrying current I perpendicular to the direction of the magnetic field of strength B . The force acting on the conductor is given by

$$F = BIL$$

Therefore
$$B = \frac{F}{IL}$$

B can thus be defined as the force acting per unit current length of a conductor placed in a magnetic field. When B is defined this way, the Tesla is equivalent to one Newton per ampere metre.

Force on a current carrying conductor

Consider a conductor of length L making an angle θ with the direction of the magnetic field of strength B and carrying current I as shown in **Fig. 6.1**.

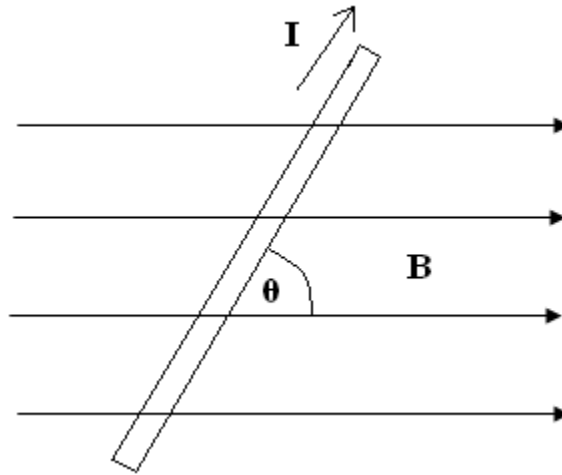


Fig. 6.1: A Conductor moving in Magnetic Field

The force on it is given by

$$F = BIL \sin \theta$$

This force tends to perpendicularise the conductor. F is maximum when $\sin \theta = 1$ i.e. when $\theta = 90^\circ$ therefore $F_{\max} = BIL$

When the conductor lies parallel to the direction of the magnetic field, i.e. $\theta = 0^\circ$ or 180° and $\sin \theta = 0$ then F equals to zero. So when a conductor carries current and lies parallel to the direction of the magnetic field it experiences **no** force.

Force on a moving charge

Consider a charged particle of charge Q moving with velocity v at an angle θ to the direction of the magnetic field of strength B as shown in **Fig. 6.2**.

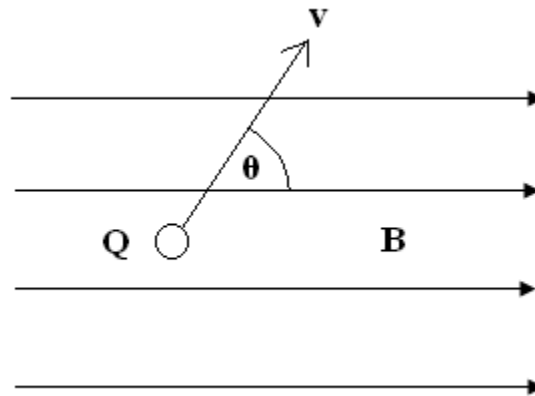


Fig. 6.2: Force on a Moving Charge in a Magnetic Field

The force acting on it is given by

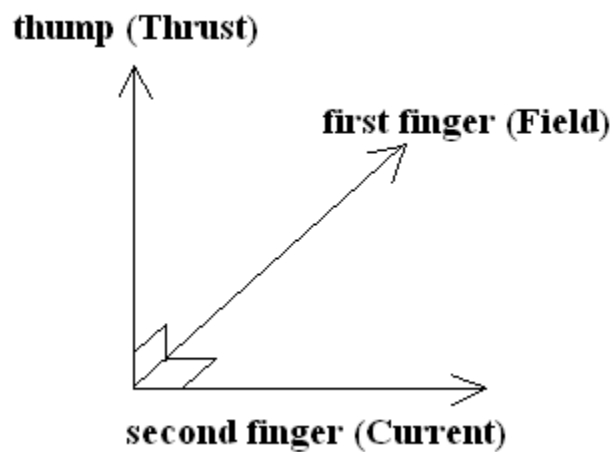
$$F = BQv \sin \theta$$

F is maximum when $\sin \theta = 1$ i.e. when $\theta = 90^\circ$

The charged particle experiences no force in a magnetic field if

- (i) it enters the magnetic field parallel to it i.e. when $\theta = 0^\circ$ or 180° because $\sin \theta = 0$
- (ii) the charged particle is stationary in the field i.e. when $v = 0$

Fleming's left hand rule (The motor rule)

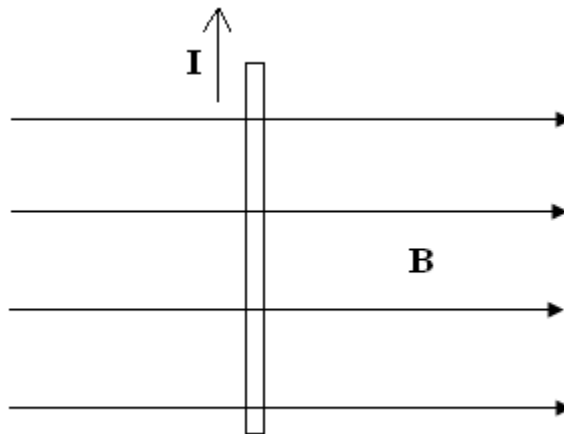


When your left hand fingers are arranged mutually perpendicular to each other, the thumb points in the direction of the thrust or force on the conductor. The first finger points in the direction of the magnetic field. The second finger points in the direction of the current (conventional current)

Using the motor rule

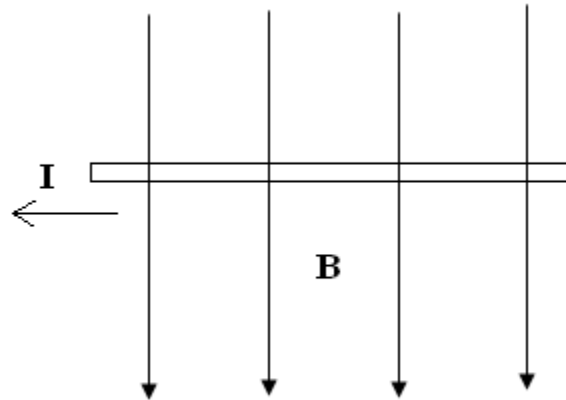
We desire to predict the direction of the force on the conductor given the direction of current flow and the direction of the magnetic field.

Case 1



In the diagram, the current is upwards while the field is directed to the right. The conductor moves into the plane of paper or away from you.

Case 2

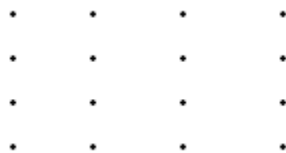


In the diagram, current is directed to the left and the field is downwards. The conductor moves out of plane of paper or towards you.

Other conventional directions of B .

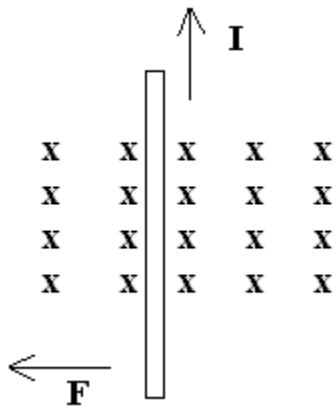


B is directed into the plane of paper (away from you)



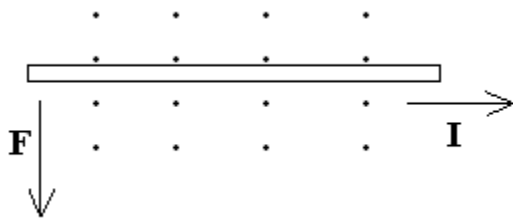
B is directed out of the plane of paper (towards you)

Case 3



In this case the conductor moves to the left

Case 4

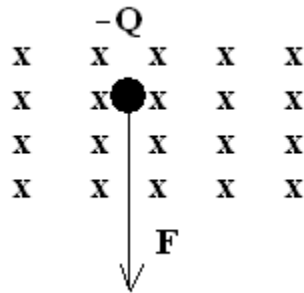


Here the conductor moves downwards

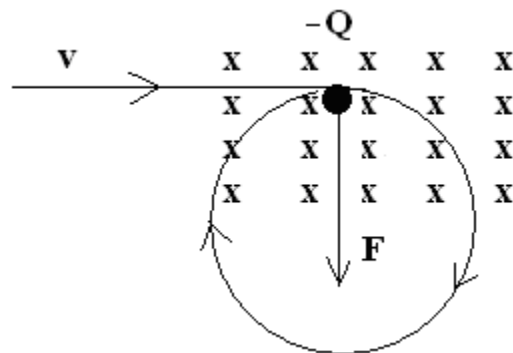
When the motor rule is applied to moving charges, the direction of v , the velocity of the charged particle is that of the motion of a positive charge. In the case of a negative charge, we imagine the motion of a positive charge in the opposite direction and apply the motor rule. A charged particles describe circular motion in a magnetic field. Recall that the motion of a charged particle in an electric field is parabolic.

Worked Examples

Case 1

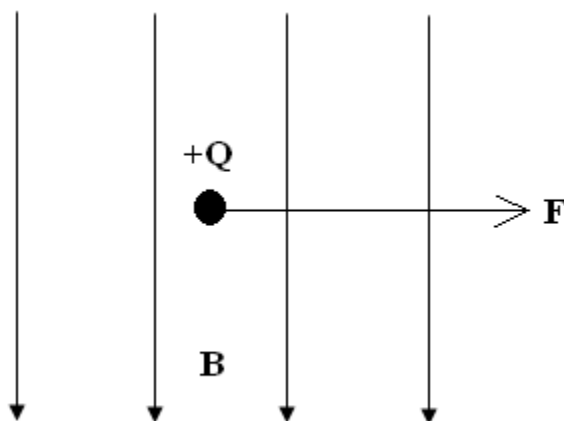


Solution

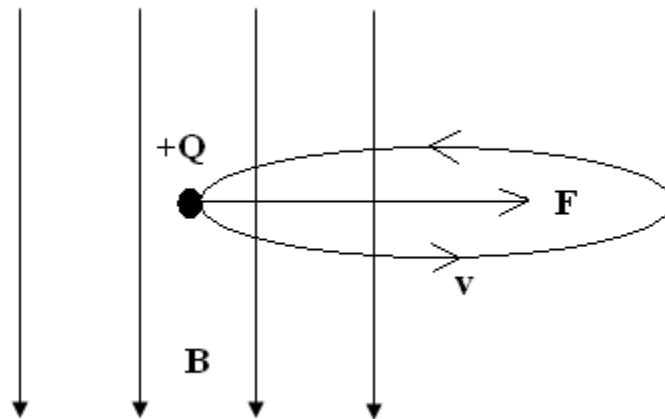


Particle enters the field from the left and describes vertical circular motion in a clockwise sense.

Case 2



Solution

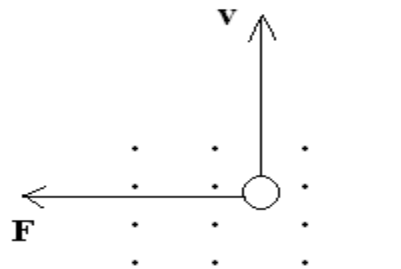


This particle enters the field directed towards you or out of plain of paper and describes horizontal circular motion counter clockwise.

We can also use the rule to predict the sign of charge on a given particle

S.A.Q

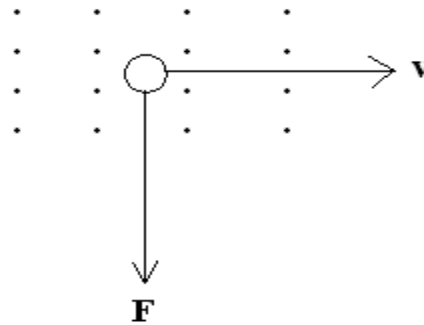
(a) Predict the sign of charge



Solution

The charge is negative

(b) Predict the sign of charge

**Solution**

The charge is positive

Magnetic flux patterns**Long straight wire**

We use the Right Hand Grip Rule. The rule works as follows: When your right hand grips a wire, the thumb points in the direction of the current while all other fingers point in the direction of the magnetic field.

(a)

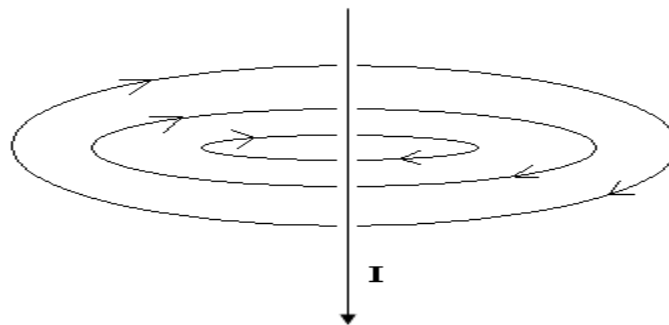


Fig. 6.3: Long Straight Wire with Current inwards

In this case the magnetic fields are concentric circles and their direction is clockwise

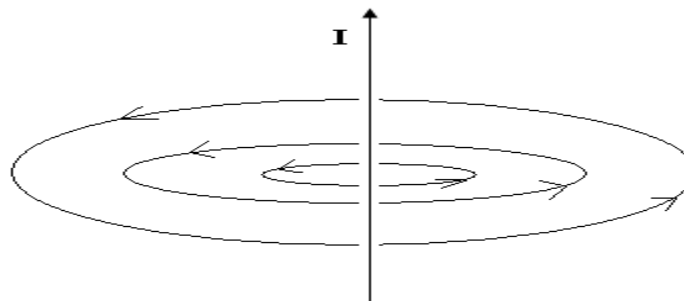
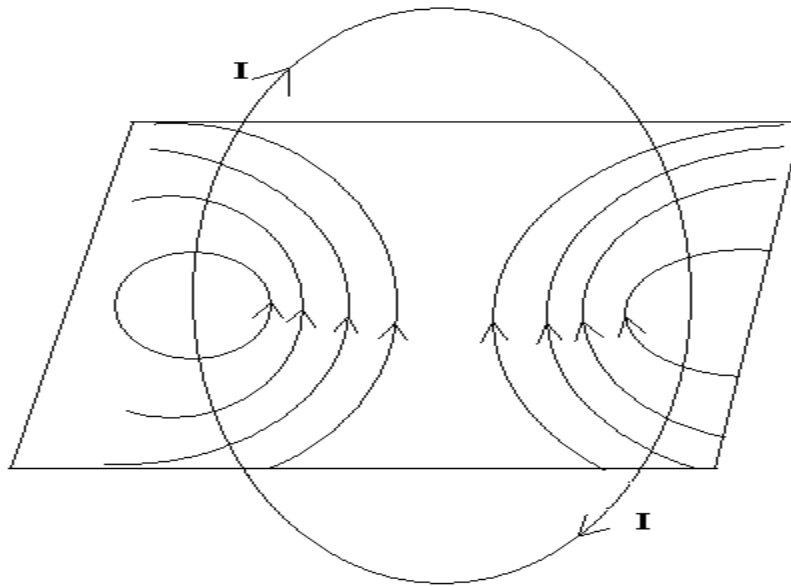


Fig. 6.4: Long Straight Wire with Current outwards

In this case the direction is counter clockwise

A flat circular coil (loop of wire)

**Fig. 6.5: Flux Pattern in a Loop of wire**

A long solenoid (spring-like coil)

If you look at the coil from the left hand side, current flows in an anticlockwise direction. Looking at the coil from the right, current flows in a clockwise sense. The solenoid behaves as a bar magnet with its north pole on the left hand side and South Pole on the right hand side. To remember this we use the curly N and curly S in the diagram. N stands for North Pole and the arrows on it indicate the direction of current flow. The same applies with the curly S.

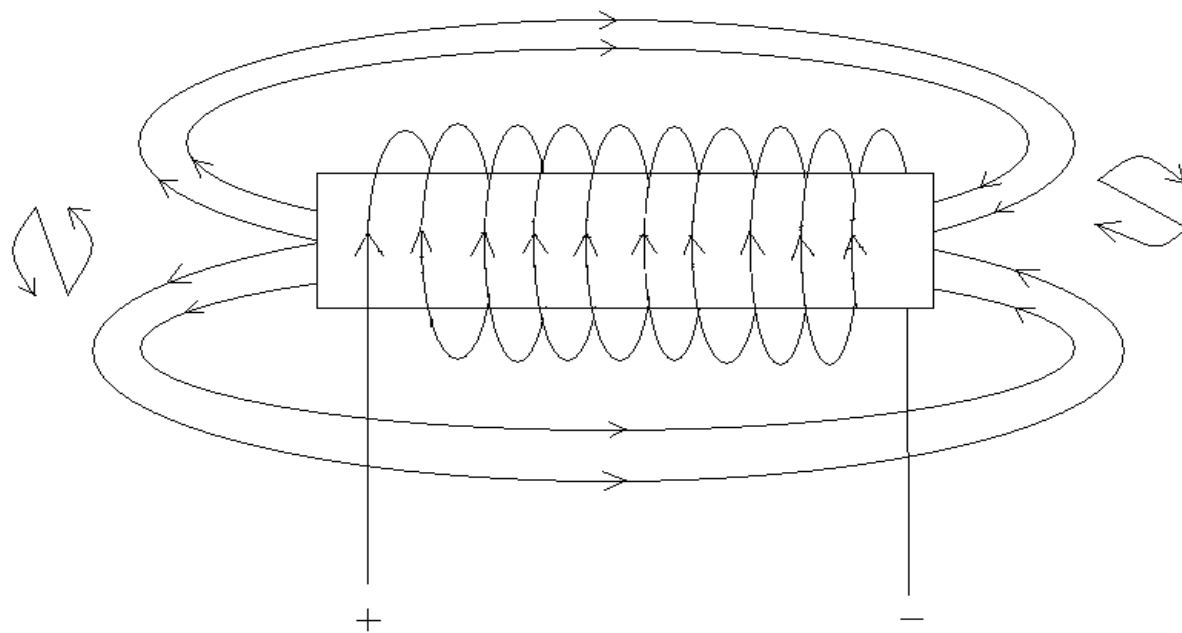


Fig. 6.6: Flux Pattern in a Long Solenoid

Force between current carrying conductors

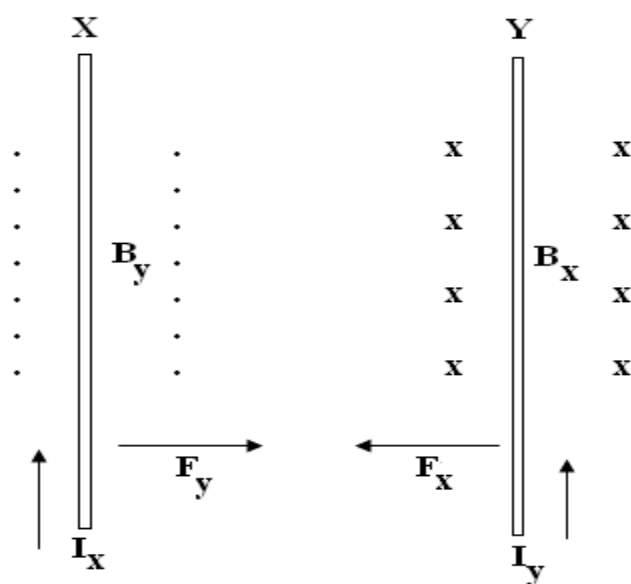
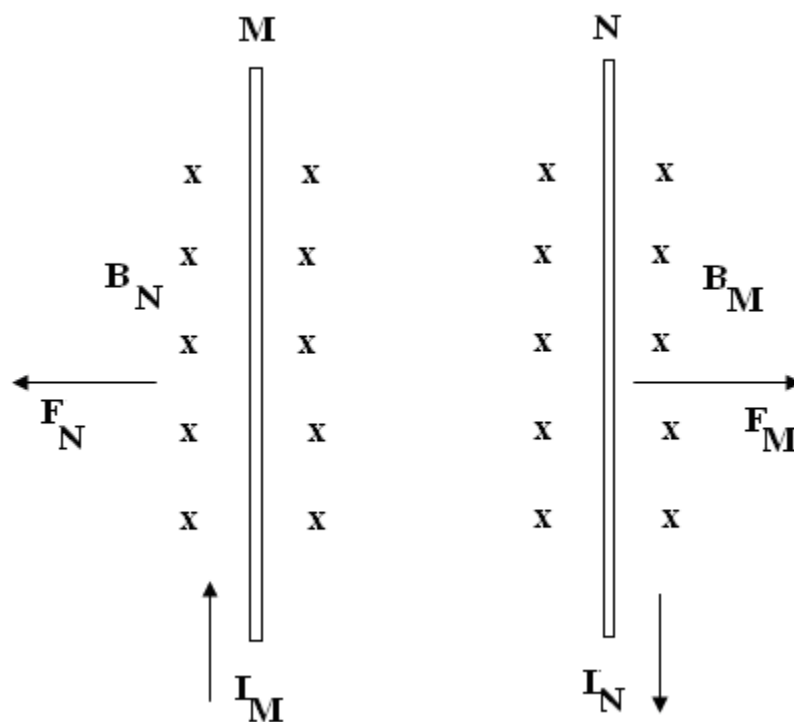


Fig. 6.7: Force when Current is in the same directions

In **Fig. 6.7** the dots show the magnetic field on conductor X carrying current I_x due to conductor Y carrying current I_y . We use the right hand grip rule on conductor Y to show that the magnetic field around X due to Y is directed out of paper as indicated by the dots. The same rule applied on conductor X shows that the magnetic field it produces around wire Y is directed into the plane of paper. We then apply Fleming's left hand rule on both conductors to show that the force F_y exerted by conductor Y on X is directed towards Y and the force F_x exerted by conductor X on conductor Y is directed towards X. The forces are attractive.

**Fig. 6.8: Force when the Current is in the opposite direction**

When the conductors carry current in opposite directions as shown in fig the rules indicate that the conductors exert repulsive forces on each other. We therefore conclude that like currents attract while unlike current repel.

4.6 :ELECTROMAGNETIC INDUCTION

We are now aware that a current carrying conductor or a moving charge produces a magnetic field which causes attractive or repulsive forces on other conductors or charges respectively. So an electric field generates a magnetic field. The question now is: starting with a magnetic field, can we generate an electric field i.e. can we generate electricity from magnetism? The answer is yes and an electromotive force is generated in two ways:

By relative motion (the generator effect)

When a conductor moves inside a magnetic field, cutting across the magnetic field lines or magnetic flux an e.m.f. is generated in it which drives current through the conductor. The same effect is observed when a magnetic field moves relative to a stationary conductor. It should be noted that an e.m.f. is generated **only** when there is relative motion. The motion of a conductor inside a magnetic field is called the generator effect.

Factors affecting the magnitude of the e.m.f and induced current

The magnitude of the e.m.f. and the current that flows is influenced by the following factors:

The strength of the magnetic field

The greater the value of B the greater the magnitude of the e.m.f. and current

The number of turns if the conductor is a coil

The greater the number of turns the greater the e.m.f. and current

The speed of motion

The greater the speed the greater the e.m.f. and current

Changing of magnetic flux linkage (The transformer effect)

An e.m.f. is generated in a coil whenever there is a change in the number of magnetic field lines (flux) linking or passing through the coil. The number of magnetic field lines linking the coil is called magnetic flux linkage.

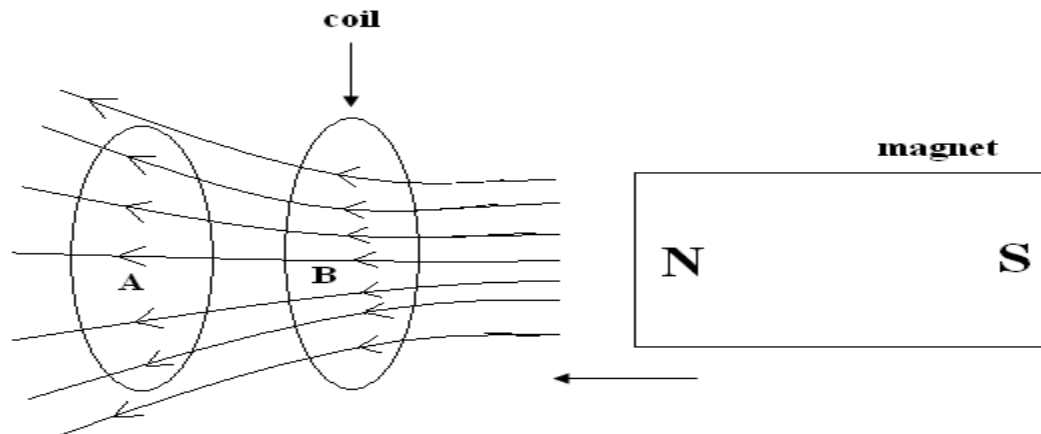


Fig. 6.9: The Transformer Effect

In the **Fig. 6.9**, if the coil has two turns then the change in the number of field lines linking the coil at **B** (7) to the number linking the coil at **A** (5) generates an e.m.f. in the coil.

Magnetic flux linkage

The magnetic flux passing through a given area is given by

$$\phi = BA$$

This equation applies for an area A whose normal is parallel to the direction of the magnetic field.

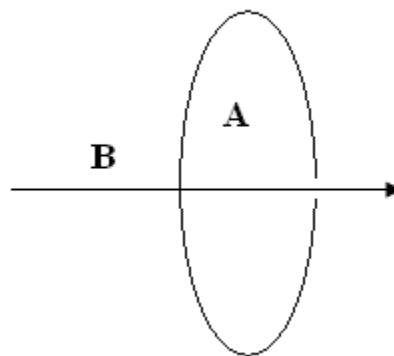


Fig. 6.10: Area Parallel to the direction of B

In general, if the normal to the area makes an angle θ with the direction of the magnetic field, then

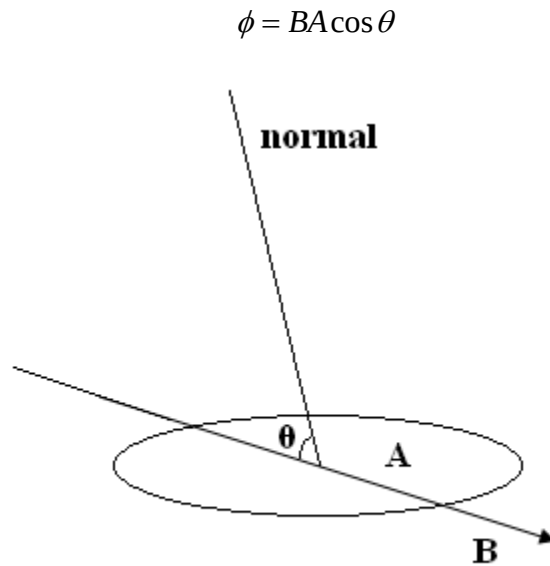


Fig. 6.11: Area at an angle to the direction of B

$B \cos \theta$ is a component of B resolved along the normal (remember B is a vector quantity).

If a coil has N number of turns then the total flux linkage is $N\phi$ i.e. $\phi_T = N\phi$

Therefore $\phi_T = NBA \cos \theta$

Laws of electromagnetic induction

Faraday's Law

It states that the induced e.m.f. is directly proportional to the rate of magnetic flux linkage i.e.

$$E \propto \frac{d\phi_T}{dt}$$

The constant of proportionality is unity (one) therefore

$$E = \frac{d\phi_T}{dt}$$

Lenz's law

It states that the induced e.m.f. or current produces a magnetic field that opposes the inducing magnetic field.

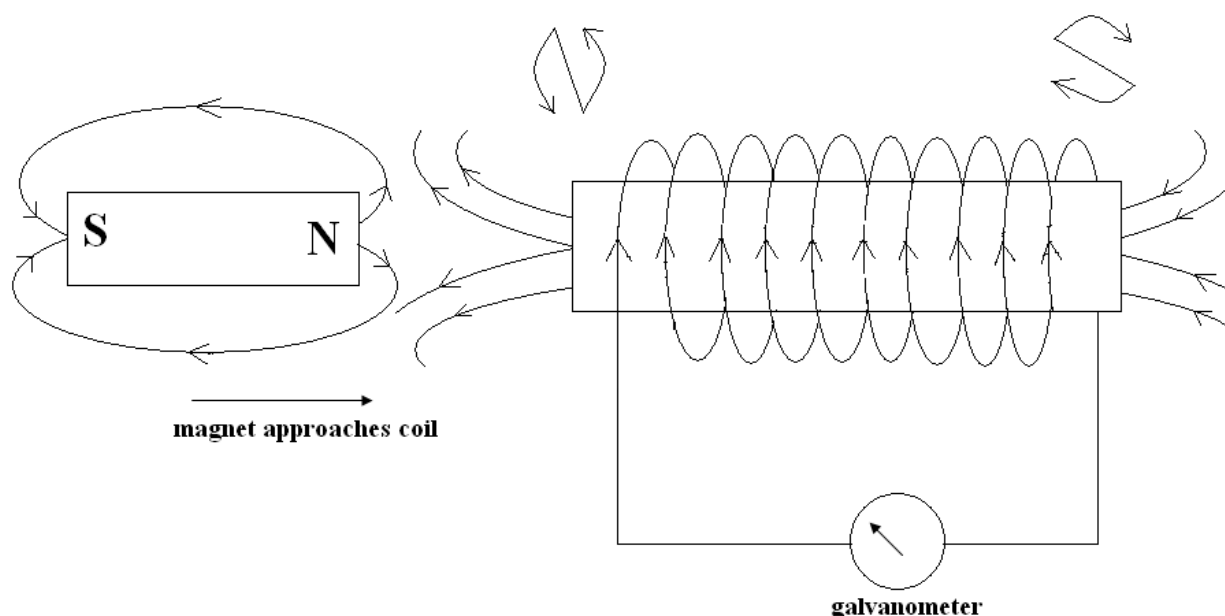


Fig. 6.12: Representation of Lenz's Law

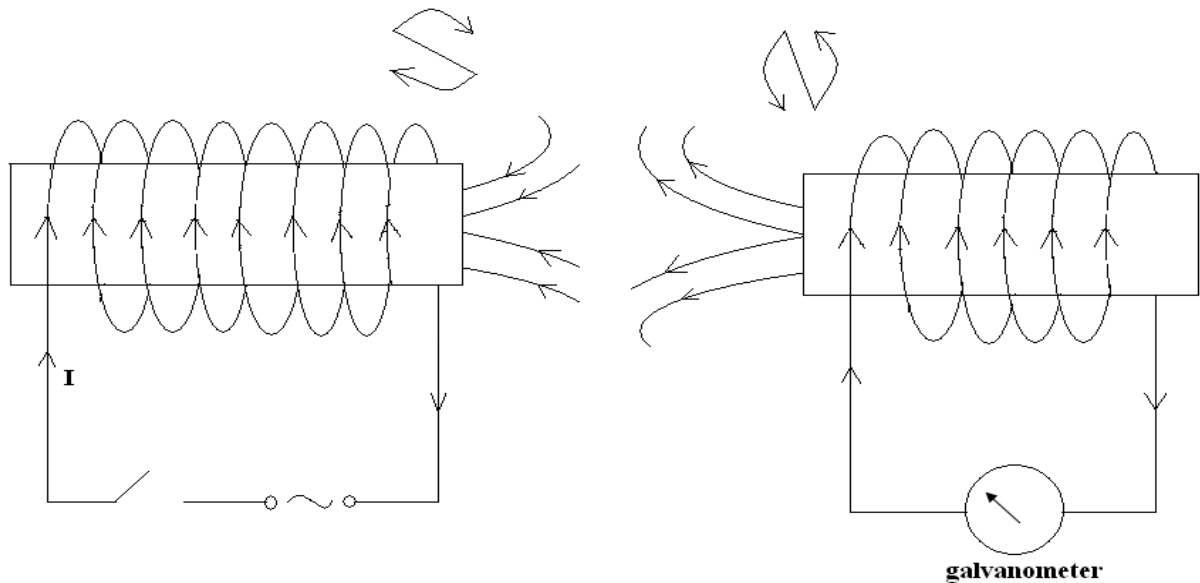
In **Fig. 6.12** the magnet approaches the coil with its north pole. The current that flows in the coil is in such a direction that the coil behaves as a bar magnet with its north pole on its left hand side thus causing repulsion or opposition.

When the bar magnet is withdrawn from the coil, the coil behaves as a bar magnet with its south pole on its left hand side thus causing attraction or opposition again.

Lenz's law is based on the principle of conservation of energy. When an e.m.f. is induced in a coil in this way, there is conversion of energy from mechanical to electrical. When there is energy conversion, some work is done. The current that flows opposes the flux change causing it so as to allow work to be done against resistive forces thus converting mechanical energy into electrical energy. If for instance a south pole was created on the left hand side of the coil in fig then the magnet would accelerate towards the coil i.e. no work is done and energy is converted from nothing thus violating the law of conservation of energy. In order to account for Lenz's law, Faraday's law is written as

$$E = - \frac{d\phi_T}{dt}$$

The negative sign shows that the e.m.f. generated opposes the flux change causing it

Mutual induction (The transformer effect)**Fig. 6.13: Mutual Induction**

Suppose that an a.c. source is used as shown in **Fig. 6.13**. When the switch is closed, alternating current flows in the left hand side coil. This causes a changing magnetic field to link the right hand side coil and so an e.m.f. is generated in this coil. This e.m.f is also alternating. Note that the two coils are not joined together but they are separate. If the frequency of the inducing e.m.f. is adjusted to allow the pointer on the galvanometer to move slowly then this pointer is observed to deflect in both directions continuously.

E.m.f. induced in a straight conductor

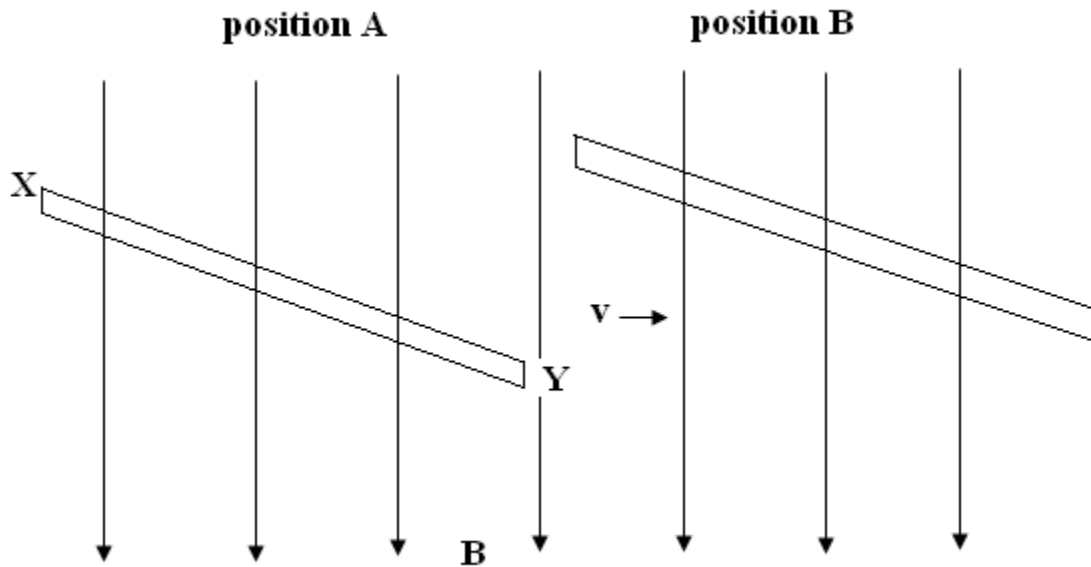


Fig. 6.14: Induced e.m.f in a straight conductor

Consider a straight rod XY of length L moving with constant velocity v at right angles to the direction of the magnetic field of strength B . Suppose that in a time interval Δt , the rod moves from position A to position B . The distance it moves is $v\Delta t$. The small area swept by the rod ΔA is given by

$$\Delta A = Lv\Delta t \dots\dots\dots (0)$$

By Faraday's law, the e.m.f. induced in the conductor is given by

$$E = \frac{d\phi}{dt} \dots\dots\dots (1)$$

$$\Rightarrow E = \frac{\Delta\phi}{\Delta t}$$

But

$$\phi = BA \dots\dots\dots (2)$$

$$\Rightarrow E = \frac{\Delta(BA)}{\Delta t} \text{ and since } B \text{ is a constant we have:}$$

$$E = \frac{B\Delta A}{\Delta t} \dots\dots\dots (3)$$

Substituting (0) in (3)

$$\Rightarrow E = \frac{BLv\Delta t}{\Delta t}$$

$$\Rightarrow E = BLv$$

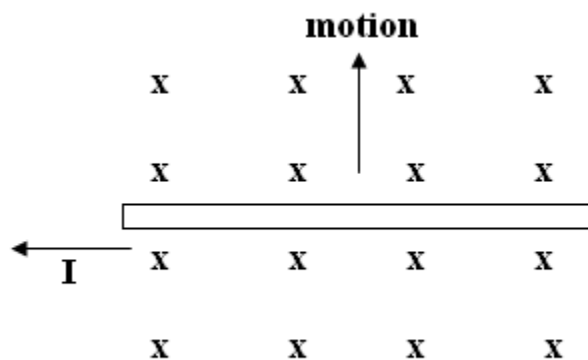
Fleming's right hand rule (The dynamo rule)

This rule is similar to the left hand rule but in this case the thumb points in the direction of motion of the conductor rather than the force on the conductor. The following examples illustrate how the rule can be used to predict the direction of current flow in a conductor moving in a magnetic field.

Worked Examples

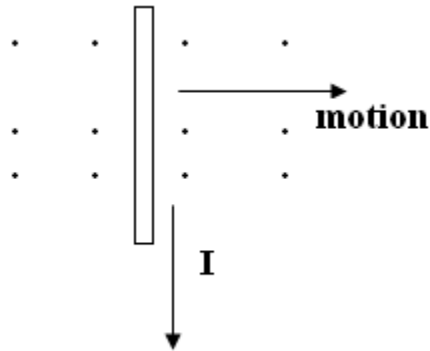
Example 1

It is desired to predict the direction of current flow given the direction of the magnetic field and the direction of motion of the conductor.



In this example current flows to the left

Example 2



In this example, current flows downwards

Example 3

A straight wire of length 50 cm and resistance $10\ \Omega$ moves sideways with velocity 15 ms^{-1} at right angles to a uniform magnetic field of flux density $2.0 \times 10^{-3}\text{ T}$. What current would flow if its ends were connected by leads of negligible resistance?

Solution

$$E = BLv$$

$$\Rightarrow E = 2 \times 10^{-3} \times 0.5 \times 25$$

$$\Rightarrow E = 0.015\text{ V}$$

$$I = \frac{E}{R}$$

$$\Rightarrow I = \frac{0.015}{10}$$

$$\Rightarrow I = 1.5\text{ mA}$$

The a.c. generator

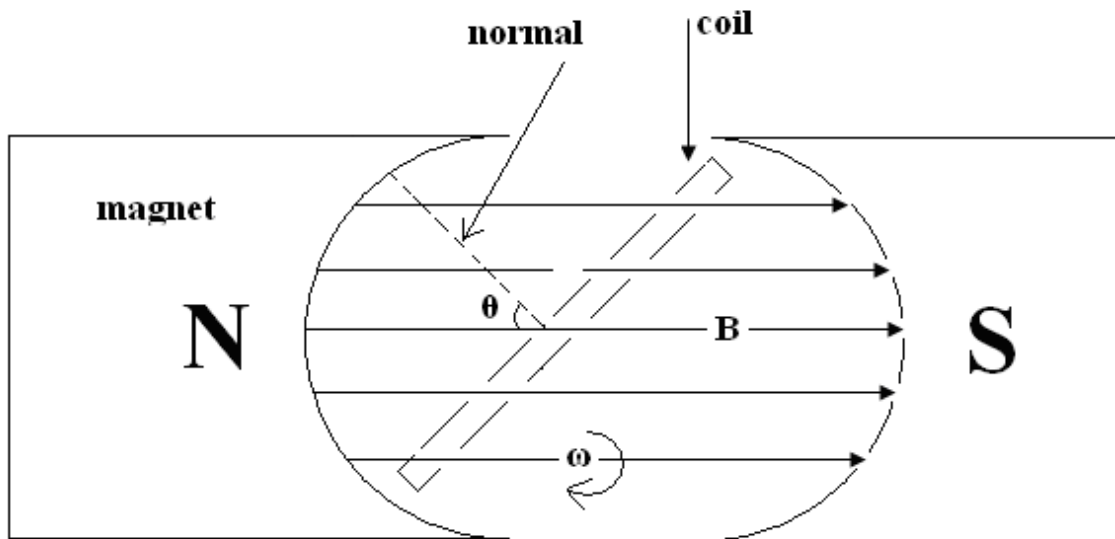


Fig 6.15: Structure of an a.c. generator

Consider a coil of N turns each of area A rotating in a magnetic field of strength B with constant angular velocity ω . The normal to the area of the coil makes an angle θ with the direction of the magnetic field. We recall that the total magnetic flux linkage is given by

$$\phi_T = NBA \cos \theta$$

But

$$\theta = \omega t$$

$$\Rightarrow \phi_T = NBA \cos \omega t$$

ϕ_T is maximum when $\cos \omega t = 1$

$$\text{i.e. } \phi_{T_{\max}} = NBA$$

If we define $\phi_{T_{\max}} = \phi_o = \text{constant}$ then

$$\phi_T = \phi_o \cos \omega t$$

The e.m.f. E generated is by Faraday's law given by

$$E = -\frac{d\phi_r}{dt}$$

$$\Rightarrow E = -\frac{d}{dt}(\phi_o \cos \omega t)$$

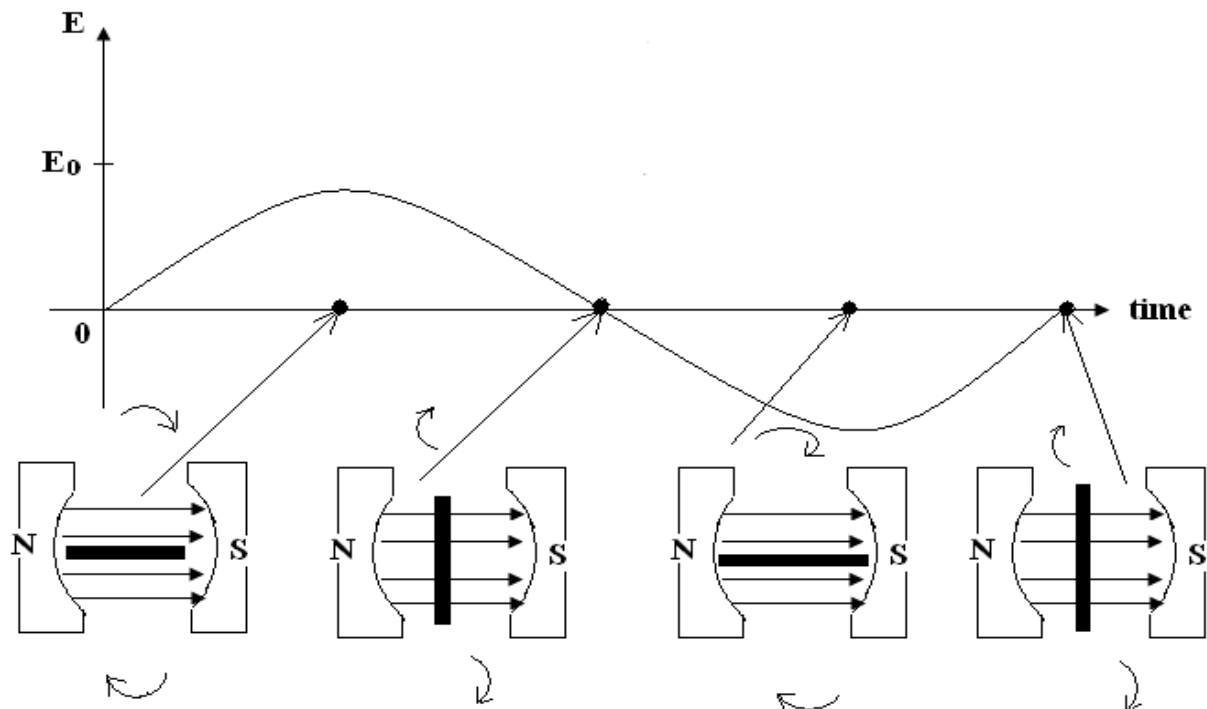
$$\Rightarrow E = -\phi_o \frac{d \cos \omega t}{dt}$$

$$\Rightarrow E = \omega \phi_o \sin \omega t$$

If we define $\omega \phi_o = E_{\max} = E_o = \text{constant}$

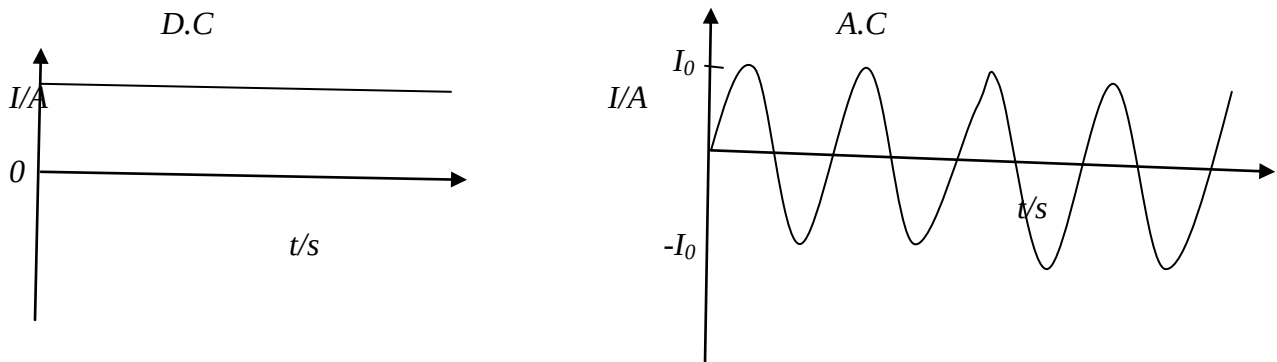
Then $E = E_o \sin \omega t$

E_o is called the peak value or the maximum value of the induced e.m.f. When the coil lies parallel to the magnetic field, $\theta = \omega t = 90^\circ$ and $\sin \omega t = 1$ the normal makes 90° with flux lines $E = E_o$ corresponding to the maximum value of the induced e.m.f. When the coil is perpendicular to the direction of the magnetic field, $\theta = \omega t = 0$ and $\sin \omega t = 0$ and $E = 0$ i.e. there is no e.m.f. induced in the coil. Each cycle of the e.m.f. corresponds to one turn of the coil in the magnetic field. The variation with time of the e.m.f. generated is shown in Fig. 6.16. Fig 6.16 also show the various positions of the coil



4.7: ALTERNATING CURRENT

An a.c current or voltage reverses its direction regularly and is usually sinusoidal.



Frequency is the number of complete cycles per unit time.

Period is the time taken to complete one cycle.

Peak value is the maximum value of current or voltage.

Root mean square value is the steady voltage or current which dissipates the same energy in a resistor as the alternating voltage or current of the same value.

We can represent current or voltage by the equations below.

$I = I_0 \sin \omega t$ Or $I = I_0 \cos \omega t$ depending on the origin of the graph. Where I is the current at any given time t , I_0 is the maximum current, ω is the angular frequency.

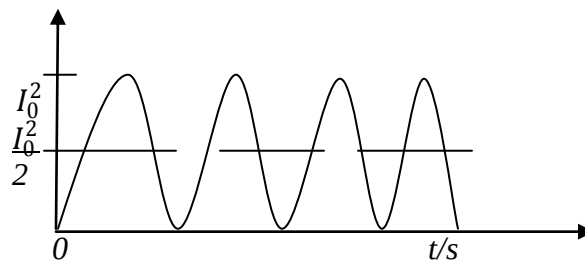
$$V = V_0 \sin \omega t$$

$$\text{Or } V = V_0 \cos \omega t.$$

$$\text{Peak to peak value} = 2I_0 \text{ or } 2V_0.$$

The average value of a.c current or voltage is zero. However, this does not mean that when a.c source is connected to a resistor, no power is generated in the resistor. The power generated is given by $P = I^2 R$.

The graph of I^2



Average or mean power $\langle P \rangle = \frac{RI_0^2}{2} = \frac{RV_0^2}{2}$

The average value of the square of current and voltage are:

$\langle I^2 \rangle = \frac{I_0^2}{2}$ and $\langle V \rangle = \frac{V_0^2}{2}$

The root-mean square current and voltage, which is the square root of the mean square current or voltage, are.

$I_{rms} = \sqrt{\langle I^2 \rangle} = \sqrt{\frac{I_0^2}{2}} = \frac{I_0}{\sqrt{2}}$ and

$V_{rms} = \frac{V_0}{\sqrt{2}}$

EXAMPLE

$V = 200 \cos 20\pi t$

Determine

1. Frequency
2. Peak value
3. r.m.s value

Solution

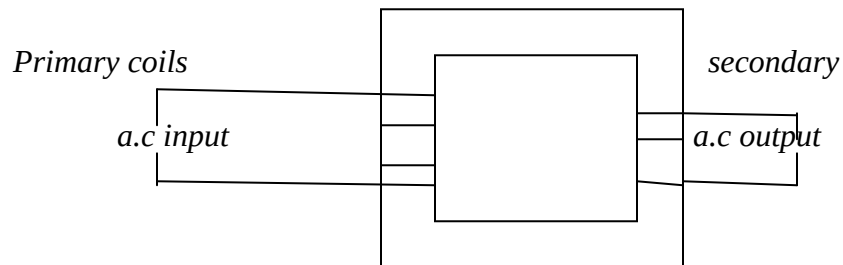
$V = V_0 \cos \omega t$

1. $\omega = 2\pi f = 20\pi$. Therefore $f = 10\text{Hz}$

2. Peak value = 200V

3. $V_{rms} = \frac{200}{\sqrt{2}} = 141.4\text{V}$

TRANSFORMER



A transformer consists of primary and secondary coils. The ratio of the output emf to the applied emf is dependent on the ratio of the number of turns of the coils.

PRINCIPLE OF OPERATION

1. Alternating/changing current in the primary coil;
2. Sets up a changing magnetic field;
3. The field is channelled from primary to secondary by the soft iron core;

4. The secondary coil cuts the field, then;
5. Emf is induced at the secondary coil.

NB A transformer only works on a.c because only a.c is able to set up a changing magnetic field.

If a transformer has more turns in the primary than the secondary, it is called a step down transformer. If the secondary has more turns than the primary it is a step up transformer.

An ideal transformer is 100% efficient and there are no power or energy losses.

Power in the primary = power in the secondary.

$V_p I_p = V_s I_s$ And also V_s is proportional to N_s and V_p is proportional to N_p

Then $\frac{V_s}{V_p} = \frac{N_s}{N_p}$. Combining these two equations we get.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

ENERGY LOSSES IN TRANSFORMER AND WAYS OF REDUCING THE ENERGY LOSSES

1. Heating due to eddy currents
-Reduced by laminating the core
2. Heating of the core due to repeated magnetisation and demagnetisation.
-Reduced by using a soft iron.
3. Resistivity heating in the primary and secondary coils.
-Reduced by using coils made of low resistivity material.
4. Loss of magnetic flux between the primary and the secondary coils. (Hysteresis)
-The core is designed for maximum flux linkage. The iron core forms a closed loop and the primary and the secondary coils are wound on each other

USE OF TRANSFORMER OIL

It is insulating oil, oil that is stable at high temperatures and has excellent electrical insulating properties. Its functions are to insulate, suppress corona discharges and arcing, and to serve as a coolant

TRANSMISSION OF ELECTRICAL ENERGY

It is transmitted as high voltage and as alternating voltage

Advantages of high voltage transmission

High voltage implies low currents hence less energy losses ($P = I^2 R$) due to joule heating. Thinner cables are cheap (economic advantages).

Advantages of a.c voltage transmission

1. a.c can be stepped up and down .

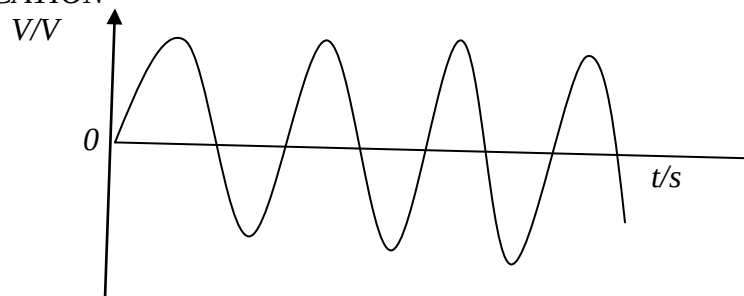
RECTIFICATION

Changing a.c to d.c

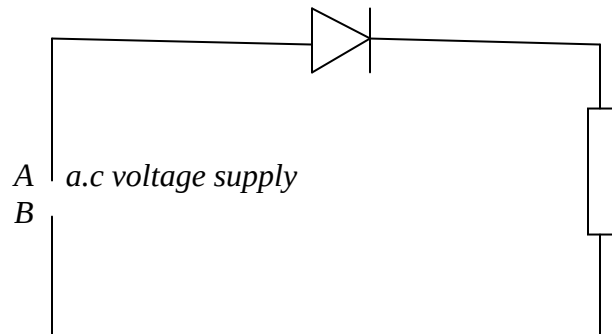
Diode is an electric device which allows current to flow in one direction.

HALF WAVE RECTIFICATION

Input voltage diagram



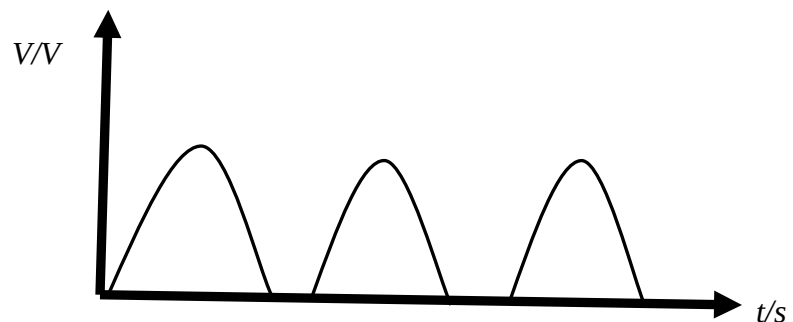
Circuit diagram



EXPLANATION

During the first half cycle when A is positive, the diode is forward biased and it conducts. During the second half-cycle when B is positive the diode is reverse biased it will not conduct

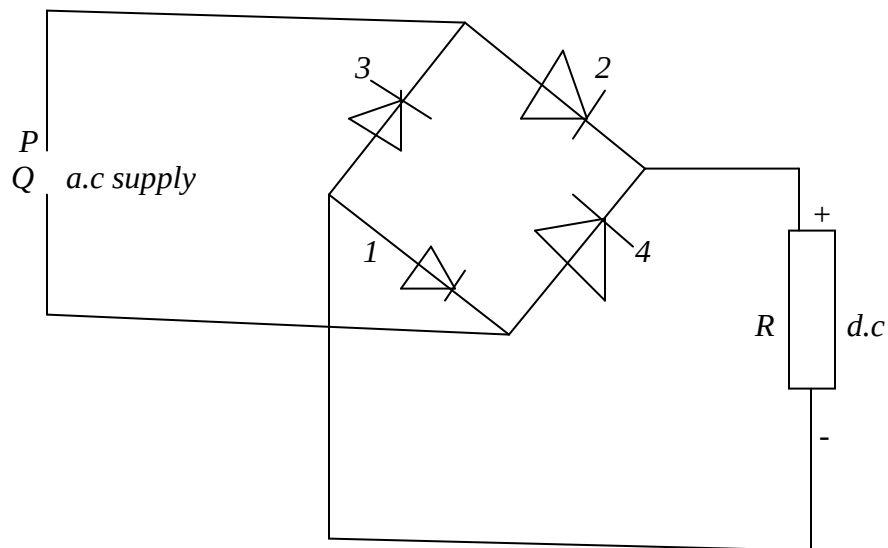
Output voltage diagram.



FULL WAVE RECTIFICATION

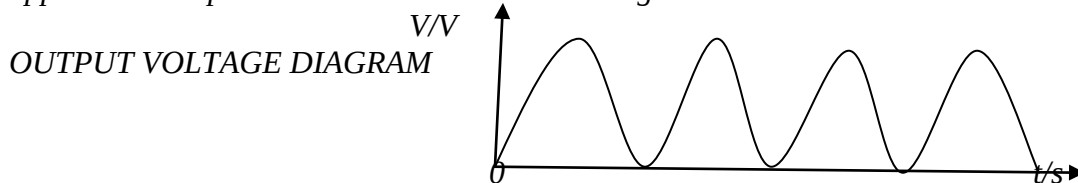
The input diagram is the same as for half wave

THE CIRCUIT DIAGRAM (BRIDGE).



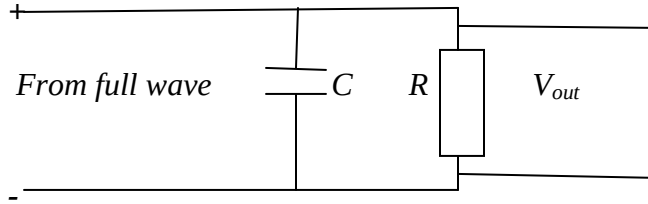
EXPLANATION

If P is positive during the first half cycle, diodes 1 and 2 conduct. In the next cycle Q is positive, and diodes 3 and 4 conduct. The resistor acting as a load will always have its upper terminal positive and its lower terminal negative



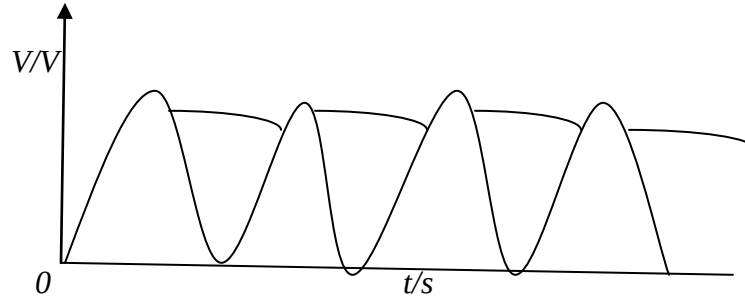
SMOOTHING

The output above is not a good approximation to steady direct voltage, as required by most electric equipment. We can improve the situation by inserting a capacitor across the output terminal of the bridge.



The capacitor charges up on the rising part of the half-cycle and then discharges through the resistor as the output voltage falls to reduce the fluctuations. This process is called smoothing

OUTPUT AFTER SMOOTHING



NB The important factor is the time constant of the resistor-capacitor circuit. If the product of the capacitance C and the load resistance R is much larger than half-period of the original supply to the rectifier circuit, the ripple on the direct voltage or current will be small.

SECTION 5: ELECTRONICS

5.1: ANALOUGE ELECTRONICS

ELECTRONIC COMPONENT

TRANSDUCERS

Transducers are electrical devices which change a physical quantity into an electrical signal or vice-versa.

INPUT TRANSDUCERS

Input transducers change a physical quantity into an electrical signal.

EXAMPLES

1. Light Dependent Resistor.(L.D.R)

2. Thermistor
3. Strain gauge
4. Microphone

OUTPUT TRANSDUCERS

Output transducers change an electrical signal into a physical quantity.

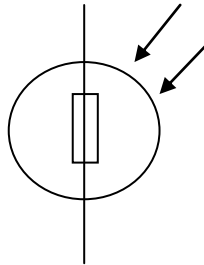
EXAMPLES

1. Light Emitting Diodes (L.E.D)
2. Loudspeaker
3. Buzzer
4. Relay

INPUT TRANSDUCERS

1. L.D.R

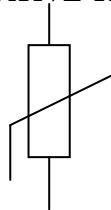
The symbol



In the dark, its resistance increases. In bright light its resistance decreases.

2. THERMISTOR (NEGATIVE TEMPERATURE COEFFICIENT N.T.C)

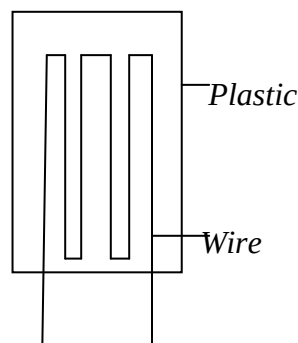
The symbol



When temperature increases the resistance decreases and when temperature decreases its resistance increases.

3. STRAIN GAUGE

The symbol

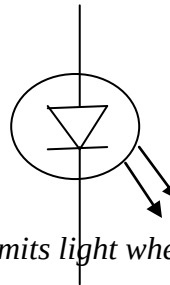


A strain gauge is made by sealing a length of very fine wire in a small rectangle of thin sheet in such a way that if the plastic is stretched (That is put under strain), then the wire will also be stretched. When this happens, the wire length increases and its cross-sectional area decreases. Both these effects cause the wire's resistance to increase.

OUTPUT TRANSDUCERS

1. L.E.D

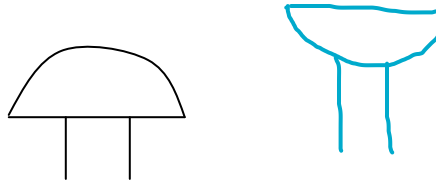
The symbol



This is a one-way conduction device which emits light when current passes through it.

2. THE BUZZER

The symbol.



A buzzer is an electronic oscillator circuit connected to a solid-state sounder which is wired and sealed inside a small plastic case. The leads which emerge from the case must be connected to a power supply for the buzzer to sound.

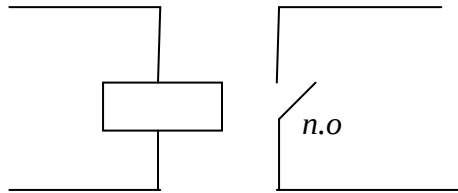
4. THE RELAY.

Relays are used when a small current in one circuit has to switch on or off a much larger current on a quite separate circuit. The relay is made up of one or more sets of switch contacts which are opened or closed by means of an electromagnet. The electromagnet is simply a coil of wire wound around a soft iron core. When current is passed through the coil, the relay is said to be energised and switch contacts move.

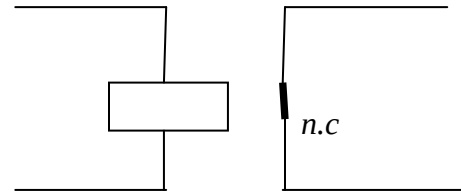
The symbols

1. Open contact (close when relay is energised)
2. Closed contact (open when relay is energised)

1. OPEN CONTACT



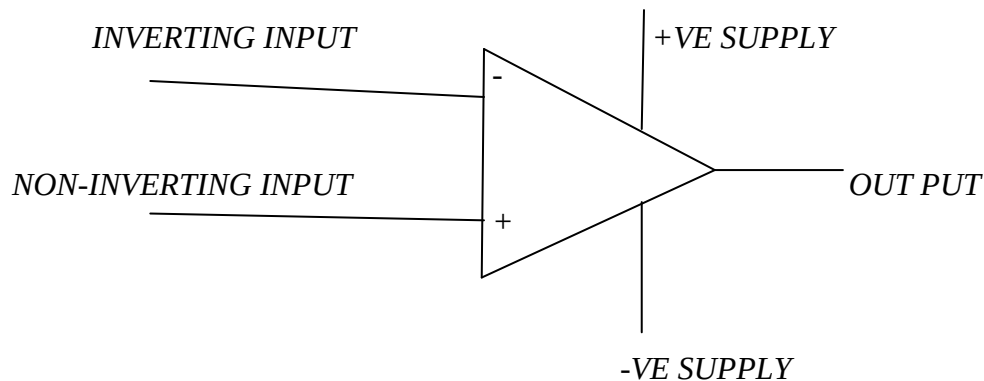
2. CLOSED CONTACT



ANALOUGE ELECTRONICS

OPERATIONAL AMPLIFIER (OPAMP)

OPAMP is an integrated circuit (I.C) that consists of transistors and resistors.



PROPERTIES OF AN IDEAL OPAMP

1. An infinite input impedance (that is no current flows into either of the two inputs).
2. Infinite open loop gain (that is if there is only the slightest difference between the two input voltages, the output will be saturated).
3. Zero output impedance (that is the amplifier can provide the current for any load, no matter how small its resistance).
4. Infinite bandwidth (That is it amplifies all frequencies by the same amount)
5. Infinite slew rate (That is if the input changes, the output suddenly changes in step without any time decay.)

PROPERTIES OF A REAL OMPAMP

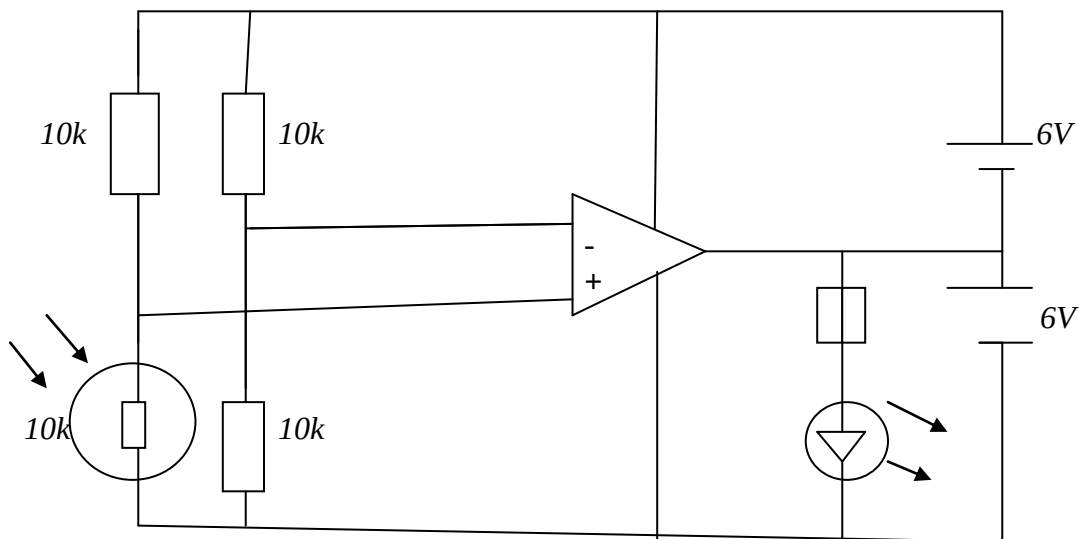
1. It has high input impedance usually between $10^6\Omega$ to $10^{12}\Omega$
2. It has high open loop again usually $10^5\Omega$ for D.C signals.

3. It has low output impedance usually about 100Ω .
4. There is limited bandwidth.
5. There is a limited slew rate.

OPAMP AS A COMPARATOR.

It has no feedback. Therefore it has an open loop gain.

Output voltage (V_O) is given by: $V_O = A_O[V^+ - V^-]$. Where A_O is the open loop gain. V^+ is the non-inverting input and V^- is the inverting input.



EXPLANATION ON HOW THE CIRCUIT WORKS

$$V^- = \text{Constant} = \frac{10k}{10k + 10k} \times 6V = 3V$$

IN THE DARK

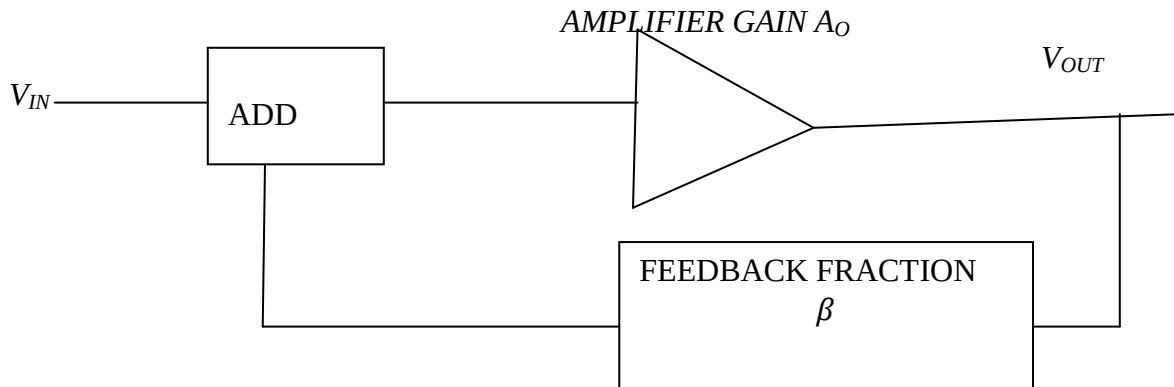
Resistance of the L.D.R increases, and V^+ increases. $V^+ > V^-$, the opamp will saturate positively. The diode is forward biased, and it conducts and emits light.

IN THE LIGHT

Resistance of the L.D.R decreases, potential at V^+ decreases. $V^+ < V^-$. The opamp will saturate negatively. The diode is reverse biased and it will not conduct and will not emit light.

FEEDBACK

Feedback is taking part or all of the output signal and feed it to the input.



$$V_{OUT} = A_O [\text{Input of amplifier}]$$

$$= A_O [V_{in} + \beta V_{OUT}]$$

$$V_{OUT} [1 - A_O \beta] = A_O V_{in}$$

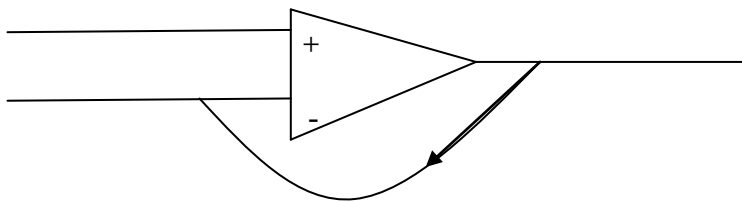
$$\frac{V_o}{V_{in}} = \frac{A_o}{1 - \beta A_o} = \text{CLOSED LOOP GAIN}$$

?

$$\left(\frac{A_o}{1 + \beta A_o} \right)$$

NEGATIVE FEEDBACK

Is taking part or all of the output signal and feed it back to the inverting input



ADVANTAGES OF NEGATIVE FEEDBACK

1. An increase in bandwidth
2. Less distortion
3. Greater operating stability

DISADVANTAG

1. Reduce the gain.

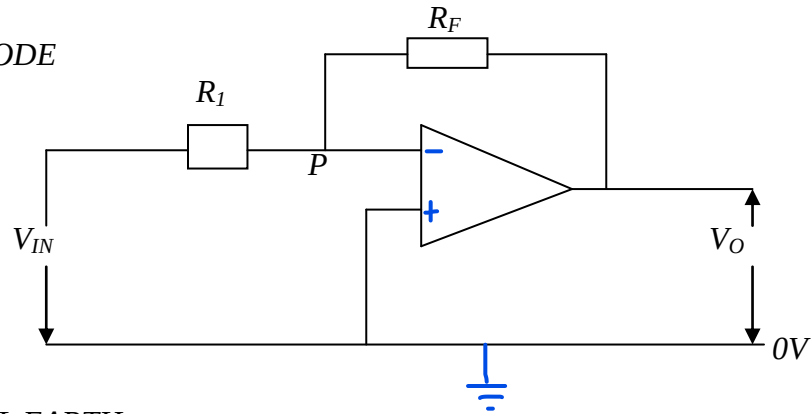
POSITIVE FEEDBACK

Is taking part or all of the output signal and feed it back to the non-inverting input

If the fraction β is positive, then the denominator in the above equation can be made to be zero. This produces an amplifying system with an infinite gain which means an output

is produced for no input. This is the operating principle of signal generators that is oscillators are simply amplifiers with positive feedback.

THE INVERTING MODE



POINT P IS VIRTUAL EARTH

For the Opamp not to saturate, the two inputs must be at the same potential. Since V^+ is at zero potential so must also be point P.

DERIVATION OF THE GAIN

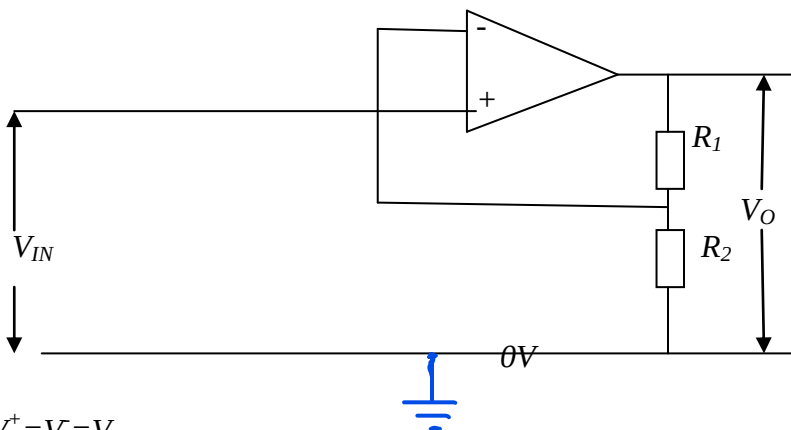
The opamp has high input impedance.

Current in R_1 = Current in R_F

$$\frac{V_{IN} - 0}{R_1} = \frac{0 - V_O}{R_F}$$

$$\frac{V_O}{V_{IN}} = \frac{-R_F}{R_1} = \text{Gain of the opamp.}$$

NON -INVERTING MODE



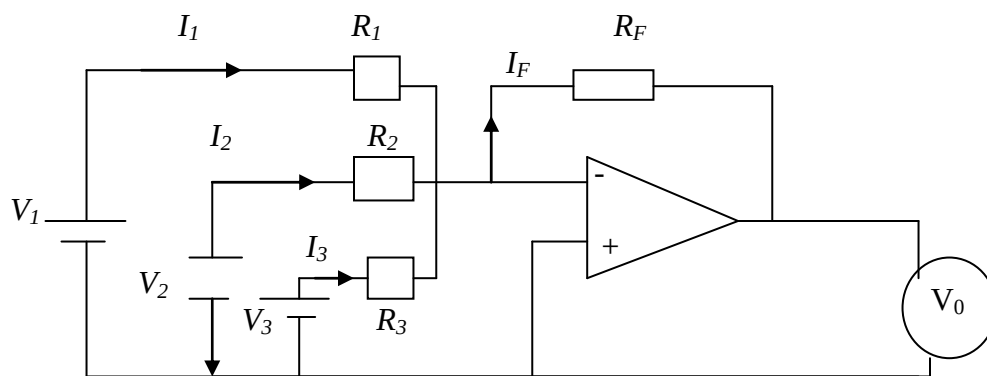
For the opamp not saturate $V^+ = V^- = V_{IN}$

Using potential divider. $V = \frac{R_1}{R_1 + R_2} V_O = V^+ = V_{IN}$

$$V_{IN} = \frac{R_1}{R_1 + R_2} V_O$$

$$\frac{V_O}{V_{IN}} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} = \text{Gain of the opamp.}$$

SUMMING OPAMP



The opamp has high impedance

$$I_1 + I_2 + I_3 = I_F$$

$$\frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_O}{R_F}$$

$$V_O = - \left[\frac{R_F V_1}{R_1} + \frac{R_F V_2}{R_2} + \frac{R_F V_3}{R_3} \right]$$

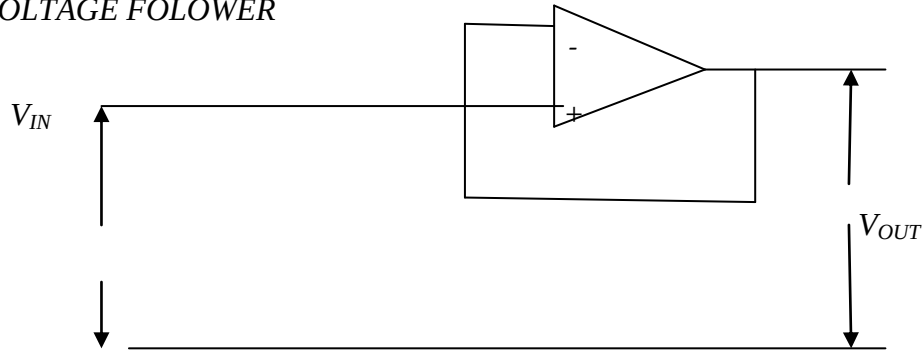
If, $R_1 = R_2 = R_3 = R_F$, Then,

$$V_O = -(V_1 + V_2 + V_3)$$

APPLICATION OF SUMMING OPAMPS

1. In music mixer

VOLTAGE FOLLOWER



From Non-inverting mode $\frac{V_O}{V_{IN}} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} = \text{Gain of the opamp}$

Then, for a voltage follower $R_2 = 0$ and $R_1 = \infty$ (OPEN CIRCUIT)

Therefore the gain $= \frac{V_O}{V_{IN}} = 1$

USES OF VOLTAGE FOLLOWER

1. To measure the voltage of a source with high internal resistance.

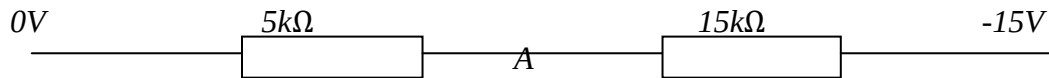
SCHMITT TRIGGER

It has a positive feedback. Which means the gain is very high and can be made to be infinite (this was shown under positive feedback).

Consider the wiper starting at -15V. This makes the non-inverting input A negative and thus less than the inverting input marked B, which is held at zero. The output voltage will saturates negatively and $V_O = -15V$. The opamp easily saturates because it has a positive feedback and its gain is very high.

If the moveable contact on the potentiometer is moved upwards slightly V_{IN} becomes less negative but this does not affect the output voltage because the feedback through the $15k\Omega$ remains A at a lower potential than B.

When the contact is now at 0V, it can be shown that the output voltage will be negative.



Potential difference between -15V and 0V = $0V - (-15V) = 15V$

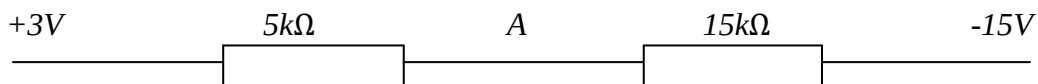
p.d across the $5k\Omega$ resistor is given by the potential divider rule = $\frac{5k\Omega}{5k + 15k} \times 15V = 3.75V$

Potential at A: $0 - V_A = 3.75$

$V_A = -3.75V$

This shows that potential at A is still less than that at B, Then the opamp will saturates negatively.

It can also be shown that, when the sliding contact is now at +3V, the opamp still saturates negatively.



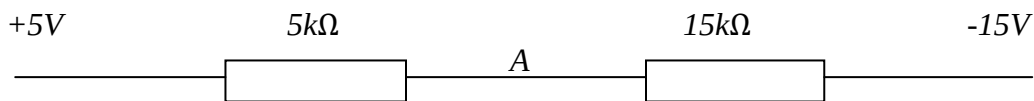
p.d between -15V and +3V = $+3 - (-15) = 18V$

p.d across the $5k\Omega$ resistor = $\frac{5k}{5k + 15k} \times 18 = 4.5V$

Potential at A: $3 - V_A = 4.5$.

$V_A = -1.5V$. Potential at A is still less than that at B, the opamp will saturate negatively.

When the sliding contact is now at 5V, it can now be shown that the output will saturate positively. This is the switch over voltage.

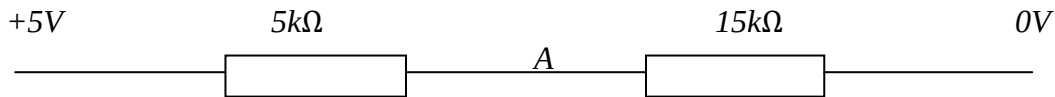


p.d between -15V and +5V = $+5V - (-15V) = 20V$.

p.d across the $5k\Omega$ resistor = $\frac{5k}{5k + 15k} \times 20 = 5V$.

Potential at A: $+5 - V_A = 5$, $V_A = 0V$.

When $V_{IN} = +5V$, then potential at A = 0V and potential at A = potential at B, hence $V_{OUT} = 0V$. But if $V_{OUT} = 0V$, it can be shown that, the potential at A cannot be equal to zero(0V)



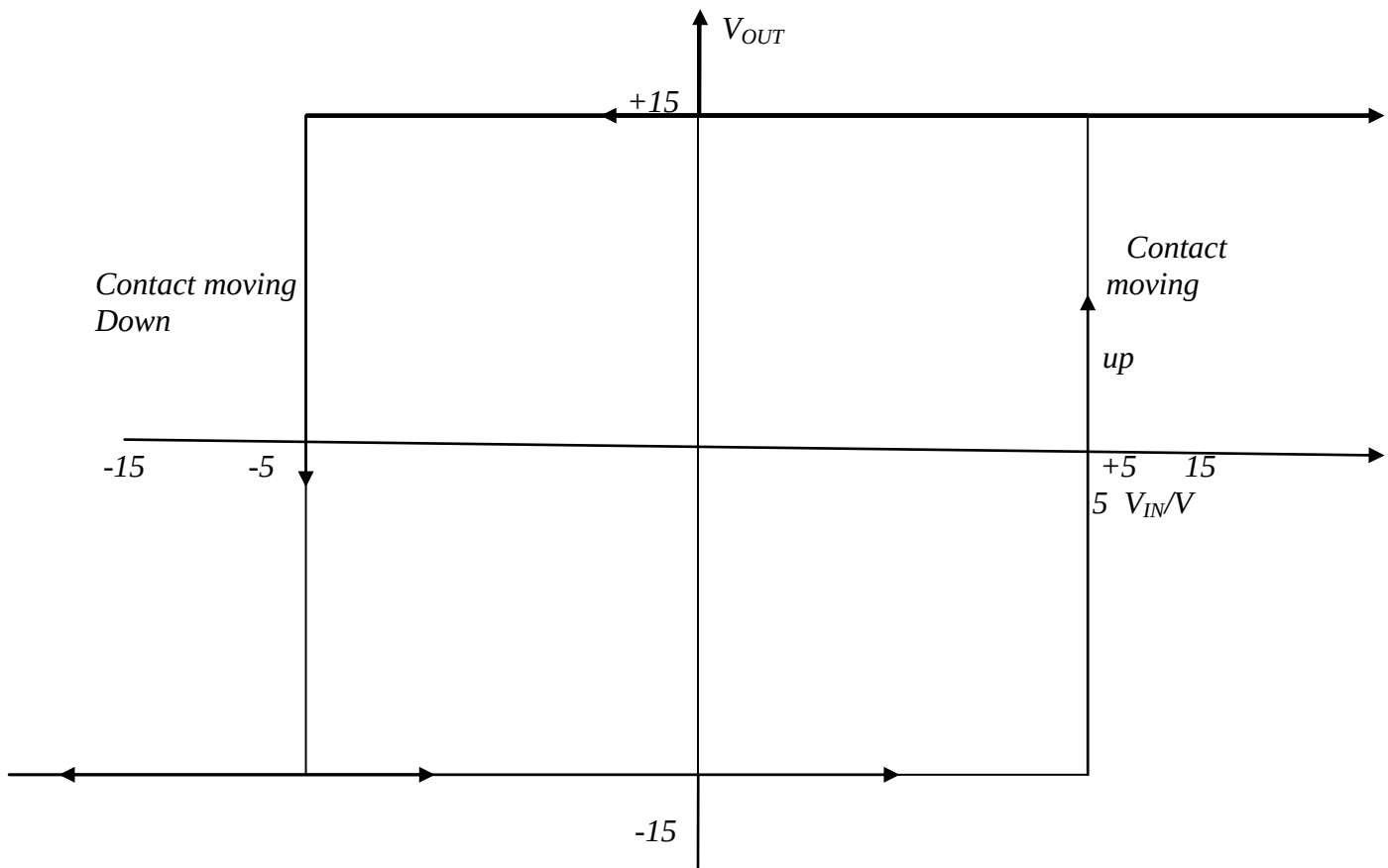
p.d between +5V and 0V = +5-0 = +5V

p.d across the $5k\Omega$ resistor = $\frac{5k}{5k+15k} \times 5 = 1.25V$

Potential at A: $5 - V_A = 1.25$: $V_A = +3.75V$.

This shows that potential at A is now greater than the potential at B. The opamp will saturate positively.

All these can be shown when the sliding contact is now moving from +15V to -15V, the opamp will switch over from positive at -5V.



NOTE: The switch over voltages can easily be changed by a suitable choice of resistors. The output of Schmitt trigger is a perfectly sharp digital signal. A digital signal cleans noisy of a signal.

5.2: DIGITAL ELECTRONICS

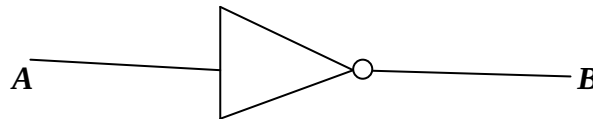
LOGIC GATE

A logic gate is high speed integrated switch with two levels of voltage for input and output.

1. NOT GATE [INVERTING GATE]

A NOT gate is an inverting gate

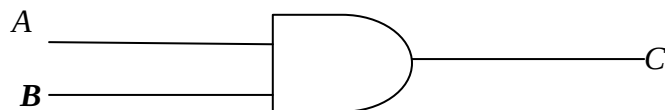
Symbol



Truth table

INPUT	OUTPUT
A	B
0	1
1	0

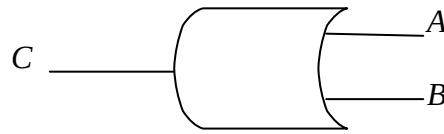
2. AND GATE [MULTIPLICATION GATE] $A \times B = C$



TRUTH TABLE

INPUT		OUTPUT
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

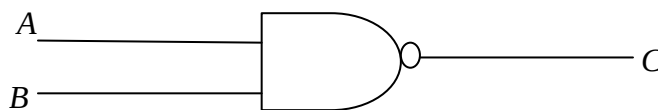
3. OR GATE [ADDITION GATE] $A + B = C$



TRUTH TABLE

INPUT		OUTPUT
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

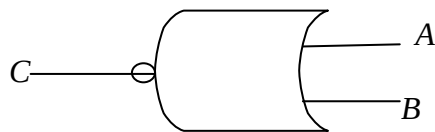
4. NAND GATE=NOT AND



TRUTH TABLE

INPUT		OUTPUT
A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

5. NOR GATE=NOT OR

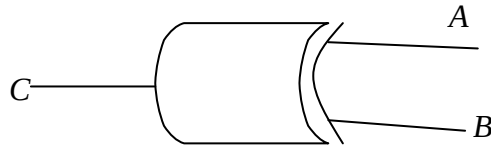


TRUTH TABLE

INPUT		OUTPUT
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

6.EX-OR EXCLUSIVE OR= $[A \text{ AND NOT } B] \text{ OR } [\text{NOT } A \text{ AND } B]$

Is the difference gate because the output is logic 1 when inputs are different

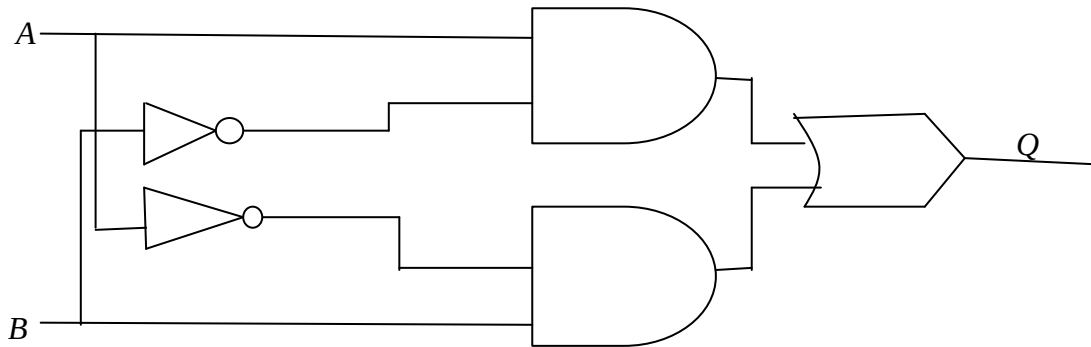


TRUTH TABLE

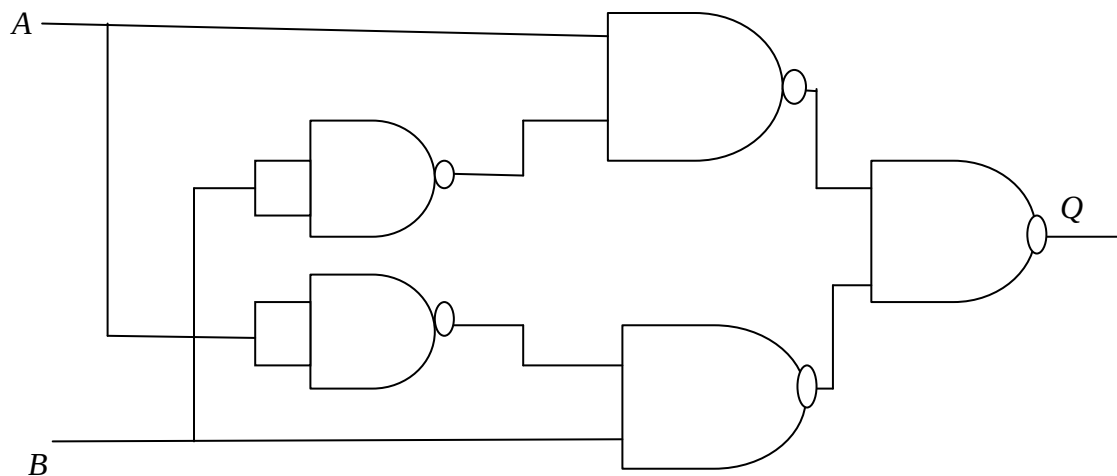
INPUT		OUTPUT
A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

It is found using combinations of NAND GATES OR [AND, NOT AND OR]

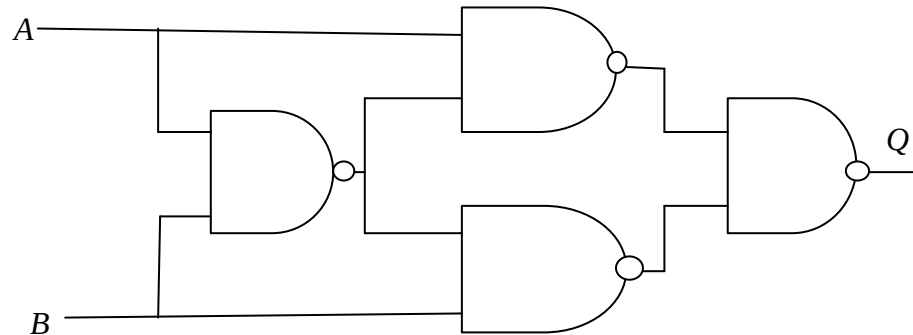
1.



2.



3



7. THE EX-NOR



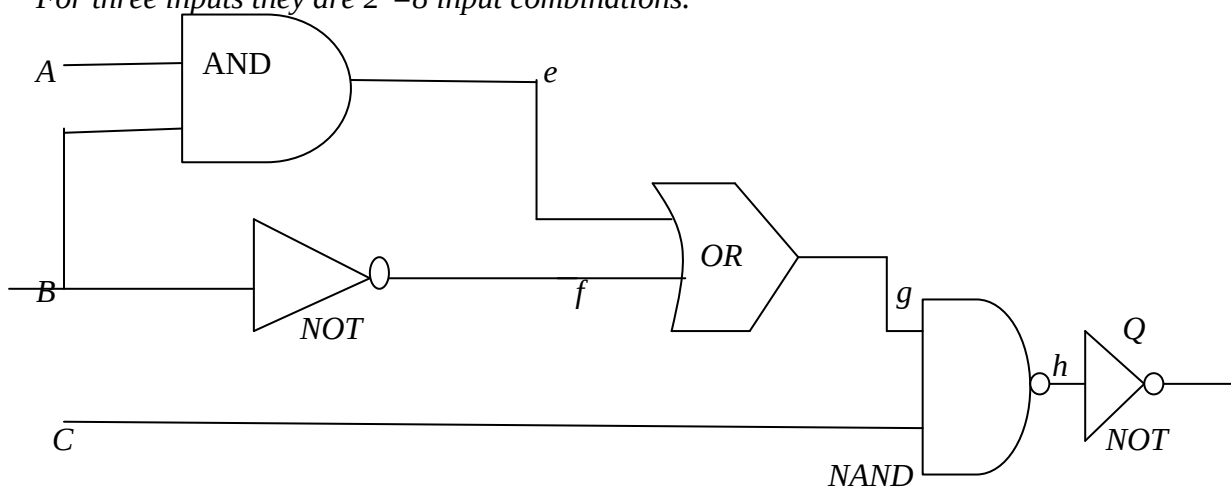
TRUTH TABLE

INPUT		OUTPUT
A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

COMBINATIONS OF LOGIC GATES

For two inputs, they are $2^2=4$ input combinations

For three inputs they are $2^3=8$ input combinations.



Draw the truth table.

They are three inputs A, B and C. Therefore they are 8 inputs combinations.

A	B	C	e	f	g	h	Q
0	0	0	0	1	1	1	0
1	0	0	0	1	1	1	0
0	1	0	0	0	0	1	0
1	1	0	1	0	1	1	0
0	0	1	0	1	1	0	1
1	0	1	0	1	1	0	1
0	1	1	0	0	0	1	0
1	1	1	1	0	1	0	1

There is a systematic manner of presenting these states in the truth table so that no state is missed out.

A	0	1	0	1	0	1	0	1
B	0	0	1	1	0	0	1	1
C	0	0	0	0	1	1	1	1

SECTION 5: MATTER

5.1: PHASE OF MATTER

Density is the mass per unit volume. It is a measure of the degree of compactness of a material. It can also be defined by the equation.

$$\text{Density} = \frac{\text{mass}}{\text{density}}$$

A material that has a high density has a very closely packed a regular structure, for example most solids.

THE PROPERTIES OF THE STATES OF MATTER

PROPERTY	SOLID	LIQUID	GAS
Arrangement of particles	Set in regular arrangement called lattice. Particles are held strongly in position. Solids have definite volume and definite shape.	Regular structure but spaces between molecules allow for free movement. As such liquids have definite volume but no definite shape. Liquids take the shape of the containers.	Irregular arrangement of particles with no arrangement of particles, gases have neither a definite volume nor a definite shape. Gases take the shape of the container and fill all spaces.
Spacing between particles	Particles are closely packed very little space between particles.	Small spaces between molecules which allow for free (not unlimited) movement, hence liquids flow.	Particles are widely spaced (infinite spaces between particles).
Forces of attraction	Very strong forces of attraction.	Relatively strong forces of attraction between molecules	Negligible forces of attraction. No potential energy
Motion of particles	Vibrate around mean fixed lattice position. Little kinetic energy.	Particles vibrate and move within ordered clusters with limited freedom	Particles move randomly (translational and rotational) with high speeds in all direction.

KINETIC THEORY OF MATTER

Matter is anything that possesses a mass and occupies space. Mass is the amount of matter in a substance. Matter exists in three main states solids, liquids and gases. A fourth state is plasma, is a system of charged particles interacting electromagnetically. A fifth state is a super fluid, is a liquid that occurs at temperature close to absolute zero.

The kinetic theory of matter states that all matter is made up of tiny particles in a state of constant random motion.

The kinetic theory is often called the kinetic theory of gases because from the definition of the kinetic theory it is obvious that only gases can actually obey this theory.

Since a theory must link theoretical assumptions and physical observations. We look at the assumptions of the kinetic theory and the phenomena that provide physical evidence.

ASSUMPTIONS OF THE KINETIC THEORY.

1. All matter consists of particles that are in state of constant random motion.
2. The forces of attraction between the particles are negligible very small forces of attraction between the particles.
3. Collisions between the particles are perfectly elastic. This means that when the particles collide between themselves and with the walls of the container, they do not coalesce (stick together). This is because the forces between the particles and between the walls of the container are insignificant.
The individual kinetic energy and individual momentum of the particles are all conserved. The speed of approach of a particle is equal to the relative speed of separation. Since there is no change in the kinetic energy of particles.
4. Volume of particle is negligible compared to the volume of gas. Since gases fill the whole space of a container, there are infinite (very very big) spaces between particles. The size of the gas molecule though does not change; hence the size of the molecules becomes so small it is insignificant compared to the volume of the gas.
5. Time of collision is negligible compared to the time in between collision. With infinite spaces between particles the probability (chance) of particles actually colliding is very small. Since the collisions of the particles are perfectly elastic the time that the particles are actually in contact is very little.

EVIDENCE OF THE KINETIC THEORY

There are two key pieces of evidence that support the kinetic theory of matter

1. Brownian motion
2. Diffusion

Brownian motion

Is the random motion of every small particle in a liquid or gas due to uneven bombardments (collisions) by the smaller surrounding molecules of the liquid or gas

Observation

- (i) Zig-zag movement of smoke particles in air.
- (ii) Erratic motion of pollen grain in water.

Diffusion

Movement of particles from a region of higher concentration to a region of lower concentration. Diffusion occurs down a concentration gradient until equilibrium is achieved.

Observation

(i) Spread of perfume, cooking aromas and scents in air.

(ii) Spread of tea and coffee grains in water.

NOTE: The rate of diffusion is affected by three main factors

1. Molecular mass- the smaller the M_r (size of the gas, the faster the diffusion).
2. Temperature- the higher the temperature, the faster the diffusion (particles have greater kinetic energy, movement is more erratic and covers a great area)
3. Density- the lower the density, the faster the diffusion.

MELTING, BOILLING AND EVAPORATION

Melting- the change of state from solid to liquid at constant temperature.

DIFFERENCES BETWEEN BOILING AND EVAPORATION

BOILING	EVAPORATION
Occurs at constant temperature	Occurs at any temperature
Occurs throughout the liquid	Occurs at the surface of the liquid only
Caused by gain in thermal energy of molecules(by heating)	Maybe caused by draught, wind or heating
Particles possess the same energy throughout the liquid	The most energetic particles are located at the surface of the liquid
Results in a change in state of the whole liquid	Only the most energetic molecules change state
Affected by atmospheric pressure	Not largely affected by pressure but mainly by humidity, surface area and temperature of the surrounding.

PRESSURE

Pressure is the force per unit cross-sectional area.

PRESSURE IN GASES

According to the kinetic theory, gas particles (molecules or atoms) are in constant random motion at high speeds in all directions. If a gas is confined in a container, the particles collide between themselves and with the walls of the container. When the particles collide with the walls of the container, they experience a change in momentum and therefore they exert a force on the container by these means, a gas exerts a pressure on the walls of the container that it is kept.

Factors affecting gas pressure.

1. Changing the temperature eg increasing the temperature will cause the gas particles to move faster and collide harder with the walls, thereby increasing pressure.
2. Changing the volume eg. Decreasing the volume of the gas increases the frequency of collision of the gas molecules with the walls of gas container and again, increases the gas pressure.

PRESSURE IN LIQUIDS

$$\text{Pressure} = \frac{\text{Force (weight)}}{\text{area}}$$

Weight = mass $\times$ acceleration due to gravity

$$\text{Pressure} = \frac{\text{mass} \times \text{acceleration due to gravity}}{\text{area}}$$

But mass of liquid = density $\times$ volume

$$\text{But } \frac{\text{volume}}{\text{area}} = \text{height}$$

Pressure = density $\times$ acceleration due to gravity $\times$ height.

$$P = \rho gh$$

Note When calculating total pressure in liquids. It is important to add atmospheric pressure. Therefore. Total pressure = ρgh + atmospheric pressure.

6.2: DEFORMATION OF SOLIDS

Deformation is caused by a force and in one dimension; the deformation can be tensile or compressive.

ELASTICITY OF METALS.

Elasticity is that property of a body to regain its original dimensions when the deforming force acting on the body is removed.

Not all materials return to their original dimensions when the deforming force acting on the body is removed. These materials are said to be inelastic eg plasticine, clay.

Most materials are elastic up to a certain limit known as elastic limit.

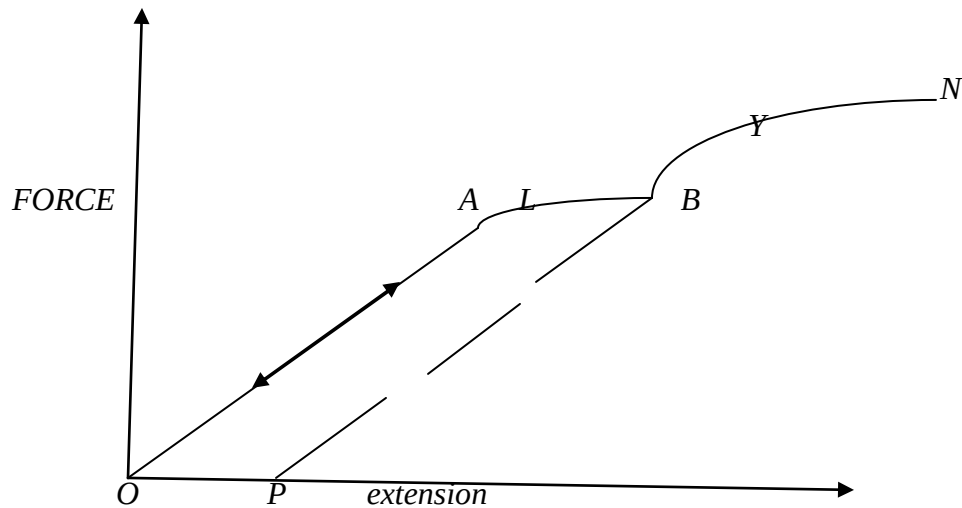
Beyond the elastic limit material will not return to its original dimensions when the deforming force is removed.

HOKE'S LAW

The extension and compression of a spring is directly proportional to the applied force. When a spring is stretched or compressed, the extension or compression is directly proportional to the applied force.

Hooke's law states that within elastic limits the extension or compression is directly proportional to the force applied if the proportional limit is not exceeded

(i)



(ii)

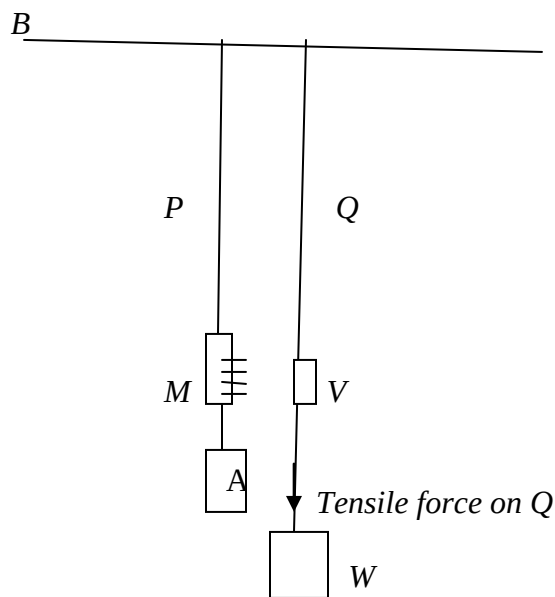


Fig (ii) illustrates a simple laboratory experiment. Method of investigating the property of steel.

Two thin wires P and Q are suspended beside each other from a rigid support B. The wire P is kept taut by weight A attached to its end and carries a small scale M graduated in millimetres. The wire Q carries a vernier scale V which is alongside the scale m.

When load W is attached to the end Q the wire increases in length by a small amount which can be read from the change in the reading on the vernier V . If the load is taken off, the reading on V returns to its original value. The wire is said to be elastic. Plotting extension e against force F a graph fig(i) is obtained, which is a straight line OA followed by an arc ABY . Along OA , and up to L just beyond A , the wire returns to its original length when the load is removed. The point L is called elastic limit. Along OL the metal is said to undergo elastic deformation. Beyond the elastic limit, however, the wire has a permanent extension such as OP when the force is removed at B .

Beyond L the wire is no longer elastic. The extension increases rapidly along the curve ABY as the force on the wire is further increased and at N the wire thus breaks. The theory of plastic deformation explains this.

STRESS, STRAIN AND YOUNG MODULUS

Stress is force per unit area.

Strain is the extension per unit original length.

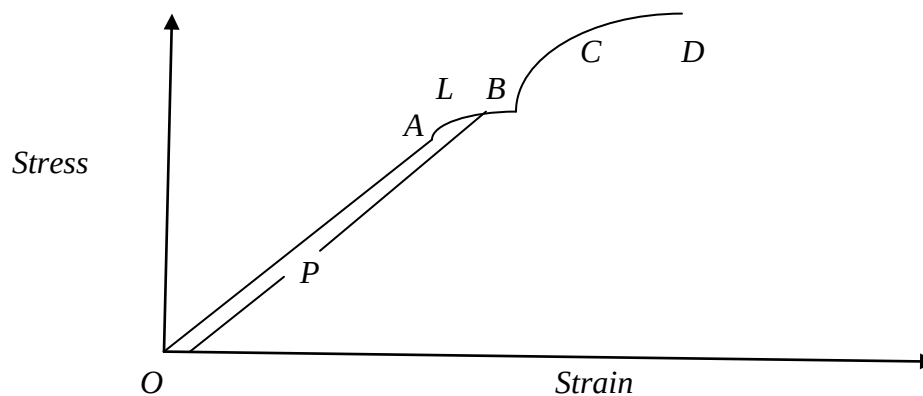
$$\text{Young modulus} = \frac{\text{stress}}{\text{strain}}$$

THE YIELD POINT: DUCTILE AND BRITTLE SUBSTANCES; BREAKING STRESS.

A careful experiment show that for mild steel and iron for example, the molecules of the wire begin to slide across each other soon after the load exceeds the elastic limit that is the material becomes plastic. This is indicated by a slight kink at B beyond L in fig(i), and it is called the yield point of the wire.

The change from elastic to plastic state is shown by a sudden increase in the extension. In the plastic stage, the energy gained by the stretched wire is dissipated as heat and unlike the elastic state; the energy is not recovered when the load is removed. As the load is increased further the extension increase rapidly along the curve YN and the wire then becomes narrower and breaks.

Substances that undergo plastic deformation until they break are known as ductile substances, eg lead, copper and wrought iron. Other substances eg glass and high carbon steels.



A = proportional limit
 L = elastic limit
 B = yield point
 C = breaking (max) stress
 D = wire breaks
 OL = elastic deformation
 BL = plastic deformation

Under elastic condition, a modulus of elasticity of the wire, called the Young modulus (E), is defined as

$$E = \frac{\text{tensile stress}}{\text{tensile strain}}$$

$$E = \frac{Fe}{AL}$$

EXPERIMENT TO DETERMINE THE YOUNG MODULUS OF A METAL WIRE

The young modulus of a metal wire is found with the apparatus illustrated on fig (ii) previously.

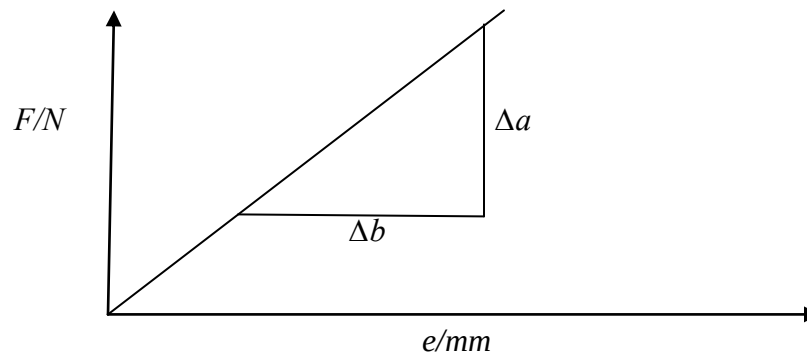
The following practical points should be specially noted, remembering that elastic limit should not be exceeded.

1. The use of two wires P and Q of the same material and length eliminates the correction for (a) the yielding of the support when the loads are added to Q (b) changes in temperature.

2. The wire is made thin so that a moderate load of several kilograms produces a large tensile stress. The wire is also made **long** so that a measurable extension is produced.
3. Both wires should be free of kinks, otherwise the increase in length cannot be accurately measured. The wires are, straightened by attaching suitable weights to their ends.
4. A vernier scale is necessary to measure the extension of the wire since it is always small. The original length of wire is measured from the top B to the vernier V by a ruler, since an error of 1mm is negligible compared with an original length of several metres. For very accurate work, the extension can be measured using a spirit level between the wires.
5. The diameter of the wire must be found by a micrometer screw gauge at several places, and the average value then calculated. The area of cross-section $A = \pi r^2$.
6. The readings on the vernier are also taken when the load is gradually removed in steps of 1kg, they should be very nearly the same as the readings on the vernier when the weights were added, showing that the elastic limit was not exceeded.

DETERMINATION OF THE YOUNG MODULUS.

A graph of F against average extension is plotted. Straight line AB passing through the origin is drawn.



$E = \frac{F}{A} \div \frac{e}{L} = \frac{F}{e} \times \frac{L}{A}$. The value $\frac{F}{e}$ is the gradient $\frac{\Delta a}{\Delta b}$. L is known as the original length and $A = \pi \frac{d^2}{4}$. E can be calculated.

ELASTIC AND PLASTIC DEFORMATION

As discussed, the atoms are slightly displaced at low tensile stress and return to their normal positions when the stress is released, if this is less than the elastic limit of the metal.

If the metal is stretched beyond the elastic limit, it does not return to its original length. The metal is then said to have undergone plastic deformation. This is caused by movement of crystal planes called slip.

STRAIN ENERGY IN DEFORMED MATERIAL FROM FORCE, EXTENSION GRAPH
[AREA UNDER F, e graph]

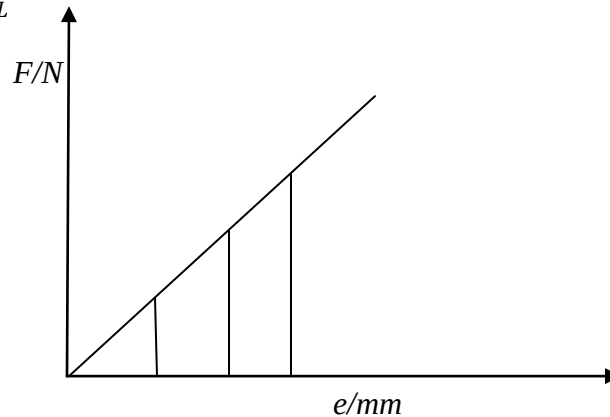
The work done = Force $\times$ displacement

$$\text{Work} = \text{average force} \times \text{extension} = \frac{1}{2} Fe.$$

This is the amount stored in the wire. It is the gain in molecular potential energy of the molecules due to their displacement from their mean positions

$$\text{Work} = \frac{1}{2} Fe \text{ but } F = \frac{EAe}{L}$$

$$\text{Energy } W = \frac{1}{2} EA \frac{e^2}{L}$$

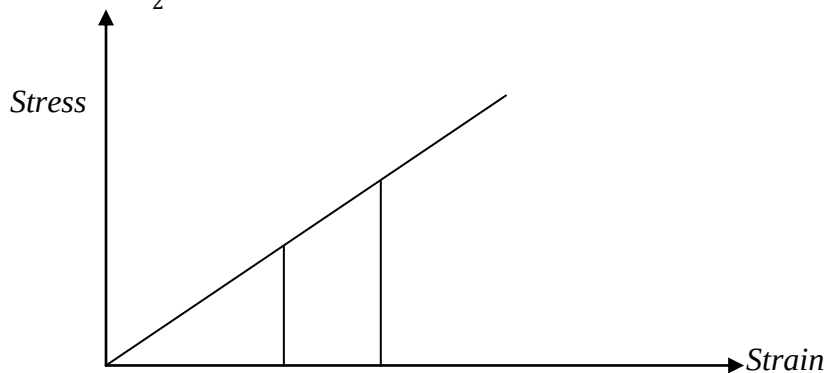


$$\text{Energy stored} = \frac{1}{2} Fe$$

ENERGY STORED PER UNIT VOLUME OF A WIRE

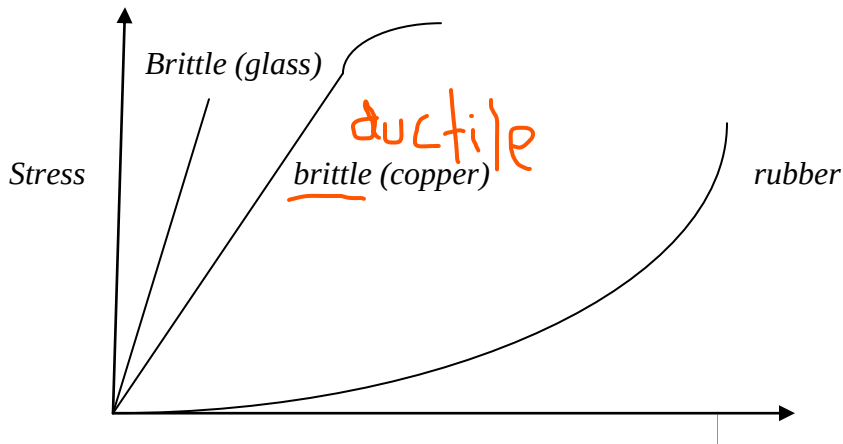
Energy stored $= \frac{1}{2} Fe$ and the volume of the wire $= AL$, where A is the cross-sectional area and L is the length of the wire

$$\begin{aligned} \text{Energy per unit volume} &= \frac{\frac{1}{2} Fe}{AL} \\ &= \frac{1}{2} \times \left(\frac{F}{A} \right) \times \left(\frac{e}{L} \right) \\ &= \frac{1}{2} \times \text{Stress} \times \text{Strain}. \end{aligned}$$



The area under the stress- strain graph is the energy per unit volume of the wire.

STRESS, STRAIN GRAPHS FOR DUCTILE, BRITTLE AND POLYMERIC MATERIALS



RUBBER- stretches easily as compared to metals. It is less stiff than metals. Rubber is elastic at high strain whereas copper is elastic at relatively low strains. The reason for the large stretching of rubber without breaking lies in its molecular structure. Unstretched rubber has coiled molecules. When stretched the molecules unwind and become straight.

METAL

When a metal is slightly stretched, the inter atomic spacing increases. The spacing returns to its original to its original value when the force is removed. When the metal is stretched beyond the elastic limit, slip occurs and metal undergoes plastic deformation.

GLASS

Glass is very stiff at room temperature. Its stiffness is then greater than that of steel. It has a small elastic region and fractures easily. This behaviour of glass is due to the existence of cracks in its surface.

FATIGUE AND CREEP

FATIGUE

Fatigue is the failure due to cyclic stress below yield stress

It is a consequence of cyclic stress. This may cause fracture, often with little or no warning, and happens when a metal is subjected to a large number of cycles of varying

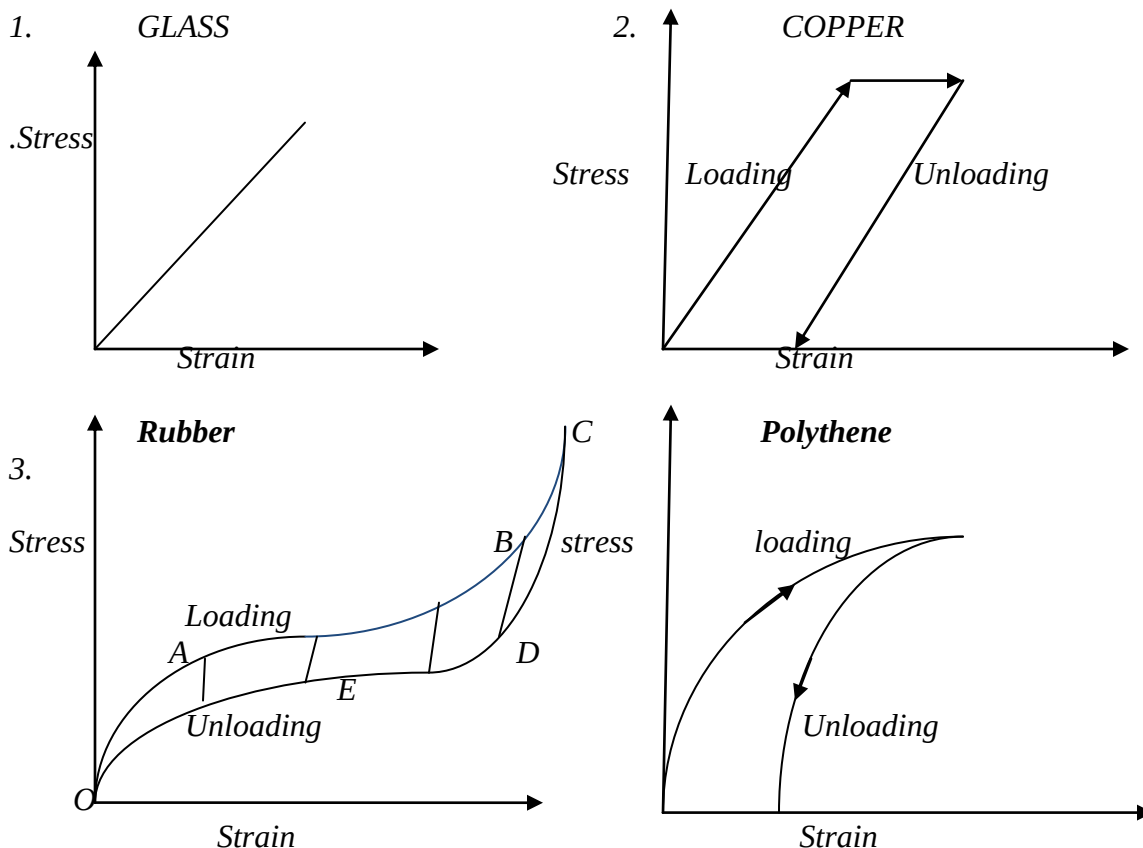
stress, even is the maximum value of the stress could be applied steadily with complete safety. It is estimated that about 90% of all metal failures are due to fatigue, it occurs in aircraft parts, in engine connecting rods, axles.

CREEP

Creep is the failure due to sustained stress, below yield stress combined with high temperatures.

In general this occurs at high temperatures and results in the metal continuing to deform as time passes, even under constant stress.

LOADING AND UNLOADING STRESS, STRAIN GRAPHS



EXPLANATION FOR RUBER

If a stress-strain curve is plotted for the loading and unloading of a piece of rubber the two parts do not coincide.

OABC is for stretching and CDEO for contracting. The strain for a given stress is greater when unloading than when loading. The unloading strain can be considered to lag behind the loading strain, the effect is called elastic hysteresis. Hooke's law is not obeyed.

The area enclosed by OABC and the strain axis represents the energy supplied to cause stretching. Area under CDEO represents the energy given up by the rubber during contraction. The shaded area is called a hysteresis loop; it is a measure of the energy lost as heat during one expansion-contraction cycle.

6.3: TEMPERATURE

Temperature is the measure of the difference in thermal energy of two objects. It is the degree of hotness or coldness of a body.

Heat is the transfer of thermal energy.

THERMOMETRIC PROPERTIES

A thermometric property is a physical property of a material that varies linearly with changes in temperature.

A thermometric property may be defined as a measurable, observable, state variable which responds to temperature.

Thermometers must use materials that have a thermometric property so that they can be used to accurately measure the temperature of an object or system.

THERMOMETRIC PROPERTY	TYPES OF THERMOMETERS
1.Length of liquid in capillary tube	Liquid in glass thermometer
2.Pressure of a fixed mass of gas at constant volume	Constant volume gas thermometer
3.Resistance of coil of wire	Resistance thermometer
4.E.m.f of a thermocouple	Thermoelectric thermometer

FIXED POINTS

A fixed point is a temperature where all thermometers show the same reading.

When measurement of temperature is done, a temperature scale must be established. This involves the use of fixed points.

THE ICE POINT: is the constant temperature at which ice melts/water solidifies.

THE STEAM POINT: is the constant temperature at which water vaporises (boils and changes into vapour).

The Celsius (also called centigrade) and Fahrenheit scales uses these two points as the basis of their measurement.

A third point called the triple point of water is a temperature at which ice(solid), water(liquid), and steam(gas) coexist. This temperature is 0.01°C . This point is the basis of the absolute (also called Kelvin) temperature.

CALIBRATING A THERMOMETER

To describe how a temperature scale maybe established in terms of an arbitrary thermometric property.

(i)Some property of matter is selected whose value varies continuously with the degree of hotness {thermometric property}.

(ii)Two standard degree of hotness are chosen called fixed points and numbers (arbitrary constants) are assigned to them eg on the Celsius method of numbering known as centigrade method, the lower fixed point ice point that is the temperature of pure ice in equilibrium with air- saturated water at standard a.t.m pressure and is designated as 0°C . The upper fixed point is the steam point that is temperature at which steam and pure boiling water are at equilibrium at standard a.t.m and is taken at 100°C .

(iii)The values X_{100} and X_0 of the temperature measuring property are found at the steam and ice point respectively. $(X_{100}-X_0)$ gives the fundamental interval of the scale. If X_{θ} is the value of the property at some other temperature θ which we wish to know then the values of θ in $^{\circ}\text{C}$ is given by the equation.

$$\frac{\theta}{100} = \frac{X_{\theta} - X_0}{X_{100} - X_0}$$

The temperature scale is defined so that the property varies uniformly or linearly with temperature measured on its own scale. And also the equation above has been derived assuming that the thermometric property X varies linearly with temperature.

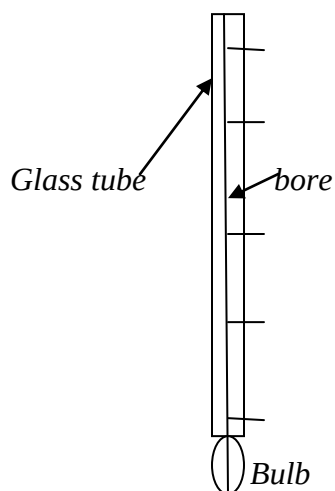
THERMOMETERS

Properties of thermometers

1. Sensitivity. The size of movement or change in the thermometer gives a reading even when there is a very small temperature change.
2. Range. The difference between the minimum and maximum temperatures that can be measured by a thermometer. A thermometer with a wide range can give readings over a wide spectrum of temperatures.
3. Linearity. Change is the same for all degrees , if a thermometer has a change in its thermometric property (length of liquid column, volume, potential difference or current) that is not uniform, it is considered to be non linear

TYPES OF THERMOMETERS

The liquid in glass.



This is the most common thermometer and it exists in two main forms
(a)Alcohol thermometer (b) Mercury thermometer

The liquid in glass thermometer uses thermal expansion of the liquid. When the bulb of the thermometer is placed in a hot substance, the ethanol or mercury expands and rises up the bore.

Properties

Sensitivity-they have low sensitivity. This can be improved by using, a thin glass; a narrow bore and a large bulb.

Range- have small range, eg alcohol thermometers contain ethanol which boils at 78°C and therefore cannot measure temperatures beyond this.

Linearity-the linearity of the expansion decreases as the measured temperatures approach the boiling point of the liquid.

ADVANTAGES

1. Easy to read off and use
2. Cheap
3. Direct reading
4. Very portable
5. Suitable for normal use

DISADVANTAGES

1. Low sensitivity
2. Very fragile (glass can break and crack)
3. Not accurate
4. Limited range 23K to 633K

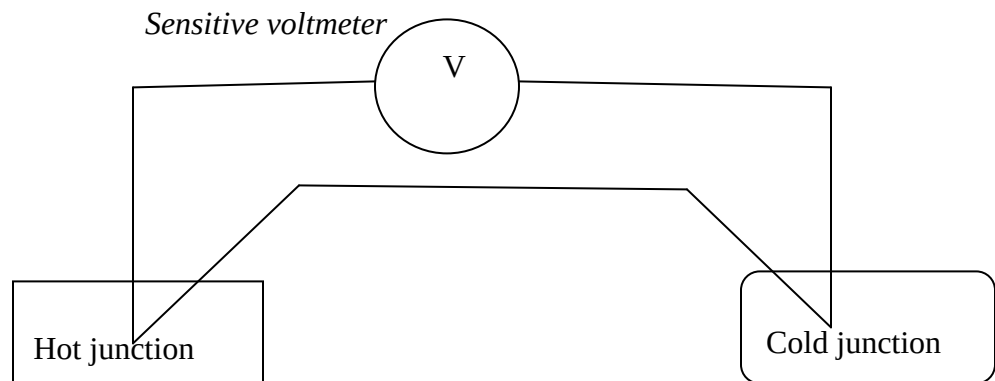
5. Glass expansion and contraction results in spoiled calibration.

USES

1. Clinical
2. Laboratory
3. Domestic uses

$$\theta^{\circ}C = \frac{L_{\theta} - L_0}{L_{100} - L_0} \times 100^{\circ}C$$

THERMOCOUPLE THERMOMETER



A thermocouple thermometer is made of two different metal wires of different electrical conductivity. The commonly used wires are iron and copper. One end of the wire is connected to the cold junction at ice point, and the other is placed at the point whose temperature is to be determined called the hot junction. The cold and hot junctions are joined by another wire.

OPERATION OF A THERMOCOUPLE THERMOMETER

The difference in temperature causes an electric current to flow from the hot junction to the cold junction. The current is very small and can only be detected by a galvanometer or mill ammeter. The size of the current is proportional to the temperature difference between the two junctions. The temperature is obtained by graduating the mill ammeter to temperature or comparing with a standard chart and calculating the temperature using proportion.

$$\theta^{\circ}C = \frac{V_{\theta} - V_0}{V_{100} - V_0} \times 100^{\circ}C$$

ADVANTAGES

1. Wide range 25K-1400K
2. Very sensitive
3. Fast in response and is used to measure rapidly changing temperature
4. Readings can be done remotely (at a distance and stored in computer (can measure very low or very high temperatures)
5. Very accurate.
6. Able to measure temperature of small and isolated places.

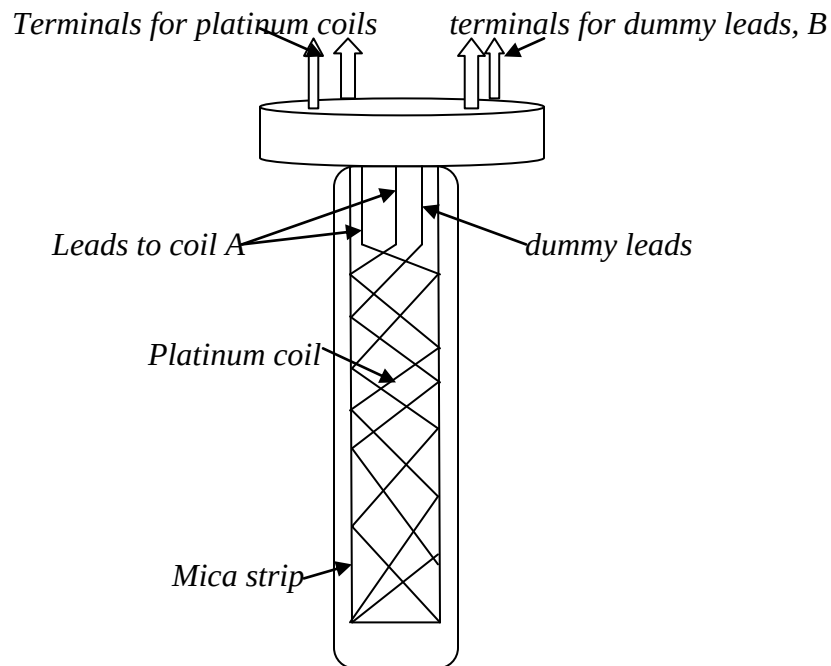
DISADVANTAGES

1. Expensive
2. Less accurate if milli voltmeter is used to measure e.m.f. (a potentiometer used to measure emf would be more accurate)

USES

Used to measure very high temperature eg in a furnace.

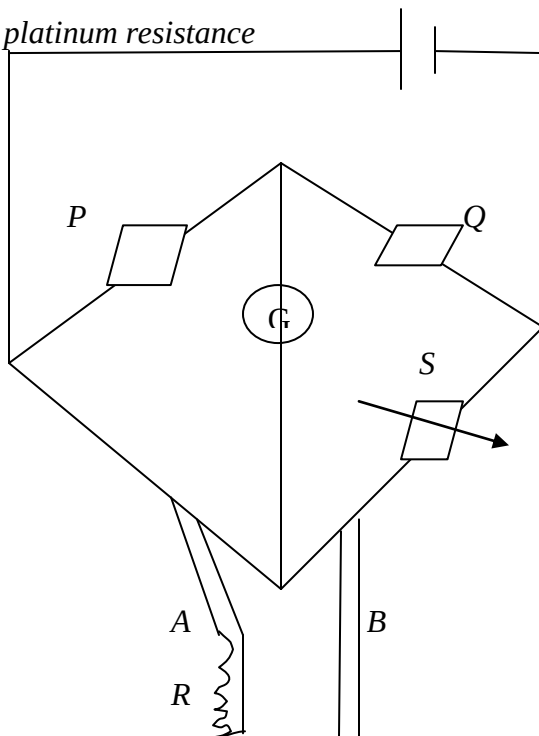
RESISTANCE THERMOMETER



The thermometric property used in the platinum resistance thermometer is the resistance of a coil of platinum wire. The platinum wire is wound on a strip of mica. Two other terminals of the platinum resistance thermometer are connected to dummy leads that run along the leads to the platinum coil.

The dummy leads are to compensate for any change in resistance in the leads to the platinum coil when temperature changes.

The circuit for platinum resistance



The resistance R of the platinum coil is measured using the Wheatstone bridge circuit as shown above. P and Q are standard resistors. The variable resistance S is varied until the galvanometer is balanced, then

$$\frac{R}{S} = \frac{P}{Q} \text{ and } R = \left(\frac{P}{Q}\right)S$$

The resistance scale of temperature is then defined by the equation

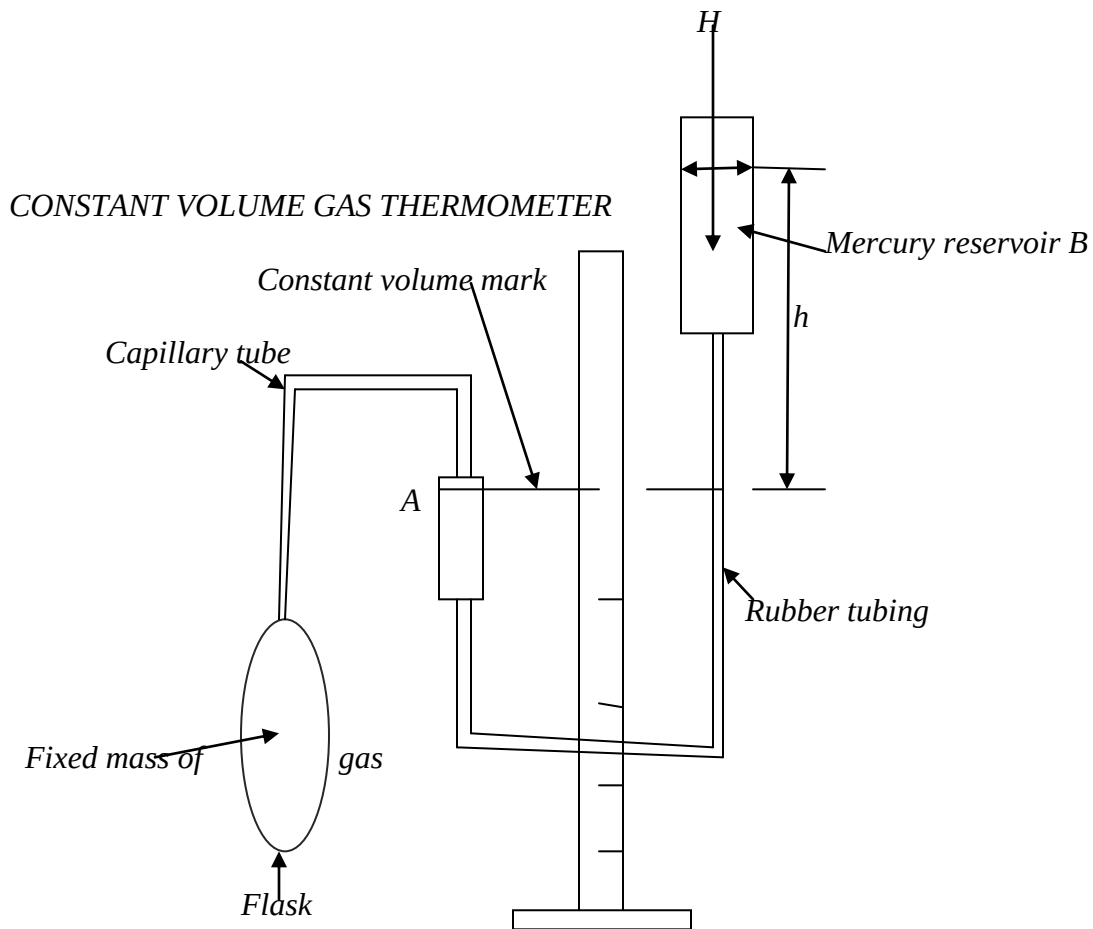
$$\theta^{\circ}\text{C} = \frac{R_{\theta} - R_0}{R_{100} - R_0} \times 100^{\circ}\text{C}$$

ADVANTAGES

1. Accurate
2. Wide range
3. Most suitable for measuring constant temperatures accurately.
4. Used as standard thermometer for temperatures. Between -185°C to 630°C

DISADVANTAGES

1. Not suitable for rapid changes in temperature, because it takes time to attain thermal equilibrium with the surrounding.



The constant volume gas consists of a flask containing dry air or nitrogen connected to a capillary tube to a manometer. The mark A is the constant volume mark. To measure the pressure of the gas in the flask, the height of the mercury reservoir B is adjusted until the level of mercury in the capillary tube reaches the level A, the constant volume mark. If h/cm = difference in height between the mercury meniscus in the manometer when thermal equilibrium is reached. And H/cm = height of mercury barometer (atmospheric pressure)., then the pressure of the gas in the flash

$$P = (H + h) \text{ cm mercury}$$

Hence if $h = h_0$ at ice point, $h = h_{100}$ at the steam point, $h = h_\theta$ at the temperature to be measured

$$\begin{aligned}\text{Then } \theta^{\circ}\text{C} &= \frac{P_{\theta} - P_0}{P_{100} - P_0} \times 100^{\circ}\text{C} \\ &= \frac{h_{\theta} - h_0}{h_{100} - h_0} \times 100^{\circ}\text{C}\end{aligned}$$

ADVANTAGES

1. Sensitive
2. Accurate
3. Large range
4. Easily constructed
5. Used for calibrating other types of thermometers
6. Used as standard thermometer below -183°C

DISADVANTAGE

1. Difficult to use and time consuming
2. Volume of flask is large, therefore unable to measure temperature at isolated confined space.

THERMODYNAMIC SCALE OF TEMPERATURE

The thermodynamic scale of temperature is not an empirical scale of temperature. It is a scale of temperature that is not based on any thermometric property and on experimental results. The thermodynamic scale of temperature is a scale of temperature which is based on the property of ideal gas.

An ideal gas only exists in theory. The property of an ideal gas used to set up the thermodynamic scale of temperature is the product of the pressure P and volume V of a fixed mass of ideal gas ie PV .

The two fixed points in the thermodynamic scale of temperature are.

1. Absolute zero-which is the temperature at which the pressure of an ideal gas becomes zero and is given the value of 0K (zero Kelvin). Equivalent to -273.15°C .
2. The triple point of water-which is the temperature at which ice, water and water vapour are in thermal equilibrium.

Thermodynamic temperature scale is defined by.

$$T = \frac{PV_T}{PV_{tr}}$$

PV_T = value of PV for a fixed mass of ideal gas at temperature T

PV_{tr} = value of PV of gas at the triple point.

6.4: THERMAL PROPERTIES OF MATERIAL

Heat capacity (C) is the quantity of heat required to raise the temperature of the body by a unit temperature.

Specific heat capacity (c) is the quantity of heat required to raise the temperature of a unit mass by a unit temperature.

$C=mc$ where C is heat capacity, c is the specific heat capacity and m is the mass of the body.

Water has a relatively high specific heat capacity of $4180 \text{ JKg}^{-1} \text{ K}^{-1}$. A relatively small amount of water absorbs a large amount of heat for a corresponding small temperature rise. This property of water is a reason for its use as a cooling agent eg in cars.

SPECIFIC HEAT CAPACITY BY ELECTRICAL METHOD**SOLID**

Electrical heat supplied by heater $= VIt$ is equal to heat gained by the aluminium block $= mc(\theta_2 - \theta_1)$

$$\text{Therefore } c = \frac{VIt}{m(\theta_2 - \theta_1)}$$

FOR A LIQUID

Heat supplied by heater (VIt) = Heat gained by liquid ($mc(\theta_2 - \theta_1)$) + heat gained by calorimeter ($Mc_1(\theta_2 - \theta_1)$)

From the equation $VIt = mc(\theta_2 - \theta_1) + Mc_1(\theta_2 - \theta_1)$: c can be calculated.

Where c = specific heat capacity of liquid, c_1 = specific heat capacity of calorimeter, m is the mass of water and M is the mass of calorimeter.

LATENT HEAT

This is the energy required to change the state of pure substance at constant temperature.

While heat capacity involves a change in temperature, latent heat involves a change in state. Latent means hidden.

MELTING AND BOLING POINTS

These are two main fixed temperatures at which phase changes occurs.

1. Melting point-is a constant temperature at which a solid changes to a liquid. The reverse is called freezing point.
2. Boiling point- is the constant temperature at which liquid changes to gas. If we continue adding energy when water is at 100°C, the temperature will remain constant. The energy supplied is used to boil water this changes its state from liquid to gas. The energy is used in weakening of the intermolecular forces called Van der Waals forces and breaking of bonds between water molecules

When latent heat is supplied, it increases the potential energy of molecules and supplies energy for work to be done against the attractive forces and bonds between particles. There is no change in kinetic energy of the molecules since the temperature is constant.

SPECIFIC LATENT

This is the energy required to change the state of a unit mass of a substance without a change in temperature.

If a body of mass m and specific latent heat of L needs a total energy E to change its state at a constant temperature then $E = \pm mL$

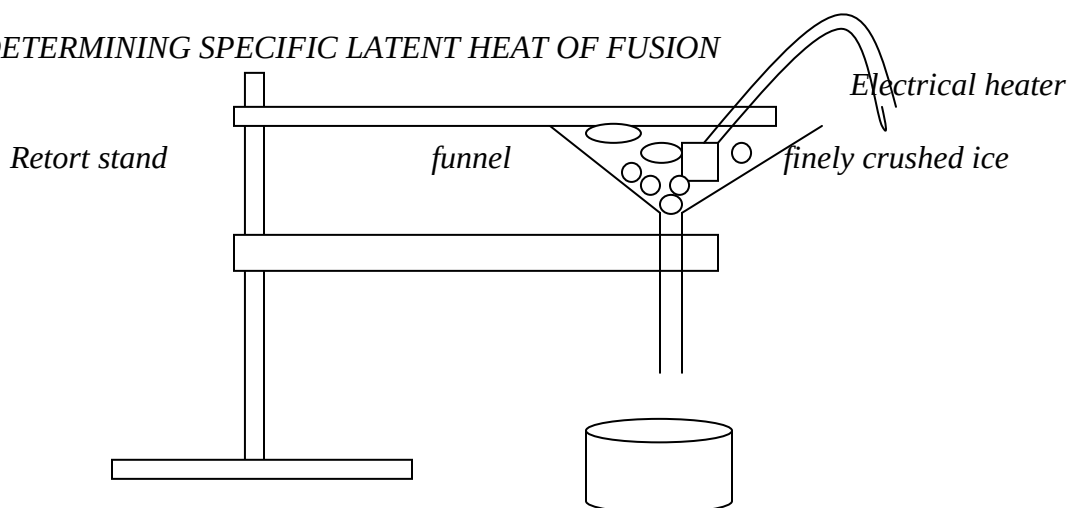
If energy is absorbed the latent heat is positive endothermic and the state change is melting or boiling. If energy is evolved latent heat is negative (exothermic) since energy is lost and the phase change is solidification commonly called freezing or condensation.

TWO TYPES OF SPECIFIC LATENT HEAT

1. SPECIFIC LATENT HEAT OF FUSION (MELTING)

This is the energy required to change a unit mass of a pure solid into a liquid state at a constant temperature.

DETERMINING SPECIFIC LATENT HEAT OF FUSION



METHOD

When water starts coming out of the funnel, start the stopwatch. After some time, switch off the heater and record the time for which the heater was on as well as the mass of the water collected

Electrical energy = thermal energy

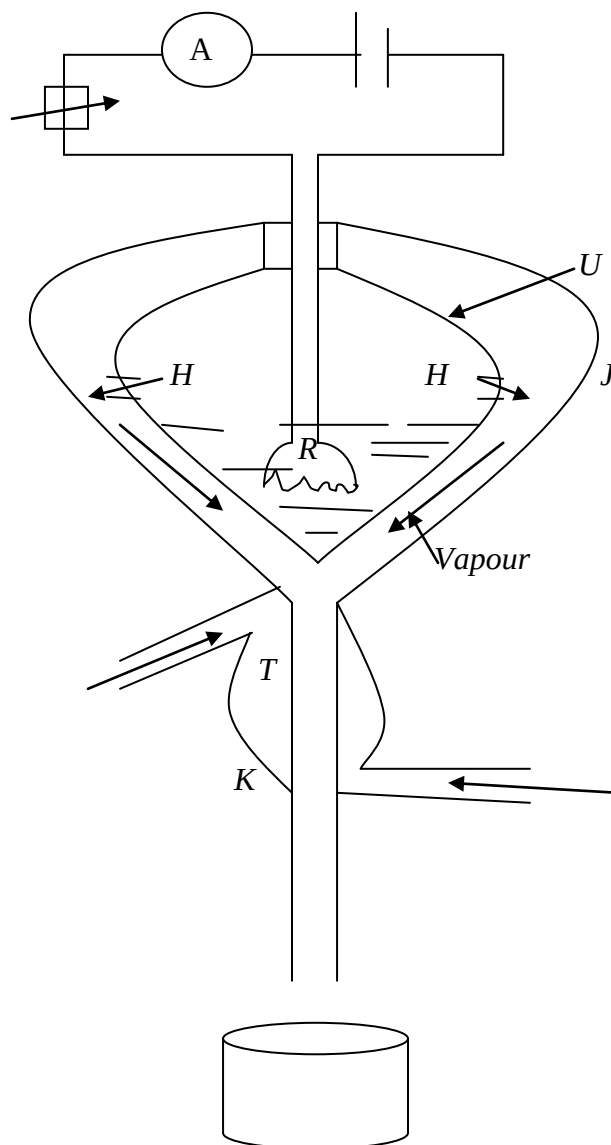
$$VIt = ml$$

Then l can be determined

2. SPECIFIC LATENT HEAT OF VAPOURISATION

The energy required to change a unit mass of a pure liquid into its gaseous phase at constant temperature.

SPECIFIC LATENT HEAT OF VAPOURISATION OF A LIQUID BY ELECTRICAL METHOD.



EXPLANATION

The liquid is heated in the vessel U by the heating coil R. The vapour escapes through H at the top of U into a surrounding jacket J and the passes down to tube T. Here the vapour is condensed by cold water flowing through the jacket K. When the apparatus has reached its steady state the liquid is at its boiling point, and the heat supplied by the coil is used in evaporating the liquid and in offsetting the loses.

The heat losses are considerably reduced by the use of the vapour jacket. The liquid emerging from the condenser is collected for a measured time and weighed. And if h is the heat lost from the vessel per second, and m the mass of the liquid collected per second, then.

$$VI = mlt + h \quad \text{equation 1.}$$

The heat losses h are determined by the temperature of the vessel, which is fixed at the boiling point of the liquid. Therefore they may be eliminated by a second experiment with different rate of evaporation if I^1 and V^1 are the new current and potential difference and m^1 . Then

$$V^1 I^1 t = m^1 l t + h \quad \text{equation 2.}$$

Subtracting equation 2 from 1. Yields

$$L = (VI - V^1 I^1) / (m - m^1)$$

NB It may be noted that a much higher power supply is needed for determining the specific latent heat of vaporisation of water than for alcohol, water has much bigger value ($2.26 \times 10^6 \text{ J Kg}^{-1}$)

COOLING EFFECTS BY EVAPORATION

Since the more energetic molecules escape, molecules that remain in the liquid have lower kinetic energy. A liquid with less kinetic energy have less temperature. Thus evaporation produces cooling. Evaporation of the sweat removes heat from the body.

THE INTERNAL ENERGY

The internal energy of a system is the sum of the kinetic and potential energy of the particles in the system. This energy is randomly distributed throughout the system and is in effect the total energy of the atoms and / molecules of the system

We derive this energy from the kinetic theory in that we know that kinetic energy is associated with the.

- (i) Translational (from fixed point to another)
- (ii) Rotational (spinning)

(iii) Vibrational (revolving around a mean fixed lattice position) movement of particles.

The kinetic energy component is affected by temperature that is the thermal energy transferred between the system and the surroundings.

The potential energy component is associated with the bonding and forces of attraction between particles

Internal energy = kinetic + potential. According to the equation, an increase in either kinetic or potential will increase the internal energy (U). This means an increase in temperature increases kinetic energy or a change in state increases potential will increase internal energy.

LAWS OF THERMODYNAMICS

THERMODYNAMICS

Is the branch of physics that describe and correlates the physical property of macroscopic systems of matter and energy. In other words, it is concerned with heat and related thermal phenomena. The various laws of thermodynamics enable us to accurately describe processes involved in heat energy.

THE ZEROth LAW OF THERMODYNAMICS

States that if two systems A and B are in thermal equilibrium with a third body C, then they are in equilibrium with each other ie A is in equilibrium with C or B is in equilibrium with C. We can then determine the direction of heat flow when two systems (A and B) are put in contact. One system A is said to be hotter than B if heat flows from the former A to the latter B when they are in thermal contact (thermal contact- bodies can exchange heat energy)

This then allows us to introduce a parameter, called an empirical temperature, which is the same for all bodies that are in thermal equilibrium with each other. This is done by constructing a system, called a thermometer which allows us to ascribe a number to the temperature.

THE FIRST LAW OF THERMODYNAMICS

The laws tell us that heat is the form of energy which is conserved. The first law applies to closed systems, ie for a given amount of matter, and is expressed in the form of energy balance equation.

States that, the heat supplied to a system equals the sum of the increase in internal energy of the system and the external work done by the system.

$$\Delta Q = \Delta U + \Delta W$$

When a quantity ΔQ is supplied to a fixed mass of gas, its temperature may increase causing an increase in internal energy ΔU . The gas may expand and if the external work done by the gas is ΔW .

Sign conversions for $\Delta Q = \Delta U + \Delta W$

	POSITIVE (+)	NEGATIVE (-)
ΔQ	Heat supplied to the system	Heat loss by the system
ΔU	Increase in internal energy	Decrease in internal energy
ΔW	Work done by the system	Work done on the system

NB The first law of thermodynamics can be stated in the form.

$$\Delta U = \Delta Q + \Delta W$$

ΔU = is the increase in the internal energy of the system

ΔQ = is the heat supplied to the system.

ΔW = is the external work done on the system.

In words: The increase in internal energy of the system equals to the sum of the heat supplied to the system and the external work done on the system.

Sign conversion of $\Delta U = \Delta Q + \Delta W$

	POSITIVE (+)	NEGATIVE (-)
ΔQ	Heat supplied to the system	Heat loss by the system
ΔU	Increase in internal energy	Decrease in internal energy
ΔW	Work done on the system	Work done by the system

Simple example to emphasis on the difference between the two equations.

Heat supplied to the system $\Delta Q = 20J$

Increase in internal energy $\Delta U = 8J$

Work done **by the system** $\Delta W = 12J$

$$1. \Delta Q = \Delta U + \Delta W$$

$$\Delta Q = +; \Delta U = + \text{ and } \Delta W = +$$

$$20J = 8J + 12J$$

$$2. \Delta U = \Delta Q + \Delta W$$

$$\Delta U = +, \Delta Q = + \text{ and } \Delta W = - (\text{work done by the system})$$

$$8J = 20J + (-12J)$$

6.5: IDEAL GASES

Assumptions of ideal gas

1. A gas consist of a large number of molecules
2. The gas molecules are constantly in rapid, free random motion. Newtonian mechanics may be applied to the motion of the gas molecules
3. Gas molecules collide elastically with one another and the walls of the container. {The molecules are like perfectly elastic spheres}
4. They is no intermolecular forces except during collision {the attraction between molecules is negligible}.
5. The volume of the molecules is negligible compared with the volume occupied by the gas.

6. The duration of collision is negligible compared to the time ~~interval~~ between collisions.

The equation of ideal

$$PV_m = RT$$

P = Pressure of gas

V_m = Volume of 1 mole of gas

R = Molar gas constant = $8.31 \text{ K}^{-1} \text{ mol}^{-1}$

The equation of state can be written in different form eg. If the volume of n mole of an ideal gas is V , then volume of 1 mole, $V_m = \frac{V}{n}$

The equation above can be written as $PV = nRT$

If the mass of a gas of volume V is m , and M is the mass of 1 mole of the, gas then the number of mole $n = \frac{m}{M}$

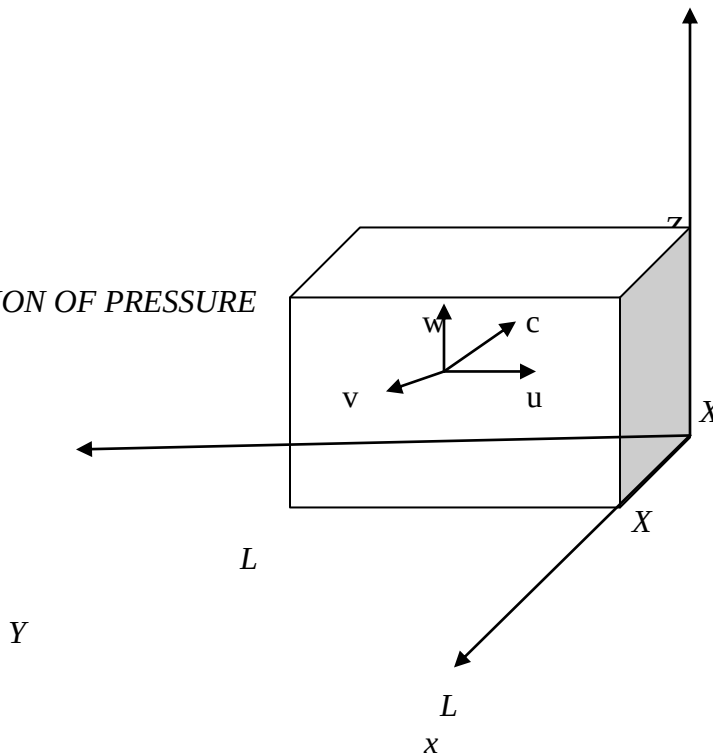
$$PV = nRT$$

Can be written as $PV = \frac{m}{M}RT$

Also if N = number of molecules in the volume V . And N_A = Avogadro's number of molecules in a mole of gas, then the number of mole $n = \frac{N}{N_A}$. The equation above becomes

$$PV = \frac{N}{N_A} RT$$

CALCULATION OF PRESSURE



Consider for convenience a cube of side L , containing N molecules of gas each of mass m . A typical molecule will have a velocity c at any instant and this will have components u, v, w respectively in the direction of the three perpendicular axes OX, OY and OZ . Thus $c^2 = u^2 + v^2 + w^2$.

Consider force exerted on face X of the cube to the component u . Just before impact, the momentum of the molecule due to u is mu .

Momentum after impact $= -mu$

Since momentum reverses. Speed before collision = speed after collision, since collision is elastic.

Momentum change $= mu - (-mu) = 2mu$

The time taken for the molecule to move across the cube to the opposite face and back to X is $\frac{2L}{u}$, which is also the time to make one impact. So the number of impacts per

second n^1 on a given face $= \frac{u}{2L}$

So at face X the momentum change per second $= n^1 \times \text{one momentum change}$

$$= \frac{u}{2L} \times 2mu = \frac{mu^2}{L}$$

Force on $X = \frac{mu^2}{L}$

Therefore pressure on $X = \frac{\text{Force}}{\text{Area}} = \frac{mu^2}{L} \div L^2 = \frac{mu^2}{L^3}$ Eq(i)

We now take into account of N molecules in the cube. Each has a different velocity and hence a component of different magnitude in the direction Ox , if these are $u_1, u_2, u_3; \dots u_N$. It follows from Eq(i) that the total pressure on X , P , is given by.

$$P = \frac{mu_1^2}{L^3} + \frac{mu_2^2}{L^3} + \frac{mu_3^2}{L^3} + \dots + \frac{mu_N^2}{L^3}$$

$$= \frac{m}{L^3} \{ [u_1^2] + [u_2^2] + [u_3^2] + \dots + [u_N^2] \}$$

Let $\langle u^2 \rangle$ represent the average or mean value of all the squares of the components in the direction, that is

$$\langle u^2 \rangle = \frac{u_1^2}{N} + \frac{u_2^2}{N} + \frac{u_3^2}{N} + \dots + \frac{u_N^2}{N}$$

$$\text{Then } N \langle u^2 \rangle = u_1^2 + u_2^2 + u_3^2 + \dots + u_N^2$$

$$\text{Hence } P = \frac{Nm \langle u^2 \rangle}{L^3}$$

Now with large number of molecules of varying speed in random motion, the mean square of the component of speed in any one of the three axes is the same.

$$\langle u^2 \rangle = \langle v^2 \rangle = \langle w^2 \rangle$$

$$\text{But for each molecule, } \langle c^2 \rangle = \langle u^2 \rangle + \langle v^2 \rangle + \langle w^2 \rangle$$

$$\text{Therefore } \langle u^2 \rangle = \frac{1}{2} \langle c^2 \rangle$$

$$P = \frac{1}{3} \frac{Nm \langle c^2 \rangle}{L^3}$$

$$\text{The number of molecules per unit volume } n = \frac{N}{L^3}. \text{ Thus } P = \frac{1}{3} nm \langle c^2 \rangle$$

ROOT MEAN SQUARE SPEED

From the equation $P = \frac{1}{3} nm \langle c^2 \rangle$ the factor nm is the product of the number of molecules per unit volume and the mass of one molecule. It is therefore that the total mass of the gas per unit volume, its density ρ . Thus $P = \frac{1}{3} \rho \langle c^2 \rangle$.

$$\langle c^2 \rangle = \frac{3P}{\rho}. \text{ The square root of } \langle c^2 \rangle \text{ is called the root-mean square that } \sqrt{\langle c^2 \rangle}$$

$$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3P}{\rho}}$$

INTRODUCTION OF TEMPERATURE TO KINETIC THEORY

Consider a volume V of gas, containing N molecules. The number of molecules per unit volume $n = \frac{N}{V}$

So pressure of the gas $P = \frac{1Nm\langle c^2 \rangle}{3V}$

$$PV = \frac{1Nm\langle c^2 \rangle}{3}$$

But from ideal gas equation for 1 mole
 $PV = RT$

$$PV = \frac{1Nm\langle c^2 \rangle}{3} = RT$$

TO SHOW THAT THE AVERAGE KINETIC ENERGY IS PROPORTIONAL TO THERMODYNAMIC TEMPERATURE

$RT = \frac{1Nm\langle c^2 \rangle}{3}$. Multiplying the equation by $\frac{3}{2}$ yields

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3RT}{2N}$$

Thus, the average kinetic of translation of a molecule is proportional to the absolute temperature of gas

The ratio $\frac{R}{N}$ is a universal constant, since $R =$ molar gas constant ($8.31 \text{ J mol}^{-1} \text{ K}^{-1}$) for all gases, and $N = N_A$, the Avogadro constant ($6.02 \times 10^{23} \text{ mol}^{-1}$) for all gases. Thus $\frac{R}{N_A} = k$, is called the Boltzmann constant $= 1.38 \times 10^{-23} \text{ JK}^{-1}$

In terms of k
 $\frac{1}{2}m\langle c^2 \rangle = \frac{3kT}{2}$

6.6: NON VISCOUS FLUID FLOW

DEFINITION OF TERMS

1. Lamina flow: in a laminar flow each fluid particle follows a smooth path (streamline)

2. *Incompressible flow*: in incompressible flow a change in pressure produce no change in density.
3. *Ideal fluid*: a fluid which is non-viscous and incompressible.
4. *Turbulent flow*: irregular or non steady (laminar) flow.
5. *Streamline*: is the path taken by a fluid in a steady flow.
6. *Non –viscous fluid*: is a fluid with no viscous forces which cause resistance to relative motion of two adjacent layers of the fluid.
7. *Tube of flow*: is a bundle of streamlines.

The flow of a fluid is said to be steady (or laminar or streamline) if each particle of the fluid follows a smooth path and the path of each particle do not cross each other. Above a certain critical speed, the fluid flow becomes irregular and turbulent (or non steady).

In describing the motion of a fluid, it is helpful to make two simplifying assumptions.

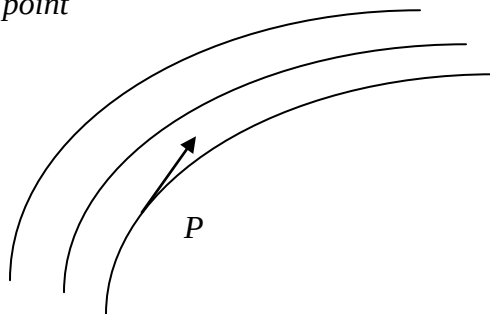
1. The density of liquid is assumed to remain constant, so that the fluid is incompressible.
2. The liquid is non viscous.

In real fluids, viscous forces cause resistance to the relative motion of two adjacent layers of the fluid. In non viscous fluid, this internal resistance is neglected and an object moving in non viscous fluid would experience retarding force.

A fluid which is non viscous and incompressible is said to be ideal. The assumptions allow us to develop a picture of such a fluid.

STREAMLINES

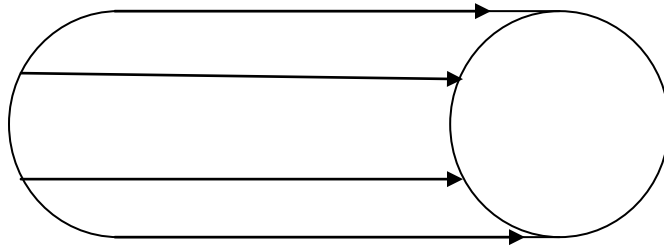
The path taken by a fluid in a steady flow is called streamline. Fig below shows a set of streamlines. A particle P follows one of those streamlines and its velocity vector is a tangent at that point



Note that, in steady flow, no two streamlines can cross each other. If they did, a particle of the fluid could follow either line at that cross over point, and the flow would not be steady.

A TUBE OF FLOW

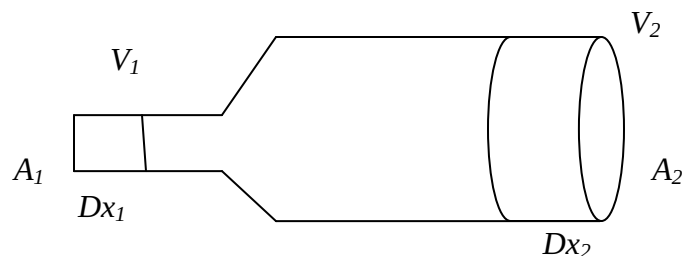
A bundle of streamlines is used to define what is called a tube of flow. Although a tube of flow is not a real tube with solids sides, it acts like one. Each streamline represents a path of fluid particles. Particles cannot move into or out of the sides of the tube as this would mean that the streamlines would be crossing each other.



DERIVATION OF THE EQUATION OF CONTINUITY

$AV = \text{constant}$. [The equation of continuity for the flow of an ideal, incompressible liquid]

We now consider the motion of an ideal, incompressible fluid in a tube of flow. In fig below the velocity of a fluid at the cross – sectional area A_1 is V_1 and that at the cross-sectional area A_2 is V_2 . Because we are dealing with ideal fluid, the velocity V_1 is the same at any point in the cross-sectional area A_1 . And V_2 is the same at any point in the cross-sectional area A_2 . [in real fluids which is viscous, the internal friction between different layers would mean that different parts of the fluids would move with different speed]



In a small time dt , the fluid at the left hand end of the tube moves a distance $Dx_1 = V_1 dt$. The mass contained in this length of the tube $dm_1 = \rho A_1 dx_1 = \rho A_1 V_1 dt$.

Similarly the fluid which moves through the right hand end of the pipe in the same time has mass $dm_2 = \rho A_2 V_2 dt$. Because fluid is incompressible, the density ρ is constant everywhere in the tube of flow. Provided that there are no leaks from the tube of flow. The mass which enters the tube at one end must be equal to the mass which leaves at the other end. So that

$$Dm_1 = dm_2$$

$$\rho A_1 V_1 dt = \rho A_2 V_2 dt$$

$$A_1 V_1 = A_2 V_2$$

The same results applies for any cross-section and can be written in the for $AV =$ constant.

This results that, the product of the area of cross-section and the fluid speed at all points along a tube of flow is a constant, is the equation of continuity. The equation tells us that as we would expect, the speed is high where the tube is narrow and low where it is wide. AV is called the volume the flow rate.

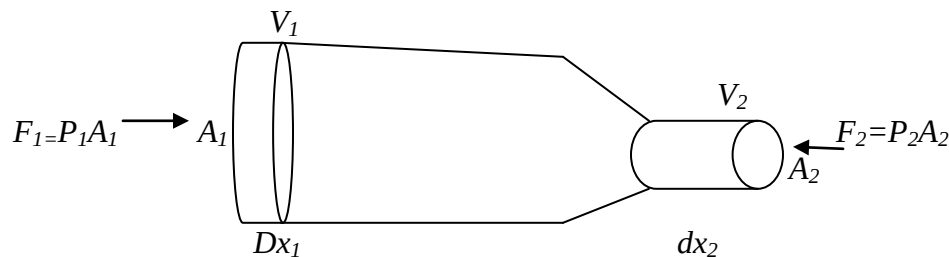
The equation of continuity is a form of the principle of conservation of energy. The derivation of the equation of continuity was based on the fact that mass of fluid which enters a tube at one end must leave the tube at the other end. Formally, this is a statement that the mass of fluid remain constant and is an example of the principle of conservation of energy.

THE BERNOULLI EFFECT

The equation of continuity relates the volume flow-rate in a tube to the area of cross-section and the speed of fluid particles. In addition to changes of particles speed in a tube of varying cross-section, the fluid pressure varies along the tube. The pressure differences can arise from different rates of flow of fluid (Bernoulli Effect).

Derivation of Bernoulli equation.

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$



At the left hand end of the tube the cross-sectional area A_1 and the fluid pressure is P_1 . The fluid particles are moving with speed V_1 . We consider the fluid in the small area dx_1 of the tube.

At the right hand end, the equivalent quantities are A_2 , P_2 , V_2 and dx_2 respectively. The force F_1 acting at the at the left hand end of the tube is P_1A_1 . The work done by this force in pushing the fluid contained in the shaded volume of the tube, of the length dx_1 in the time dt is $W_1 = F_1 dx_1 = P_1 A_1 dx_1 = P_1 dV$. Where dV is the shaded volume.

Similarly, the work done on the fluid at the right hand side end of the tube in time dt is $W_2 = -P_2 A_2 dx_2 = -P_2 dV$. As discussed before, the volume which enters is equal to the volume which leaves in the same time. The expression for the work W_2 is negative because the force exerted on the fluid is opposite to the direction of motion.

The net work done on the fluid by the force F_1 and F_2 in time dt is
 $W = (P_1 - P_2) dV$.

This work done goes into changing the kinetic energy of the fluid. If dm is the mass passing through the pipe in time dt , then the change in its kinetic energy is

$$dE_k = \frac{1}{2} dm V_2^2 - \frac{1}{2} dm V_1^2$$

The change in potential energy of the fluid is zero because the tube of flow is horizontal. By the principle of conservation of energy. $W = dE_k$, so

$$(P_1 - P_2) dV = \frac{1}{2} dm (V_2^2 - V_1^2)$$

The density ρ of the fluid is given by $\frac{dm}{dV}$; dividing left and right hand sides of the equation by dV , and using the density relation we obtain

$$P_1 - P_2 = \frac{1}{2} \rho (V_2^2 - V_1^2) \quad \text{Thus}$$

$$P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2$$

This is the Bernoulli equation for the case of a horizontal tube of flow. The equation shows that where the velocity is high, the pressure is low.

The derivation of the Bernoulli equation was based on the fact that the net work done on the fluid goes into changing the kinetic energy of the fluid. Formally this is a statement that energy is conserved.

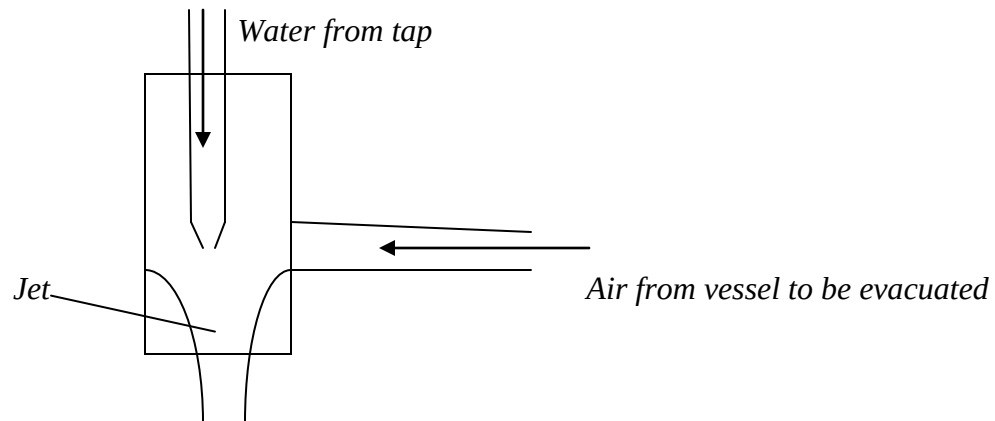
APPLICATION OF THE BERNOULLI EFFECT IN THE FILTER PUMP, VENTURI-METER, ATOMIZERS AND IN AEROFOILS.

(a). Jets (filter pumps) and nozzles (atomizer) or paint spray.

Bernoulli's equation suggests that for fluid flow where the potential energy change is very small or zero, as in horizontal pipe, the pressure falls when the velocity rises. The velocity increases at a constriction- a slow stream of water from a tap can be converted

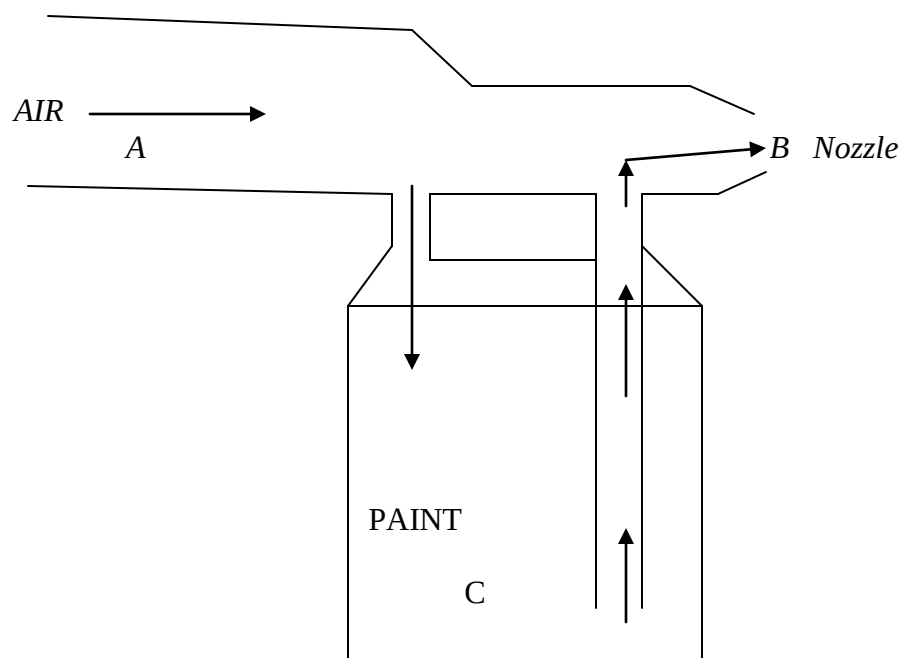
into a fast jet by narrowing the exit with a finger and the greater the change in cross-sectional area the greater is the increase in velocity and so the greater is the pressure drop. Several devices with jets and nozzles use this effect.

THE FILTER PUMP



A filter pump has a narrow section in the middle, so that a jet of water from the tap flows faster here, fig above. This causes a drop in pressure near it and air therefore flows in from the other side to which a vessel is connected. The air and water together are expelled through the bottom of the filter

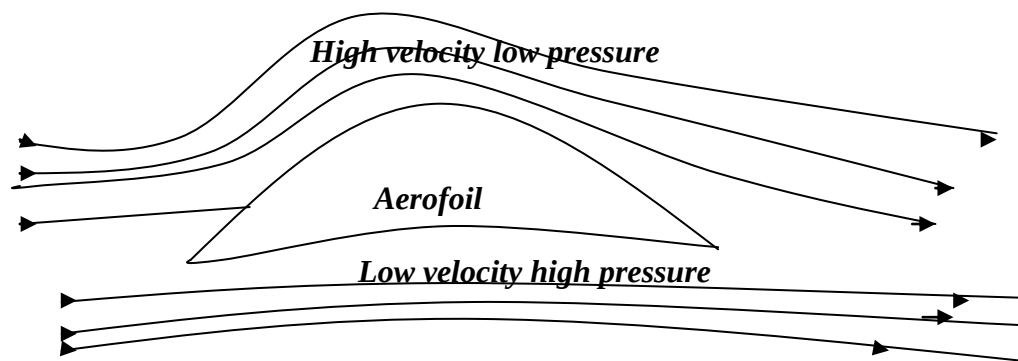
The spray paint (Atomizer)



For a spray paint air moves from A to B. At B it will move with high speed because of the constriction, making pressure at B lower than in C. Liquid or paint is forced out due to pressure difference.

(b) The Aerofoil

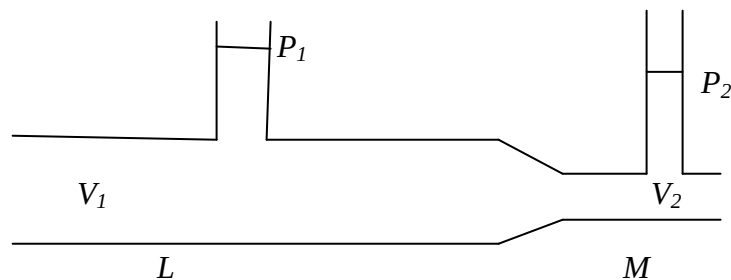
The curved shape of an aerofoil creates a faster flow of air over its top than the lower one, fig below. This is shown by the closeness of the streamlines above the aerofoil compared with those below. From the Bernoulli principle, the pressure of air below is greater than that above, and this produces the lift on the aerofoil. Example of aerofoil is aircraft wings.



(c) The venture meter

This measures the rate of flow of a fluid through a pipe

This consists of a horizontal tube with a constriction and replaces part of the piping of the system fig below. The two vertical tubes record the pressures (above atmospheric) in the fluid flowing in the normal part of the tube and constriction



The pressure P_1 and P_2 are the pressures and V_1 and V_2 the velocities of the fluid of density ρ at L and M on the same horizontal level, then assuming Bernoulli's equation holds

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$

Therefore:

$P_1 - P_2 = \frac{1}{2}\rho(V_2^2 - V_1^2)$ is the pressure difference.

If A_1 and A_2 are the cross-sectional areas at L and M and if the fluid is incompressible, the same volume passes each section of the tube per second.

$$A_1V_1 = A_2V_2$$

Hence

$$P_1 - P_2 = \frac{1}{2}\rho V_1^2 \left[\frac{A_1^2}{A_2^2} - 1 \right]$$

Knowing A_1, A_2, ρ and $(P_1 - P_2)$, V_1 can be found and also rate of flow A_1V_1 .

(d) A SUCTION EFFECT

A suction effect is experienced by a person standing close to the platform at a station when a fast train passes. The fast moving air between the person and the train produces a decrease in pressure and the excess air pressure on the other side pushes the person towards the train.

6.7: TRANSFER OF THERMAL ENERGY

Thermal conduction

There are two ways by which heat is transferred.

1. When one end of a crystalline solid is heated, atoms in the crystal lattice at the heated end vibrate more vigorously. These atoms vibrate against neighbouring atoms, which in turn vibrate against their neighbouring atoms and so on. The lattice vibrations in solid produce waves which are propagated in solid.

2. In metals, heat is transferred through the diffusion of free electrons. In good conductors such as metals, there are many free electrons which are responsible for electrical conduction. When one end of the metal is heated, the atoms in the heated part vibrate more. The free electrons that collide with these atoms gain kinetic energy and move faster, diffusing into the colder part of the metal where collisions with other free electrons and atoms in the lattice result in the transfer of energy.

CONVECTION

Heat transfers through the movement of substance itself. Convection occurs in all fluids. When a fluid is heated, it becomes less dense and rises bringing heat with it.

RADIATION

No material medium is required unlike in conduction and convection. Radiant energy is the form of electromagnetic radiation. Thermal radiation is emitted by all objects whose temperature is above zero Kelvin (0K). The thermal radiation consists of a mixture of wavelengths. Objects at low temperature emit waves of long wavelengths. Higher temperature objects emit waves of shorter waves. This is explained by Wien's displacement law.

$$\lambda_{\max} = \text{constant} = k$$

$$\lambda_{\max} = \frac{k}{T} \quad [k = 2.93 \times 10^{-3} \text{ milli Kelvin}]$$

When temperature increases, wavelength of the emitted electromagnetic wave decreases. This explains why a white hot object is hotter than a red hot object.

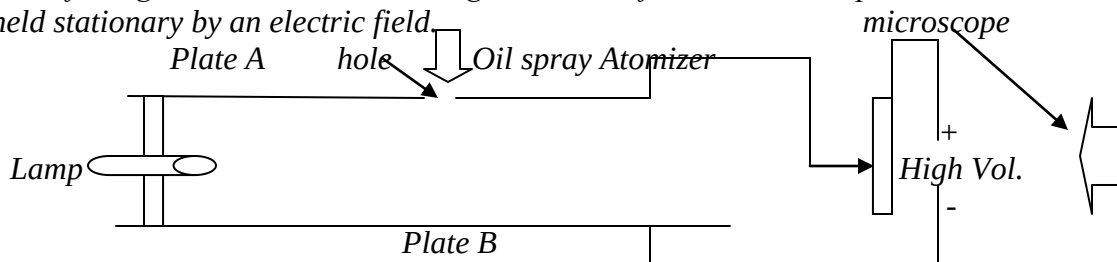
SECTION 7: MODERN PHYSICS

7.1: CHARGED PARTICLES

Determination of electric charge

Millikan's oil drop experiment

The principle of the method is to observe small oil drops, charged either positive or negative falling in air under the earth's gravitational field. The oil drop is either raised or held stationary by an electric field.



Fine drops of oil are sprayed into the space between the parallel plates A and B through a central hole. The drops are strongly illuminated by the lamp and appear as bright specks on a dark background when viewed by the microscope. The oil drops are charged by friction as they come out of the atomizer. The electric field between the plates is adjusted using a potentiometer until a particular oil droplet remains stationary.

Electric force = weight

$$qE = mg \quad \text{But } E = \frac{V}{d}$$

$$q = \frac{mgd}{V} \quad \text{but } V = \frac{4\pi r^3}{3} \quad \text{and } m = \frac{4\pi r^3}{3} \rho$$

$$q = \frac{4\pi d \rho r^3}{3V} \quad \rho \text{ is the density of oil.}$$

TERMINAL VELOCITY WITH ZERO ELECTRIC FIELD

To determine the radius of the oil droplet, r , the electric field is switched off and the oil droplet allowed to fall with its terminal velocity. For a spherical drop of radius r , moving with velocity through a homogeneous fluid having coefficient of viscosity η , Stokes law states that the viscous force retarding its motion is

$F_v = 6\pi\eta rv$ which is equal to weight of the droplet (mg)

$$6\pi\eta rv = mg$$

$$\text{But } m = \frac{4\pi r^3}{3} \rho$$

$$\text{Combining these two equations } r = \sqrt{\frac{9\eta v}{2\rho g}}$$

$$\text{Substituting } r \text{ in the equation } q = \frac{4\pi d \rho r^3}{3V}$$

$$\text{Yields } q = \frac{4\rho\pi dg}{3v} \left[\frac{9\eta v}{2\rho g} \right]^{3/2}$$

The terminal velocity v is measured by recording the time t for the droplet to fall through measured distance y . Hence terminal velocity $v = \frac{y}{t}$

RESULTS OF THE EXPERIMENT

Millikan found the value of $e = -1.6 \times 10^{-19} \text{C}$. All charges are integral multiples of this fundamental charge or charge is quantized.

Precautions taken to improve accuracy of results during experiment

1. Millikan used a non-volatile oil to eliminate the error due to evaporation (altering mass of oil drop during observation)
2. Convection currents between the plates and changes in viscosity of air due to temperature variation were eliminated by enclosing the apparatus in an isothermal oil bath.
3. Stoke's law assumes fall in a homogeneous medium. The air consists of molecules and, as Millikan put it, very small drops "fall freely through the holes in the medium". He investigated this effect and corrected the law to allow for it. (Millikan took into consideration the density of air and the effect of air molecules.

Determination of charge after an experiment

For example

An experiment, working in non-S.I unit obtained the following series of values for the magnitudes of the charges on the oil droplets:

19.64, 39.28, 9.82, 39.28, 34.37, 19.64, 29.46, 19.64, 19.64, 39.28

What the magnitude of the charge of the electron is as measured in the units used.

Solution

First arrange the experiments in order of magnitude of the charges obtained. Let's say in ascending order. The same or very close values are considered to be the same experiment. eg (3.1 and 3.2)

Step 1: arranging experimental values

9.82 is considered to be the first experiment.

19.64 All four are considered to be the second experiment

29.46 is the third experiment

34.37 is the fourth experiment

39.28 All three are the fifth experiment.

Step 2: Calculating the differences: between experiment 1 and 2, 2 and 3, etc to find the smallest difference.

$19.64 - 9.82 = 9.82$

$29.46 - 19.64 = 9.82$

$34.37 - 29.46 = 4.91$

$39.28 - 34.37 = 4.91$

Therefore the smallest difference = 4.91

Step 3: Divide every value; that is all the ten values by 4.91 to find the number of electrons from each value. If the answer is not exact round off to the nearest whole number.

Values	Divide by 4.91	Number of electrons
19.64	Divide by 4.91	4
39.28	Divide by 4.91	8
9.82	Divide by 4.91	2
39.28	Divide by 4.91	8
34.37	Etc	7
19.64	Etc	4
29.46		6
19.64		4
19.64		4
39.28	etc	8

Total=270.05		Total=55
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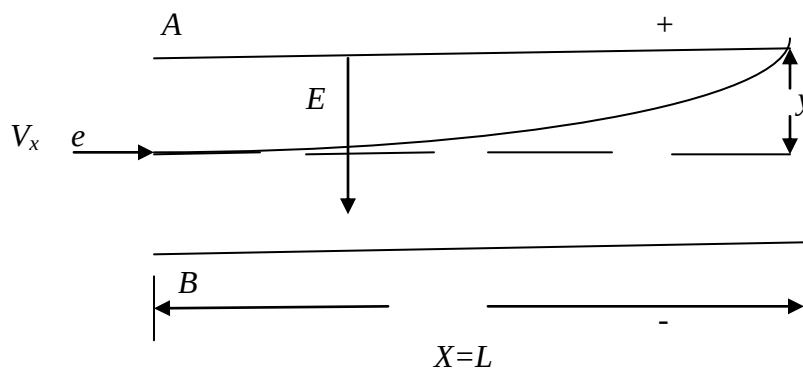
Step 4. Calculating charge

$$\text{Charge} = \frac{\text{Total of all the values}}{\text{Total number of electrons}} = \frac{270.05}{55} = 4.91$$

NOTE: It is by coincidence that from the above example the smallest difference is the charge. It is not always the case.

Note do not forget to write the multiplying factor of the charge, if it is given in the question eg 10^{-19} .

MOTION OF CHARGED PARTICLE IN A UNIFORM ELECTRIC FIELD



$$F_E \gg mg$$

A negatively charged particle of mass m and charge e is deflected upwards in the downwards electric field between two charged parallel plates A and B. The weight of the particle is small relative to the electrostatic force acting on the particle and can be neglected.

To determine the vertical deflection y

Electric force of magnitude $F = Eq$ acts upwards. The particle travels parallel to the x -axis at a constant speed V_x , it accelerates upwards with a constant acceleration a_y .

Applying Newton's second law $F = ma$

$$\text{Since } a_y = \frac{F}{m} = \frac{qE}{m} \quad \text{eqn 1}$$

Let t represent the time required for the particle to pass through the plates. During t the vertical and horizontal displacements of the particle are:

$$\text{Vertical; } y = \frac{1}{2}a_y t^2 \text{ and horizontal } X = V_x t = L$$

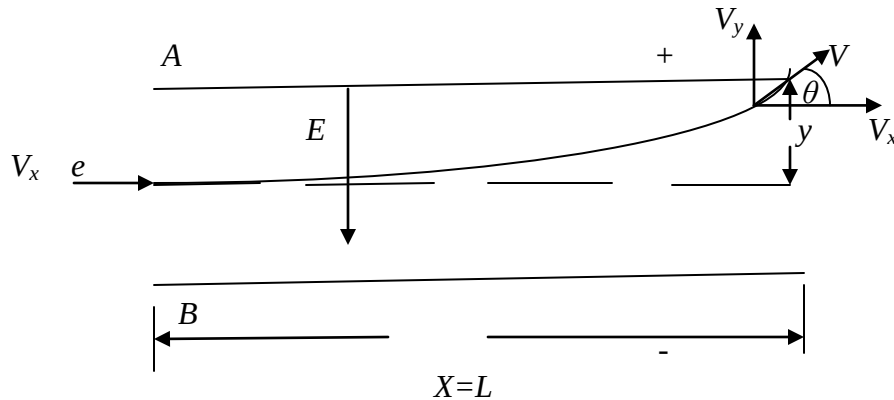
Eliminating t yields.

Since $a_y = \frac{F}{m} = \frac{qE}{m}$; $V_x t = L$, $y = \frac{1}{2} a_y t^2$:

This implies that: $y = \frac{qEL^2}{2mV_x^2}$

This is the vertical deflection of the particle at the far edge of the plates.

THE VELOCITY WITH WHICH THE ELECTRON LEAVES THE PLATES



$$F_E \gg mg$$

V_x is constant; the horizontal velocity. The vertical velocity V_y is given by the equation $v = u + at$: $V_y = a_y t$ because $u = 0$. There is no initial vertical velocity. But $t = \frac{L}{V_x}$

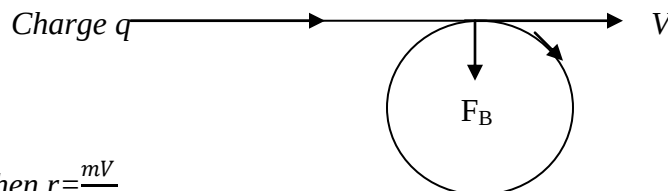
$$V_y = \frac{La_y}{V_x} \text{ but } a_y = \frac{qE}{m}$$

$$\text{Then } V = \sqrt{V_x^2 + V_y^2}$$

$$\text{Direction } \tan \theta = \frac{V_y}{V_x}$$

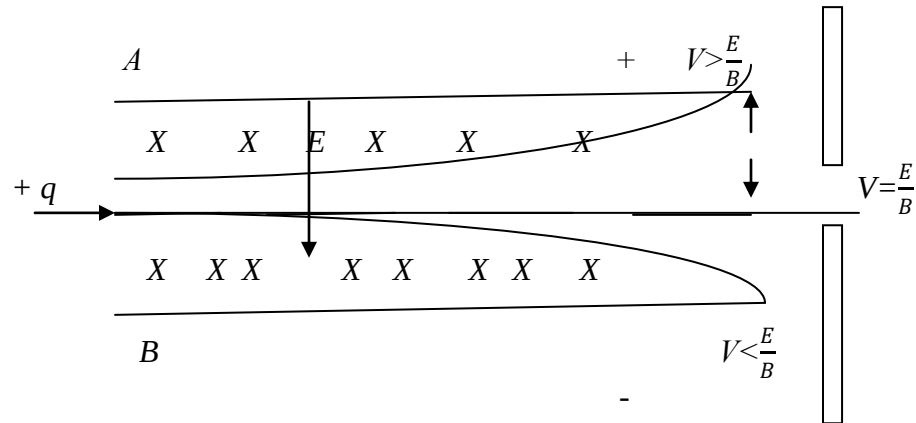
MOTION OF A CHARGED PARTICLE IN A UNIFORM MAGNETIC FIELD

Consider a particle of charge q and mass m moving with velocity V in a magnetic field of strength B normal to V . A magnetic force continuously deflecting the charge because V and B are normal to each other, this deflection causes the charge to follow a spiral path. The magnetic force provides the centripetal force.



$$F_B = \frac{mV^2}{r} = BVq \text{ Then } r = \frac{mV}{qB}$$

VELOCITY SELECTION



Two crossed fields set as shown on the diagram above. The magnetic field is into the page.

A positive charge is attracted downwards by the electric field and upwards by the magnetic field. One of the fields is adjusted so that the magnetic force equals the electric force. Magnetic force $F_B = BqV$ and electric force $F_E = Eq$.

Now $BqV = Eq$ Then $V = \frac{E}{B}$

For the value $V = \frac{E}{B}$, the particle will not be deflected and pass through the slit. Particles with other velocities will be deflected. $V > \frac{E}{B}$ are deflected upwards and $V < \frac{E}{B}$ are deflected downwards.

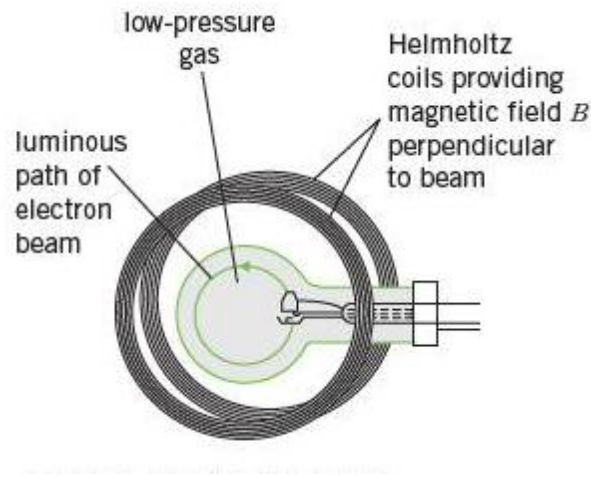
DETERMINATION OF SPECIFIC CHARGE

Specific charge is the name given to the ratio of the charge q on a particle and its mass m . Specific charge $= \frac{q}{m}$.

We have seen that a particle of mass m and charge q moving with speed V at right angles to a uniform magnetic field of flux B experiences a force F given by $F = BeV$

$BeV = \frac{mV^2}{r}$. Then $\frac{e}{m} = \frac{V}{Br}$

SPECIFIC CHARGE MAYBE DETERMINED USING A FINE BEAM TUBE



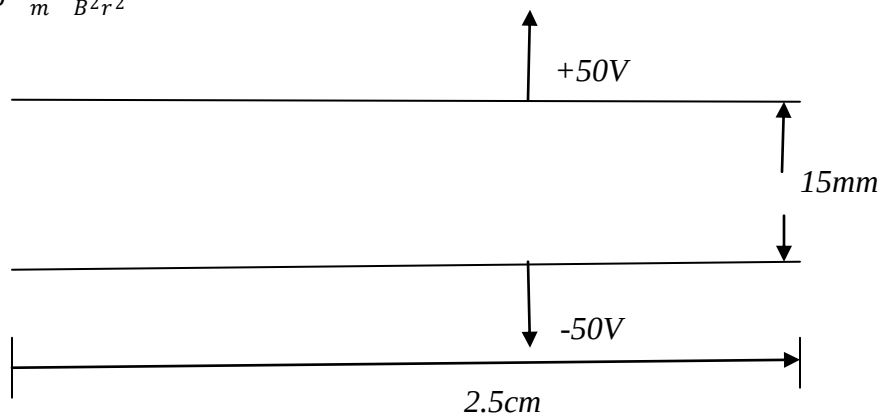
The path of electrons is made visible by having low pressure gas in the tube and, thus, the radius of the orbit may be measured. By accelerating the electrons through a known potential difference V , their speed v on entry into the region of the magnetic field may be calculated. $eV = \frac{1}{2}m_e v^2$

The magnetic field is provided by a pair of current-carrying coils (Helmholtz coils)

Combining $\frac{e}{m} = \frac{V}{Br}$ and $eV = \frac{1}{2}m_e v^2$

Specific charge $\frac{e}{m} = \frac{2V}{B^2 r^2}$

QUESTION



A beam of electrons enters the electric field at right angles to the field at a speed of $5.5 \times 10^7 \text{ m/s}$. Calculate

- The time spend by the electron between the plates
- Electric field strength between the plates
- Force on each electron
- Acceleration of the electron

(v) Velocity of the electron as it leaves the plates.

SOLUTION

$$(i) t = \frac{\text{distance}}{\text{speed}} = \frac{0.025}{5.5 \times 10^7} = 4.55 \times 10^{-10} \text{ s}$$

$$(ii) E = \frac{v}{d} = \frac{100}{0.015} = 6667 \text{ V/m}$$

$$(iii) F = Eq = 6667 \times 1.6 \times 10^{-19} = 1.07 \times 10^{-15} \text{ N}$$

$$(iv) \text{ Acceleration} = \frac{F}{m} = \frac{1.07 \times 10^{-15}}{9.11 \times 10^{-31}} = 1.17 \times 10^{15} \text{ m/s}^2$$

$$(v) \text{ Horizontal velocity } V_x = 5.5 \times 10^7 \text{ m/s and } V_y = u + at = 1.17 \times 10^{15} \times 4.55 \times 10^{-10} = 5.32 \times 10^5 \text{ m/s}$$

$$V = \sqrt{V_x^2 + V_y^2}$$

$$V = 5.50 \times 10^7 \text{ m/s}$$

$$\text{Direction: } \tan \theta = \frac{V_y}{V_x} = \frac{5.32 \times 10^5}{5.5 \times 10^7} \quad \theta = 0.55^\circ \text{ above the horizontal.}$$

NB (i) when students are asked to calculate velocity they forget to calculate direction.

(ii) When calculating the vertical velocity V_y , student must use vertical acceleration a_y instead of acceleration due to gravity g , because this is not a free fall.

7.2: QUANTUM PHYSICS

Quantum physics is the branch of science that deals with discrete indivisible units of energy. Electromagnetic waves such as light exhibit a dual nature as they possess both wave properties and particle like properties. Conversely, particles like electrons have wave-like properties as well as particle-like properties. This is referred to as wave-particle duality.

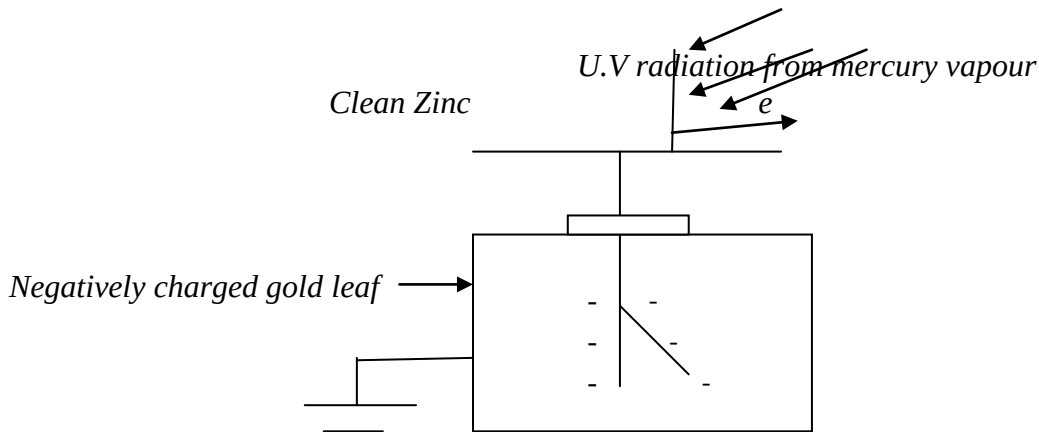
QUANTUM THEORY

Einstein presented the idea of light energy consisting of packets of electromagnetic energy. According to Planck, light energy has indivisible packets of energy called quanta. The energy is proportional to the frequency of the radiation. Quantum energy $= hf$, where h is the Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$.

PHOTOELECTRIC EFFECT

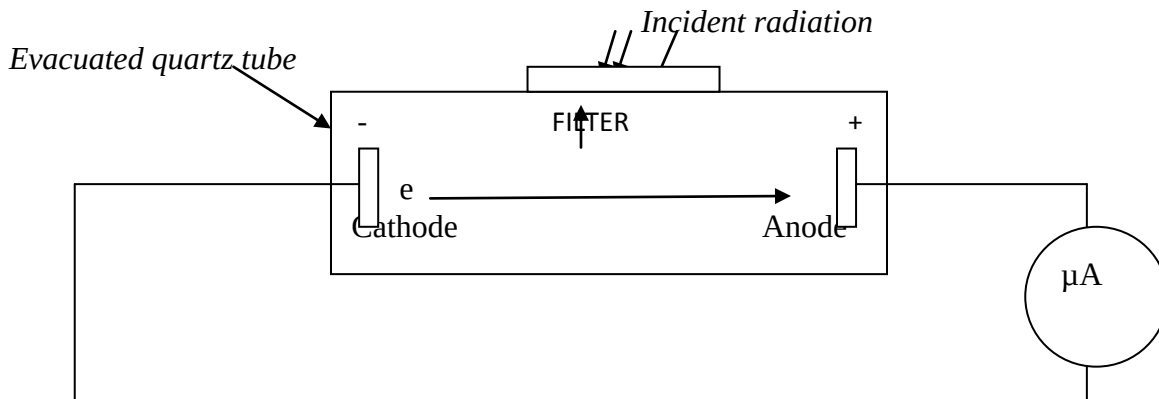
Is the ejection of electrons from a cold, clean metal surface when electromagnetic radiation of sufficiently high frequency falls on it.

SIMPLE EXPERIMENT TO DEMONSTRATE PHOTOELECTRIC EFFECT.



A clean zinc plate is placed on the metal cap of a gold leaf electroscope which is then charged negatively. U.V. radiation from mercury vapour lamp is allowed to fall on the zinc plate. The divergence on the gold leaf of the electroscope decreases. This shows that the electroscope loses its negative charge through the emission of the electrons.

ANOTHER EXPERIMENT TO DEMONSTRATE PHOTOELECTRIC EFFECT.



Monochromatic radiation usually from the U.V. part of the spectrum, falling on a clean metal plate liberates photoelectrons that flow through the circuit consisting of photocurrent. Current is recorded with a micro-ammeter because it is very small.

THRESHOLD FREQUENCY

There is a certain minimum frequency, characteristic of the metal, below which no photoelectrons are liberated. This is called threshold frequency f_0 .

From the experiment to demonstrate photoelectric effect using a gold leaf, if the intensity of U.V radiation is increased, the leaf falls more quickly, showing that the rate of emission of electrons has increased. If visible light of any colour is shone on to the plate, the leaf does not move. Even when the intensity (brightness) of the light is increased, the leaf remains in its deflected position. The difference between U.V and visible light is that U.V has a higher frequency than visible light.

ACCORDING TO QUANTUM THEORY.

1. If photoemission takes place, it does so instantaneously. There is no delay between illumination and emission.
2. Photoemission takes place only if the frequency of the incident radiation is above a certain minimum value called threshold frequency.
3. Different metals have different threshold frequencies
4. Whether or not emission takes place depends only on whether the frequency of the radiation used is above the threshold for that surface. It does not depend on the intensity of the radiation.
5. For a given frequency, the rate of emission of photoelectrons is proportional to the intensity of radiation.

NOTE: All the above theories were proved using the gold leaf experiment.

ACCORDING TO THE WAVE THEORY.

1. Photoemission is independent of the frequency, but increases with increase in the intensity of the incident radiation. That is, photoelectric effect should occur for any frequency of radiation provided the radiation is intense enough.
2. They also said that, increasing the intensity of the incident radiation, increases the kinetic energy of the maximum ejected electron. That is the kinetic energy of electrons is independent of frequency but depends with the intensity of the incident radiation.

NOTE. They failed to prove this experimentally.

Einstein (1905), looked upon the photoelectric problem as one of particle collisions. He regarded radiation as bundles of energy called photons, the quanta. (Photon is a **quantum of electromagnetic radiation**). Einstein's argument as follows: When an incident photon collides with an electron in a metal surface, all the energy of the photon is transferred to the electron. The energy is used in the following two ways.

1. A part is used to free the electron from the atom. To overcome the attractive forces of the metal. This energy is called the photoelectric work function of the metal, ϕ .

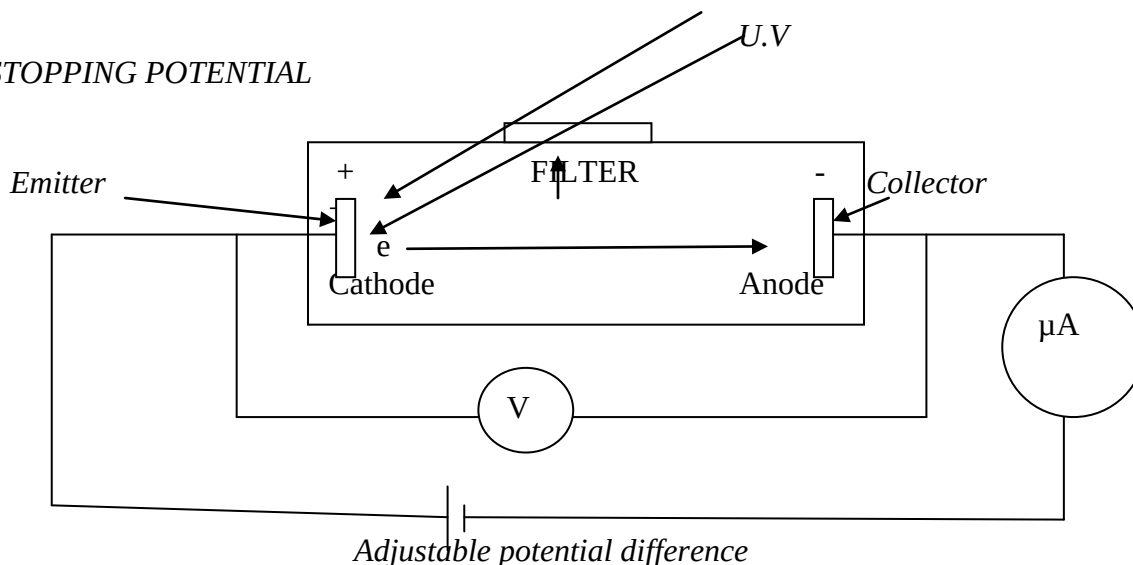
2. The other part gives the kinetic energy to the electron, $E_{kmax} = eV$

From the principle of conservation of energy.

$(hf = \phi + E_{kmax})$. This is Einstein's photoelectric equation.

It can be written in different form $(hf = hf_0 + eV_0)$ or $(hf = hf_0 + \frac{1}{2}mv_{max}^2)$

STOPPING POTENTIAL



If we set our circuit as shown on diagram above, reversing the polarity of V , the photo electric current does not immediately drop to zero, suggesting that electrons are emitted from the metal with a finite velocity. However, if this reversed potential is made large, a value V_0 (called stopping potential) is reached at which the photoelectric current drops to zero. This condition is expressed mathematically as $E_{kmax} = V_0 e$. E_{kmax} is the maximum kinetic of the fastest ejected photoelectron.

Maximum kinetic energy is independent of radiation intensity but depends on the frequency of the incident radiation (according to quantum theory). However wave theory predicts that kinetic energy of photoelectrons should increase with an increase in radiation intensity.

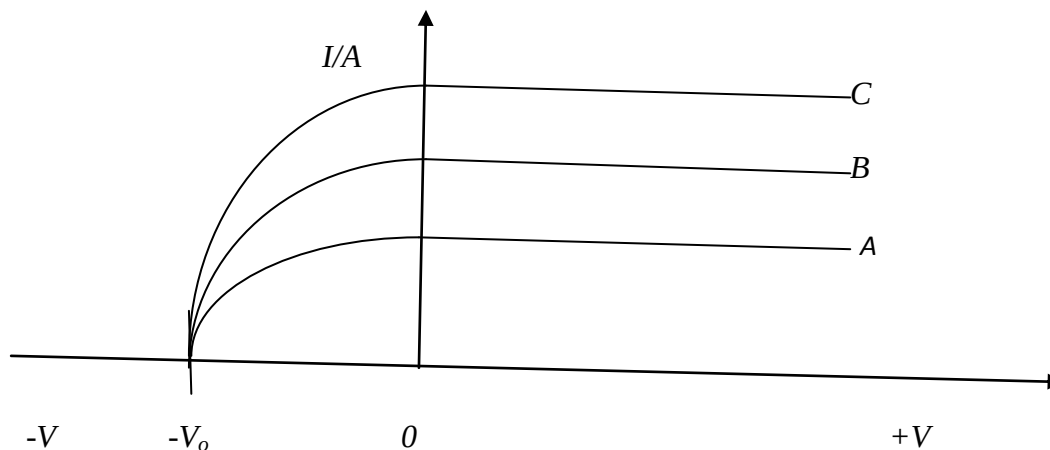
Stopping potential is the minimum value of the potential difference necessary to stop photoelectron flow.

HOW DOES PHOTOELECTRIC EFFECT PROVIDES EVIDENCE FOR THE PARTICULATE NATURE OF LIGHT.

1. No time delay observed between illumination and emission (instantaneous emission of photoelectrons).
2. Each electron absorbs one photon.
3. The existence of threshold frequency.

4. Light is concentrated into energy packets called photons
5. Changing frequency changes energy of photoelectrons.
6. Intensity does not increase the energy of photoelectrons.

GRAPH FOR CURRENT AGAINST POTENTIAL DIFFERENCE



EXPLANATION OF THE GRAPH ABOVE

1. The graphs have the same stopping potential V_0 , because (A,B and C are from the same incident radiation of the same frequency, therefore they have the same energy.
2. The intensity of $C > \text{Intensity of } B > \text{Intensity of } A$. This is causing the current of $C > \text{Current of } B > \text{Current of } A$. [greater intensity means more photoelectrons are emitted]
3. From zero to the negative of the graph, current decreases to zero. This is because electrons are moving towards the negative plate. As the plate becomes more negative, the kinetic energy of electrons further decreases, until no electrons reach the collector. The current becomes zero. The value of negative potential reached is called, stopping potential.
4. From zero to the positive of the graph, the value of current is constant. This is because all electrons ejected reach to the collector, because electrons are now moving towards the positive plate. All ejected electrons reach the collector. However the constant values of current are different due to the differences in intensities.

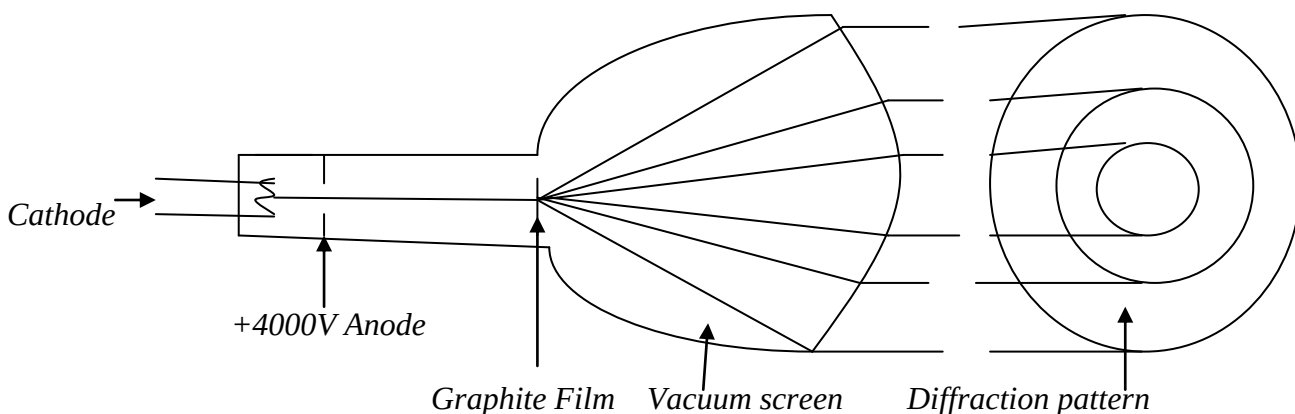
WAVE NATURE OF LIGHT

Phenomena such as interference and diffraction, provides the evidence for the wave nature of light.

ELECTRON DIFFRACTION

A beam of electrons from an electron gun is directed at a graphite film in an evacuated space. The electrons are diffracted into various directions as shown by the pattern obtained on the screen.

To demonstrate the wave properties of electrons

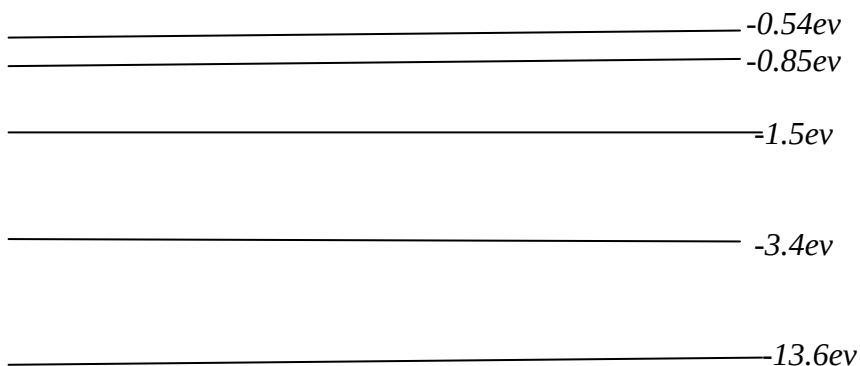


De Broglie adopted the concepts of a fundamental duality and postulated the wave theory of matter (1924): A material particle having mass m and speed v , appears in some experiments as a wave whose wavelength is given by

$\lambda = \frac{h}{mv}$ λ is called the De Broglie wavelength. $P = mv$, is the particle momentum.

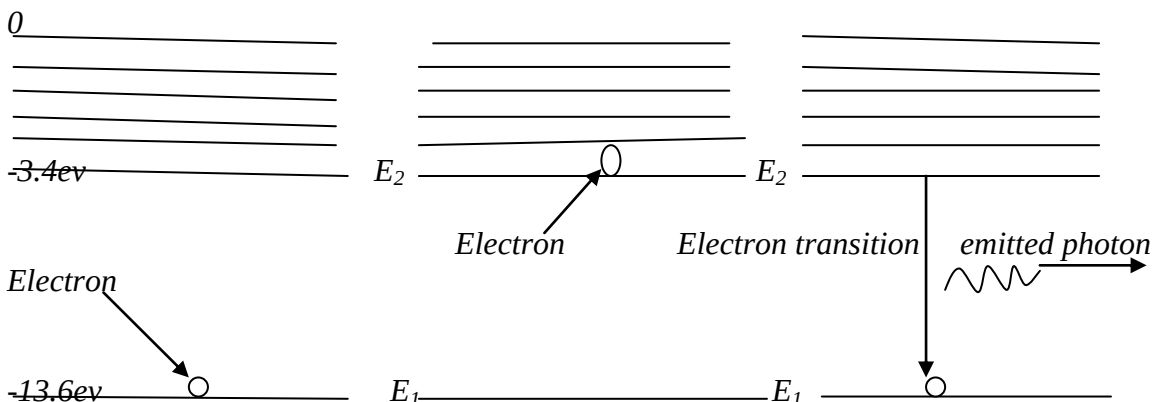
ENERGY LEVELS

Electrons in an atom can only have certain specific energies. These energies are called the electron energy levels of the atom. The energy levels may be represented as a series of lines against vertical scale of energy.



The electron in the hydrogen atom can have any of these energy values, but cannot have energies between them. The lowest energy level -13.6eV is called the ground state. Electron may absorb energy, perhaps by being heated or by collision with another

electron; it may be promoted to a higher energy level. The energy absorbed is exactly equal to the difference in energy of the two levels. The atom is described as being in excited state. An excited atom is unstable. After a short time, excited electron will return to a lower level. The electron loses energy. It does so by emitting a photon of electron radiation.



Energy of the emitted photon is given by $hf = E_2 - E_1$

$$\lambda = \frac{hc}{\Delta E}$$

The movement of an electron between energy levels is called an electron transition.

SPECTRAL LINES

CONTINUOUS SPECTRA

Whilst the light emitted by isolated atoms such as those in low pressure gases produces line spectra, the light emitted by atoms in a solid, liquid or gas **at high pressure** produces continuous spectrum.

ABSORPTION LINE SPECTRA.

Absorption line spectrum is when dark lines are shown on a bright background and part of the radiation is absorbed between the source and the observer.

When white light passes through **a low pressure gas** and the spectrum of white light is then analysed, it is found that light of certain wavelengths is missing. In their places are dark lines. As the white light passes through the gas, some electrons absorb energy and make transitions to higher energy levels. The wavelengths of the light they absorb correspond exactly to the energies needed to make the particular upward transitions.

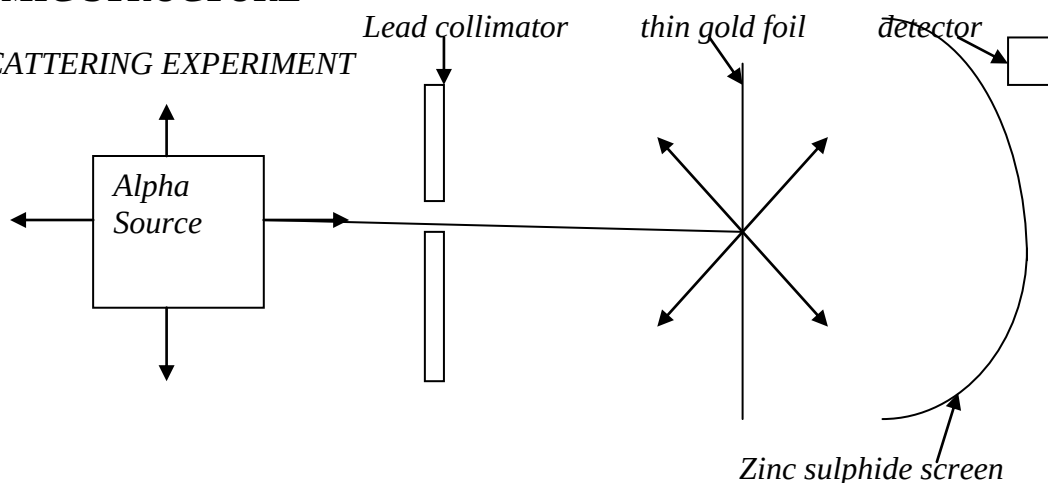
EMISSION LINE SPECTRA

Emission line spectrum is when bright lines appear on a dark background, when photons emitted by an excited gas are analysed.

When the excited electrons in absorption spectrum return to lower levels, the photons are emitted in all directions, rather than in the original direction of the white light, thus, some wavelengths appear missing. It follows that the wavelengths missing from absorption spectrum are those present in the emission spectrum of the same element.

7.3: ATOMIC STRUCTURE

ALPHA SCATTERING EXPERIMENT



Geiger and Madsen's set up was as shown in figure above. The beam of alpha particles was directed to a thin gold foil. A zinc sulphide screen which gives off a visible flash of light when struck by alpha particles was set on the other side of the foil.

OBSERVATIONS OR RESULTS FROM THE ALPHA SCATTERING EXPERIMENT

1. Most of the alpha particles pass through the gold foil undeflected.
2. A few alpha particles were slightly deflected away from the nucleus.
3. A few alpha particles were deflected at acute angles, returning along lines nearly parallel to incident path.

DEDUCTIONS OR CONCLUSIONS OF THE EXPERIMENT

1. The volume of the nucleus is negligible compared to the volume of the whole atom. An atom consists of a tiny nucleus. That is the majority of the mass of the nucleus is concentrated at the centre.
2. The volume of the nucleus is small (few particles were deflected) and it has a positive charge (because of the repulsion shown in the observation.). That is the centre (or nucleus) of an atom is charged
3. The nucleus is positive (repulsion between it and the alpha particles). The nucleus is massive. Most of the mass and density of the atom is in the nucleus (shown by how the alpha particles were repelled backwards). That is alpha particles that pass very close to the nucleus, almost striking it head on, will experience large enough repulsive forces.

THE DISTANCE OF THE CLOSEST APPROACH OF THE ALPHA PARTICLE AND THE GOLD NUCLEUS.

The kinetic energy of the alpha particle is converted to the electric potential energy. Let r be the nearest distance that the alpha particle comes to the gold nucleus.

Kinetic energy of alpha particle $E_{\alpha} = \frac{1}{2}mv^2$

Electrical potential energy of alpha particle at a distance r from the gold nucleus is

$$U = \frac{Qq}{4\pi\epsilon_0 r} \quad (q = \text{charge of alpha particle and } Q = \text{charge of gold nucleus})$$

From the principle of conservation of energy. Kinetic energy = Electrical potential energy. $U = E_{\alpha}$

$$\text{Distance } r = \frac{qQ}{4\pi\epsilon_0 E_{\alpha}}$$

NOTE $q = (2 \times 1.6 \times 10^{-19})$ and $Q = (79 \times 1.6 \times 10^{-19})$ $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$

The mass of the alpha particle is given and the speed of alpha particle.

ASSUMPTIONS ABOUT THE PATH OF THE ALPHA PARTICLE TO THE GOLD NUCLEUS.

1. The inverse square law holds for the electrostatic force between the alpha particle and the nucleus. That is $F = \frac{1}{r^2}$
2. The path of alpha particle is a straight line directed towards the centre of the gold nucleus.

MASS DEFECT

Mass defect is the difference between the mass of an atom and the total mass of the particles in the atom taken separately

Eg ${}^4_2\text{He}$. If 2 protons, 2 neutrons and 2 electrons.

$$2 \times 1.00728u = 2.01456u$$

$$2 \times 1.00867u = 2.01734u$$

$$2 \times 0.0055u = 0.0110u$$

$$\text{Total mass} = 4.03300u$$

However the mass of ${}^4_2\text{He}$ atom is 4.00260u

$$\text{Mass defect} = (4.03300u - 4.00260u) = 0.03040u.$$

The mass defect exists because the nuclear forces binding the nucleons together in the nucleus put the nucleons at a lower energy level than when the nucleons are separated.

BINDING ENERGY

Binding energy is the difference in energy between the nucleons in the nucleus and the separated nucleons. Binding energy is the energy released when an atom is formed from its constituent particles.

According to Einstein's theory of relativity, the increase in energy of the individual nucleons appears as an increase in the mass of the particles.

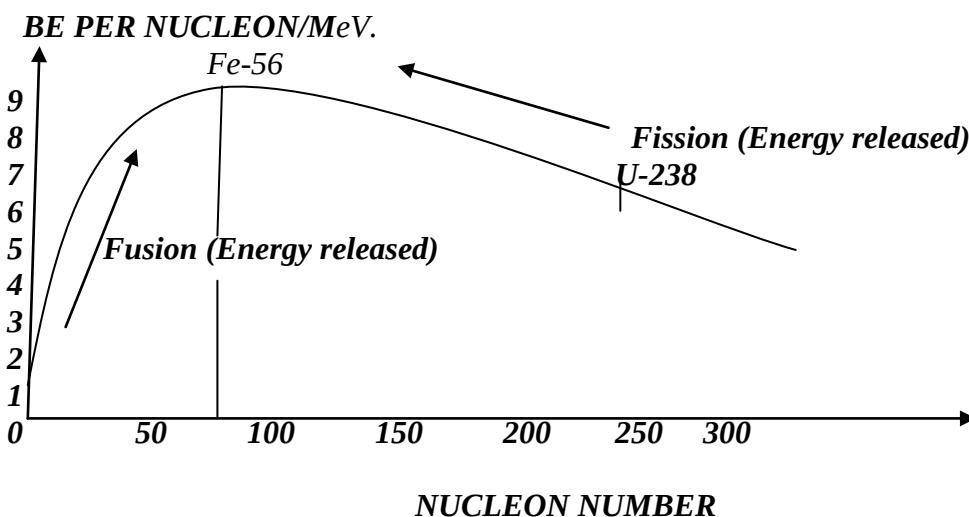
$$\text{Binding energy } (\Delta E) = \text{Mass defect } (\Delta m) \times (\text{velocity of light in a vacuum } c)^2$$

$$\Delta E = \Delta mc^2$$

Note that one unified atomic mass unit (1u) or (1m_u) = 1.66x10⁻²⁷kg.

$$\text{BINDING ENERGY PER NUCLEON} = \frac{\text{Binding energy}}{\text{Nucleon number}}$$

VARIATION OF BINDING ENERGY PER NUCLEON WITH NUCLEON NUMBER



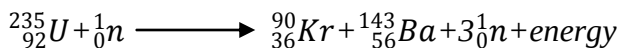
NOTE: The vertical axis is labelled binding energy per nucleon and is in MeV. The pick value is (56; 9MeV) for iron Fe-56. When you are asked to sketch this graph, make sure the axis are well labelled and the pick values shown.

When nuclei combine to form a heavier nucleus, the binding energy increases. This means that energy is released in the reaction. Similarly, when a large nucleus breaks up into smaller nuclei, the smaller nuclei have higher binding energies. Again energy is released. Nuclear fusion and fission increases the binding energy per nucleon.

NUCLEAR FISSION

Nuclear fusion is the splitting of a larger unstable nucleus to form separate and stable nuclei.

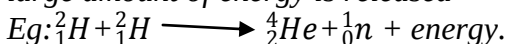
Eg: The splitting of Uranium-235



The uranium undergoes nuclear fission when bombarded by neutrons. (Neutrons are used because they are neutral, they can reach a charged nucleus). This reaction releases large amount of energy. The products of nuclear fission have more binding energy per nucleon.

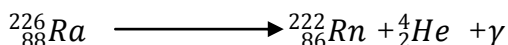
NUCLEAR FUSION

Nuclear fusion is the joining of small unstable nuclei to form large stable nucleus. Small nuclei joined together to form a larger nucleus with more binding energy per nucleon. A large amount of energy is released



For fusion to take place, the two nuclei must approach each other with sufficient speed to overcome the electric force of repulsion between them. To achieve this, the temperature of the ingredients is raised. It's difficult to carry out nuclear fusion reactions on earth because it needs high temperatures.

NUCLEAR TRANSFORMATIONS



Mass of Ra = 226.0254u

Mass of Rn = 222.0175u

Mass of He = 4.0026u

Mass defect = $[226.0254\text{u}] - [222.0175\text{u} + 4.0026\text{u}] = 0.0053\text{u}$

$$\text{Energy} = mc^2 = [0.0053 \times 1.66 \times 10^{-27}] \times [3.0 \times 10^8]^2 = 7.92 \times 10^{-13} \text{J}.$$

Changing the energy to MeV; divide the energy in joules by $[10^6 \times 1.6 \times 10^{-19}]$.

$$= \frac{7.92 \times 10^{-13}}{10^6 \times 1.6 \times 10^{-19}} = 4.95 \text{MeV}$$

OR Multiplying mass defect in unified atomic mass by 934MeV

$$= 0.0053 \text{u} \times 934 \text{MeV} = 4.95 \text{MeV}$$

This energy appears as the kinetic energy of the alpha particle and the daughter nucleus, Rn, and the energy of a photon of gamma rays.

NOTE: For a nuclear transformation to occur, there must be mass defect. The greater the mass defect, the more likely the reaction will occur. Part of the energy equivalent of mass defect appears as the kinetic energies of the daughter nucleus and the emitted particles.

THE FOLLOWING ARE CONSERVED AFTER A NUCLEAR REACTION

- 1.Proton number or (charge)
- 2.Nucleon number or mass number.
- 3.Mass-energy.

NOTE: Mass is not conserved and energy is not conserved but mass-energy is conserved.

7.4: RADIOACTIVITY

Is the random and spontaneous disintegration of unstable nucleus.

RANDOM.

There is no pattern to the decay and so we cannot predict accurately when or how much of the substance would have decayed in a given period of time. This random nature explains the fluctuations observed in radiation meters during experiments.

SPONTANEOUS

It occurs without cause. There is no reason of emission. It is not affected by physical conditions such as temperature or pressure.

BACKGROUND RADIATION

Background radiation is the Nuclear radiation always present in the environment and atmosphere due to cosmic radiation or radiation from the sun, space, earth or rocks.

Sometimes when a radiation counter called a Geiger- Muller tube (GM) is used, it detects a reading even in the absence of a radioactive sample in the area. This is due to the presence of background radiation. This occurs naturally. It is because of the emissions of plants, animals and other organisms in our environment.

SOURCES OF BACKGROUND RADIATION

- 1.Cosmic rays and microwave radiation from the universe.
- 2.Rocks
- 3.Decaying plant and animal matter
- 4.Radon gas
- 5.Nuclear fallout.

RADIATION HAZARDS

- 1.Dangerous of radioactive wastes.

2. Alpha radiation causes living cells to mutate and possibly cause cancer. Alpha also cause burns on the skin.

3. Beta and gamma rays emission can cause damage to internal organs.

SAFETY PRECAUTIONS WHEN HANDLING RADIOACTIVE MATERIALS.

1. Keeping radiative material in lead box under lock and key.

2. Using radiation counters to show the level of radioactivity for workers

3. Wearing protective clothing. Lead-lined clothing that protects the body from radiation and also notifies the level of exposure.

4. People working with ionisation in industry should wear photographic film badges, which records the amount of radiation a person has been exposed to.

ACTIVITY (A)

Is the decay rate.

$$A = \lambda N$$

DECAY CONSTANT (λ)

Is the probability of decay per unit time.

Rate of decay, $-\frac{dN}{dt}$ of a radioactive sample, is directly proportional to N , the number of radioactive nuclei present.

$$-\frac{dN}{dt} \propto N$$

$$\frac{dN}{dt} = -\lambda N$$

Intergrating from $t=0$, when $N=N_0$, $t_0=t$ when $N=N$

$$\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt$$

$$\ln N \Big|_{N_0}^N = -\lambda t \Big|_0^t$$

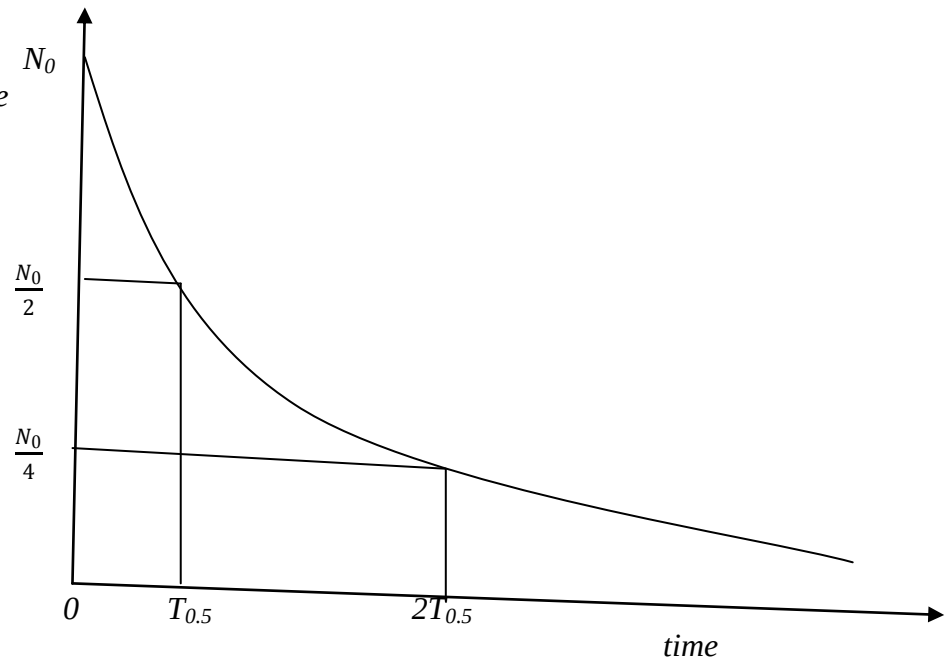
$$\ln \frac{N}{N_0} = -\lambda t$$

$$N = N_0 e^{-\lambda t}$$

Since $A = \lambda N$

Therefore $A = A_0 e^{-\lambda t}$

Number of radioactive
Nuclei in sample



$t_{\frac{1}{2}}$ = Half life.

Is the time taken by a sample of radioactive nuclide to decay to half its initial number of radioactive nuclei; OR

It is the time taken for a radioactive material to decay to half its original number of atoms. OR

It is the time taken for half the number of atoms in a given sample to decay.

Now $N = N_0 e^{-\lambda t}$, But for half life $t = t_{\frac{1}{2}}$ and $N = \frac{N_0}{2}$

$$\text{Then } \frac{N_0}{2} = N_0 e^{-\lambda t_{\frac{1}{2}}}$$

$$\frac{1}{2} = e^{-\lambda t_{\frac{1}{2}}}$$

$$\ln\left(\frac{1}{2}\right) = -\lambda t_{\frac{1}{2}}$$

$$\ln 1 - \ln 2 = -\lambda t_{\frac{1}{2}}$$

$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

USES OF RADIOACTIVITY

1. Biochemical tracers
2. Detecting leaks in underground pipes.

3. Radiography
4. Power generation (Uranium)
5. Determination of thickness of materials.
6. Carbon -14 dating
7. Sterilisation

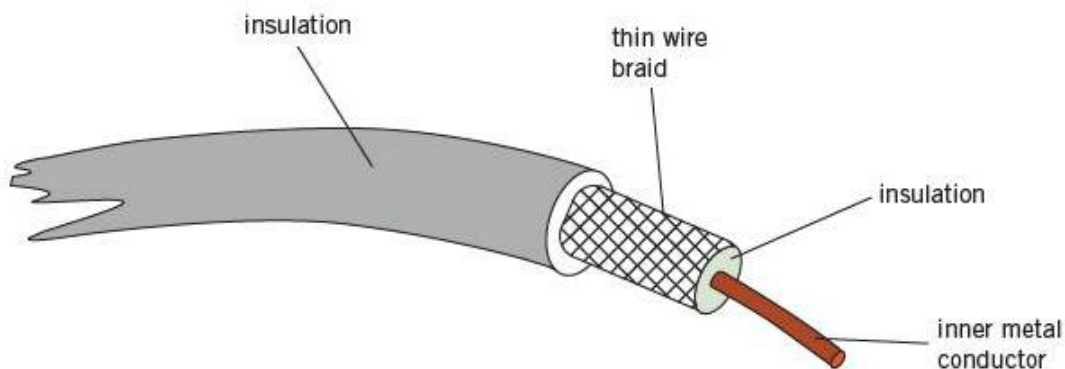
7.5: TELECOMMUNICATIONS

For communication to take place, information is carried from one place to another. This information is carried in the form of signal which may be for example, changes in the frequency or amplitude of a wave or changes in the intensity of light. The signal may be transferred in various ways using different channels of communication including wire-pairs, coaxial cables, radio and micro wave links, and optic fibres.

WIRE PAIRS

Provide a very simple link between a transmitter of information and the receiver. Wire pairs are used mainly for short distance communication at relatively low frequency for eg linking telephone to the nearest exchange.

COAXIAL CABLES



Coaxial cables, consists of two wire conductors. Electrical signal is transmitted along a central inner conductor that is covered by an insulator. The second conductor is in the form of thin wire braid that completely surrounds the insulator. This wire braid acts as the return for the signal and is earthed. The braid is covered by a protective layer of insulation. The earthed outer braiding shields the inner conductor from external interference. eg, other stray electrical signals.

Signals in coaxial cables are used for eg to connect an aerial to television receiver.

RADIO WAVES

Alternating current in the wire acts as an aerial. Energy is radiated from an aerial as electromagnetic radiations. These waves travel outside with the speed of light. E.M waves in the frequency 30Hz to 3GHz are generally referred to as radio waves. Radio waves improved from low frequency, high wavelength to high frequency low wavelength. This enabled higher quality communication and also more radio stations to operate in the same area. The intensity of the radio waves will be reduced (attenuated) as the distance from the transmitter increases.

MICROWAVE

Microwave is E.M waves that can be considered alongside other radio waves. They are in the frequency range 3GHz to 30GHz.

OPTIC FIBRE

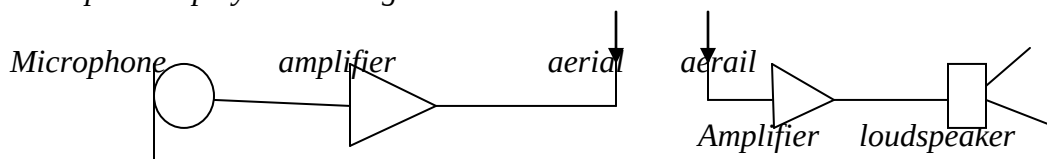
Optic fibre consists of a fine strand of very pure glass, surrounded by protective covering. The glass fibre itself is thinner than a hair. Pulses of light or infra-red radiation travel along the fibre as a result of total internal reflection. These pulses carry digital information along the fibre.

MODULATION

Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal, with a modulating signal that typically contains information to be transmitted.

Modulation is when a high frequency wave known as the carrier wave has either its amplitude or frequency altered by information signal in order to carry information.

Note: Communication of speech could be achieved using radio waves. The signal from the microphone would be amplified and applied to an aerial. The radio waves produced would then be picked up by a receiving aerial.



This simple system has two disadvantages

1. Only one system could operate in an area because the receiving aerial would not be selective, it picks up any signal.
2. The aerial required for the transmission of low frequency of sound waves (about 20Hz to 15Hz) would be very long and the electrical power required for transmission over long distance would be very large.

These problems are solved by the process known as modulation.

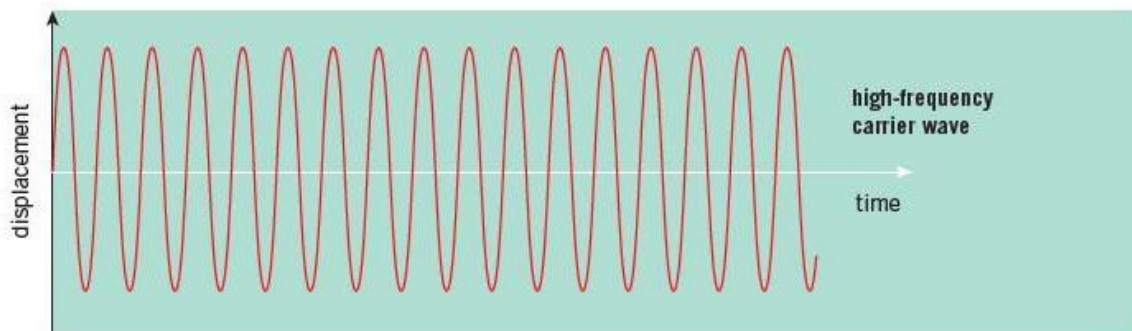
In modulation, a high frequency wave, known as the carrier wave, is transmitted. This high frequency carrier wave has either its amplitude varied, or its frequency varied as to carry information.

AMPLITUDE MODULATION

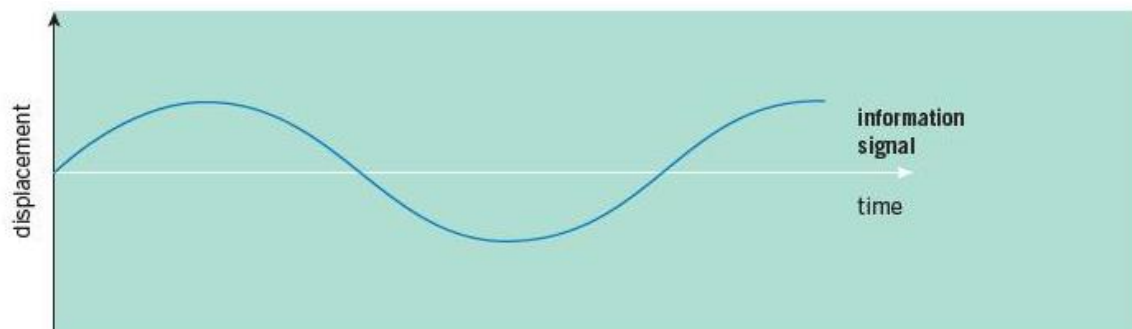
The carrier wave has constant frequency. The amplitude of the carrier wave is made to vary. These variations are in synchrony with the displacement of the information signal.

High frequency carrier wave

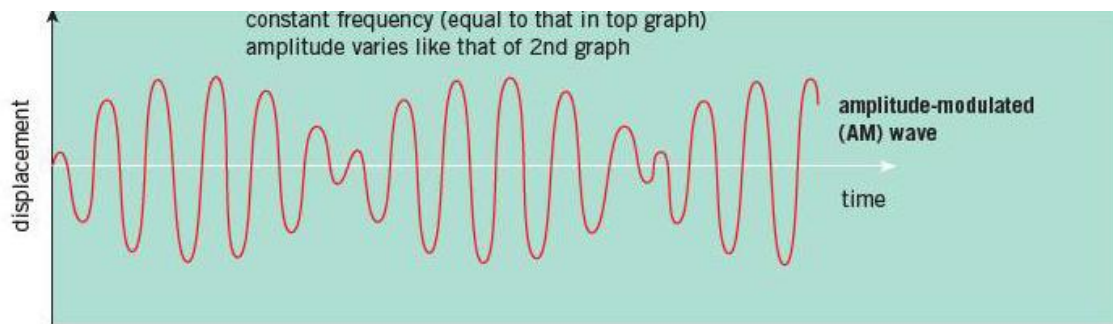
Displacement



Information signal



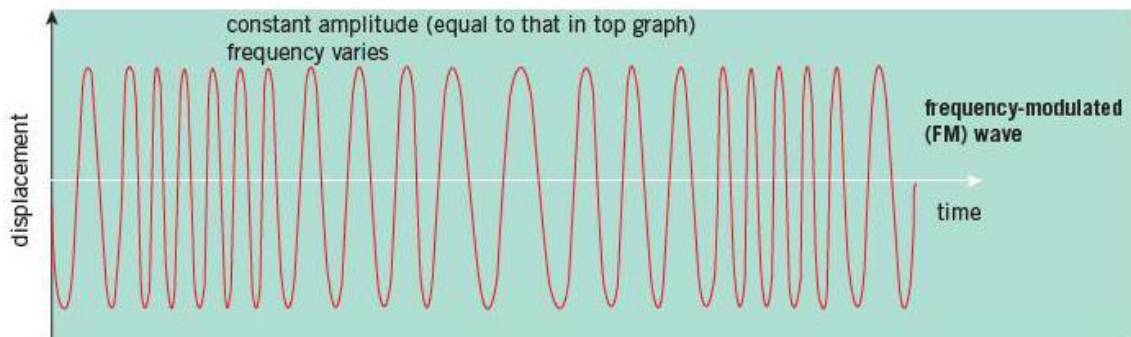
AMPLITUDE MODULATED WAVE



FREQUENCY MODULATION

The amplitude of the carrier wave remains constant. The frequency of the carrier wave is made to vary in synchrony with the displacement of the information signal.

A frequency modulated wave



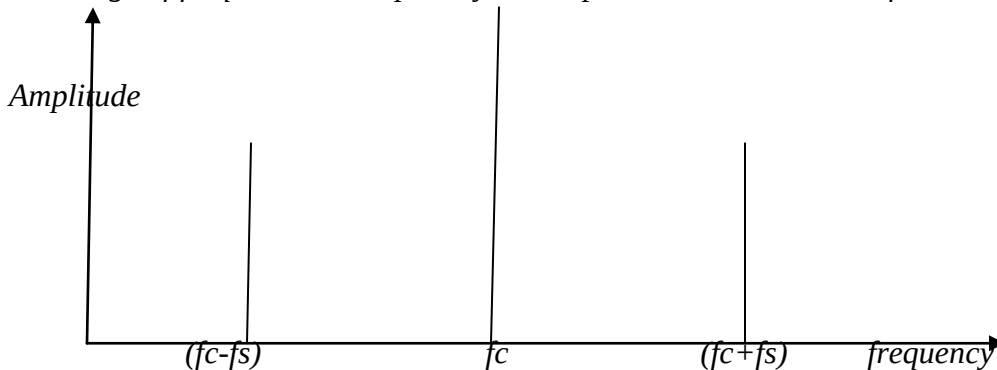
Note: Information signal varies the frequency of the carrier, which increases if the signal is positive and decreases when the signal is negative.

The use of a carrier wave allows different radio stations in the same area to transmit at the same time. Each radio has a different carrier wave frequency. The receiver is adjusted, or tuned, to the frequency of whichever transmitter is desired. The receiver accepts the signal transmitted on that particular carrier wave and rejects other carrier-wave frequencies.

BANDWIDTH

This is the range of frequencies a signal occupies.

The range of frequencies occupied by the amplitude modulated waveform.



$$\text{Bandwidth} = 2f_s$$

The carrier wave frequency f_c , that has been amplitude modulated by a signal having only one frequency f_s . If this waveform is analysed, it is found to be made up of three frequencies f_c , $(f_c - f_s)$ and $(f_c + f_s)$. The central frequency f_c is the frequency of the carrier wave and it has the largest amplitude. The other two frequencies, $(f_c + f_s)$ and $(f_c - f_s)$, are known as the side bands frequencies, f_s is the frequency of the information signal.

RADIO WAVE FREQUENCIES

E.M waves in frequency range 30 kHz to 3GHz are generally referred to as radio waves.

NAME OF RADIO WAVE	FREQUENCY RANGE	DISTANCE
Space wave	Greater than 30MHz	Line of sight between the transmitter and receiver, plus satellite communication.
Sky wave	3MHz to 30MHz	Worldwide, as a result of multiple reflections from the ionosphere and the ground or sea
Surface wave	Below 3MHz	Up to 1000km

AMPLITUDE MODULATION**ADVANTAGES**

1. Greater area covered by one transmitter.
2. Cheaper radio sets.
3. Smaller bandwidth so more stations available in frequency range

DISADVANTAGES

1. Requires a high power transmitter.
2. More electrical noise and interference.

FREQUENCY MODULATION

ADVANTAGES

1. Less electrical noise and interference.
2. Greater bandwidth produces better quality sound.

DISADVANTAGES

1. Shorter range
2. More complex circuitry
3. More expensive

ANALOUGE AND DIGITAL SIGNALS

ANALOGUE SIGNAL

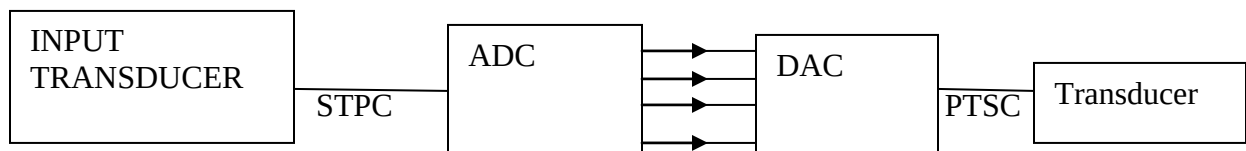
An information signal that has the same variations with time as the information itself is known as analogue signal. Much of the information that we wish to transmit and communicate is analogue in nature (speech, music, television pictures). When any signal is transmitted over a long distance, it will pick up noise.

DIGITAL SIGNAL

Digital signal is a signal that consists of a series of high and lows with no values between the highs and lows. The signal will still suffer from noise and attenuation. However, on amplification, the noisy can be reshaped or regenerated to return to its original form. Such amplifiers are called regenerator amplifiers. The amplifiers filter out noise. Unlike an analogue signal, a digital signal can be transmitted over a long distance with regular regenerations without the signal becoming degraded. Modern digital electronic circuits are, in general, more reliable and cheaper to produce than analogue circuits. An additional advantage of digital system is that extra information, or data can be added to the transmissions. These extra data are a code for the receiving system so that the corrected before the signal is finally reproduced.

THE TRANSMISSION OF A SIGNAL

In an analogue signal, such as speech, or music, the generated voltage signal varies continuously. For digital transmission, this analogue signal must be converted into a digital signal. This is achieved using an analogue to digital converter. (ADC).



ADC-Analogue to Digital converter.

DAC-Digital to Analogue Converter

STPC-Serial to Parallel Converter

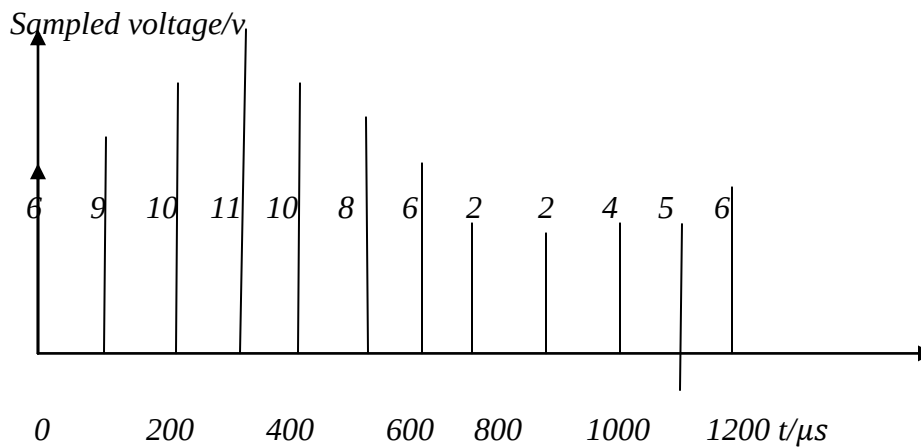
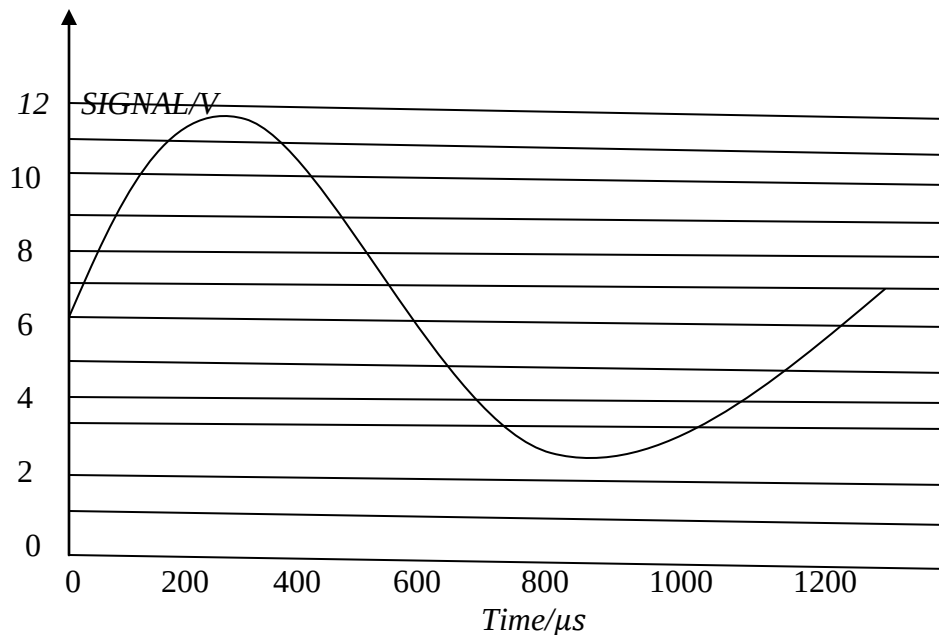
PTSC-Parallel to Serial Converter

NB Serial transmission is often used because it is less expensive because only one cable is used. It can therefore be used for long distance transmission

In an analogue to digital converter, the analogue voltage is sampled at regular intervals of time, at what is known as the sampling frequency or sampling rate. Sampling time is the time chosen to convert a particular voltage levels to digital pulse.

The value of the sample voltage measured at each sampling time is converted into digital (binary) number that represents the voltage value.

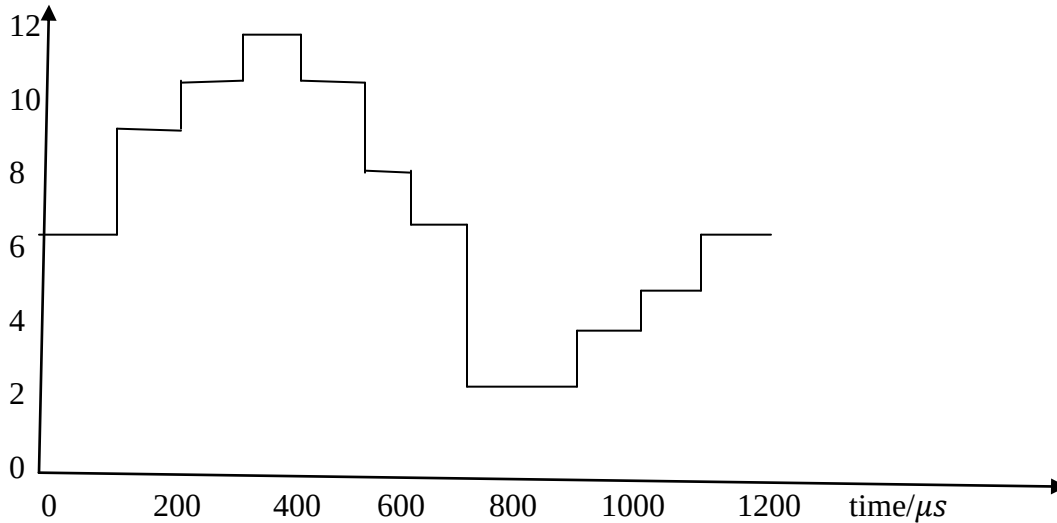
NOTE: When sampling, the number representing the sample would be the **whole number below the actual value of the sampled voltage**



Digital signal

0110 1011 1010 1011 1010 1000 0110 0010 0010 0100 0101 0110

RECEIVED ANALOUGE SIGNAL



These sample voltages are converted into a digital signal by analogue to digital converter (ADC). After this digital signal had been transmitted, it is converted back into an analogue signal using a digital to analogue converted (DAC).

DAC

The recovered signal consists of large steps. The size of these steps, and hence the faithfulness of the reproduction of the initial analogue signal can be improved by using more voltage levels and sampling at a higher frequency.

CHANNELS OF COMMUNICATIONS.

WIRE PAIRS

Wires pairs are used mainly for short distance communication.

Cause high attenuation of signal

Easily pick up noise.

Suffer from cross-talk and are of low security

Have limited bandwidth of about 500 kHz.

NB In the case of an electrical signal in a metal wire, signal power is lost as heating in the wire.

COAXIAL CABLES {Compared to wire pairs}

Has a larger bandwidth of about 50MHz. Much information can be carried along a coaxial cable. Is more costly. Causes less attenuation. Is less noisy and more secure.

RADIO AND MICROWAVE LINKS

Has a large bandwidth of about 1GHz. This large bandwidth means that the microwaves beam has a large capacity for transmitting information. These are electromagnetic waves with a wide range of frequencies.

OPTIC FIBRE

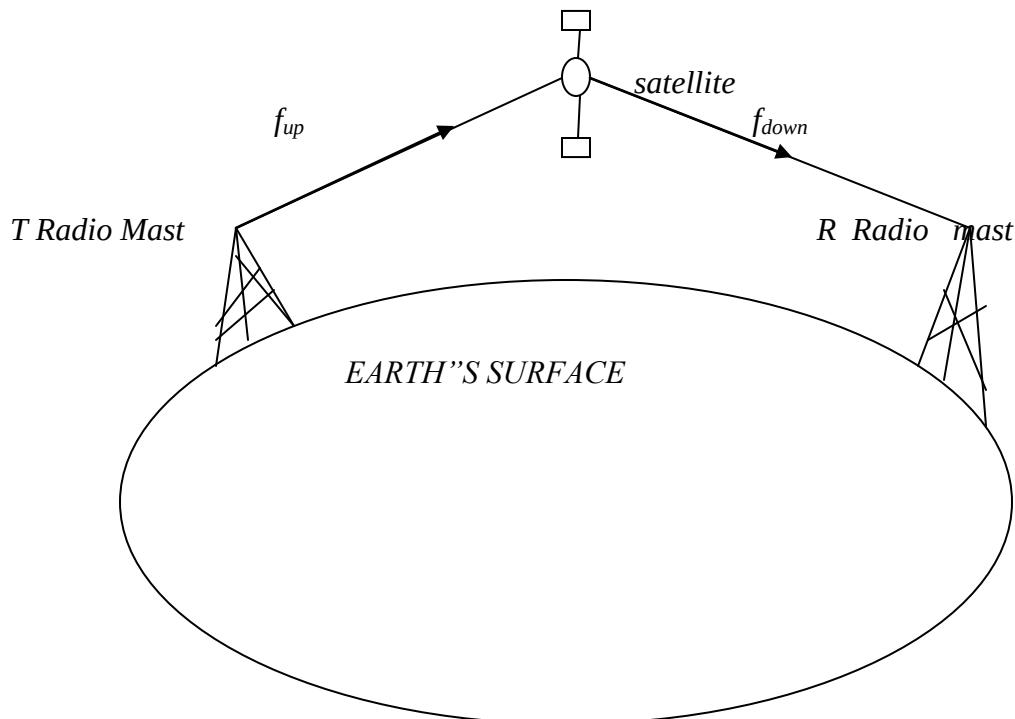
Advantages refer notes from oscillations and waves.

NB In optic fibres, light power is lost as a result of absorption in impurities in the glass and also scattering due to imperfections.

A beam of electromagnetic waves travelling through air also loses power as a result of absorption and scattering.

COMMUNICATION SATELITES

Satellite communication enables more wavebands to be made available and at much higher frequencies, thus giving rise to a much greater data carrying capacity.



A carrier wave f_{up} is sent from a transmitter on earth to a satellite. The satellite receives the greatly attenuated signal. The signal is amplified and the carrier frequency is changed to a lower value f_{down} . The carrier wave is then directed back to receiver R on Earth. The carrier wave frequency f_{up} and f_{down} are different so that the very low power signal received from earth is not swamped by (can be distinguished from) the high power signal that is transmitted back to earth. The value of the uplink frequency f_{up} and the downlink frequency f_{down} might be 6GHz and 4GHz respectively (the 6/4GHz band). Other bands are 14/11 GHz and 30/20GHz.

GEOSTATIONARY SATELLITE

Orbits the earth with a period of 24hrs at a height of $3.6 \times 10^4 \text{ km}$ above the earth's surface.

Orbits in the same direction as the rotation of the earth (from west to east) and orbit above the equator.

ADVANTAGES

1. Remains in fixed position above point on equator so no need to move dish
2. Can be used for continuous communication.
3. Continuously monitors climate change.
4. Easy to track down the satellite
5. Allows international television broadcasting to be possible, allowing live events in one country to be viewed in another.

DISADVANTAGES

1. Communication in polar regions is not possible because geostationary satellites are in equatorial orbits, satellite will not be in line of sight.
2. Another problem results because of the height of the orbit. Between transmission and receipt of signal (there is time delay). However to avoid the time delays, geostationary satellites may be used in conjunction with optic fibres.

POLAR SATELLITES

Polar satellites are satellites that have low orbits and pass over poles. The orbit period of polar satellites is about 100m. Since the earth rotates below such satellites, then in any 24hrs the satellite will pass over every region of the earth's surface.

ADVANTAGES

1. Their orbital height is only of the order of 10^5 m (a few hundred km) and thus delays in telephone conversations are not noticed.
2. since polar satellites pass over the whole of the earth in any 24hr period, they are used for remote sensing. Such uses include military, geological prospecting and weather forecasting.
3. At smaller height above the earth and can detect objects of smaller detail.

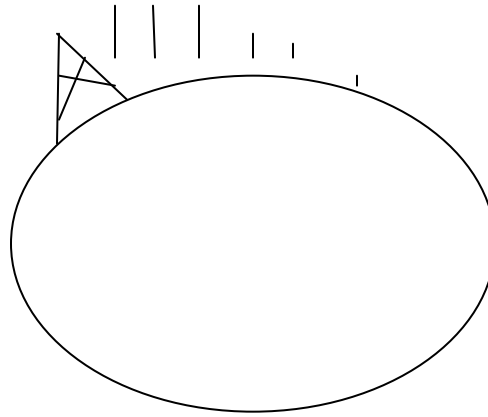
DISADVANTAGES

1. Continuous communication with a single polar satellite is not possible.
2. Not always in the same positions relative to earth so dishes must be moved.
3. Difficult to track down the satellite.

SURFACE WAVES (Frequency below about 3MHz)

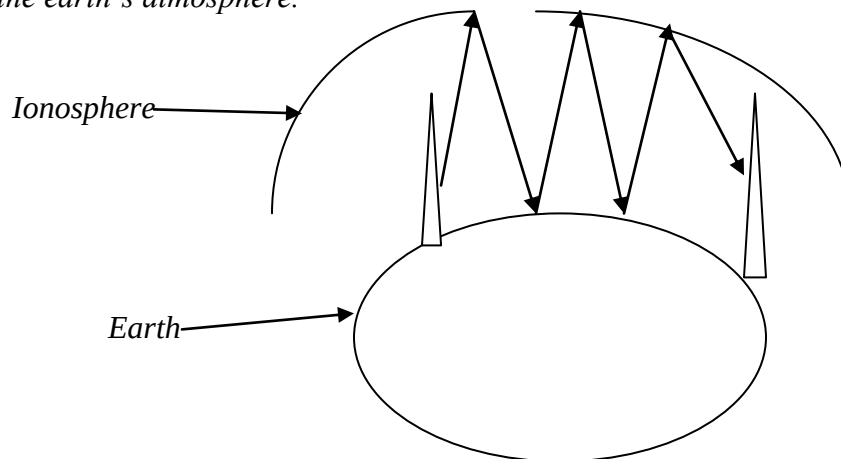
Travels close to the surface of the earth and diffract around it due to long wavelength.

A vertically polarised wave travelling near the Earth's surface will be affected by the conductivity of the earth. This causes the wave to tilt forward and, in combination with some diffraction, increases the effective range well beyond the horizon. Maximum distance for this method of propagation is 1000km.



SKY WAVE (Frequency 3MHz to 30MHz)

Travel in the atmosphere in the atmosphere in straight line, reflecting back and forth between the ionosphere and the earth's surface hence can go a long distance. Sky waves are transmitted and they are multiple reflected by both the earth's surface and the ionosphere. Ionosphere is essentially a complex series of layers of charged particles high up in the earth's atmosphere.



ATTENUATION

Attenuation is the loss of power of a signal when the signal travels through a cable. Amount of attenuation increases as the distance that the signal travels increase. In the case of an electrical in a metal wire, signal power is lost as heating of the wire.

In optic fibres, light power is lost as a result of absorption in impurities in the glass and also scattering due to imperfections. A beam of electromagnetic waves travelling in air also loses power as a result of absorption and scattering. In order that a signal may be detected the power of the signal must be a minimum number of times greater than the noise power. This signal to noise ratio, may be for example 50 to 100. Signals are

amplified and the output of the amplifier is a certain number of times greater than the input. The amplifier gain could be 10^5 . It can be seen that the ratio of the two powers may be very large. Consequently an extremely convenient unit by which power levels (or, in fact, any other quantities may be compared is the bell (B). In practice, these ratios are usually expressed in decibels (dB)

Where 10 decibels (dB) = 1 bel (B). The number of decibels is related to the ratio of two powers P_1 and P_2 by the expression.

$$\text{Number of decibels} = 10 \log \left(\frac{P_1}{P_2} \right)$$

Note. For attenuation P_1 is power in and P_2 is power out $P_1 > P_2$. For amplification P_1 is power out and P_2 is power in and $P_1 > P_2$

$$\text{Attenuation in line} = 10 \log \left(\frac{P_1}{P_2} \right)$$

$$\text{Attenuation per unit length} = \frac{1}{L} 10 \log \left(\frac{P_1}{P_2} \right)$$

EXAMPLE

The gain of an amplifier is 45dB. Calculate the output power of the amplifier for an input power of $2.0 \times 10^{-6} \text{ W}$

SOLUTION

$$\text{Number of (dB)} = 10 \log \left(\frac{P_1}{P_2} \right)$$

$$45 = 10 \log \left(\frac{P_{\text{out}}}{2 \times 10^{-6}} \right)$$

$$P_{\text{out}} = 6.3 \times 10^{-2} \text{ W}$$

EXAMPLE

The input power to a cable of length 25km is 500mW. The attenuation per unit length of the cable is 2 dB km^{-1} . Calculate the power of the signal which emerges from out end of the cable.

$$\text{Total signal loss in cable} = 2 \times 25 \text{ dB} = 50 \text{ dB}$$

$$50 = 10 \log \left(\frac{500 \times 10^{-3}}{P_{\text{out}}} \right)$$

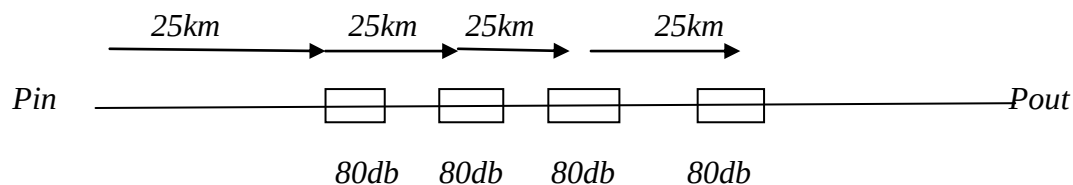
$$P_{\text{out}} = 5 \times 10^{-6} \text{ W}$$

REPEATER GAIN

The power gain of an amplifier is defined to be the ratio of the power output to the power input. When there are several repeater amplifiers along a transmission line. In order to calculate the total gain of a number of amplifiers, it is only necessary to add the gain in (dB)

EXAMPLE

A cable of length 100km has an attenuation of 3dBkm^{-1} . Every 25km along its length, there is a repeater amplifier with a gain of 80decibels, as shown below. Calculate the overall gain or loss in this system.



Total amplifier gain = $4 \times 80 = 320\text{db}$

Total cable loss = $100 \times 3 = 300\text{db}$

Overall gain in the system = $+320 - 300 = +20\text{db}$

MAXIMUM UNINTERRUPTED LENGTH IN CABLES

Total signal loss = $10\text{Log} \left[\frac{\text{Power input}}{\text{power output}} \right]$

Minimum to noise ratio = $10\text{Log} \left[\frac{\text{Minimum effective signal power in cable}}{\text{Background noise or Average noise power in cables}} \right]$
 $= 10\text{Log} \left[\frac{P_{\min}}{P_{\text{noise}}} \right]$

Maximum loss in cables/Total attenuation of signal to reach minimum = $10\text{Log} \left[\frac{\text{Input power}}{\text{Minimum power}} \right] = 10\text{Log} \left[\frac{P_{\text{input}}}{P_{\text{minimum}}} \right]$

Maximum uninterrupted

length = $\frac{\text{Total attenuation of signal to reach minimum (maximum loss in cables)}}{\text{Attenuation per unit length}}$

Example

The signal input to an optic fibre is 7.0mW. The average noise power in the fibre is $5.5 \times 10^{-19}\text{W}$ and the signal-to-noise ratio must not fall below 24dB. The fibre has an attenuation of 1.8dBkm^{-1} . Calculate

- The minimum effective signal power in the optic fibre.
- The maximum uninterrupted length of the optic fibre through which the signal can be transmitted.

Solution

(a) Number of decibels = $10\text{Log} \left[\frac{P_2}{P_1} \right]$; $24 = 10\text{Log} \left[\frac{P_{\min}}{5.5 \times 10^{-19}} \right]$; $P_{\min} = 1.38 \times 10^{-16}\text{W}$

(b) Total attenuation of the signal to reach

$$\text{minimum} = 10 \log \left[\frac{P_{\text{input}}}{P_{\text{min}}} \right] = 10 \log \left[\frac{7 \times 10^{-3}}{1.38 \times 10^{-16}} \right] = 137 \text{ Db}$$

$$\text{Maximum uninterrupted length} = \frac{137}{1.8} = 76 \text{ km.}$$